The Standard Model from a Cubic Lattice

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Abstract

Observer geometry and physical carrier. SM §4.1 distinguishes two claims. First, finite equivariant projection is classical and exact: cubic symmetry is inherited by the projected Markov part and every memory kernel, forbidding quadratic anisotropy in the scalar symbol — so whatever nondegenerate quadratic structure survives is isotropic — while allowing cubic anisotropy at quartic order. The particular normalized nearest-neighbor operator $A/(2d)$ additionally requires an **observer-level** center-free kernel; microscopic center independence alone does not imply that after correlated hidden variables are marginalized. Second, the six signed simple-cubic links and their exact $3 \oplus 2 \oplus 1$ representation are unconditional geometry, but identifying that link space with the complete physical carrier is **H-link**; the single-copy clause gives $K = 6$, and the custodial condensate-stabilizer reading is **H-cust**; the component-blind reading of the observer's trace-out — that the observation commutes with signed permutations of the six carrier components — is **H-blind**; at the symmetric point it holds for the branch's observer class, the component-complete site observer whose visible variable is the six-component field on its sites, and it is carried as a named condition for other observer classes and for extensions of the rule that couple components through a hidden sector (§4.4). Accordingly, any downstream statement here that consumes $K = 6$, $N_f = 6$, the physical SM gauge carrier, or a fixed three-sector count inherits those named bridge conditions unless it is explicitly referring only to the six geometric links.

The Standard Model arises as a distinguished low-freedom representation of a finite bijection on a $d = 3$ simple-cubic lattice governed by the lattice wave equation. Two qualifications are stated at the outset because they bound every claim below. First, the cubic lattice is not itself derived: among the fourteen Bravais lattices, simple cubic is selected as the one compatible with an observed SU(3) factor (only the cubic crystal system supplies a three-dimensional point-group irrep; §4.6, Theorem 7b), so the lattice is an empirically-anchored structural commitment, forced by the observed gauge group rather than by the substratum alone. Second, $d = 3$ is fixed by empirical filters (propagating gravity, stable matter, boundary-entropy concordance; §3.2), not derived from the construction. *Given* those commitments, an extensive amount follows by representation theory with no further freedom, and that is this paper's content: the gauge group SU(3) × SU(2) × U(1), three generations (physical reading under H-spin', §4.7; chirality conditional on H-χ' in its sharpened form (the

minimal embedding certified vector-like), §4.8), the Higgs as a composite scalar in the singlet staggered taste, anomaly cancellation with the observed hypercharges, and — conditional on H-spin' for the physical reading — a candidate generation count of three from the link count under H-link jointly with the condensate-stabilizer accounting of Theorems 5 and 7 (what is unconditional is the $1 \oplus 3$ decomposition: one singlet plus one symmetry-related triplet of candidate matter sectors) — each step a stated theorem on the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$, with the named open inputs flagged where they enter (the chirality hypothesis H-χ', §4.8; the physical-spin identification H-spin', §4.7 Remark; the carrier identification, §4.7 Remark). The foundational result — every finite visible-sector law admits a universal fixed-basis unitary quantum representation ($S \iff D \iff Q_{\text{fb}}$), with the C1/C3/C4 conditions diagnosing the memory-bearing sector and the coherent operational lift still to be established — is established in a companion paper [Main]; the representation is internal (the realization's own permutation unitary), the imported Barandes correspondence retained for compression and interpretation, not as the gate to existence. The nearest-neighbor **reference** lattice is Bell-local under measurement independence for spacelike-separated local settings and readouts; the state-dependent graph $G(x)$ supplies a native preparation-indexed route to ontic parameter dependence, while [Substratum §3.3] carries H-Bell as the open requirement that such prepared long-edge graphs preserve the metric/Ollivier–Ricci continuum limit used by the gravitational reconstruction. The gauge coupling strengths follow quantitatively: the structural induced coupling $1/\alpha_0 = 23.25$ (computed analytically from the one-loop staggered vacuum polarization), combined with a universal threshold $\delta_0 = 10.0$ (determined by the U(1) row; a 2-loop staggered VP rebuild indicates δ_0 is not reproduced by strict 2-loop PT and is accordingly treated as empirical input — see §6.2) and a C_2-dependent threshold parameterized by a logarithmic resummation, reproduces all three SM couplings at M_Z via standard SM renormalization group running. The structural predictions are the specific value $1/\alpha_0 = 23.25$, the universality of δ_0 across gauge groups, and the analytic 1-loop computation of the $A \cdot B$ coefficient, consistent with the fit under the $1/N^2$ finite-size assumption and inconclusive across finite-size models (§6.2.1). Deriving the C_2-dependent threshold parameters from first-principles lattice perturbation theory is an open computational task that would upgrade the SU(2) and SU(3) couplings from retrodictions to strict predictions. Quantitative predictions for SM observables include the Cabibbo angle $\lambda = 1/(\pi\sqrt{2})$ to 0.04%, the Wolfenstein $A = \sqrt{2/3}$ to 0.23σ, the Koide angle $\theta_0 = 2/9$ to 0.02%, all three PMNS angles within 1.1σ — with $\sin^2\theta_{12}$ confirmed by the post-JUNO global fit at 0.07σ — six fermion masses from a single empirical input to better than 1%, and the top mass matching the weak-scale relation $m_t = v/\sqrt{2}$ ($y_t(m_t) \approx 1$) to 0.9% — a retrodiction whose fixed-point mechanism is open: one-loop running shows $y_t(M_{\text{Pl}}) = 1$ yields $m_t \approx 207$ GeV, so the weak-scale coincidence is not delivered by the compositeness-scale normalization (§7.2). These predictions are forced by the cubic-lattice geometry and contain no parameters fit to the predicted quantities; they are best understood as a tight compression of the Standard Model onto the cubic-lattice commitment — a claim about the non-independence of the SM's structural facts — rather than as a derivation of the Standard Model from first principles (see §8.3, *Derivation status: the three tiers*).

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1. Introduction

The Standard Model contains approximately nineteen free parameters: three gauge couplings, nine fermion masses, four CKM angles, two Higgs parameters, and the QCD vacuum angle. None of these are derived from a deeper principle within the SM itself, and despite five decades of effort, no successful explanation of their values has emerged from grand unification, supersymmetry, string compactifications, or other extensions. This paper presents a derivation of the Standard Model that fixes the gauge group, the matter content, the discrete symmetries, and the gauge coupling values from a single structural input — a finite bijection on a $d = 3$ cubic lattice — with the primary chain a stated sequence of theorems — the named open inputs (H-χ', §4.8; H-spin', §4.7; the carrier, §4.7) flagged where they enter — and the quantitative predictions matching observation with no parameters fit to the predicted quantities.

Prerequisites. This paper treats as established the following result from a companion work [Main]: an observer embedded in a deterministic bijection on a finite configuration space, coupled to a hidden sector that is non-trivial (C1: hidden influence at some accessible step), record-persistent (C2 — the slow horizon $\tau_S \ll \tau_B$ our universe's realization), of sufficient memory capacity (C3: $\log_2 |\mathcal{C}_H| \geq I^*$), and read back (C4: history-sensitive hidden mediation — in the realization, the interaction reads what it writes), necessarily sees non-Markovian visible dynamics on accessible timescales, with the unitary quantum representation attaching to processes in the correspondence's class and the fixed-$\widehat{H}$ form constructive for realized processes. The present paper consumes only the emergent quantum description and its Born-rule statistics.

The present paper reproves none of [Main]'s foundational results. Where a theorem below invokes a specific input — the P-indivisibility of reduced dynamics on finite bijections (used in §2.1), the Born rule at the ancilla-marginal level, whose trivial-ancilla form is $T_{ij}(t) = |U_{ij}(t)|^2$ (used in §5's detailed-balance chain), or the C2-necessity argument (used in §8.2's genericity theorem) — the invocation is stated as a known result and the reader is directed to [Main] for the proof. The SM-derivation chain developed here depends on these inputs but is otherwise self-contained.

This paper develops a cubic-lattice route toward the Standard Model, but the derivation is now explicitly layered. The exact finite-observer result added in §4.1 is independent of QM: equivariant projection preserves cubic symmetry through every memory kernel, and cubic symmetry forbids quadratic anisotropy in the scalar symbol. The stronger normalized nearest-neighbor operator follows only with observer-level center-freedom. Separately, the six signed cubic links decompose exactly as $3 \oplus 2 \oplus 1$ (Theorem 7), but the identification of that geometric link space with a single physical gauge carrier is the named bridge **H-link**, and the custodial condensate-stabilizer reading is **H-cust**. Consequently the Standard-Model gauge/matter chain, generation count, and any numerical result using $N_f = 6$ inherit H-link (and where relevant H-cust, H-spin', or H-χ'). The paper retains those conditional calculations because they are sharp tests of the branch. Coupling-degree minimization is not a proof that the physical carrier must have $K = 6$.

The status of these results admits a useful three-tier classification.

Tier 1 — Structural foundations. The minimal mathematical object is established (a finite set with a deterministic bijection of bounded coupling degree, partitioned into visible and hidden sectors), and the spatial dimension $d = 3$ is shown to be the unique self-consistent dimension by three independent filters: propagating gravity ($d \geq 3$), stable matter ($d \leq 3$), and the boundary entropy concordance $\rho_s/\rho_{\text{crit}} = 2/(d-1)$ (which equals unity only for $d = 3$; its coefficient inherits the horizon conditions of [GR §2, §8.5] through $\epsilon = 2\,l_P$). Renormalizability of the emergent Yang-Mills coupling at $d = 3$ is a downstream consequence rather than an independent filter (§3.2). The structural object is a theorem about the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$, and the $d = 3$ selection holds at self-consistency strength (§3.2; Tier 2 in §8.3); neither depends on which specific dynamics is selected.

Tier 2 — Conditional physical-carrier chain. The observer-level isotropy theorem and the geometric six-link decomposition are proved. The physical identification then has named bridges: observer-level center-freedom for the exact normalized $A/(2d)$ Markov part; H-link for identifying the physical carrier with $V_{\text{link}} \otimes \mathbb{C}^m$; the single-copy clause $m = 1$ for $K = 6$; H-cust for the custodial condensate-stabilizer route; H-blind for the component-blind trace-out, which the symmetric-point rule supplies for the branch's component-complete site observer and which names the condition under which the universal cone of §4.4 extends to other observer classes and to inter-component hidden couplings; and the previously named H-spin'/H-χ' for physical-generation and weak-chirality readings. Downstream Standard-Model and numerical calculations are results *on that branch*, not unconditional consequences of finite observation alone.

Tier 3 — Redundant confirmations. Asymptotic freedom requiring $N_c \geq 3$, the minimal chiral group constraint, and the Jordan-Chevalley projection (Appendix A) provide independent second-route checks on the Tier 2 results without being load-bearing in the primary chain.

The paper is organized as follows. §2 sets out the minimal structure on which the framework's results depend. §3 establishes background independence and derives $d = 3$ from the three self-consistency filters. §4 develops the emergent quantum field theory through the bottom-up, gauge-emergence, and top-down chains. §5 shows that T-invariance and detailed balance give $\theta \in \{0, \pi\}$ and reciprocal transition counts but not $\bar{\theta} = 0$, leaving strong CP open under two named conditions (H-top, H-det); §5.5 sets out what is and is not established, and records a reported, unverified tension in the octahedral-equivariant reconstruction — reciprocity itself leaves the Jarlskog invariant free. §6 extends the chain quantitatively through the gauge coupling prediction. §7 collects the quantitative predictions for the CKM matrix, the lepton sector, the fermion mass chain, the PMNS angles, and the Higgs mass. §8 discusses structural realism, observer genericity, the structural-vs-bijection-level distinction, and the neutrino sector. §9 concludes — including a synthesis with the cosmological application developed in [GR] and the substratum-level reconstruction theorem developed in [Substratum].

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2. The Minimal Structure

2.1 What the theorems require

Deterministic bijectivity. The P-indivisibility result used below (see Prerequisites) requires a bijection on a finite set: bijectivity guarantees recurrence (φ^N = id), and recurrence produces the non-monotonic distinguishability that defines P-indivisibility. The entire framework rests on the memory-bearing description — with its admitted quantum representation — emerging from marginalizing deterministic dynamics — if the dynamics were already stochastic, there would be nothing to derive. Determinism is the core of the framework and cannot be weakened. However, determinism does not require a lattice. A continuous Hamiltonian system on a compact phase space is also deterministic and bijective (by Liouville's theorem). The lattice is one way to implement determinism on a finite set; it is not the only way.

Finite boundary entropy. The gap equation [GR, §3] requires a finite number of degrees of freedom across the partition boundary: S = A/ε^2. This follows from the holographic entropy bound [1] applied to any finite-area surface, not from lattice structure. What matters is that the boundary carries finitely many modes — not that the bulk is a lattice.

Bounded coupling degree (locality). This assumption does the most work in the reference equilibrium sector. The mod-q area-law lemma (§3.1) uses range-1 dynamics together with **uniform initial conditions** to obtain the spatial Markov property; it is therefore a theorem for that state class, not a universal entropy theorem for every admissible prepared state. Locality does not require a regular lattice. Any bounded-degree graph with range-1 dynamics has the same Markov argument in the uniform class. The state-dependent formulation below permits preparation-dependent changes of $G(x)$ while retaining bounded degree. Such changes preserve the graph-boundary form $S(V) \lesssim |\partial V|$ when the lemma's state conditions apply, but non-geometric long edges are outside the random-geometric-graph hypothesis used in the continuum curvature step. The emergent-QM, dimension, and gravity theorems (Theorems 1–6) therefore require separate bookkeeping: bounded degree controls local graph entropy and dynamics, while continuum GR additionally requires a geometric-limit condition on the relevant state class. The gauge-group derivation is where the substratum's crystallography becomes visible, the $(3, 2, 1)$ multiplicities reading out the octahedral point group of the simple-cubic structure (§2.2, §4.6).

Statistical homogeneity and isotropy. The Myrheim-Meyer dimension [2] requires a well-defined notion of spacetime dimension, which emerges from the causal order statistics on a homogeneous, isotropic structure. The dispersion relation (§3.1) uses translation invariance. The dynamics selection (§4.1) uses isotropy. Exact lattice regularity is sufficient but not necessary for these. A random graph with statistical homogeneity and isotropy at large scales produces the same emergent-QM, dimension, and gravity results (Theorems 1–6); the $(3, 2, 1)$ gauge decomposition is the direct read-out of the cubic point group, which statistical isotropy alone does not supply (§2.2, §4.6).

Non-trivial coupling (C1), record persistence, and sufficient capacity (C2, C3). These are conditions on the partition, not on the lattice. They hold for any system with the right partition geometry, regardless of whether the underlying space is a lattice, a graph, or something else.

2.2 What the theorems do not require

A *regular lattice.* Any bounded-degree graph with statistical isotropy suffices for emergent QM, the emergent dimension, and the gravitational sector (Theorems 1–6). Within OI's universality class, the *gauge-group structure* and the quantitative predictions of §7 additionally require the simple-cubic Bravais lattice: the multiplicities $(3, 2, 1)$ are fixed by the octahedral point group on the six link directions (§4.5–4.6, §8.3), and a generic isotropic graph preserves only the spectral mode count $2d_s = 6$, not the cubic decomposition. The cubic Bravais type is an empirically-selected input, forced by the observed SU(3) factor (§8.3). The gauge group itself is a universality-class invariant, shared with any Standard-Model-reproducing class through different machinery ([Structure §2.3, §14.4]); the cubic-group commutant of §4.6 is OI's specific realization of it. A *specific alphabet size q.* All proofs work for any $q \geq 2$; no prediction depends on q. A *specific dimensionality d.* The proofs work for any $d \geq 1$; self-consistency selects $d = 3$ (§3.2). The *wave equation.* It is derived from center independence, isotropy, and linearity (§4.1), each of which follows from the structural properties listed above.

2.3 The minimal object

The theorems require exactly: a deterministic bijection on a finite set, whose state space factors into local degrees of freedom coupled by a bounded-degree graph with statistical isotropy, partitioned into visible and hidden sectors satisfying C1–C4. Everything else — the regular lattice, the alphabet, the dimensionality, the wave equation, quantum mechanics, general relativity — is either derived or irrelevant. The minimal object for the local lattice/gauge chain is the triple (S, φ, V): a finite set S, a bijection φ: S → S with bounded-degree coupling, and a partition V ⊂ S of the degrees of freedom into visible and hidden sectors. This local object does not by itself supply the ontic parameter dependence required for quantum Bell violation under measurement independence; the joint-completion issue is H-Bell in [Substratum §3.3].

2.4 Definition and the factorization problem

Let S = S_1 × S_2 × … × S_N be a product of finite sets (the local state spaces), and let φ: S → S be a bijection. The *coupling graph* G_φ is the undirected graph on vertices {1, …, N} where i and j are connected if the i-th component of φ(s) depends on the j-th component of s for at least one state s ∈ S. The coupling graph is not an additional structure imposed on the system — it is read off from φ.

What determines the product decomposition? An abstract finite set S does not come with a canonical factorization. The answer is that the factorization is determined by φ itself, as the one that *minimizes the coupling degree*. For the wave equation, numerical verification confirms this: the natural spatial factorization is the *unique minimizer* of coupling degree — 0% of random factorizations achieve it, and 100% have strictly higher coupling.

Theorem (factorization uniqueness for translation-invariant dynamics). *Let φ be a bijection on $S = (\mathbb{Z}/q\mathbb{Z})^N$ satisfying: (a) translation invariance, (b) range-1 coupling on the natural factorization, and (c) genuinely bidirectional coupling. Then the natural factorization is the unique minimizer of coupling degree among all product decompositions $S = S_1 \times \cdots \times S_M$ with $|S_k| = q$ for all k.* The key step is that the wave equation's addition $x_{i-1} + x_{i+1} - x_i^{\text{prev}}$ creates irreducible algebraic dependence among all input components (for $q > 2$), so any relabeling that splits or merges natural sites increases the coupling degree.

Scope of $K = 2d$. Coupling-degree minimization does not establish $K = 2d = 6$: diagonalizing the internal matrix decouples eigenvectors but does not assign an eigenvector to a single signed spatial neighbour, since every component receives the whole neighbour sum. What holds exactly is the six-link decomposition $V_{\text{link}} = T_1 \oplus E \oplus A_1 = 3 \oplus 2 \oplus 1$; conditional is the further claim that those geometric links are the physical gauge carrier, which is H-link ($V_{\text{physical}} \cong V_{\text{link}}^{\oplus m}$, giving $K = 2dm$, with single-copy $m = 1$ a premise rather than a result).

General-graph carrier extension — not established. Polynomial/spectral dimension controls large-scale heat-kernel scaling, but it does not canonically provide $2d_s$ signed internal propagation channels. The earlier inference from quasi-isometry or spectral dimension to a unique $K = 2d_s$ physical carrier is therefore unavailable. The exact link representation and H-link are presently specific to the simple-cubic branch; a broader universality-class carrier theorem would require a separate construction.

2.5 Space as coupling structure

Every "spatial" property used in the derivation chain is a property of G_φ. The *area* of a region V is the number of edges crossing from V to its complement. The *dimension* is extracted from the causal partial order via the Myrheim-Meyer estimator [2]. The *area law* follows from the spatial Markov property on any graph with range-1 dynamics. The *dispersion relation* requires translation invariance. The regular cubic lattice is one graph with these properties; it is not the only one.

2.6 The dissolution of the container problem

The coupling graph is not embedded in a pre-existing space. There is no ambient manifold in which the graph "lives." The vertices are labels for degrees of freedom; the edges record which degrees of freedom are dynamically coupled. Any spatial embedding is a representation for human convenience, not a physical fact. The coupling graph is not made of material. The vertices are not atoms of space, and the edges are not physical connections. A degree of freedom is a component of the state — it is an abstract mathematical entity, not an object. Asking "what is site i made of?" is a category error: it is a label for a variable, not a name for a thing.

The most common objection to discrete models of spacetime is the container problem: if space is a lattice, what does the lattice sit in? On the structural reading, the container problem does not arise. The coupling graph is defined by the bijection φ, not by embedding in a manifold. Two sites are neighbors because φ couples them — because the state at one affects the future of the other. This is a dynamical fact, not a spatial margin. Locality is defined by the dynamics, and space emerges from locality. At no point does anything need to be "inside" anything else. The relationship is analogous to a social network: two people are "connected" not because they occupy adjacent points in physical space, but because they interact. The topology is defined by the interactions, not by geography.

2.7 The alphabet as gauge freedom

Every prediction of the OI framework is independent of the alphabet size q. The gap equation $\hbar = c^3\varepsilon^2/(4G)$ contains no q. The Bekenstein-Hawking formula contains no q. The dispersion relation, area law, P-indivisibility, center independence, and Einstein's equations are all q-independent. Numerical confirmation: on a 1D ring of L = 40 sites, the wave equation produces significant P-indivisibility at every q tested from 2 to 64. No experiment, even in principle, could measure q.

Two systems (S, φ) and (S', φ') with different alphabet sizes q and q' are *physically equivalent* if they produce the same emergent transition probabilities. The analogy to electromagnetic gauge invariance is precise: the gauge potential A_μ is not physical; the field strength F_$\mu\nu$ is. In the OI framework, the mod-q state space is not physical; the coupling structure and its emergent consequences are. Operationally this means the *absolute scale* of the field value is unobservable — only the coupling structure and emergent transition probabilities are (the discrete analogue of field-normalization redundancy in ordinary field theory). The transformations realizing this are the alphabet's **additive-group automorphisms** — the size q, rescalings $x \mapsto \lambda x$ (for λ a unit), and shifts $x \mapsto x + c$ — not arbitrary relabellings of $\mathbb{Z}/q\mathbb{Z}$. A nonlinear relabelling preserves the orbit structure (hence the transition probabilities under relabelling) but changes the fluctuation dispersion, so it is not a symmetry of the physical observables; the gauge group is the additive-automorphism group, a proper subgroup of all permutations. This amplitude-scale (additive-automorphism) gauge is what the §4.1 linearity argument invokes; whether it is entailed by the q-size statement above, or is an additional principle, is an identified open structural question.

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3. Background Independence and the Selection of d = 3

3.1 Background independence

The companion paper [Main] treats the coupling graph as fixed. In general relativity, the spacetime geometry is dynamical. If space is the coupling graph, background independence requires the graph to evolve with the state: s(t+1) = φ_{s(t)}(s(t)), where each φ_s is a bijection but G_{φ_s} varies with s.

Bijectivity is automatic. The wave equation is second-order: x_i(t+1) = f(neighbors of i at time t) − x_i(t−1) mod q. The phase-space map F: (x(t−1), x(t)) → (x(t), x(t+1)) has an explicit inverse that uses x(t) to determine which graph to apply. Bijectivity of the phase-space map is automatic for any second-order reversible dynamics, regardless of whether the coupling is state-dependent. The P-indivisibility proof applies without modification.

The reference derivation chain survives under three constraints. (i) *Local graph-dependence:* in the reference vacuum/uniform state class, $G(x)$ at site i depends only on x_j within a bounded range of i. (ii) *Center-independent graph-dependence:* $G(x)$ at site i does not depend on x_i, preserving center independence. (iii) *Statistical isotropy:* $G(x)$ is statistically isotropic on large scales for typical reference configurations.

Under these constraints, the uniform-class Markov area law holds, center independence is preserved, the gap equation is unchanged, and the dynamics selection theorem applies at each step. Prepared states may carry additional bounded-degree, preparation-indexed edges. Those edges are compatible with the graph-level area statement because $|\partial V|$ is defined in $G(x)$ itself; their unresolved effect is instead on the continuum identification of graph distance and Ollivier–Ricci curvature with a Riemannian metric and Ricci curvature.

Explicit construction. On a ring of $L \geq 4$ sites with alphabet $\mathbb{Z}/q\mathbb{Z}$, define the right-neighbor assignment $R(i,x) = i + 2 \bmod L$ if $x_{(i+1) \bmod L} = 0$, and $R(i,x) = i + 1 \bmod L$ otherwise, with left neighbor $L(i) = i - 1 \bmod L$ fixed. The dynamics is $x_i(t+1) = (x_{L(i)}(t) + x_{R(i,x(t))}(t) - x_i(t-1)) \bmod q$. This satisfies: constraint (i) with range 1 ($R(i,x)$ depends only on x_{i+1}); constraint (ii) exactly ($R(i,x)$ does not depend on x_i); constraint (iii) by translation invariance. Bijectivity is verified by explicit inverse: $u_i = (v_{L(i)} + v_{R(i,v)} - w_i) \bmod q$, since v determines all neighbor assignments. Center independence, P-indivisibility, and graph regularity (undirected degree bounded in [3, 2] independently of the state) all hold for any L and q.

Local gauge invariance from background independence — conditional carrier reading. On the H-link + H-cust single-copy branch, the six-link carrier has the exact $(3, 2, 1)$ cubic decomposition of Theorem 7 and the corresponding generic condensate stabilizer of Theorem 5. Background independence is then the premise promoting the surviving global stabilizer to the local gauge reading developed below. Without H-link, Theorem 7 remains a theorem about geometric links rather than a proof of the physical gauge carrier.

$$\phi(\mathbf{n}) \to G(\mathbf{n})\,\phi(\mathbf{n}), \qquad M(\mathbf{n}, \hat{e}_j) \to G(\mathbf{n})\,M(\mathbf{n}, \hat{e}_j)\,G(\mathbf{n} + \hat{e}_j)^{-1}$$

The wave equation is invariant. This is local gauge invariance. The link variable $M(\mathbf{n}, \hat{e}_j)$ transforms as a gauge connection: the plaquette product $P(\mathbf{n}, j, k) = M(\mathbf{n}, \hat{e}_j)\,M(\mathbf{n} + \hat{e}_j, \hat{e}_k)\,M(\mathbf{n} + \hat{e}_k, \hat{e}_j)^{-1}\,M(\mathbf{n}, \hat{e}_k)^{-1}$ transforms in the adjoint ($P \to G(\mathbf{n})\,P\,G(\mathbf{n})^{-1}$), so $\operatorname{Re}\operatorname{Tr}(P)$ is gauge-invariant — it is the Wilson plaquette action [4], now derived rather than postulated.

Einstein's equations as self-consistency. The Jacobson thermodynamic argument [5], applied to local causal horizons on the state-dependent graph, requires three inputs — each proved here for the wave equation.

Lemma (Area law). *The entanglement entropy of a spatial region V in the wave equation scales as $S(V) = \eta\,|\partial V|$. This holds for both the linear wave equation over $\mathbb{R}$ and the mod-q wave equation over $\mathbb{Z}/q\mathbb{Z}$.*

Proof. Over $\mathbb{R}$: the wave equation is a system of coupled harmonic oscillators with a quadratic Hamiltonian determined by the adjacency matrix. The area-law theorem for Gaussian systems applies directly [3]. Over $\mathbb{Z}/q\mathbb{Z}$: the wave equation has range $R = 1$, so for any $i \in V^\circ$ (interior of V), all neighbors lie inside V. With uniform initial conditions: $V^\circ \perp V^c \mid \partial V$ (spatial Markov property). By the chain rule: $I(V; V^c) = I(\partial V; V^c) \leq H(\partial V) \leq |\partial V| \cdot \log_2 q$. The mutual information scales as boundary area, not volume. □

Lemma (Relativistic dispersion, one spatial dimension). *In $d = 1$ the wave equation has exact dispersion relation* $\cos\omega = \cos k$. The restriction is essential:

on the normalized $d = 3$ observer branch of §4.1 the relation is $\cos\omega(\mathbf{k}) = \frac{1}{3}\sum_{i=1}^{3}\cos(ak_i)$, which reduces to $\cos\omega = \cos(ak)$ only where all three cosine arguments coincide — the body diagonals. Along a lattice axis it is $\cos\omega = (2+\cos ak)/3$, since setting the transverse momenta to zero leaves their two $+1$ contributions inside the three-dimensional operator; recovering $\cos\omega = \cos k$ there would require each component to read a single signed link, the operator substitution Lemma 6a rejects.

Proof. Substituting $x_j(t) = A\exp(i(kj - \omega t))$: $e^{-i\omega} + e^{i\omega} = e^{-ik} + e^{ik}$, giving $\cos\omega = \cos k$. For small k: $\omega = |k| + O(k^3)$ — relativistic propagation with $v = 1$. $\square$

Scope of the emergent-Lorentz claim (what is and is not established). The dispersion lemma establishes that the leading low-energy behavior is relativistic, with the first correction entering at $O(k^3)$ in the frequency (equivalently $O(\varepsilon^2 k^2)$ in ω^2); there is no linear Lorentz-violating term. This is necessary for emergent Lorentz invariance but is not the full statement. Three points. (i) *Rotational sector.* The leading rotational-symmetry-breaking operator on the cubic lattice transforms in a nontrivial representation of the octahedral group O_h (the E_g irrep of the symmetric rank-2 tensors), so it is forbidden in any cubic-invariant action; the lowest *allowed* anisotropy is the $O(k^4)$ operator, which is irrelevant by power counting (dimension six in $d = 3 + 1$). Spatial isotropy at leading order is therefore protected by the cubic symmetry already required for the gauge derivation (§4.5), not an independent assumption — though it remains a statement about the bare action. (ii) *Boost sector.* The lattice possesses a preferred frame (the rest frame in which update steps are simultaneous), and the lattice dispersion is not boost-invariant at the cutoff scale; boost invariance can hold only as an emergent infrared property. The deviation of the dispersion surface from boost-invariance is $O(k^4)$ (dimension six, irrelevant) with no lower-order term, so the boost sector is, at tree level, in the same position as the rotational sector: Lorentz violation is irrelevant and suppressed by $(E/M_{\mathrm{Pl}})^2$. (iii) *Radiative stability (open).* The tree-level irrelevance of both the rotational and boost Lorentz-violating operators does not by itself guarantee emergent Lorentz invariance in the interacting theory: radiative corrections can in principle generate a lower-dimension (relevant or marginal) Lorentz-violating operator that is not Planck-suppressed — the Lorentz fine-tuning problem familiar from effective-field-theory treatments of Planck-scale Lorentz violation. The standard field-theoretic resolution (supersymmetric suppression of the offending divergences) is not available here, as the framework is not supersymmetric. Cubic symmetry must be checked at loop level, not assumed: the lattice-field-theory analysis of Davoudi and Savage (Phys. Rev. D **86**, 054505) shows that on a hyper-cubic lattice the renormalization mixing into lower-dimensional operators with power-divergent coefficients is mixing among *angular-momentum-covariant* operators of definite J — an operator-identification effect — while the genuinely rotational-symmetry-violating contributions remain suppressed by $\mathcal{O}(a^2)$ at every order (unconditionally in $\lambda\phi^4$; in gauge theory when the gauge fields are smeared comparably to the matter fields). Since the E_g operator is an octahedral non-singlet, its forbiddenness *does* extend to all orders for any dynamics that preserves the octahedral symmetry of the action and measure — radiative corrections of an octahedral-invariant theory generate only octahedral-invariant operators. The loop-level protection of the rotational sector therefore reduces to a single question: whether the emergent theory preserves the octahedral symmetry exactly (treated in §3.1 (*A present-day constraint, and the rotational sector's distinct status*) and [Ch. 19 §19.3.7]). The mechanism by which rotational-symmetry restoration is known to hold at all orders is operator *smearing* — a definition of the effective operators that filters ultraviolet modes over a physical region — under which the rotational-violating contributions remain suppressed by $\mathcal{O}(a^2)$ at every order in $\lambda\phi^4$, and in gauge theory provided the gauge fields are likewise smeared. Whether the framework's substratum-to-observer coarse-graining map (§4.7.1.1) realizes a sufficient smearing of this type is now characterized, in the negative: the map's role in assigning gauge representations is fixed (§4.7.1.2), and an exact surface-Green's-function extraction of its boundary spectral measure on the linear substratum yields a power-suppressed local exponent ($dN/d\omega \propto \omega^4$; the $+1/+2$ estimator controls validated), invariant over thirty orders of magnitude in the slow-bath ratio and under generic quenched deformation — the map is neither an ultraviolet filter nor an infrared amplifier, the spectral-measure counterpart of the $R \approx \varepsilon$ kernel-width result of §3.1 (*A present-day constraint, and the rotational sector's distinct status*). The emergent-Lorentz claim should therefore be read as: relativistic to leading order with no linear violation (established at tree level, consistent with the one-loop and two-loop staggered checks of §6), with radiative stability resting not on operator smearing — which the characterized map does not supply — but on loop-level octahedral invariance of the emergent theory (the mechanism developed below and in [Ch. 19 §19.3.7]); the interacting (condensate-dressed) extension of the characterization is the stated residual. A second, structural ingredient is available within the framework itself, and it has a matching part and a running part. *Matching.* At $M = \mu I$ the six carrier components are dynamically identical and decoupled, so the substratum dynamics, the threshold nonlinearity of the finite alphabet and the selected measure all commute with the signed permutations of the components — a symmetry larger than the cubic group acting on the six links, and one whose six-dimensional representation is irreducible. For the branch's observer — the component-complete site observer whose visible variable is the six-component field on its sites — the trace-out commutes with it too (H-blind, supplied there; §4.4), and the complete memory-resummed quadratic response of the carrier is then custodially scalar: every massless block shares one cone at the matching scale (§4.4, Proposition 4b). The custodial U(6) of the real lift (Theorem 5, §4.6) is broken by the threshold only down to this signed-permutation group, which suffices, and it is broken spontaneously by the equivariant condensate, so sector-differential Lorentz-violating coefficients admit no condensate-independent counterterm at the regulator level; condensate-dependent derivative operators are allowed and decouple with the order parameter. *Running.* Conditional on that matching, the sector-differential boost-violation is controlled by the heavy cross-charged mass M_X generated at the condensate gap. A one-loop lattice calculation of the boost-projected self-energy curvature, whose integrand, driver and numerical output are not available for verification, reported that, after absorbing the universal tree-level cone, the differential coefficient obeys a quadratic-with-logarithm decoupling law of the form $\delta c \sim \Delta C_2\,(\alpha_0/4\pi)\,(M_X/M_{\rm Pl})^2\,[\,c_L \ln(M_{\rm Pl}/M_X) + c_0\,]$ rather than a non-decoupling constant, so that the heavy sector decouples as M_X rises; the same calculation reported that the law survived both a scalar- and a fermion-line check and reverted to a non-decoupling logarithm when the cone was not absorbed. The form, the constants, the controls and every window derived from them below are reported results of that calculation and are not independently verified; they are retained as illustrative and carry no weight as predictions, bounds, or inputs to any other result. What is certified independently of it is the normalization's bound from the framework's own representation content: ΔC_2 is a difference of total gauge Casimirs across species, giving $\Delta C_2 \in [0.58,\, 2.08]$ — order unity. The representation-side point values are fixed per bound: the clock-comparison pair (nucleon–electron, chirality-averaged) gives $\Delta C_2 \approx 0.9$–1.1 across the two U(1)-normalization conventions, and the electron coefficient — measured against the *induced* photon, whose boost coefficient is the vacuum-polarization-weighted average of the charged fermions', $w_f \propto N_c Q_f^2$ — gives $\Delta C_2 \approx 0.6$–0.7. The induced architecture produces one structural alignment worth recording: the photon coefficient lies within $\Delta C_2 \lesssim 0.03$ of the loop-weighted species mean (a weighted average cannot stray far from itself), so the photon-sector channel is suppressed roughly two orders below generic when referenced to that mean, with its residual sensitivity confined to the absorbed-cone reference definition — the one place the reference choice materially matters, owned by the kernel calculation. With the reported kernel constants ($c_L \simeq -9.4$, $d \simeq -0.88$) and the point values, the tightest ceiling would refine from the worst-case figures below to $M_X \lesssim 4 \times 10^7$ GeV — an illustrative figure carrying the provenance of those constants. What remains open on the normalization side is the absorbed-cone reference (photon channel only). The non-logarithmic constant receives no separate condensate-dependent contribution from the substratum itself: the threshold nonlinearity of the finite alphabet acts per component, so every operator it generates is component-local, and a component-local operator contracted with an O_h-equivariant condensate is sector-common — the link-basis diagonal of $aP_{T_1} + bP_E + cP_{A_1}$ is the constant $a/2 + b/3 + c/6$ — so such terms shift only the universal cone. Block-differential derivative terms require inter-component interactions, which on this branch are the induced gauge couplings of §5 and enter first at one loop, in the calculation reported above; c_0 is that diagram's own constant. The mechanism's remaining quantitative item is the condensate-gap determination of M_X itself.

Baryon safety of the cross-charged sector, the pinned reference, and the closed form of the window. Three further results complete the accounting. First, the cross-charged sector is baryon-safe at the gauge level, structurally: all matter carriers are mixed-index objects in $V_K \otimes V_K^*$ (§4.7.1.2), on which the U(6) adjoint action conserves index flow per block — baryon number, realized as V_3-flow/3, is conserved identically by all thirty-six generators, and the diquark vertices that gauge-mediated proton decay requires are absent because they need same-index carriers ($V \otimes V$), which the framework does not have. The cross-charged bosons are Pati–Salam-class vector leptoquarks, not SU(5)-class X/Y: the $\mathrm{Hom}(V_3, V_1)$ modes carry exactly the U_1 vector-leptoquark quantum numbers $(\mathbf{3}, \mathbf{1}, 2/3)$ — an identity of the forced hypercharge assignment, $Y_Q - Y_L = Y_u = 2/3$, obtained by two independent routes. The M_X floor is therefore leptoquark phenomenology rather than proton decay: $\sim 1.5 \times 10^6$ GeV ($K_L \to \mu e$ class) if the cross couplings are generation-off-diagonal, 10^3–10^4 GeV (collider and precision) if generation-diagonal — the generation structure being solution-level, like M_X itself. On the reported constants the illustrative window is $[10^3$–$10^6,\ \sim 4 \times 10^7]$ GeV: between one and a half and four and a half decades. Scope: this argument covers the gauge sector; baryon violation through non-gauge operators (the condensate and Yukawa sectors, higher-dimension operators) is a separate accounting item, not excluded here. Second, the absorbed-cone reference is pinned by the framework's own architecture rather than left as a scheme choice: spacetime geometry is classical-substratum-level ([GR §6.2]) and takes no emergent-loop corrections, so the gravitational-wave cone *is* the tree cone — gravity-referenced bounds (the GW170817 class) therefore constrain each species' full coefficient ($\Delta C_2^{\text{eff}} = \sum_f w_f C_2^f \approx 1.6$ for the induced photon, ceiling 4×10^{11} GeV, still the loosest), while matter-matter bounds constrain differences, as computed above. The same classical priority of geometry that dissolves the cosmological-constant problem pins the Lorentz-violation reference. Third, with M_X classified as solution-level (set by the equivariant block splittings, §4.7.1.1, not fixed at the lattice scale), the boost conditional has two parts of different standing: its matching part is the certified symmetric-point theorem of §4.4, and its running part rests on the reported, unverified decoupling calculation. On that reported form the running part would be discharged into one solution-level parameter confined to a one-and-a-half-to-four-and-a-half-decade window, baryon-safe, reference-pinned, and falsifiable from both sides — the same closure the framework accepts for the mass eigenvalues. With $\alpha_0 = 1/23.25$ and the worst-case ΔC_2, the reported form gives ceilings of $M_X \lesssim 9 \times 10^7$ GeV against the tightest (clock-comparison) multi-species bound, 7.5×10^9 GeV against the electron SME coefficient, and 1.1×10^{12} GeV against photon-sector bounds, while the floor is set by cross-charged (leptoquark-like) phenomenology at 10^3–10^6 GeV: a window of two to five orders of magnitude. These figures are illustrative consequences of the reported calculation, not verified results; on them the mechanism would require the cross-charged condensate scale to be intermediate, $M_X \lesssim 10^8$ GeV rather than GUT- or Planck-scale, and an independent determination of M_X from the condensate-gap chain above that ceiling would falsify it. The bound-to-coefficient mapping and the cross-charged phenomenology floor are treated here at order of magnitude only. We situate this in the literature. The radiative Lorentz-fine-tuning problem is general to Planck-scale Lorentz violation (Collins, Perez, Sudarsky, Urrutia, and Vucetich, Phys. Rev. Lett. **93**, 191301), and its standard field-theoretic resolution — supersymmetric suppression of the offending divergences — is unavailable here, as the framework is not supersymmetric. The one non-supersymmetric protection mechanism consistent with the framework's structure is renormalization-group emergence: Lorentz-violating couplings flowing to zero at an infrared-attractive Lorentz-invariant fixed point (Chadha and Nielsen, Nucl. Phys. B **217**, 125 (1983); Bednik, Pujolàs, and Sibiryakov, JHEP **11** (2013) 064). This mechanism is natural in the present setting because the trace-out of [Main] is itself a coarse-graining toward the infrared, and it is the same mechanism on which emergent-quantum-mechanics-from-lattice constructions rely (e.g. arXiv:2207.09465); its documented limitation is that the inter-species velocity flow is only logarithmically fast, leaving a residual fine-tuning against the tightest multi-species bounds. Independently, the causal-set discreteness results (Bombelli, Henson, and Sorkin, *Discreteness without symmetry breaking*) establish that a finite-valency, locally-coupled discrete structure cannot carry exact Lorentz invariance microscopically — only a random, statistically isotropic structure can, and then only non-locally — which is consistent with the framework's concession of a preferred lattice frame and reinforces that Lorentz invariance here must be wholly emergent (an infrared property of the coarse-grained theory), never microscopic.

A present-day constraint, and the rotational sector's distinct status. The radiative-stability discussion above acquires a mild empirical handle from ultra-high-energy data, but the handle applies unevenly across the two subsectors, and the distinction is important. Ultra-high-energy observations now constrain the quadratic ($n = 2$, dimension-six) Lorentz-violating coefficient in the *isotropic* dispersion $E^2 = m^2 + k^2[1 + c_6(k/M_{\rm Pl})^2]$: the absence of neutrino splitting for KM3-230213A gives $\Lambda_{n=2} \gtrsim 1.1 \times 10^{19}$ GeV $\approx M_{\rm Pl}$ (Satunin, arXiv:2502.09548), and the GZK consistency of the Auger/HiRes proton spectra bounds $\delta_{n=2} \lesssim 10^{-23}$ at Lorentz factor $\sim 2 \times 10^{11}$. In the framework's parameterization these require the *isotropic / sector-differential* coefficient to satisfy $c_6 \lesssim \mathcal{O}(1)$, consistent with an order-unity coefficient at the lattice scale; this is genuine present-day exposure for the boost-adjacent sector treated above. It does **not**, however, control the rotational E_g anisotropy, and that sector is in a more precarious position. The rotational operator is a custodial-U(6) singlet — it carries spatial indices but is a gauge-sector singlet — so the power-law $(M_X/M_{\rm Pl})^2$ decoupling that suppresses the sector-differential boost violation does not apply to it; its protection is instead the cubic symmetry itself, operating at the level of operator *dimension* rather than coefficient. The leading rotational-violating operator is the E_g ($\ell = 2$) anisotropy, an octahedral non-singlet; the lowest octahedral-invariant anisotropy is the $\ell = 4$ cubic harmonic, dimension six. A direct computation of the emergent tree-level dispersion confirms this places the violation at dimension six: the fractional anisotropy $[\omega^2(\hat{n}_1) - \omega^2(\hat{n}_2)]/|k|^2$ scales as $(\varepsilon k)^2 = (E/M_{\rm Pl})^2$, with no dimension-four (k-independent) term. Whether this survives loops reduces, as in §3.1 (*Scope of the emergent-Lorentz claim*, point (iii)), to whether the emergent theory preserves the octahedral symmetry: octahedral-invariant dynamics cannot generate the E_g operator at any order. This corrects an earlier reading of Davoudi and Savage: the power-divergent operator mixing that arises without ultraviolet smearing is mixing among *angular-momentum-covariant* operators of definite J, not the generation of a relevant rotational-violating operator, which they find remains suppressed by $\mathcal{O}(a^2)$ at every order. The kernel width R/ε therefore sets the *coefficient* of the dimension-six rotational operator — an additional $(\varepsilon/R)^2$ suppression when $R \gg \varepsilon$ — and not its dimension; even $R/\varepsilon = \mathcal{O}(1)$ leaves the rotational violation Planck-suppressed. The differentiating angular signature of any residual — the $\ell = 4$ cubic harmonic aligned to the cosmic rest frame ([Main §3.5]), correlated across species — is recorded as the conditional forward signature F14 ([appendix-A §A.5]); being dimension six, its amplitude is $\sim (E/M_{\mathrm{Pl}})^2$ and unobservable at accessible energies. One structural consideration bears on the coefficient. The smearing kernel is the traced-out hidden-sector propagator, and a direct computation of its position-space width gives $R \approx \varepsilon$, flat as the slow-bath capacity ratio τ_B/τ_S is varied over thirty orders of magnitude: the framework's (large) observational incompleteness is incompleteness about the hidden *state*, not about spatial scale, and a state-average does not damp ultraviolet spatial modes. This fixes the dimension-six operator's coefficient at order unity rather than further-suppressed, but does not lift it from dimension six, so it is not by itself an exposure. The interpretation — that embedded observation does not actively *smear* the rotational sector, for the same reason the trace-out selects a rest frame and yields the quantum-representable reduced description — is developed in [Ch. 19 §19.3.7]; on the present reading it does not need to, because octahedral symmetry and locality already place the violation at dimension six. The open question that remains is correspondingly narrower, and is not a sidereal-bound fine-tuning: whether the emergence preserves octahedral symmetry *exactly* at loop level, or introduces an octahedral-breaking effect — for instance an effective momentum-shell structure in the trace-out — not visible in the octahedral-invariant checkerboard partition.

The octahedral-invariance premise has a second dependency, more direct than the trace-out. The effective operator is octahedral-invariant only if the condensate Σ is, and the O-invariance of the dynamics does not descend to the solution: the framework's own gauge mechanism is spontaneous breaking of a dynamical symmetry by that condensate (§4.6). Theorem 5 takes Σ to be O-equivariant, and that equivariance is a hypothesis on the solution rather than a consequence of the dynamics. It is not secured by the observed gauge group alone: a condensate with the same block multiplicities $(3, 2, 1)$ — hence the same stabilizer $\mathrm{U}(3)\times\mathrm{U}(2)\times\mathrm{U}(1)$ — but with eigenspaces not aligned to the O-irreps generates an E_g anisotropy at dimension four rather than dimension six. What secures the alignment is the geometric identification of the blocks with the link-direction irreps, used independently in the link-carrier construction (§4.7.1.2) and in the inter-generation prescription of §7.1. In the other direction the premise is robust: an equivariant condensate leaves the E_g channel exactly null at every block splitting, and a staggered spatial modulation of the condensate does not lift it, so the rotational sector is insensitive to the condensate's spatial structure and turns only on its internal alignment. That loop-level question, not the kernel width, is the arbiter. Two results bear on it. If the microscopic action, the measure, and the partition are each exactly O_h-invariant about the observer's centre, the effective action obtained by marginalizing the hidden sector is O_h-invariant to all orders (substitute $H \to g^{-1}H$ in the defining integral for $g \in O_h$; the Jacobian is unity), and every ingredient of the checkerboard trace-out — the dispersion, the position-space partition, and the timescale-selected momentum region $\{\omega(k) < \omega_{\text{cut}}\}$ — is O_h-invariant, so no direction-dependent shell is available to break the symmetry. A numerical check on the rounded interacting dynamics is consistent with this: with the extraction validated against the analytic tree coefficients ($1/3$, $-1/135$, $-1/36$) to the percent level and the allowed $\ell = 4$ cubic harmonic detected at high significance, the E_g coefficient is zero within errors at the 2×10^{-5} level in the perturbative regime — a discrimination of order 10^3 against the allowed anisotropy — and consistent with zero in structure-factor measurements of the thermalized strong-coupling state. What remains is larger-volume strong-coupling statistics and the check that the $O(\varepsilon/R_H)$ observer-position boundary anisotropy is $\ell = 4$-aligned. This tree-level premise — that the trace-out commutes with the lattice octahedral group — is itself grounded in the factorization criterion that fixes the gauge content ([Main §3.1, §3.5]): an observer-centered partition breaks translations, boosts, and rotations about distant points (each maps visible boundary sites into the hidden sector), but the octahedral group about the observer's own centre is not of this kind, and the horizon-isotropic visible region is spherical and hence octahedrally symmetric, so this O_h preserves the visible/hidden factorization and descends to an exact symmetry of the effective theory. The criterion that selects the gauge symmetries (factorization-preserving visible-internal symmetries), the one that protects the rotational sector (a factorization-preserving spatial symmetry, hence dimension six), and the one that exposes the boost sector (boosts are factorization-breaking) are a single criterion; the rotational-protected/boost-exposed asymmetry is therefore structural rather than incidental.

Clarification of the protection mechanisms (relationship and remaining gap). Two of the ingredients above are not independent: operator smearing is the coarse-graining-toward-the-infrared realization of the renormalization-group-emergence mechanism, so the question "does the substratum-to-observer map smear sufficiently?" and the question "does the theory flow to a Lorentz-invariant infrared fixed point?" are two descriptions of the same requirement. The custodial ingredient is logically separate and stronger where it applies: at the symmetric point the signed-permutation symmetry of the carrier components makes the quadratic response custodially scalar under H-blind (§4.4, Proposition 4b), which removes condensate-independent sector-*differential* Lorentz-violating counterterms at the regulator level, and, conditional on that matching, the resulting differential coefficient obeys the reported power-law-with-logarithm decoupling in M_X — the matching is certified, the running is the reported, unverified calculation — a power-law suppression rather than the logarithmically-slow RG flow. The remaining gap is sharper than “radiative stability in general”: the custodial argument controls the *sector-differential* (inter-species) boost-violation that the multi-species bounds constrain most tightly, but the *overall* (sector-common) Lorentz-violating operator in the rotational sector has no analogous power-law ally and rests on the smearing/RG-emergence mechanism alone, whose inter-species flow is only logarithmically fast. The rotational sector is therefore the less-protected of the two sub-problems. Finally, the preferred frame conceded in (ii) is not free-floating: on identification with the observed universe it is the frame in which the substratum update steps are simultaneous, which is operationally the cosmic rest frame (the CMB frame); the framework makes no claim to derive this identification, but records that the preferred frame, if physical, is not an additional tunable structure but the cosmological rest frame already present in the data.

Lemma (Lattice Bisognano-Wichmann). *For the wave equation, whose low-energy effective theory is Lorentz-invariant, the entanglement Hamiltonian for a half-space bipartition has the Bisognano-Wichmann form. The reduced state for a Rindler observer is thermal at* $T_U = \hbar\kappa/(2\pi c k_B)$*, with corrections at* $O(\varepsilon^2\kappa^2/c^2)$*.*

Proof. The mod-q wave equation is a coupled harmonic oscillator system — the class for which the lattice BW theorem is proved analytically [6]. Its low-energy dispersion is relativistic (the preceding lemma). Thermality of the Unruh effect is robust against UV modifications of the dispersion relation, with corrections scaling as $O(\varepsilon^2\kappa^2/c^2)$. □

Theorem (Discrete Einstein equation). *For the state-dependent wave equation on a bounded-degree graph G(x) satisfying constraints (i)–(iii), the Jacobson thermodynamic argument produces:*

$$\kappa_{OR}(i, j; G(x)) = \alpha \cdot \mathcal{T}_{ij}(x, G(x))$$

where κ_{OR} *is the Ollivier-Ricci curvature [7],* $\mathcal{T}_{ij}$ *is the discrete stress-energy, and* $\alpha = 8\pi G/(c^4\eta)$*.*

Proof. (i) *Area law.* In the reference state class used by the mod-q proof, the entropy measure scales as $S(V) = \eta\,|\partial V|$ (area-law lemma above); the real Gaussian statement has its own ground-state/regularity hypotheses. The graph boundary is evaluated in the state-dependent $G(x)$, so preparation-indexed edges contribute to $|\partial V|$ rather than invalidating the graph-level form by definition. (ii) *Unruh temperature.* The lattice Bisognano-Wichmann lemma gives $T = \hbar\kappa/(2\pi c k_B)$ at local causal horizons. (iii) *Clausius relation.* Graph deformation is measured by the MESOSCOPIC Ollivier curvature $\kappa_{OR}^{(\delta_N)}(x, y)$ at separation $d(x, y) \le \delta_N$ with measures of radius δ_N, giving $\delta A/A = -\kappa_{OR}^{(\delta_N)}\,\delta\lambda$ and hence $\kappa_{OR}^{(\delta_N)} = \alpha \cdot \mathcal{T}_{xy}$. The scale is not cosmetic: the convergence result of

(iv) is stated for the mesoscopic generalization, and one-link Ollivier curvature on sparse unweighted graphs is not what it controls, so an edge-wise reading of this step would not be supported by the theorem it invokes. (iv) *Continuum limit — with the reference debt stated.* Van der Hoorn et al. [8] prove $\kappa_{OR}^{(\delta)} \to \mathrm{Ric}$ for arbitrary manifold dimension $N \geq 2$, but for the MANIFOLD-DISTANCE-WEIGHTED graph metric d_G^w; their intrinsic rescaled hop-count version $d_G^* = \varepsilon_n d_G^s$ is established in the cited work for $D = 2$ only. That split is load-bearing here. The substratum of §2 has intrinsic adjacency and no ambient manifold, so supplying $d_{\mathcal{M}}(u, v)$ as an edge weight would import precisely the geometric data this chain is meant to recover; the arbitrary-D weighted theorem is therefore not the theorem this step needs. What it would need is $\varepsilon_N d^s_{G_N} \to d_{\mathcal{M}}$ at $D = 3$ with local stretch control, followed by the mesoscopic Ollivier limit. **That first statement is false for this reference family.** On the degree-6 simple-cubic graph the hop metric is exactly the ℓ_1 metric, $a\, d_G(x, y) = \|x-y\|_1$ at every spacing a, so along a diagonal $y - x = (s, s, 0)$ the ratio of hop length to Euclidean length is $\sqrt{2}$ at every scale — the stretch does not decay, in flat space, in the continuum limit or anywhere else. Enlarging the stencil does not repair it: any FIXED finite neighbour set has a polytopal unit ball, and a polytope is never a Euclidean sphere, so 18- or 26-neighbour cubic variants merely exchange ℓ_1 for another crystalline norm. A second and independent mismatch points the same way: [8]'s hypotheses force the random-geometric average degree to grow without bound, while this reference graph has degree exactly 6 for every N, which is precisely the ultrasparse regime [8] excludes. **A future extension of [8] from $D = 2$ to $D = 3$ would therefore not rescue this step.** The hop-metric Ollivier route therefore fails at its reference instantiation, and the discrete Einstein theorem above is not established by the cited chain. **Where the geometry actually lives, and the three routes.** The failure is confined to the combinatorial metric; it is not a failure of the emergent geometry. This paper's own dispersion relation already supplies a Euclidean quadratic form: $\omega^2(k) = \sum_i (2 - 2\cos k_i a)/a^2 = |k|_2^2 - (a^2/12)\sum_i k_i^4 + \cdots$, whose leading principal symbol is isotropic with anisotropy vanishing as $O(a^2)$. So the substratum has two distinct emergent geometries — a PROPAGATION/spectral geometry that is ℓ_2, and an intrinsic SHORTEST-PATH geometry that is ℓ_1 — and the curvature functional must be attached to the first, not the second. Three repairs are logically available. (A) Re-found the curvature functional on the low-energy wave/Laplacian operator rather than on hop count, seeking a generator-based curvature (Γ_2/Bakry-Émery is the obvious candidate class) — this fits the dispersion result already in hand, but it is a NEW gravity route and [8], [9] would no longer be the proof. (B) Compute curvature on an auxiliary mesoscopic graph with densifying directions while keeping degree-6 as the causal coupling graph — which then requires an argument that the Jacobson area/deformation variable belongs to the auxiliary graph, whereas the proof above identifies them. (C) Give the reference adjacency increasing angular resolution, which contradicts the bounded-degree range-1 architecture that the area law, the Bell-locality analysis and the gauge derivation all rest on, and is by far the most expensive. Route (A) is the one this paper's own results most nearly support. Until one is carried out, the continuum step is open at the level of its construction. Kelly et al. [9] control the integrated curvature limit, but under a Gromov–Hausdorff closeness hypothesis on the whole graph sequence, which a macroscopic metrically cheap preparation-indexed edge destroys outright — so [9] does not transfer to a prepared family at all, independently of sparsity. Consequently the continuum Einstein conclusion is conditional: on the $D = 3$ intrinsic-hop convergence above for the reference geometric state class, and separately on a metric/curvature convergence for any prepared family, which no cited theorem supplies. □

Sparse preparation-indexed edges: local-contamination diagnostic. Let $G_N^{(0)}$ be a geometric reference graph on N vertices and let a preparation add L_N long edges while keeping the added degree bounded. For a mesoscopic Ollivier radius r_N, write $m_N = \max_v |B_{G_N^{(0)}}(v, 3r_N)|$. At most $2L_N m_N$ vertices have a $3r_N$ reference ball containing an endpoint of an added edge, so a uniformly sampled mesoscopic neighborhood is uncontaminated with probability at least $1 - 2L_N m_N/N$. Thus

$$\frac{L_N m_N}{N} \longrightarrow 0$$

is a sufficient sparsity condition for *typical local neighborhoods* to retain the reference graph geometry. In a three-dimensional geometric scaling with $m_N \asymp N\delta_N^3$, this diagnostic becomes $L_N \delta_N^3 \to 0$. This is not a theorem that κ_{OR} converges everywhere: it does not control curvature at shortcut endpoints, global shortest-path distortions, or the integrated curvature sum. It isolates the exact remaining Bell–gravity question for the parameter-dependent branch: whether preparation-indexed Bell edges can satisfy a curvature-stability condition strong enough to extend step (iv) while being available to the prepared entangled pairs.

Every orbit is self-consistent by construction: the state determines the graph, which determines the entropy and temperature, and the Clausius relation constrains the graph's response to energy flux — paralleling continuum GR where the Einstein equations are the equations of motion, not constraints on separate dynamics. The mathematical framework is unchanged: the full dynamics F(u, v) = (v, WE_{G(v)}(v) − u) is a single bijection on the finite phase space. The ontological object is still (S, F, V).

Lattice corrections to Einstein's equations. The derivation invokes continuum results in the final step (Jacobson). Decomposing Jacobson's argument into its four inputs and checking each independently: (i) $\delta S = \eta\, \delta A/\varepsilon^2$ (area-law entropy) — proved for Gaussian systems over $\mathbb{R}$ [3] and for any nearest-neighbor dynamics over $\mathbb{Z}/q\mathbb{Z}$ via the spatial Markov property (area-law lemma above). No Lorentz invariance enters. **Exact on the lattice.** (ii) $T = \hbar\kappa/(2\pi c k_B)$ (Unruh temperature) — requires the Bisognano-Wichmann theorem. On the lattice, the BW form is proved analytically for coupled harmonic oscillator systems [6] and shown robust against UV modifications of the dispersion relation with corrections at $O(\varepsilon^2\kappa^2/c^2)$. **Approximate.** (iii) $\delta Q = \int T_{\mu\nu} k^\mu d\Sigma^\nu$ (energy flux) — a definition on the emergent manifold. No Lorentz invariance required. **Exact.** (iv) The Raychaudhuri equation — kinematic: it holds on any pseudo-Riemannian manifold as a consequence of the definition of curvature. **Exact.** Since inputs (i), (iii), and (iv) are lattice-exact, corrections enter only through input (ii):

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \times \left[1 + \mathcal{O}\left(\frac{\varepsilon^2\kappa^2}{c^2}\right)\right]$$

For the cosmological horizon, $\varepsilon\kappa/c^2 = 2l_P \cdot H/c \sim 10^{-61}$, giving corrections of order 10^{-122}. For solar-mass black holes, $\varepsilon\kappa/c^2 \sim 10^{-38}$, giving corrections of order 10^{-76}. The corrections are negligible for any horizon much larger than the Planck scale.

The $\eta = 1/4$ coefficient. The Bekenstein-Hawking formula $S = A/(4l_P^2)$ follows from $\varepsilon = 2l_P$ (the gap equation [GR, §3]): one entropy unit per cell of area $\varepsilon^2 = 4l_P^2$, giving $S = A/\varepsilon^2 = A/(4l_P^2)$. This cannot be confirmed via entanglement entropy because of the species problem — the entanglement entropy coefficient is species-dependent (Srednicki found $\eta_{\text{ent}} \approx 0.295$ per scalar field), while the BH entropy is species-independent. The thermal matching route avoids this by counting classical boundary degrees of freedom (one per ε^2 cell, independent of field content) rather than any particular field's vacuum entanglement.

3.2 Why d = 3

The theorems work for any d $\geq$ 1. Three independent self-consistency filters converge on d = 3.

Propagating gravity requires d $\geq$ 3. The Weyl tensor vanishes for d $\leq$ 2. The Jacobson derivation requires propagating gravitational degrees of freedom.

Stable matter requires d $\leq$ 3. The Coulomb potential in d dimensions gives unstable atoms for d $\geq$ 4 (Ehrenfest [10]; the non-existence of normalisable bound states of the d-dimensional Coulomb Hamiltonian for d $\geq$ 4 is the corresponding quantum-mechanical statement). Gravitational orbits are unstable for d $\geq$ 4 [11]. Without stable matter, no observers exist to define the partition.

The intersection of d $\geq$ 3 and d $\leq$ 3 is d = 3, already uniquely. The remaining filter is a further consistency check rather than a logically independent constraint.

On circularity. The stable-matter filter invokes the existence of observers ("without stable matter, no observers exist to define the partition"), and a reasonable objection is that this imports a known fact about our d = 3 world — that stable matter exists — to select d = 3, rather than deriving d = 3 from the substratum. If the selection rested *only* on this filter, the objection would have force: it would be an anthropic selection presented as a derivation. The objection is answered by the third filter, which is purely structural and observer-independent. The boundary-entropy concordance below is a property of the partition geometry alone — it is computed from the Friedmann equation, the shell theorem, and horizon-mode counting ([GR §7.2]), none of which mention observers or matter stability — and it equals unity only at d = 3. This filter selects d = 3 without invoking the existence of observers at all. The logical structure is therefore: propagating gravity ($d \geq 3$) and stable matter ($d \leq 3$) together fix $d = 3$, and the circularity worry attaches only to the anthropic-flavored stable-matter step. That worry is answered by the boundary-entropy concordance — a non-anthropic filter computed from partition geometry with no reference to observers or matter stability — which independently excludes $d \geq 4$ by equalling unity only at $d = 3$. A referee who rejects the stable-matter filter as circular is therefore left with propagating gravity plus the boundary-entropy concordance, which still yield $d = 3$: the anthropic step is *substitutable*, not indispensable. This does not make the selection a pure a priori derivation — the concordance filter uses the observed near-flatness, so $d = 3$ remains a self-consistency selection that consumes d-dependent emergent physics (§8.3), independently anchored by the observed gauge group — but it does defeat the specifically anthropic form of the circularity objection.

Boundary-entropy concordance (d = 3 gives $\rho_s = \rho_{\mathbf{crit}}$). In d spatial dimensions, the boundary entropy ratio is:

$$\frac{\rho_s}{\rho_{\text{crit}}} = \frac{2}{d-1}$$

This equals unity only for d = 3. For d = 2: overclosure. For d ≥ 4: gravitational deficit with no source.

d	ρ_s / ρ_crit	Status
2	2	Overclosure
3	**1**	**Exact concordance**
4	2/3	Deficit

The d = 3 case is derived explicitly in [GR §7.2] from the Friedmann equation, shell theorem, and horizon-mode counting. The general-d formula is obtained by running the same construction with Friedmann in d spatial dimensions, $H^2 = 16\pi G_d \rho/[d(d-1)]$.

Self-contained statement of the concordance ratio. Because the circularity rebuttal above relies on this filter being structural and observer-independent, the origin of the ratio $\rho_s/\rho_{\text{crit}} = 2/(d-1)$ is stated here rather than only cited. Two densities are compared, both computed from the partition geometry alone. (a) The *critical density* is read off the d-dimensional Friedmann equation $H^2 = 16\pi G_d\, \rho/[d(d-1)]$ at spatial flatness, giving $\rho_{\text{crit}} = d(d-1)H^2/(16\pi G_d)$ — this is the energy density at which the spatial sections are flat, a property of the gravitational dynamics with no reference to observers or matter content. (b) The *boundary (horizon) entropy density* ρ_s is the energy density associated with the horizon-mode count via the holographic area law: the number of boundary modes scales as the horizon area $A \sim r_H^{d-1}$, the energy per mode is set by the horizon temperature $T_H \sim 1/r_H$, and dividing the resulting boundary energy by the enclosed volume $V \sim r_H^d$ gives $\rho_s \sim r_H^{d-1} \cdot r_H^{-1}/r_H^d = r_H^{-2} \sim H^2$ with a d-dependent geometric coefficient. Carrying the coefficients (horizon area formula, mode counting, and the Friedmann normalization, all in d spatial dimensions) yields the ratio $\rho_s/\rho_{\text{crit}} = 2/(d-1)$. Neither density uses the existence of observers, stable matter, or any anthropic input; both are computed from the gravitational dynamics and the holographic mode count, which is why this filter is independent of the stable-matter filter and answers the circularity objection. *What is imported from [GR]:* the horizon-mode counting is derived in [GR §3] and the area-law coefficient in [GR §5], the latter conditional on the horizon conditions H-slope, H-frame and H-Hawking of [GR §2, §8.5]; this paragraph states the structure of the ratio and its observer-independence but relies on [GR] for the coefficient derivation, which a referee checking the concordance filter would verify there. The qualitative result — that the ratio depends only on d and equals unity solely at $d = 3$ — does not depend on the coefficient details, only on the scaling $\rho_s \sim H^2$ and $\rho_{\text{crit}} \sim H^2$ with the stated d-dependence.

Downstream consequence on the H-link gauge branch: renormalizability at $d = 3$. If the H-link/H-cust construction yields the stated Yang–Mills sector, its gauge coupling is dimensionless in 3+1 dimensions, $[g] = (3-d)/2 = 0$. This is a property of that conditional emergent QFT branch; it is not an independent proof of the carrier count or of H-link.

The intersection of the three filters — d $\geq$ 3, d $\leq$ 3, and d = 3 concordance — is d = 3 uniquely. Removing the concordance filter still gives d = 3 from the other two; adding concordance sharpens to a consistency check that the framework passes exactly rather than by a numerical coincidence.

Integer dimension is derived, not presupposed. The self-consistency argument might appear to presuppose that the coupling graph has a well-defined integer dimension. In fact, every bounded-degree graph falls into one of three growth classes: exponential ($|B(r)| \sim k^r$), fractal ($|B(r)| \sim r^\alpha$ with non-integer α), or integer-polynomial ($|B(r)| \sim r^d$ with integer d). Self-consistency excludes the first two.

Exponential growth fails because the wave equation on such graphs does not produce Lorentz-invariant dispersion (the spectral gap doesn't close correctly), so Jacobson's derivation of Einstein's equations fails. Additionally, $d_{\text{eff}} \to \infty$ is incompatible with stable matter ($d \leq 3$).

Fractal growth fails because Jacobson's argument requires the Raychaudhuri equation, which requires geodesic congruences on a manifold — fractal spaces don't support them. The staggered fermion decomposition requires a hypercubic BZ structure ($\eta \in \{0, 1\}^d$), which doesn't exist on fractal graphs. The cubic group decomposition of $2d$ link directions — which constrains the gauge structure (§4.6) — requires cubic lattice symmetry, absent in fractals.

Only *integer-polynomial growth* survives. By Gromov's theorem (finitely generated groups of polynomial growth are virtually nilpotent) and statistical isotropy, the graph is quasi-isometric to $\mathbb{Z}^d$ for integer d. The three filters then select $d = 3$.

Corollary ($d = 3$ is necessary, not contingent). *Any bijection whose coupling graph has $d \neq 3$ produces internal contradictions in its emergent description. The coupling graph dimension is not a free parameter or a property of a specific φ — it is the unique value compatible with self-consistency.*

Relation to the anthropic principle. This is not a standard anthropic argument. The standard argument invokes a multiverse of dimensions and selection bias: many d exist; we observe $d = 3$ because only there can observers exist. The present argument is different: the framework's definition requires an observer as a constitutive element (the partition is defined relative to an observer). The requirement that such an observer exist, combined with the framework-internal concordance calculation, constrains d. The observer is not selecting from a pre-existing landscape but is a structural prerequisite for the definition to have content.

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4. The Emergent Quantum Field Theory

This section studies the observer-level field-theory branch associated with the $d = 3$ cubic substratum. The new projection theorem below fixes its quadratic spatial isotropy directly from finite equivariant observation, without assuming QM. The stronger identification of the observer kernel with the normalized nearest-neighbor wave operator is conditional on observer-level center-freedom, and the identification of the six geometric links with the physical gauge carrier is the named H-link bridge. The subsequent Standard-Model calculations are therefore layered: exact representation-theoretic statements are separated from the physical-carrier assumptions they use.

4.1 Observer-level propagation and the lattice Klein–Gordon branch

The OI substratum is represented here by a d-dimensional hypercubic coupling graph $\Lambda = \mathbb{Z}^d$ with microscopic spacing $\epsilon = 2l_p$. The finite bijection is fundamental; a real/complex wave operator is an observer-level lift whose derivation must be stated separately. The following selection statement therefore classifies the *form* of a nearest-neighbor center-free linear rule, while Theorem 1a and its corollaries determine what finite equivariant observation forces at low momentum.

Lemma (form of the center-free isotropic linear rule). *At either the substratum-symbolic level or an observer-level linear lift, a second-order reversible nearest-neighbor rule that is center-free, cubic-isotropic, and linear has the form $x_i(t+1) = \alpha \sum_{j\sim i} x_j(t) - x_i(t-1)$ for one coefficient α. These conditions fix the form, not the observer-level normalization of α.*

Remark (scope). The nearest-neighbor restriction is an assumption beyond bounded coupling degree (A3): center-independent families with next-nearest-neighbor couplings exist and tune the low-energy speed. The assumption is not rescued by equivariance: the range and the self-weight of the projected Markov part are set by the hidden conditional law rather than by the microscopic radius (a correlated hidden preparation of range two makes the visible one-step law range two), and a block's low-energy speed is the second moment of its memory-resummed kernel rather than its range — a range-two kernel with the nearest-neighbor second moment keeps the cone exactly, while a gapped hidden sector coupled through a derivative lowers it by the squared coupling over the gap. None of these moves one block's cone relative to another's unless the signed-permutation symmetry of the components is explicitly broken (§4.4, Proposition 4b). Nearest-neighbor is adopted as the minimal local branch used by the observer-level spectral calculation; the dispersion — and the 4/45 coefficient of §6.4 — is conditional on that branch and on the coarse-graining bridge of §4.1.

Proof. The update function has the form $x_i(t+1) = (f(\text{neighbors of } i \text{ at time } t) - x_i(t-1)) \bmod q$. Center independence requires that f not depend on x_i itself: f reduces to a function of the $2d$ neighboring values only. Spatial isotropy requires f to be symmetric under all permutations of the neighbors corresponding to lattice symmetries. Linearity over $\mathbb{Z}/q\mathbb{Z}$ requires $f(a+a') = f(a)+f(a')$. The unique function satisfying all three is $f = \alpha(x_1 + x_2 + \cdots + x_{2d}) \bmod q$ for some constant α. This is the discrete wave equation. $\square$

Remark (the coefficient's reading). α is neither the propagation speed nor free to be set to 1. First, microscopic causal speed is not set by the coefficient: by [Main]'s cone theorem, $(\varphi^k(s))_i$ depends only on initial variables within graph distance k, so one update propagates dependence at most one edge whatever α multiplies the neighbour sum. Conversion of that cone into a physical c belongs to the emergent clock/length calibration, not to α. Second, the real harmonic lift used throughout §4 is UNSTABLE at $\alpha = 1$: $x(t+1) = \alpha \sum_{\text{nbrs}} x(t) - x(t-1)$ gives $\cos\omega = \alpha \sum_j \cos(k_j a)$, so at $k = 0$ and $d = 3$, $\cos\omega = 3$ and ω is complex. The lift is stable exactly at $\alpha \le 1/d$, and $\alpha = 1/d$ is also what makes the leading symbol isotropic: $\cos\omega = 1 - (a^2/2d)|\mathbf{k}|_2^2 + O(a^4k^4)$, giving $\omega/(a|\mathbf{k}|) \to 1/\sqrt{d}$ along axis, face- and body-diagonal directions alike (check 13). The normalised operator $A/(2d)$ is already what the spectral-measure computation (`grr_spectral_measure.py`) uses.

Scope (which level $\alpha = 1/d$ belongs to, and what is not yet proved). The fundamental object is the finite bijection φ; [Main §3.2] states that a continuous one-parameter interpolation of a finite permutation is additional structure rather than something the permutation determines, and [Methodology Part V] warns specifically against demanding emergent objects at substratum level. The normalisation therefore most naturally belongs to an OBSERVER-LEVEL operator L_{obs} obtained after the trace-out, not to the substratum update — which also removes an arithmetic obstruction, since $1/d$ need not exist in $\mathbb{Z}/q\mathbb{Z}$ ($1/3$ does not when $3 \mid q$) if it never acts there. **But the map $\varphi \to L_{\text{obs}} \to \Delta_g$ is not proved anywhere in this framework.** [ch19] names preservation of low-energy locality, isotropy and spectral mode count under the substratum-to-observer coarse-graining as open, and that is precisely this bridge. Until it is supplied, §4's dispersion results — including the 4/45 coefficient, the Lorentz-invariance argument and Theorem 6's mode count — rest on an operator whose relation to φ is assumed rather than derived, and the q-independence claim of §2.6 must be read as applying to the observer-level operator rather than to a substratum coefficient.

Each requirement has a physical justification.

Center independence: exact scope after projection. Center independence is a *functional* statement about the microscopic update: the next value at a site contains no explicit copy of that site's present value. It does **not** by itself imply statistical independence of successive visible variables after the hidden sector is marginalized. The reversible two-bit swap $\varphi(x, h) = (h, x)$ is the smallest counterexample: the visible update $x' = h$ contains no x, yet on a correlated ensemble with $h = x$ one has $x' = x$ with certainty. Thus the earlier inference "CI $\Rightarrow X_V(t+1)$ carries no information about $X_V(t)$" is not established. Main's C4/readback and finite-recurrence theorems establish the memory/P-indivisibility content without that inference. CI remains a legitimate structural condition on the center-free/chiral branch (§4.3); whether it descends to a zero observer-level self-weight is an additional coarse-graining condition, isolated below.

Spatial isotropy is required for Lorentz invariance: without rotational symmetry, the dynamics has a preferred spatial direction, breaking the isotropy that Lorentz invariance demands.

Linearity and coefficient selection: revised scope. The strongest non-QM argument retained here is the amplitude-scale-gauge lemma: if the embedded observer has no access to an absolute field-amplitude scale, base-point-independent first differences force an affine rule, and the center-free branch reduces it to a linear one. The earlier coefficient arguments based on "maximum propagation speed" and on $\alpha = 1$ maximizing P-indivisibility do not hold as selectors of the $d > 1$ real lift: microscopic causal speed is fixed by dependency support, while $\alpha = 1$ is outside the stable $d = 3$ harmonic branch. The Unruh/KMS and Lorentz arguments are consequently downstream tests of the *observer-level* continuum operator, not proofs of a substratum coefficient. What is needed is a derivation of the projected operator from finite observation itself.

Theorem 1a (equivariant finite projection preserves spatial symmetry). *Let S be finite, let U be the Koopman operator of a bijection $\varphi : S \to S$ on real observables, and let P be a linear observer projection with $Q = I - P$. For a spatial symmetry representation R_g, suppose $[U, R_g] = [P, R_g] = 0$ for every g in the symmetry group. Define $A = PUP$, $B = PUQ$, $C = QUP$, and $D = QUQ$. Then the exact projected evolution is*

$$p_{t+1} = Ap_t + \sum_{s=0}^{t-1} BD^{t-1-s}C\, p_s + BD^t q_0,$$

and every memory kernel $K_m = BD^mC$ commutes with every R_g.

Proof. The P/Q decomposition gives $p_{t+1} = Ap_t + Bq_t$ and $q_{t+1} = Cp_t + Dq_t$. Iterating the second equation gives $q_t = D^t q_0 + \sum_{s=0}^{t-1} D^{t-1-s}Cp_s$, which yields the displayed identity. Since $Q = I - P$ commutes with R_g, so do A, B, C, D, and hence every product BD^mC. □

Attribution. The exact projected identity above — a Markov term plus memory kernels acting on the projected history, with an initial-condition term from the unobserved sector — is standard Mori–Zwanzig/Nakajima–Zwanzig structure and is not claimed as new. What is specialized to this framework is the combination: a FINITE deterministic bijection in place of a continuous flow, an observer projection that is equivariant for the substratum's point group, cubic invariant theory applied to the resulting kernels, and the operationally finite reading of what the observer can resolve. The conclusion those support is that neither fundamental quantum structure nor a microscopic Riemannian metric is needed for quadratic observer-level isotropy. Theorem 1a is an exact projection result specialized to this setting.

Lemma 1c (canonical projection and the resummed kernel). *Let μ be the selected invariant measure — the counting measure of [Main] Lemma 3, or a stated structured prior — and $P_\mu = \mathbb{E}_\mu[\cdot \mid \sigma(X)]$. (i) For every measure-preserving visible-fixing relabelling S of the total system — $X \circ S = X$ and $S_*\mu = \mu$ — P_μ commutes with the Koopman operator R_S, so the conjugate dynamics $U' = R_S U R_S^{-1}$ has the same canonical Markov part, $P_\mu U' P_\mu = P_\mu U P_\mu$; the coordinate-block projection of Theorem 1a has no such invariance — the shear $S(x, y) = (x,\, y - Lx)$ conjugates the second-order rule to one whose block Markov part vanishes while the visible recurrence $x_{t+2} = Lx_{t+1} - x_t$ is unchanged. (ii) The visible poles are the poles of the reduced resolvent $P(\lambda - U)^{-1}P = \mathcal{D}(\lambda)^{-1}$, with $\mathcal{D}(\lambda) = \lambda - A - B(\lambda - D)^{-1}C$ the memory-resummed Schur complement continued meromorphically through the spectrum of D; the observer-level dispersion is the zero set of that continued $\mathcal{D}$, and it depends only on the visible σ-algebra, not on the coordinate split. (iii) On the standard second-order companion realization of Theorem 1 — current field visible, retarded register hidden,*

$U = \left(\begin{smallmatrix} A & -I \\ I & 0 \end{smallmatrix}\right)$ — *with the Corollary-1b kernel A, $K_0 = -I$ and $K_{m\geq 1} = 0$, so $\mathcal{D}(\lambda) = \lambda - A + \lambda^{-1}$ and the dispersion is $\cos\omega = \widehat{A}(\mathbf{k})/2$.*

Proof. (i) Conditional expectation is covariant under a measure-preserving map that fixes the conditioning variable, and R_S acts trivially on functions of X; the counterexample is a direct computation. (ii) Block inversion of $\lambda - U$ off the spectrum of D, where $\det(\lambda - U) = \det(\lambda - D)\det\mathcal{D}(\lambda)$; the identity extends as a polynomial identity after clearing denominators, and factors cancelling against $\det(\lambda - D)$ are removable. (iii) Substitute $D = 0$, $B = -I$, $C = I$. □

The statements below are made on the companion realization of (iii), where the Markov part and the resummed kernel coincide; away from it the kernel, not the Markov part, carries the dispersion.

Corollary 1a (cubic symmetry FORBIDS quadratic anisotropy). *Suppose a scalar projected kernel is translation invariant and equivariant under the full cubic point group O_h. (i) If it has finite second absolute moment, then near $\mathbf{k} = 0$ its Fourier symbol has the form*

$$\widehat{K}_m(\mathbf{k}) = a_m - b_m|\mathbf{k}|_2^2 + o(|\mathbf{k}|^2).$$

(ii) If it has finite fourth absolute moment — in particular if it has finite range — the remainder sharpens to $O(|\mathbf{k}|^4)$, and that quartic term need not be rotationally invariant.

Scope. The content is negative: equivariant finite observation FORBIDS quadratic spatial anisotropy. It does not by itself produce a Euclidean metric, because the corollary does not fix the sign or nonvanishing of b_m. If $b_m = 0$ the quadratic term is absent and the kernel defines no metric at that order at all; a nondegeneracy condition $b_m \neq 0$ (with the sign fixed by stability) is an ADDITIONAL requirement, not a consequence of O_h equivariance. The theorem's honest form is therefore: whatever quadratic spatial structure survives projection is isotropic — not that quadratic structure survives. The two moment conditions are not interchangeable: a symmetric kernel can have finite second moment and infinite fourth moment, and its remainder is then nonanalytic between quadratic and quartic order, so clause (i) cannot be strengthened to $O(|\mathbf{k}|^4)$ without the further hypothesis of clause (ii). The negative content is carried by (i) alone; clause (ii) is what licenses reading the surviving anisotropy as quartic.

Proof. Inversion in O_h removes odd powers. Permutations and sign changes of the coordinate axes force the quadratic tensor to be proportional to δ_{ij} — equivalently, the space of O_h-invariant symmetric quadratic forms on $\mathbb{R}^3$ is one-dimensional, spanned by δ_{ij}. For (i), $|\cos x - 1 + x^2/2| \leq \min(x^2, |x|^3/6)$ and dominated convergence against the finite second moment give the $o(|\mathbf{k}|^2)$ remainder. For (ii), the finite fourth moment dominates the quartic Taylor remainder directly. At quartic order the independent cubic invariants $\sum_i k_i^4$ and

$\sum_{i<j} k_i^2 k_j^2$ remain, so exact microscopic rotational symmetry is neither assumed nor obtained. $\square$

This is the observer-level isotropy result needed by the framework: discrete cubic symmetry is inherited through *exact projected memory*, and any nondegenerate leading scalar propagation geometry is therefore isotropic, without assuming quantum mechanics or unlimited experimental resolution; the coefficient b_m may vanish, in which case no metric arises at that order, so nondegeneracy is supplied separately by the normalized canonical branch. Finite observation does not excuse an $O(1)$ anisotropy in the quadratic term; it only makes higher-order lattice corrections operationally suppressible on a bounded low-momentum envelope.

Corollary 1b (when the normalized adjacency is forced). *Let the observer-level Markov part T be scalar, translation invariant, nearest-neighbor local, O_h-equivariant, and constant-preserving. Then*

$$(Tf)(n) = p_0 f(n) + p \sum_{j=1}^{d} [f(n+\hat{e}_j) + f(n-\hat{e}_j)], \qquad p_0 + 2dp = 1.$$

*If, in addition, the **observer-level** kernel is center-free ($p_0 = 0$), then $p = 1/(2d)$ and $T = A/(2d)$.*

This derivation is classical. It uses symmetry and normalization of the observer kernel, not the Born rule. The extra center-free clause is load-bearing: microscopic CI alone does not imply $p_0 = 0$ after correlated hidden degrees of freedom are traced out, as the swap counterexample above shows. Thus $A/(2d)$ is a theorem **conditional on observer-level CI**, while the descent of CI through the substratum-to-observer map remains open.

Theorem 1 (conditional observer-level lattice Klein–Gordon branch). *If the scalar observer-level branch satisfies Corollary 1b and the reversible second-order temporal update, then*

$$\phi(\mathbf{n}, t+1) = \frac{1}{d} \sum_{j=1}^{d} [\phi(\mathbf{n}+\hat{e}_j, t) + \phi(\mathbf{n}-\hat{e}_j, t)] - \phi(\mathbf{n}, t-1)$$

has a massless low-energy Klein–Gordon limit. Its harmonic dispersion is $\cos\omega = (1/d)\sum_j \cos(k_j a)$, so $\omega^2 = (a^2/d)|\mathbf{k}|_2^2 + O(a^4|\mathbf{k}|^4)$ after the corresponding clock calibration.

4.2 Factorization into Dirac operators

Theorem 2 (Susskind [12], conditional branch). $D_{st}^2 = -\frac{1}{4}\Box_{lat}$. *On the observer-level lattice-wave branch of Theorem 1, this factorization gives the corresponding massless staggered Dirac operator.*

The bosonic and fermionic descriptions are related by an exact change of variables on the lattice.

Remark (measure reduction). Theorem 2 is an operator identity: it identifies $D_{\rm st}^2$ with $-\frac{1}{4}\Box_{\rm lat}$ acting on lattice fields. For the standard staggered ChPT machinery used in §6 and §7.5 to apply with its conventional measure factors (e.g., the $1/(16\pi^2 f^2)$ measure factor of Aubin–Bernard SχPT), one needs a stronger statement: that the substratum's path-integral measure on real scalar bijections $\varphi : S \to S$ reduces, under the Susskind change of variables, to the standard Berezin pseudofermion measure on staggered fermion fields in the relevant correlator sector. This measure-equivalence statement is established at Level 2 rigor as follows.

The substratum is real-valued and, on linear dynamics with energy conservation, has a stationary distribution that is Gaussian on energy shell at large L (sphere-Gaussian projection theorem). The substratum two-point function is then $\langle\varphi(x)\varphi(y)\rangle = [\Box_{\rm lat}^{-1}]_{xy}$. The standard MC code (cf. §7.5) does not use Grassmann variables directly; it uses the *pseudofermion* representation, in which fermion determinants are rewritten as bosonic Gaussian integrals over auxiliary fields with covariance $(D_{\rm st}^\dagger D_{\rm st})^{-1}$. With $D_{\rm st}^\dagger = -D_{\rm st}$ (anti-Hermiticity of the staggered Dirac operator), $D_{\rm st}^\dagger D_{\rm st} = -D_{\rm st}^2 = +\frac{1}{4}\Box_{\rm lat}$ by Theorem 2, so the pseudofermion covariance is $4\,\Box_{\rm lat}^{-1}$. Both substratum and pseudofermion measures are bosonic Gaussian; their covariances differ by a factor of 4 absorbable into field normalization. The measure equivalence is therefore established up to (i) the standard sphere-Gaussian projection at large L (classical), and (ii) cycle ergodicity for the specific bijection determining which energy shell is sampled.

The cycle ergodicity assumption deserves explicit acknowledgment. Pure linear dynamics on a cubic lattice is *integrable*, not chaotic: the lattice frequencies $\omega(\vec{k}) = 2\sin(|\vec{k}|/2)$ are not linearly independent over $\mathbb{Q}$, so orbits fill a sub-torus of the energy shell rather than the full shell. The Gaussianization argument therefore has two natural interpretations: (way out i) the cubic-lattice + finite-set discretization introduces small but non-zero non-linearity (from rounding rules in the bijection update), providing the chaos needed for full-shell sampling; or (way out ii) the "Gaussian on energy shell" claim is shorthand for "Gaussian on the sub-torus actually sampled," which is a smaller phase space and would modify the substratum-pseudofermion covariance match. Way out (i) is the standard interpretation for lattice-substrate frameworks. Numerical verification of way out (i) for OI's specific bijection has been carried out via two-trajectory separation with per-site RMS amplitude measurement inside the spreading cone (the correct Lyapunov-relevant diagnostic, distinguishing chaotic amplitude amplification from Lieb-Robinson ballistic spreading). The measurement gives a chaotic Lyapunov exponent $\lambda_{\rm max} = 0.0092 \pm 0.0001$ per timestep in the thermodynamic limit, $N_{\rm levels}$-independent for $N_{\rm levels} \geq 16$, with exponential amplitude growth in the cone confirmed against power-law alternatives at high statistical significance (exponential-vs-power-law residual ratio ~ 2.2 across $\log(\|\Delta\phi\|/\delta) \in [0, 1.0]$). Way out (i) is therefore *empirically demonstrated* rather than merely adopted: the rounded bijection is genuinely chaotic, the integrable linear dynamics having been broken by the rounding rule's threshold-crossing nonlinearity. The cycle-ergodicity question for OI's specific bijection is closed at chat-track numerical scale; analytical derivation of the $\lambda_{\max}$ coefficient and the structural source of N_{levels}-independence remains open and would elevate this result to Level 2(a) in the bridge-gap classification of §7.5. Four further diagnostics on the rounded bijection fall on the chaotic side: spectral continuity of local observables (broadband, versus the linear control's discrete peaks), thermalization (autocorrelation decay to the noise floor, versus persistent coherent oscillation in the linear control), comparative cycle-structure consolidation in an exactly enumerable one-dimensional analog (the rounded map's dominant cycle is $\sim 400\times$ larger than the linear control's), and orbit non-fragmentation in direct three-dimensional sampling (no recurrences or trajectory merges). The quantitative rate is preparation-ensemble- and protocol-dependent: at an infinite-temperature ensemble the separation shows an incubation-then-avalanche profile with an effective rate several times the value quoted above, so the quoted $\lambda_{\max}$ is specific to its preparation ensemble and fit protocol. The qualitative chaos, which is what the ergodicity interpretation requires, is robust across protocols.

The two roles of the dynamics, stated explicitly. The substratum update carries two structurally distinct pieces, and the framework's results divide cleanly between them. The *exact* dynamics is the linear rule $f = \sum_{\text{nn}} x - x(t-1)$ over $\mathbb{Z}/q\mathbb{Z}$; the finite-state arithmetic (the mod-q wraparound / threshold-crossing at the alphabet boundary) is a bounded nonlinearity that is present in any faithful finite realization but vanishes in the continuum/large-q limit. The *structural* chain — the wave-equation uniqueness (§4.1), the Susskind factorization (Theorem 2), the $(3, 2, 1)$ decomposition and the gauge group (§4.6), and the amplitude-scale-gauge argument for linearity (§4.1) — reads only the linear part: it is exact modular-linear algebra and does not invoke the threshold nonlinearity. The *ergodic/measure* properties — full-shell sampling for the Gaussianization of §4.2, the chaotic mixing underlying operational measurement independence ([Main] §3.3), and the ETH-conditioning of the C2 necessity direction ([Main] §3.4) — read the threshold nonlinearity, which is what breaks the integrability of the strictly-linear map (way out (i) above). The two are not in tension within the local lattice sector: they are different features of one update, consumed by different parts of the framework, and the ergodic properties ride on a bounded correction that leaves the structural (continuum-limit) content untouched. The one commitment this makes explicit is that the amplitude-scale gauge of §4.1 is exact at the level of the structural chain (where the threshold nonlinearity is absent) and holds up to the bounded finite-state correction at the level of the realized bijection; a full account of how large that correction is on each ergodic observable is the open residual the $\lambda_{\max}$ measurement begins to quantify.

4.3 Center independence and chiral symmetry

Theorem 3. *Center independence of the lattice dynamics is equivalent to exact chiral symmetry of the emergent staggered fermions.*

Proof. The staggered mass term squares to $\Box_{\mathrm{lat}}\phi = -4m_{\mathrm{lat}}^2\phi$: center dependence. Conversely, center independence gives $D_{\mathrm{st}}\chi = 0$, invariant under $\chi \to \varepsilon\chi$ (chiral symmetry). $\square$

Remark (scope of the center-dependence–mass identification). The specific self-term $C = 2(1-d)$ is not a mass term: it is the diagonal of the massless 4D lattice Laplacian, whose standard 4D staggered factorization has exact chiral symmetry. The hypercubic alternative is not excluded by this lemma. The later matter/gauge reading uses the spatial cubic group and the exact six-link $(3,2,1)$ representation, but identifying that link representation with the physical carrier is H-link (§4.5–4.6).

Revised chain. Center independence implies the chiral statement of Theorem 3 on the staggered branch; it is not derived here from P-indivisibility. Main's C4/readback theorem supplies the memory/P-indivisibility content independently. Any claim that one algebraic CI condition simultaneously forces backflow and chirality is therefore unavailable.

4.4 Multi-component dynamics and gauge structure

Each site now carries a K-component vector $\phi(\mathbf{n},t) \in (\mathbb{Z}/q\mathbb{Z})^K$. The general second-order linear update is $\phi(\mathbf{n},t+1) = C\,\phi(\mathbf{n},t) + \sum_j M^{(j)}[\phi(\mathbf{n}+\hat{e}_j,t) + \phi(\mathbf{n}-\hat{e}_j,t)] + D\,\phi(\mathbf{n},t-1)$, where $C, D, M^{(j)} \in \mathrm{Mat}(K)$. Reversibility requires $D = -I_K$; center independence requires $C = 0$; spatial isotropy requires $M^{(j)} = M$ for all j. This fixes the isotropic matrix form. On the observer-level normalized branch of Corollary 1b, it is convenient to factor the scalar normalization explicitly:

$$\phi(\mathbf{n},t+1) = \frac{1}{d}M\sum_{j=1}^{d}\left[\phi(\mathbf{n}+\hat{e}_j,t) + \phi(\mathbf{n}-\hat{e}_j,t)\right] - \phi(\mathbf{n},t-1).$$

Here $M \in \mathrm{Mat}(K)$ parametrizes internal mixing relative to the normalized scalar branch. This equation is not identified with the fundamental finite bijection without the coarse-graining bridge of §4.1.

Theorem 4 (conditional observer-level mass spectrum). *For the normalized matrix branch above, each eigenvalue μ_a of M determines*

$$\cos\omega_a(\mathbf{k}) = \frac{\mu_a}{d}\sum_{j=1}^{d}\cos(k_j a).$$

The $\mathbf{k} = 0$ *mode is massless iff* $\mu_a = 1$*, and real-frequency stability for all* $\mathbf{k}$ *requires* $|\mu_a| \le 1$*.*

Proof. Diagonalize M and substitute a plane wave. At $\mathbf{k} = 0$, $\cos\omega_a = \mu_a$; the zero-frequency branch therefore has $\mu_a = 1$. Since $|(1/d)\sum_j \cos(k_j a)| \le 1$, $|\mu_a| \le 1$ is sufficient, and $\mathbf{k} = 0$ or the Brillouin-zone opposite corner shows it is necessary. $\square$

The theorem is a statement about the normalized observer-level branch. It does not derive that branch from φ; that is the projection/closure debt isolated in §4.1.

Lemma 4a (one cone on the branch). *On the branch of Theorem 4 every block with* $\mu_a = 1$ *has* $\omega_a^2 = (a^2/d)|\mathbf{k}|^2 + O(a^4k^4)$ *with the same coefficient along every direction, and a block with* $\mu_a = \cos m_a$*,* $0 < m_a < \pi/2$*, has* $\omega_a^2 = m_a^2 + (a^2/d)\, m_a \cot m_a\, |\mathbf{k}|^2 + O(k^4)$*.*

Proof. μ_a multiplies one block-independent kernel; expand $\cos\omega = \mu_a f(\mathbf{k})$ about $\omega = m_a$. $\square$

With the time step at the Planck scale the massive-block deviation $1 - c_a/c \simeq m_a^2/6$ is of order $(m_a/M_{\mathrm{Pl}})^2$, far inside every bound on species-dependent limiting speeds; massless blocks share the cone exactly. The lemma is conditional on the branch: it reads the factorized form, one spatial kernel times M, as given.

Proposition 4b (symmetric-point custodial scalarity for a component-complete site observer). *Let* $M = \mu I_6$*, let the finite-alphabet rule act identically and independently on the six components, take the selected counting measure, which is invariant under signed permutations of the components, and let the observer projection be the conditional expectation onto the* σ*-algebra generated by all six carrier components on one common set of visible sites. Then the complete memory-resummed quadratic response is* $\mathcal{D}_{\mathrm{obs}}(\omega, \mathbf{k}) = d(\omega, \mathbf{k})\, I_6$*, so the ratio of spatial to temporal quadratic coefficients is common to* T_1*,* E *and* A_1 *and the massless blocks share one cone. The same conclusion holds for extensions of the rule whose dynamics, measure and observer* σ*-algebra are equivariant under signed permutations of the six components (H-blind).*

Proof. At $M = \mu I_6$ no component's update depends on another component, so the dynamics is a product of six identical systems; the counting measure is a product of identical factors, and the visible σ-algebra is generated by the same family on every component, so the conditional-expectation projection is equivariant under signed permutations and the reduced response is six identical copies — for any site-based split the visible Schur complement is I_6 tensored with one scalar block. For the extensions the rule, its symmetric-residue real lift, the measure and the observation commute with the hyperoctahedral group by hypothesis, the response inherits the equivariance, and the signed-permutation representation on $\mathbb{R}^6$ is irreducible — its commutant is one-dimensional, against three dimensions for the cubic group acting on the six links — so by Schur's lemma the response is scalar. $\square$

The observer class of the proposition is the branch's own: Theorem 4 takes the observer-level variable to be the six-component field on the observer's sites. The cubic $(3, 2, 1)$ structure is therefore not a property of the symmetric-point dynamics; it enters through the H-link identification of components with links and through the condensate state of Theorem 5. What can split the block cones is an explicit reduction of the signed-permutation symmetry to the cubic group, either in the rule — a hidden sector coupled through one combination of links, or a direction-attached hidden coupling — or in the observation, through a readout that selects components or ties the component index to a spatial direction; the branch posits neither, and for other observer classes H-blind is the named condition. The condensate does not enter the linear transfer function; its derivative dressing of the blocks arises through the emergent interactions and belongs to the matching/running accounting of §3.1, where it enters at one loop, because the substratum's own nonlinearity is component-local and a component-local operator contracted with an O_h-equivariant condensate is sector-common.

Theorem 5 (condensate stabilizer, conditional physical reading). *Assume H-link with one carrier copy and H-cust: the observer-level kinetic structure is custodially scalar on the six-link carrier before the state/condensate is chosen (at the symmetric point $M = \mu I_6$ this clause is supplied by Proposition 4b). Let $\Sigma = \langle \phi\phi^\dagger \rangle$ be O-equivariant. By Schur's lemma, on the exact $T_1 \oplus E \oplus A_1$ link decomposition one has $\Sigma = aP_{T_1} + bP_E + cP_{A_1}$. For generic distinct block values its stabilizer in* $\mathrm{U}(6)$ *is* $\mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$.

Proof. Under H-cust the pre-condensate observer-level kinetic term is invariant under the custodial unitary action on the physical link carrier. The unbroken subgroup of a state with condensate Σ is its unitary stabilizer. The exact cubic decomposition of Theorem 7 and Schur's lemma give the three blocks; distinct block values give the product stabilizer. □

This theorem is exact **conditional on H-link + H-cust**. It does not establish that the six geometric links are the physical carrier, nor that a single copy is selected. The reduction from the natural unitary stabilizer to the Standard-Model group carries the separate amplitude-scale-gauge/background-independence premises discussed in §4.6.

4.5 The six-link carrier: exact geometry, open physical identification

The cubic lattice has an exact $2d$-dimensional signed-link representation

$$V_{\text{link}} = \operatorname{span}\{e_{+1}, e_{-1}, \dots, e_{+d}, e_{-d}\}.$$

What is not established is that the *physical internal carrier* of the observer-level field is necessarily one copy of this link representation.

Lemma 6a (operator-substitution control). *Diagonalizing the single isotropic internal matrix M in §4.4 decouples internal eigenvectors but does* *not attach eigenvector a to one signed spatial link. In the diagonal basis each component still obeys*

$$\phi_a(\mathbf{n}, t+1) = \mu_a \sum_{j=1}^{d} [\phi_a(\mathbf{n} + \hat{e}_j, t) + \phi_a(\mathbf{n} - \hat{e}_j, t)] - \phi_a(\mathbf{n}, t-1).$$

Therefore the former Step 1 equation in which component a reads only $\mathbf{n} + \hat{e}_{j(a)}$ *is not a consequence of §4.4, and the claimed* $\delta(2d) = 1$ *minimization proof is not established.*

H-link (physical-carrier bridge). The observer/gauge carrier is an O_h-equivariant copy of the signed-link representation, possibly with multiplicity,

$$V_{\mathrm{phys}} \cong V_{\mathrm{link}} \otimes \mathbb{C}^m, \qquad m \geq 1.$$

No theorem in the present framework derives $m = 1$ from finite observation, center independence, or the link count; coupling-degree minimization does not establish it either. The single-copy clause $m = 1$ is therefore a named bridge rather than a result.

Theorem 6 (conditional link-carrier count). *Under H-link,* $K = 2dm$*. Under the single-copy clause* $m = 1$*,* $K = 2d$*, and for the selected spatial dimension* $d = 3$*,* $K = 6$*.* □

This reformulation preserves the exact geometric representation theory below while removing a circular carrier derivation. It also blocks an earlier unsupported general-graph extension: a spectral dimension d_s controls heat-kernel scaling but does not canonically furnish $2d_s$ signed propagation channels. H-link is presently a simple-cubic-branch hypothesis; extending it to a broader universality class requires a separate construction.

The status distinction is load-bearing. Statements about the six **geometric link directions** are unconditional mathematics. Statements identifying those six directions with all physical gauge/matter components inherit H-link, and statements selecting a single copy inherit its $m = 1$ clause.

4.6 Cubic decomposition: multiplicities (3, 2, 1)

Theorem 7 (exact link representation; physical reading conditional). *The* $2d = 6$ *nearest-neighbor signed link directions of the* $d = 3$ *simple-cubic lattice decompose under the cubic rotation group as* $6 = T_1(3) \oplus E(2) \oplus A_1(1)$*.* This representation-theoretic statement is unconditional. **Under H-link**, the same multiplicities act on one physical carrier copy. **H-cust** denotes the additional premise used by the condensate-stabilizer route: the observer-level kinetic/condensate structure is scalar on each of these isotypic blocks and does not introduce extra copy-mixing custodial structure. Its kinetic clause is a theorem at the symmetric point (Proposition 4b); its condensate clause remains a premise. Under H-link + H-cust, a generic equivariant condensate has stabilizer $U(3) \times U(2) \times U(1)$; the further reduction to the Standard-Model group carries the amplitude-scale-gauge and background-independence premises stated below.

Proof. Characters at each conjugacy class of O (24 elements): E: $\chi = 6$; $8C_3$: $\chi = 0$; $3C_2$: $\chi = 2$; $6C_2$': $\chi = 0$; $6C_4$: $\chi = 2$. Character inner products give $n(A_1) = 1$, $n(E) = 1$, $n(T_1) = 1$, with $n(A_2) = n(T_2) = 0$. By Schur's lemma and isotropy, any equivariant operator is block-scalar; in particular the condensate is $\Sigma = \mathrm{diag}(aI_3, bI_2, c)$, while the coupling matrix itself is scalar (Remark below). □

The 6D representation in question is the natural action of O on the six nearest-neighbor link vectors $\{\pm\hat{e}_1, \pm\hat{e}_2, \pm\hat{e}_3\}$ of the cubic lattice — the same 6D space on which the coupling matrix $M(\mathbf{n}, \hat{e}_j)$ of §4.4 acts. The decomposition therefore fixes the multiplicities of M's eigenspaces uniquely; there is no freedom in "which 6D rep" is meant.

On the H-link branch the physical identifications proposed here are T_1 (spatial vector) $\to$ color, E (quadrupole) $\to$ weak isospin, and A_1 (scalar) $\to$ hypercharge. Their status is conditional on the physical-carrier bridge and the condensate/gauge premises; the exact result at this stage is the $(3, 2, 1)$ link representation itself. The earlier "max-speed $\Rightarrow \mu = 1$" argument is not established with the coefficient correction of §4.1.

Remark (scope of the former Theorems 4/6/7 chain). The previous proof combined a link-diagonal $\delta = 1$ factorization with the isotropic §4.4 matrix equation and then used a speed argument to select a scalar M. Lemma 6a shows that the link-diagonal equation was an operator substitution, while §4.1 shows the speed argument does not hold. The exact statement is Theorem 7's geometric decomposition. Any physical use of that decomposition now cites H-link, and any stabilizer use also cites H-cust.

Remark (robustness and the cubic commitment). Theorem 7 uses the character table of the octahedral group O, the exact symmetry of the simple cubic (SC) lattice.

Stability under small perturbations. Integer multiplicities cannot change continuously. Small perturbations of the coupling graph that preserve local O symmetry shift the eigenvalues of M continuously but leave the multiplicities $(3, 2, 1)$ intact. The geometric six-link representation is stable under deformations that preserve the local cubic link orbit; H-link itself is not derived from quasi-isometry or spectral dimension.

Dependence on the Bravais lattice. The $(3, 2, 1)$ decomposition depends on the discrete O action on the six NN link vectors of SC. Under the continuous group $O(3)$, the same six vectors span two copies of the vector representation — multiplicities $(3, 3)$, not $(3, 2, 1)$ — because $O(3)$ does not preserve the discrete set of six vectors as a distinguished orbit. The three-factor split requires the discrete subgroup.

Theorem 7b (Bravais-lattice uniqueness). *Among all 14 three-dimensional Bravais lattices, the simple cubic (SC) lattice is the unique candidate compatible with the Standard Model gauge group* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *under the commutant construction of Theorem 5. Furthermore, the inter-generation BZ-distance prescription of §7.1 produces the observed Cabibbo angle* $\lambda = 1/(\pi\sqrt{2})$ *uniquely on SC; BCC and FCC produce* λ *values discrepant with observation by* 42% *and* 29% *respectively.*

The proof has two parts: (a) elimination of all non-cubic Bravais lattices via the crystallographic point group's irrep dimensions, and (b) elimination of BCC and FCC within the cubic system via the explicit O-decomposition of the NN link representation on each, with the SC Cabibbo prediction as quantitative confirmation.

Among the three cubic Bravais lattices, only SC produces the SM gauge group under the commutant construction of Theorem 5. Computing the O-decomposition of the NN link representation on each:

Lattice	NN	O-decomposition of NN rep	Gauge group (commutant)
SC	6	$T_1 \oplus E \oplus A_1$	$SU(3) \times SU(2) \times U(1)$ ✓
BCC	8	$A_1 \oplus A_2 \oplus T_1 \oplus T_2$	$SU(3)^2 \times U(1)^2$
FCC	12	$A_1 \oplus E \oplus T_1 \oplus 2T_2$	$SU(3)^3 \times SU(2) \times U(1)$

SC is uniquely compatible with the observed Standard Model gauge structure. BCC gives two non-abelian $SU(3)$ factors with no $SU(2)$; FCC gives three $SU(3)$ factors and three additional $U(1)$s. Neither matches observation at the group-theoretic level, independent of any numerical prediction.

The Cabibbo angle on the three lattices, computed using the inter-generation BZ distance prescription of §7.1 on each lattice's natural triplet sector (SC: X-points; BCC: H-points; FCC: L-points), with equal-NN normalization:

Lattice	Triplet separation	Cabibbo λ
SC	$\pi\sqrt{2}$	0.2251
BCC	$\pi\sqrt{6}$	0.1299
FCC	2π	0.1592

Observed $\lambda = 0.2250 \pm 0.00067$ [20]. The SC value matches at 0.04% (0.12σ); BCC and FCC miss by 42% and 29% respectively. The SC match is tied to the specific BZ geometry corresponding to the SC gauge group.

The §2 structural reading that "any bounded-degree graph with statistical isotropy suffices" therefore refers to the existence-and-emergent-QM part of the chain (Theorems 1–6, §§3.1–3.4), not to the specific gauge group or quantitative predictions of §§4.6 and 7. These require the SC Bravais structure; the framework treats this as a structural commitment forced by the observed SM gauge group and its quantitative consequences, rather than as a free choice.

Extension to non-cubic Bravais lattices. The uniqueness of SC among cubic Bravais lattices extends to a stronger statement: **among all 14 three-dimensional Bravais lattices, SC is the unique candidate compatible with an $SU(3)$ gauge factor**. The reason is a standard crystallographic fact: among the 32 crystallographic point groups, the 3-dimensional irreducible representations appear only in the cubic crystal system (point groups T, T_h, T_d, O, O_h). Point groups of the hexagonal, trigonal, tetragonal, orthorhombic, monoclinic, and triclinic systems act on $\mathbb{R}^3$ reducibly as (axial direction) + (perpendicular plane), and therefore have irreps only of dimensions 1 and 2 — no 3-dimensional irrep. Theorem 5's commutant construction requires a 3-dimensional irreducible block in the M matrix to produce an $SU(3)$ factor, which requires a 3-dimensional irrep of the point group acting on the NN vectors. No non-cubic Bravais lattice supplies this. The binding constraint is in fact the SU(2) rather than the SU(3): a 3-dimensional irrep is necessary for the color factor but is shared by the tetrahedral group A_4 and the (non-crystallographic) icosahedral group A_5, so it does not by itself single out the cube. The full ladder $(3, 2, 1)$ is equivalent to the point group's irreducible-degree set containing $\{1, 2, 3\}$, realized among finite subgroups of O(3) only by $S_4 \cong O$ — A_4 has a 3 but no genuine 2-dimensional irrep (its apparent doublet is a complex-conjugate pair of 1-dimensional irreps, commutant $\mathrm{U}(1)^2$), and A_5 has 3-dimensional irreps but no 2-dimensional one. The octahedral group is the unique finite rotation group carrying both a genuine 2- and a 3-dimensional irrep, and is the point group of the cubic system; it is specifically the SU(2) factor that forces the octahedral structure. Combined with the result above that BCC and FCC within the cubic system produce the wrong gauge structure, SC is uniquely compatible with the SM gauge group among all 14 Bravais lattices.

Remark (completeness of the irrep catalog). The octahedral group O has five irreducible representations: A_1, A_2, E, T_1, T_2. All five are accounted for in the framework, either by identified physical role or by derived absence:

- A_1: the U(1) factor of the 6-link decomposition (Theorem 7); the singlet-taste scalar that hosts the Higgs (Theorems 8, 11); appears in $\mathrm{Sym}^2(T_1)$ in §7.2 and §8.4.
- E: the SU(2) factor of the 6-link decomposition; controls neutrino mass splittings in $\mathrm{Sym}^2(T_1)$ (§8.4); provides the Koide angle $\theta_0 = 2/9$ via the quadratic Casimir $C_2(T_1)$ normalized by the lattice bandwidth d^2 (§7.2).
- T_1: the SU(3) factor of the 6-link decomposition; the generation space from staggered tastes (Theorems 8, 10); the neutrino flavor basis (§8.4).
- T_2: appears in $\mathrm{Sym}^2(T_1) = A_1 \oplus E \oplus T_2$ and **controls the off-diagonal**

structure of the neutrino mass matrix (§8.4); the resulting mixing angles enter the structural PMNS prediction of §7.3, where TBM (from cubic-group representation theory) plus U-parity-graded charged-lepton corrections at scale λ give all three angles.

- A_2: the pseudoscalar/sign representation; does not appear in the 6-link decomposition ($n(A_2) = 0$ in the character calculation above), does not appear in $\mathrm{Sym}^2(T_1)$, and plays no structural role in the framework. Its absence reflects the parity-even nature of nearest-neighbor coupling on the cubic lattice.

No irrep of O is unaccounted for.

Remark (the substratum-emergent layer split). Part III distinguishes between two layers: the *substratum* — the bijection (S, φ) on the cubic lattice, with its coupling matrix $M(\mathbf{n}, \hat{e}_j)$ and the wave-equation dynamics — and the *emergent* (observer) level — the unitary QFT that an embedded observer reports after the trace-out of [Main]. The SM gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ lives at the substratum: it is the commutant (stabilizer) of the O-equivariant condensate Σ (Theorem 5; the coupling matrix M itself is scalar, $M = \mu I$), a literal algebraic structure on the lattice, existing before any trace-out is performed. The SM gauge *couplings* $\alpha_1, \alpha_2, \alpha_3$ at M_Z live at the emergent level: they arise from the induced coupling $1/\alpha_0 = 23.25$ produced by the fermion determinant (§6.1), followed by thresholds and RG running. The framework's structural commitment is therefore: *group structural, couplings emergent.* This split explains why the gauge group is rigid (no parameter freedom — forced by the commutant structure of Σ) while the gauge couplings admit quantitative predictions at all (they depend on fermion-loop dynamics in the emergent theory, with the substratum's role being to fix $N_f = 6$ and $T(R) = 1/2$ structurally). The same split clarifies the asymmetry noted in §4.7.1.1 between gauge bosons (substratum objects) and fermions (observer-level objects).

4.7 Staggered species, generations, and the Higgs

The Nielsen-Ninomiya theorem [15] requires 2^{d+1} Weyl species. Staggered reduction gives $N_{\text{taste}} = 2^{\lfloor (d+1)/2 \rfloor} = 4$ Dirac tastes in d+1 = 4. (At the free spatial-Schur level the kernel is not a Dirac operator at all — `hspin_kernel_probes.py`, §4.7 Remark — so the theorem's premises are vacuously evaded there; the doubling census is the taste-level statement, and the Nielsen-Ninomiya question becomes live exactly where H-spin' does.)

Theorem 8. *Under the cubic group O:* $4 = \mathbf{1} \oplus \mathbf{3}$. *One singlet taste and three triplet tastes.*

Proof. The 2^d = 8 BZ corners pair into 4 taste pairs under $\eta \leftrightarrow \mathbf{1} - \eta$. The pair {(0,0,0), (1,1,1)} is O-invariant (singlet A_1). The three axis-aligned pairs form irreducible T_1. □

Remark (carrier of the T_1 action). The irreducible T_1 is carried by the signed vector action of the spin-taste matrices: the j-th triplet pair is represented by γ^j (Theorem 9), which transforms as a genuine vector under O. The unsigned corner labels alone carry only the axis-permutation action, which decomposes as $A_1 \oplus E$, since sign flips act trivially on $\{0, \pi\}$ momentum components. The distinction is representation-theoretically real but does not affect diagonal-channel observables: both actions induce the same S_3 permutation on the axis label, so the generation-channel masses of §7.2 — and the Z_3-Fourier amplitude extraction built on them — are insensitive to it. One quantity is *not* covered by this insensitivity and should be flagged: the Koide amplitude and angle depend on the quadratic Casimir $C_2(T_1) = 2$ (§7.2), and the value of that invariant depends on which representation the mass-splitting operator transforms in. The insensitivity established here is to the *permutation* structure of the diagonal channel; the identification that the mass-splitting operator transforms in the signed T_1 (carrying $C_2 = 2$) rather than in the unsigned $A_1 \oplus E$ is a separate, load-bearing input to the Casimir chain. It is supplied by the identification of the generation triplet with the γ^j-carried tastes, and its derivation from the substrate mass operator's transformation law is the same open task as the amplitude closure of §7.2 (the $A^2 = C_2$ dynamical derivation); the pre-registered A^2 measurement tests both together, since a mis-identified carrier would present as a wrong amplitude.

Theorem 8b (Taste coupling structure). *On the $d = 3$ lattice with the normalized wave equation, each BZ corner η has coupling $\mu(\eta) = \sigma(\eta)/d$ where $\sigma = \sum_j \cos(\pi\eta_j)$:*

$$\mu_\Gamma = 1, \quad \mu_X = \tfrac{1}{3}, \quad \mu_M = -\tfrac{1}{3}, \quad \mu_R = -1$$

The staggered taste pairs have couplings: singlet (Γ, R): $|\mu| = 1$; triplet (X_j, M_j): $|\mu| = 1/3$, identical for all $j = 1, 2, 3$.

Proof. Direct computation: $\sigma(\Gamma) = 3$, $\sigma(X) = 1$, $\sigma(M) = -1$, $\sigma(R) = -3$. Dividing by $d = 3$ gives the couplings. The triplet members are related by cubic symmetry (O permutes spatial axes), hence identical. □

Theorem 9 (Spin-taste correspondence — algebraic content; free-kernel identification refuted; H-spin' open). *(a) The singlet pair's spin-taste matrices are SO(3)-scalars. (b) The j-th triplet pair's are the vector γ^j and the spin generator $-i\Sigma_j$. The identification of these labels with the physical spin of the emergent fields — singlet → spin-0, triplet → spin-1/2 — is the named hypothesis H-spin' (Remark below).*

Proof. (a) The singlet pairs $\Gamma(0{,}0{,}0) = \mathrm{I}_4$ (scalar) and $\Gamma(1{,}1{,}1) = \gamma^1\gamma^2\gamma^3$ (pseudoscalar), both SO(3)-invariant: J = 0. (b) The j-th triplet pairs γ^{j} (vector) with −iΣ_j (spin generator): the pair carries the spin-1/2 label content; its identification with physical spin is H-spin'. □

Remark (H-spin' — the physical spin identification; free kernel settled negative). What is proved, and probe-certified (`hspin_labels_probes.py`):

the O-decomposition $4 = \mathbf{1} \oplus \mathbf{3}$ (Theorem 8), the coupling structure (Theorem 8b), and the label content above — the conjugation action carries exactly the T_1 vector characters, the singlet pair is invariant, and the unsigned corner labels carry only $A_1 \oplus E$. What is settled negative (`hspin_kernel_probes.py`, exact and basis-free): on the free spatial-Schur kernel the identification fails — the triplet sector carries the Dirac mass structure exactly ($\{\alpha_j, \beta\} = 0$, $\beta^2 = P$) while the velocity algebra fails the Clifford relations, with per-direction longitudinal zero modes and cross anticommutators of spectrum $\{\pm 1/9, 0\}$; independently, $\nabla_k \omega = 0$ at every corner of the exact dispersion, so no linear cone exists for a Dirac form to inhabit. The original free-kernel form of the hypothesis (H-spin) is thereby refuted. What remains open is the sharpened **H-spin'**, now in narrowed form: whether the interacting, condensate-dressed kernel restores the Dirac form within the framework's second-order-in-time axioms. The four-dimensional space-time staggered route is itself settled negative (`hspin4d_probes.py`, exact): the on-shell corner set is $\{\Gamma, \mathrm{R}\}$ alone, with the six triplet corners gapped by 4/3; every corner linearization vanishes — a structural fact for any $2\cos k_0 - 2c(k)$ cosine-hop symbol — and the on-shell kernel's blocked linearization is identically zero. The dressed kernel is the sole surviving mechanism, and its candidate class is algebraically nonempty — the same spin-taste-locked condensate that carries H-χ''s mechanism (`hchiprime_probes.py`) — with dynamical selection the shared open question. That question has an exact finite arena: the admissible condensates (Hermitian, invariant under the joint cubic action) form a space of dimension 32 which the involution $M \mapsto \xi_5 M \xi_5$ splits evenly into sixteen taste-chirality-blind and sixteen taste-chirality-flipping directions, with M^* in the flipping half (`condensate_space_probes.py`) — so selection is a choice between two equal-dimensional sectors, with no dimensional bias in either direction. The physical spin-1/2 reading of the matter chain rides H-spin', and its discharge test is the Dirac-form test on the condensate-dressed effective kernel.

Corollary (Spin-statistics from the lattice, under H-spin'). Singlet → spin-0 → boson (Higgs). Triplet → spin-1/2 → fermion (quarks and leptons).

Theorem 10 (Three degenerate fermion sectors). *The three triplet staggered tastes provide three degenerate sectors related by cubic symmetry — spin-1/2 sectors, one per SM generation, under H-spin' (Remark, Theorem 9). The SM matter representations are placed on links (§4.7.1.2), where the link-carrier space $V_K \otimes V_K^*$ admits cross-charged tensor-product representations such as $(\mathbf{3}, \mathbf{2})$ without algebraic obstruction. Anomaly cancellation (Theorem 15) then uniquely fixes the hypercharge assignment as the sole consistent completion of the given sector pattern.*

Remark (availability, not selection). The link carrier establishes representation availability — the cross-charged representations exist and are the admissible matter carriers — and anomaly cancellation fixes charges given the sector pattern; the selection of the observed family pattern from the available link-carrier representations is not supplied by either and is carried by the chain's open links

(H-spin'; light-spectrum selection).

Proof. (i) Three sectors carrying the spin-1/2 labels and one carrying the scalar labels (Theorem 9; physical reading under H-spin'). (ii) The three sectors are related by cubic symmetry (O permutes spatial axes), hence degenerate. (iii) Theorem 15 below shows that, given a chiral $SU(3) \times SU(2) \times U(1)$ gauge structure *and the observed family pattern*, the SM hypercharges are the unique anomaly-free assignment [16] — anomaly cancellation constrains the charges given the pattern; it does not by itself select the pattern from the link carrier. The framework therefore derives three degenerate sectors as a forced consequence, with their reading as three physical SM generations conditional on the flagged inputs (H-spin' for the fermion identification; H-χ' for chirality; light-spectrum selection): cubic symmetry fixes three degenerate sectors (step ii), the link-carrier construction of §4.7.1.2 places matter on links where cross-charged representations are admissible (no phenomenological input required), and anomaly cancellation (step iii) fixes the hypercharge assignment uniquely. □

Scope. The link-representation construction contains the required tensor-product representation spaces once H-link is assumed, and anomaly cancellation then fixes hypercharges conditional on the matter assignment. The exact cubic fact is the $1 \oplus 3$ taste decomposition on the single-copy six-link branch. The statement that this is the complete physical matter carrier now inherits **H-link with** $m = 1$, and the reading of its three candidate sectors as physical generations additionally inherits H-spin'. No stabilizer theorem currently derives $m = 1$; the former claim that Theorems 5–7 fixed the copy number is not established.

Theorem 11 (Higgs sector, consistency). *Given a Higgs doublet* $(\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ *in the SM, the singlet taste provides a spin-0 sector whose* E *projection is consistent with hosting it; the* T_1 *and* A_1 *projections do not admit Yukawa couplings to the SM fermion doublets and play no role in the low-energy theory.*

Proof. The singlet taste is spin-0 (Theorem 9a, under H-spin') and decomposes under the gauge group as $T_1(\mathbf{3}) \oplus E(\mathbf{2}) \oplus A_1(\mathbf{1})$ (Theorem 7). Of these three projections, only $E(\mathbf{2})$ admits a gauge-invariant Yukawa coupling of the form $\bar{Q}_L \cdot \Phi \cdot u_R$: the T_1 projection is excluded on color grounds and the A_1 projection on weak grounds. The E projection is therefore the natural host for the SM Higgs doublet. □

Scope. This theorem does **not** derive the existence of a Higgs doublet, nor its hypercharge, from lattice structure. The Higgs doublet is taken as phenomenological input; the singlet taste is shown to accommodate it.

Consequences. The framework accommodates exactly one Higgs doublet via the singlet taste, consistent with the SM.

4.7.1 The fermion embedding question: diagnosis and closure The framework as written derives, conditionally on H-link, the gauge group $SU(3) \times SU(2) \times U(1)$ from the commutant construction (Theorems 5, 7, 14) and uses anomaly cancellation (Theorem 15) as a uniqueness check on the SM matter assignment, given a chiral $SU(2)$ structure. The matter representations and hypercharge assignments are derived — via the link-carrier construction of §4.7.1.2 and anomaly cancellation (Theorem 15) — rather than taken as input. The framework's claim on the fermion sector is therefore: *the SM gauge group is forced by first principles; the SM matter content, its gauge representations, and its hypercharges are, given the chiral structure (H-χ', §4.8), the unique anomaly-free consequence of placing matter on links in the link-carrier space $V_K \otimes V_K^*$.* The residual gap is narrower: the assignment of SM representations to specific tastes rests on a uniqueness argument via anomaly cancellation rather than a direct geometric derivation of the representations themselves, and is noted as the residual scope.

Whether a stronger derivation exists — via a geometric embedding of (taste $\times$ sublattice $\times$ spinor) indices into SM gauge representations such that $Q_L = (\mathbf{3}, \mathbf{2}, +\frac{1}{6})$ emerges as an irreducible tensor-product representation — is open. One specific candidate construction (extending V_K by an O-equivariant taste factor and redefining the gauge group as the commutant of an extended operator) is ruled out: O has no irreps of dimension > 3, so no O-equivariant operator on $V_K \otimes V_{\text{taste}}$ admits a 6-dimensional irreducible eigenspace with the required tensor-product structure. Other constructions — breaking O-equivariance on the taste factor, replacing the commutant definition with a bimodule construction, or placing fermions outside the gauge-defining carrier space — are not ruled out, but none has been produced.

The neutrino mass mechanism (Dirac vs. Majorana, presence or absence of $ u_R$) is not fixed by the lattice structure as currently understood and is left open. The candidate three-sector structure exists on the single-copy H-link branch and its physical-generation reading rides H-spin'. Because $m = 1$ is now an explicit bridge rather than a consequence of coupling-degree minimization, the absolute generation count is correspondingly conditional on H-link single-copy; the exact taste decomposition itself remains unchanged.

4.7.1.1 Gauge bosons from the Schur complement, and the level-confusion diagnosis The Schur-complement trace-out used in Theorem 13 to establish chirality contains more structural information than that theorem extracts. Pursuing it carefully yields a clean retrodiction of the gauge boson content of the Standard Model from $\mathrm{End}(V_K)$, and simultaneously sharpens the fermion-embedding question of §4.7.1 into a precise and tractable form — a form that §4.7.1.2 then answers via the link-carrier construction.

The gauge sector closes. With $D_{ee} = D_{oo} = 0$ (Theorem 13(ii)), the Schur-complement effective action takes the form $\mathcal{L}_{\text{eff}} = \bar{\psi}_+ K_{\text{eff}} \psi_+ + \cdots$ with kernel $K_{\text{eff}} = D_{eo} G_{oo} D_{oe} : W_+ \to W_+$ — W_+ the visible (even-sublattice) half of the spin⊗taste space, carrying both spin chiralities. The kernel is a linear map on the visible single-excitation space, i.e., an element of $\mathrm{End}(W_+) = W_+^* \otimes W_+$ — *not* of $\Lambda^2 W_+$ or $\mathrm{Sym}^2 W_+$. The two ψ_+ factors are distinguishable (one barred, one unbarred), and bilinears of distinguishable objects pair W_+^* against W_+ via the trace pairing.

Restricting to the internal index, the gauge content of K_{eff} is given by the decomposition of $\mathrm{End}(V_K) = \mathrm{End}(V_3 \oplus V_2 \oplus V_1)$ under $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$. The diagonal blocks give

- $\mathrm{End}(V_3) = \mathbf{8} \oplus \mathbf{1}$ — the SU(3) adjoint (gluons) plus a singlet,
- $\mathrm{End}(V_2) = \mathbf{3} \oplus \mathbf{1}$ — the SU(2) adjoint (W bosons) plus a singlet,
- $\mathrm{End}(V_1) = \mathbf{1}$ — the U(1) hypercharge boson.

The off-diagonal blocks $\mathrm{Hom}(V_i, V_j)$ for $i \neq j$ would carry cross-charged "gauge bosons" mixing different commutant factors. These are absent from the Standard Model. The substratum dynamics is locally U(6) invariant and does not exclude them; they are removed spontaneously: the O-equivariant condensate Σ breaks $\mathrm{U}(6) \to \mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$ (Theorem 5, revised), and the 22 cross-charged modes acquire masses at the condensate gap scale M_X — set by the equivariant block splittings, a solution-level quantity not fixed at the lattice scale — by the Higgs mechanism, decoupling from the low-energy spectrum: K_{eff} inherits the block structure of the stabilizer, and the off-diagonal Homs are gapped rather than absent. The Schur-complement kernel therefore retrodicts the SM gauge boson content cleanly: the adjoints of SU(3), SU(2), and U(1) together with the corresponding singlets, and nothing else.

The fermion sector does not close, and the obstruction is sharper than it looks. The natural next move is to ask whether the same Schur-complement structure also produces the SM matter content. It does not, and the failure mode is structurally informative.

The visible single-excitation space is $W_+ = V_K \otimes V_{\mathrm{taste}} \otimes S_L$, with $V_K = V_3 \oplus V_2 \oplus V_1$ by Theorem 5. Tensor products distribute over direct sums, so

$$W_+ = (V_3 \otimes V_{\mathrm{taste}} \otimes S_L) \oplus (V_2 \otimes V_{\mathrm{taste}} \otimes S_L) \oplus (V_1 \otimes V_{\mathrm{taste}} \otimes S_L).$$

A single excitation in W_+ lives in exactly one summand and therefore carries exactly one nontrivial gauge charge: it is a color triplet *or* a weak doublet *or* a hypercharge singlet, never two simultaneously. The left-handed quark doublet Q_L, by contrast, transforms as $(\mathbf{3}, \mathbf{2}, +\frac{1}{6})$ — an irreducible representation of the joint gauge group that lives in $V_3 \otimes V_2$, not in $V_3 \oplus V_2$. There is no element of a direct sum that carries simultaneous nontrivial labels under summands meant to act independently. As a statement about the bijection-level carrier V_K, this is a hard algebraic fact and it cannot be circumvented by reorganizing $\Lambda^k W_+$ for any k.

Diagnosis: a level confusion between bijection-level and observer-level structure. The obstruction is fatal only under a hidden assumption that the present framework is required to reject. The assumption is that the observer-level single-fermion Hilbert space *is* $W_+ = V_K \otimes V_{\text{taste}} \otimes S_L$, with the gauge group acting on it via the bijection-level decomposition $V_K = V_3 \oplus V_2 \oplus V_1$. This identifies two distinct objects:

1. The bijection-level carrier at a site, which is the finite alphabet of values the lattice variable φ can take, equipped after-the-fact with the complex inner-product structure $V_K = \mathbb{C}^6$ on which M acts and Theorem 5 diagonalizes.
2. The observer-level single-fermion Hilbert space, which is whatever an embedded observer's coarse-graining produces when it organizes lattice deviations from a vacuum into states it calls "one fermion."

In standard QFT, where quantum mechanics is axiomatic, these are the same object by fiat: one writes down a single-particle Hilbert space and gauge representations act on it directly. In the present framework that identification is not available. The quantum representation is *supplied* by the bijection (S, φ) itself — the universal fixed-basis form over the finite laws the marginalization produces ([Main]), with C1/C3/C4 identifying the memory-bearing sector and the common coherent operational lift stated separately — and the derivation runs through the observer's coarse-graining of the lattice, not through the lattice itself. The notion of "a state in V_K with definite gauge label" is a structure that exists at the bijection level. The notion of "one fermion as seen by an observer" is a structure that exists at the observer level. There is no theorem in the present paper, and none in [Main], asserting that these two structures coincide, and the foundational commitment to emergent QM positively requires that they need not.

Once this distinction is made, the direct-sum-vs-tensor-product obstruction reorganizes. At the bijection level, $V_K = V_3 \oplus V_2 \oplus V_1$ is correct and the commutant $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ acts block-diagonally on it. At the observer level, the single-fermion Hilbert space $\mathcal{H}_{\text{obs}}$ is the image of bijection-level configurations under the (still-to-be-characterized) coarse-graining map, and the gauge group acts on $\mathcal{H}_{\text{obs}}$ via the *induced* action of the bijection-level commutant. This induced action need not be block-diagonal in any natural basis of $\mathcal{H}_{\text{obs}}$, and a single observer-level fermion can perfectly well be a vector in $\mathcal{H}_{\text{obs}}$ whose preimage in the bijection-level structure has support in multiple V_K summands. That is exactly what the representation $(\mathbf{3}, \mathbf{2})$ encodes: a state with a definite color label and a definite weak-isospin label, which are two different observables on $\mathcal{H}_{\text{obs}}$ both inherited from the bijection-level structure but neither equal to "which V_K summand are you in."

This reframing is consistent with — and arguably required by — the way Theorem 13 already handles chirality. That proof treats the gauge group as acting on V_K *coupled to* the spinor/sublattice grading, not as a pure internal rotation on a 6-dimensional space: SU(3) is vector-like because color commutes with the grading; SU(2)‘s chirality is carried by the taste-selectivity hypothesis H-χ’ (§4.8) — the trace-out supplies the ε-grading, not spin chirality. Theorem 13 is already implicitly working at the observer level, where the gauge action is induced rather than literal. §4.7.1.1 simply makes this commitment explicit and applies it to the matter sector.

The asymmetry is structural, not embarrassing. The reframing also explains *why* the gauge sector closes from $\mathrm{End}(V_K)$ while the matter sector does not close from V_K. Gauge bosons are properties of the coupling matrix M itself — they live at the bijection level, where the direct-sum decomposition of V_K is the literal structure and $\mathrm{End}(V_K)$ is the literal kernel produced by the Schur complement. Fermions are observer-level objects: they are what an embedded observer reports when it coarse-grains lattice deviations into single-particle states. The two sectors live at two different layers of the framework, and the layer separation is exactly what "QM is emergent" means. The asymmetry between a closed gauge-boson retrodiction and a fermion-embedding question that requires a layer-corrected treatment is therefore not a defect; it is the expected signature of a framework in which gauge fields are structural and matter is emergent. §4.7.1.2 shows how the layer-corrected treatment resolves both the embedding and the generation count.

The correct successor question — characterizing the coarse-graining map from bijection-level configurations to the observer-level single-fermion Hilbert space $\mathscr{H}_{\mathrm{obs}}$ such that the induced gauge action realizes exactly one SM generation per T_1 taste — is answered in §4.7.1.2 via the link-carrier construction.

4.7.1.2 Closure: Matter on Links The closure is a two-step corollary of theorems already proved, using constructions distributed across §3.1 and §4.4.

Step 1: The link is the carrier. The local gauge invariance established in §3.1 and §4.4 establishes that the coupling matrix $M(\mathbf{n}, \hat{e}_j)$ on a nearest-neighbor link is itself the gauge connection: under a local gauge transformation G, it transforms as $M(\mathbf{n}, \hat{e}_j) \to G(\mathbf{n})\, M(\mathbf{n}, \hat{e}_j)\, G(\mathbf{n} + \hat{e}_j)^{-1}$, with one gauge index at each endpoint. The link variable therefore lives in $V_K^{(\mathbf{n})} \otimes V_K^{(\mathbf{n}+\hat{e}_j)*}$, not in a single V_K. Furthermore, Theorem 7 establishes that the multiplicities $(3, 2, 1)$ themselves come from the cubic decomposition of the *$2d = 6$ nearest-neighbor link directions* under O, not from any site-internal structure: V_K was a link-direction space from the start. The site-level reading of $V_K = V_3 \oplus V_2 \oplus V_1$ that produced the direct-sum obstruction is therefore not the framework's natural reading; the link reading is.

On a link, the relevant carrier for matter is $V_K \otimes V_K^*$, which contains the cross-charged tensor products $V_3 \otimes V_2$, $V_3 \otimes V_1$, $V_2 \otimes V_1$, and so on. The left-handed quark doublet $Q_L \in (\mathbf{3}, \mathbf{2})$ has a natural home in $V_3 \otimes V_2 \subset V_K \otimes V_K^*$, with no algebraic obstruction. The direct-sum-vs-tensor-product problem dissolves: it was an artifact of misidentifying the site-internal V_K with the matter carrier, when the framework's own gauge-connection construction places the carrier on links.

Step 2: Anomaly cancellation completes the assignment. Once the matter carrier is $V_K \otimes V_K^*$ on links, the question reduces to which subspaces of $V_K \otimes V_K^*$ host propagating chiral fermions. Theorem 13 identifies the chirality structure conditionally: taste-blind couplings are vector-like — and the delivered minimal link-current couplings are certified taste-blind for all three blocks (§4.8, `hchi_selectivity_probes.py`) — so a spin-chiral SU(2) requires the non-minimal selective mechanism that is H-χ''s sharpened content (§4.8), which is open. Theorem 15 then fixes the matter content uniquely: given chiral $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ with fermions in fundamental or singlet representations, the six anomaly conditions force exactly the SM hypercharge assignment $(Y_Q, Y_u, Y_d, Y_L, Y_e) = (1/6, 2/3, -1/3, -1/2, -1)$, with no free parameters beyond the overall normalization and the $Y_u \leftrightarrow Y_d$ relabeling (Theorem 15). Theorem 10 then distributes this content across the three T_1 staggered tastes as three identical generations, and Theorem 11 places the Higgs in the E projection of the singlet taste.

The closure, reclassified (availability established; chirality and selection open). Combining these: (i) the local gauge invariance construction establishes that gauge connections live on links with carrier $V_K \otimes V_K^*$; (ii) cross-charged SM representations such as $(\mathbf{3}, \mathbf{2})$ exist in $V_K \otimes V_K^*$ as ordinary tensor-product subspaces, with no direct-sum obstruction — representation *availability* with no algebraic obstruction; (iii) the ε-grading survives the trace-out and, on the visible sector, spin chirality and taste chirality coincide (Theorem 13(iii)), so the chirality of any gauge coupling reduces to the taste-chirality selectivity of its embedding: hypothesis H-χ', underived; (iv) *given* a chiral $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ structure with the observed family pattern, Theorem 15 fixes the hypercharges uniquely; (v) Theorems 8–11 distribute the content across the three T_1 tastes plus the singlet-taste Higgs. The §4.7.1 question is answered at the level of representation availability: every SM multiplet has a carrier, and the hypercharges are the unique anomaly-free completion given chirality. Two things are open and are stated as such: the taste-selectivity hypothesis H-χ' that would make SU(2) spin-chiral on the visible sector (§4.8), and the light-spectrum selection question — which of the available modes are light, and why mirror and exotic modes are gapped — for which no derivation exists. Earlier versions' "fully answered … no additional construction required" claims are retired accordingly.

Step 3: physical carrier multiplicity is the open H-link single-copy clause. On a single six-link carrier, the $2^3 = 8$ Brillouin-zone corners stagger-reduce to four Dirac tastes and Theorem 8 decomposes these as one scalar plus three spin-1/2 candidate tastes. That representation calculation is exact for the branch. What is **not** proved is that the observer/gauge carrier contains exactly one copy: H-link permits $V_{\mathrm{link}} \otimes \mathbb{C}^m$, and the current framework has no theorem selecting $m = 1$. Therefore "three candidate sectors" is a conditional single-copy result; reading those candidates as the three physical generations further requires H-spin'.

The earlier $\mathrm{End}(V_K)$ result for the gauge sector now fits cleanly into this picture. Gauge bosons live in $\mathrm{End}(V_K)$ because they are properties of M itself, viewed as an operator on the carrier at one endpoint of a link; matter lives in $V_K \otimes V_K^*$ because it propagates *along* a link, picking up one gauge index at each end. Both sectors live at the link level. The asymmetry between "gauge bosons close from $\mathrm{End}(V_K)$" and "matter does not close from V_K" was an artifact of asking the matter question at the wrong layer; once both questions are asked at the link level, both close, and the closure is forced by theorems already in §§4.4–4.9.

Theorem 12 (Partition = staggered ε-grading). *The OI checkerboard partition coincides with the staggered ε-grading of the spin⊗taste space: visible and hidden sectors carry the two eigenspaces of the product grading $\gamma_5 \otimes \xi_5$ (spin chirality $\times$ taste chirality), each containing both spin chiralities.*

Proof. The checkerboard assigns site $\mathbf{n}$ to the visible sector iff $(-1)^{\sum n_\mu} = +1$ (even sublattice). In the staggered-to-Dirac reconstruction, the even sublattice carries spinor components with $\Gamma(\eta)$ matrices of even rank; these satisfy $\gamma_5 \Gamma(\eta) \gamma_5 = +\Gamma(\eta)$ and therefore span the $+1$ eigenspace of the product grading $\gamma_5 \otimes \xi_5$ — not a spin-chirality eigenspace. Each sublattice contains both spin chiralities in equal dimension: within the even-corner span the left- and right-handed subspaces are each four-dimensional ($\Gamma(0) = I_4$ is even-rank and chirality-neutral; certified exactly by `papers/oi_lattice_code/foundations/chirality_grading_probes.py`). The partition structure and the ε-grading of the spin⊗taste decomposition are the same object. □

Remark (the checkerboard is not a free choice). The checkerboard partition is not selected by hand to produce chirality — it is the bipartite structure of the lattice itself. The wave equation is nearest-neighbor, so the cubic lattice is bipartite: even sites couple only to odd sites and vice versa. This bipartite structure is a property of the dynamics (range-1 coupling on a hypercubic graph), not of the partition. The staggered-to-Dirac reconstruction identifies the two sublattices with the ± 1 eigenspaces of the product grading $\gamma_5 \otimes \xi_5$ (spin chirality $\times$ taste chirality) — the standard spin-taste result in lattice QFT [12, 17], not an OI construction. Theorem 12 observes that the OI partition (visible vs. hidden) and the staggered partition (left vs. right) are the same decomposition. The physical partition (the cosmological horizon) traces out a spatial region containing both sublattices; at scales large compared to ϵ, any macroscopic region contains equal numbers of even and odd sites. The ε-grading in the emergent description is a property of how the staggered-to-Dirac reconstruction organizes the lattice variables — it follows from the bipartite structure of the dynamics, not from a choice of which sites to trace out; spin chirality proper is a further question (Theorem 13).

4.8 The trace-out grading, and chirality as a conditional identification

Theorem 13 (Chirality — conditional identification; minimal embedding certified vector-like; H-χ' sharpened, open). *On the visible sector the spin-chirality and taste-chirality operators coincide, so a gauge coupling is spin-chiral on the visible sector if and only if its embedding is taste-chirality-selective, while any taste-blind coupling is vector-like. The minimal link-current embeddings the construction delivers are certified taste-blind for all three gauge blocks — SU(3) on T1u, SU(2) on Eg, U(1) on A1g — by exact spin⊗taste classification (`hchi_selectivity_probes.py`, block-momentum method through the $\Gamma(\eta)$ map): the minimal construction is vector-like throughout. A spin-chiral SU(2) therefore requires a non-minimal selective mechanism — a taste-asymmetric structure entering the Schur complement below — which is the sharpened content of the named hypothesis H-χ'. H-χ' in this sharpened form is not derived and is recorded as open — its mechanism class is algebraically nonempty (the spin-taste-locked condensate $M^* = \sum_j \gamma_j \otimes \xi_j$ is Hermitian, jointly cubic-invariant, and taste-chirality-flipping, with the dressed Eg coupling acquiring selective structure at first order; `hchiprime_probes.py`), so the open question is dynamical selection — and the physical reading additionally rides H-spin' (§4.7).*

Proof. (i) *The grading.* The staggered-to-Dirac reconstruction [17, 12] assigns spinor components to hypercube corners $\eta \in \{0,1\}^{d+1}$ via $\chi(y+\eta\epsilon) = \frac{1}{4}\Gamma(\eta)_{\alpha\beta}\psi_\beta(y)$, where $\Gamma(\eta) = \gamma_0^{\eta_0}\cdots\gamma_d^{\eta_d}$. Under γ_5-conjugation, $\Gamma(\eta) \to (-1)^{|\eta|}\Gamma(\eta)$: the sublattice parity $\varepsilon(\mathbf{x}) = (-1)^{\sum x_\mu}$ is the product grading $\gamma_5 \otimes \xi_5$ of the spin⊗taste space, and each sublattice contains both spin chiralities in equal dimension (Theorem 12; certified exactly by `chirality_grading_probes.py`, which also records that the identification "even = left-handed" of earlier versions fails — $\Gamma(0) = I_4$ is even-rank and chirality-neutral). (ii) $D_{ee} = D_{oo} = 0$. The staggered Dirac operator couples only across sublattices: $\{D_{\mathrm{st}}, \varepsilon\} = 0$ (Theorem 3), so the even–even and odd–odd blocks vanish identically. (iii) *Visible-sector identity.* On the +1 eigenspace of $\gamma_5 \otimes \xi_5$, $\gamma_5 M \gamma_5 = M$ is equivalent to $\gamma_5 M = M\gamma_5$: the spin-chirality action and the taste-chirality action coincide there (probe P4). The Schur complement over the hidden sublattice — a standard procedure in lattice QFT [12, 17] — therefore yields an effective theory $\mathcal{L}_{\mathrm{eff}} = \bar{\psi}_+ (D_{eo} G_{oo} D_{oe}) \psi_+ + \cdots$ on a space where chirality projections and taste-chirality projections are the same operators. (iv) *Consequence.* A taste-blind embedding acts on both spin chiralities of the visible sector — vector-like. A coupling is spin-chiral on the visible sector precisely when its embedding selects a taste chirality. SU(3) acts on the internal $T_1(3)$ components, taste-blind: **vector-like**, as observed. SU(2) acts on the internal $E(2)$ components and, as constructed, is likewise taste-blind; its observed spin-chirality therefore requires taste-chirality selectivity of the physical SU(2) embedding — hypothesis H-χ' — which the present construction does not supply. (v) *Status.* The observed chiral SU(2) is carried as the named hypothesis H-χ'; the previous derivation via "trace-out removes right-handed fields" is retired (route closed and recorded, per the project conventions). One consequence is immediate and positive: right-handed fields are retained on the visible sublattice (as the $s \cdot t = +1$ partners), so the Yukawa constructions of Theorem 10 and the trace accounting of §6 use states the framework possesses. □

Remark (chirality without a continuum limit). In standard lattice QCD, staggered chirality is a property of the regularization that maps to physical chirality only via the continuum limit $a \to 0$ — a procedure that requires the lattice to be a regulator approximating an underlying continuum theory. The OI lattice has no continuum limit because $\epsilon = 2\,l_p$ is the fundamental scale, not a regulator. This raises the question of how lattice chirality becomes physical chirality without the standard universality argument.

The resolution: physical observers don't take continuum limits either. Every experimentally probed scale satisfies $\lambda \gg \epsilon$, so what observers see is the *infrared effective theory* obtained by integrating out modes near the lattice scale. This effective theory inherits the sublattice parity $\varepsilon(\mathbf{x}) = (-1)^{\sum x_\mu}$ as a $\mathbb{Z}_2$ grading because the hidden-sublattice modes do not propagate to IR scales — they are integrated out at the lattice scale via the Schur complement (step (iii) of the proof). The resulting low-energy action

$$\mathcal{L}_{\text{eff}} = \bar{\psi}_+ \left(D_{eo}\, G_{oo}\, D_{oe}\right) \psi_+ + (\text{gauge}) + (\text{Yukawa})$$

has external states confined to the visible half of the spin⊗taste space — the ε-grading, carrying both spin chiralities. SU(2) acts on internal E components; its spin-chirality on these states is the content of hypothesis H-χ' (Theorem 13). SU(3) acts on internal T_1 components and commutes with the sublattice grading (color is internal, not spatial), giving vector-like SU(3). The standard lattice-QCD objection to identifying staggered gradings with physical chirality — that the doublers carry "wrong" chirality and contaminate the continuum theory — is transformed rather than escaped in OI: the doublers are the physical generations (the $2^3 = 8$ Brillouin-zone corners stagger-reduce to 4 Dirac tastes — 3 generations plus the Higgs — Theorem 9), and the exact $\mathrm{U}(1)_\varepsilon$ symmetry is the $\gamma_5 \otimes \xi_5$ grading; physical chirality proper enters through H-χ', not through ε.

Anomaly cancellation as a consistency check. A chiral gauge theory is consistent only if its anomalies cancel. Theorem 15 below shows that, *given* the chiral structure named by H-χ' with the observed family pattern, the framework's matter content cancels all six anomalies and determines the SM hypercharges $Y_Q = 1/6$, $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$ (uniquely in the sense stated there). This is a consistency check on the hypothesis rather than a derivation of it: a generic chiral assignment on this representation content would leave non-vanishing anomalies, so the existence of an anomaly-free completion — landing uniquely on the observed hypercharges — is non-trivial evidence that H-χ' is a hypothesis the structure can consistently carry. What anomaly cancellation cannot supply is the taste-chirality selectivity itself; that remains open (Theorem 13(v)).

Remark (relation to spatial sublattice grading). The ε used in Theorem 13's proof is the 4D parity $\varepsilon_{4D}(x) = (-1)^{\sum_{\mu=0}^{d} x_\mu}$. By the Bridge Theorem of §7.5, this is equivalent to the 3D spatial sublattice grading $\varepsilon_{3D}(\vec{x}) = (-1)^{x_1+x_2+x_3}$ for static observables: $\varepsilon_{4D} = (-1)^{x_0}\varepsilon_{3D}$, with $(-1)^{x_0}$ constant on each time slice. The framework-essential identification is therefore $\gamma_5 \leftrightarrow \varepsilon_{3D}$, with the time-direction sign factor reducing to a sign convention. This is consistent with the broader pattern that time enters OI's emergent 4D structure as bookkeeping rather than as fundamental Clifford structure (cf. §7.5 *Implication for the framework's $d = 3$ commitment*).

4.8.1 Evasion of Nielsen–Ninomiya

The Nielsen–Ninomiya theorem forbids a single Weyl fermion on the lattice under four hypotheses: (NN1) **locality** of the action (couplings decay faster than any power of separation), (NN2) **hermiticity** of the lattice Hamiltonian, (NN3) **translation invariance** under the lattice group, and (NN4) the action is a **bilinear in fermionic fields carrying a definite, conserved chirality charge** $U(1)_\chi$. Under NN1–NN4 the spectrum is forced to contain equal numbers of left- and right-handed Weyl modes — the doubling phenomenon — so a chiral gauge theory cannot be regulated on a lattice in the conventional sense.

OI manifestly satisfies NN1–NN3. The wave equation is nearest-neighbor (NN1, range exactly ϵ); the bijection φ is real and the dynamics is T-invariant, so the induced operator is hermitian (NN2); and $(S, \varphi)/\mathcal{G}_{\mathrm{sub}}$ is invariant under the cubic translation group (NN3). Were NN4 also to apply, OI would inherit the no-go conclusion and chiral SU(2) would be impossible. It does not apply, and the reason is structural rather than evasive.

NN4 presupposes that the **fundamental** degrees of freedom are Grassmann fields $\psi(x)$ entering the action bilinearly, with chirality γ_5 defined ab initio on those fields. In OI the fundamental object is a *real scalar bijection* $\varphi : S \to S$ governed by the lattice wave equation. There are no fermions at the bijection level, no Grassmann variables, no γ_5, and consequently no $U(1)_\chi$ on which an obstruction could be formulated. Dirac structure is *derived*: Theorem 2 (Susskind) shows $D_{\mathrm{st}}^2 = -\frac{1}{4}\Box_{\mathrm{lat}}$, so staggered Dirac fermions appear as a *factorization* of the scalar wave operator, not as fields posited by an action principle. Theorem 12 then identifies the staggered spin⊗taste ε-grading with the bipartite checkerboard partition of the lattice (the exact chiral symmetry of Theorem 3, $\chi \to \varepsilon\chi$, being the standard staggered $U(1)_\varepsilon$ — the $\gamma_5 \otimes \xi_5$ grading). Chirality is addressed in the third step, Theorem 13: the Schur-complement trace-out of the odd sublattice produces an effective theory on the visible half of the spin⊗taste space, with $D_{ee} = D_{oo} = 0$ enforced by sublattice parity; on that space spin- and taste-chirality coincide, and the physical chirality of SU(2) is carried as the named hypothesis H-χ'.

The NN premise therefore **fails to apply** at the level on which OI is defined — it is not violated within its scope of applicability; which specific NN assumption fails for the *effective* chiral fermion construction remains to be identified once that construction exists (an open item alongside H-χ'). NN1–NN4 are conditions on lattice fermionic actions; the OI fundamental action is bosonic. By the time fermions and chirality exist as derived structures (post-Theorems 2, 12, 13), the trace-out has already broken the bilinear-fermionic-action setting that NN4 requires. The doubler-inversion remark of §4.8 makes the dual point from the opposite direction: the "wrong-chirality" doublers that NN forces upon a fermionic lattice action become, in OI, the physical generations (Theorem 9), so the very modes the no-go theorem treats as obstructions are reinterpreted as observed matter content. NN is not circumvented by a clever fermion formulation; it is bypassed because OI never enters its hypothesis space.

Remark (locality of the effective visible-sector operator, and its independence from the §3.1 (*Scope of the emergent-Lorentz claim*) smearing question). The "bypass" reading above is the strong one — that NN4 has no referent — and it is worth separating it explicitly from a weaker fallback that a referee might expect, namely that the effective theory escapes NN only by being *non-local* (the route taken by perfect-action and Ginsparg–Wilson formulations, where chiral symmetry is bought at the price of a non-local action). Whether OI needs that weaker fallback turns on the locality of the Schur-complement effective operator $K_{\text{eff}} = D_{eo}\, G_{oo}\, D_{oe}$ of Theorem 13(iii): a long-ranged G_{oo} would make K_{eff} non-local, and the evasion would then be "non-locality escapes NN" rather than "NN does not apply." On the free substratum the question resolves in favour of the strong reading. The odd-sublattice self-term in the bipartite wave operator is the time part $2\cos\omega$, not a frozen spatial constant; on the physical dispersion surface $2\cos\omega = t(\mathbf{k})/d$, the effective kernel reduces to $K_{\text{eff}}(\mathbf{k}) = t(\mathbf{k})/d = \frac{2}{d}\sum_j \cos k_j$ — a degree-one trigonometric polynomial, i.e. a strictly nearest-neighbour (finite-range) kernel in position space, with no interior poles. (An apparent interior divergence on the $t(\mathbf{k}) = \pm 4$ surface arises only in the frequency-frozen static approximation, which replaces $2\cos\omega$ by a spatial constant; it is an artifact of that approximation and is absent once the dispersion is imposed.) The free effective operator is therefore local, and the NN evasion is the strong form. The same Schur-complement structure yields a second exact property. With the site-diagonal grading $\Gamma = (-1)^{x+y+z}$ and the nearest-neighbour operator anticommuting ($\Gamma A \Gamma = -A$), the effective kernel obeys $\Gamma_V K_{\text{eff}}(E)\, \Gamma_V = -K_{\text{eff}}(-E)$ for *any* site partition — a one-line identity from the block-restriction of Γ. At the mirror point the kernel is exactly ε-graded and the staggered-mass projection vanishes identically: the trace-out preserves the bipartite grading and cannot gap the taste multiplet at the linear level, for the horizon partition or any other. This has been verified at machine precision in full $d = 3$ with two-directional controls (a grading-breaking deformation confined to the hidden sector is detected; grading-preserving disorder is not). The interacting (condensate-dressed) extension is the residual, shared with §3.1 (*Scope of the emergent-Lorentz claim*).

This locality result is a statement about the *free* effective operator, and it should not be conflated with the separate — and still open — question of whether the interacting, condensate-dressed coarse-graining map realises a sufficient ultraviolet *smearing* to protect the rotational sector at loop level (§3.1 (*Scope of the emergent-Lorentz claim*, point (iii))). The two questions share the object G_{oo} but are different computations: the locality question asks whether the free kernel is finite-range (answered here: yes), while the Lorentz-radiative-stability question asks whether the dressed induced links inherit the matter-field smearing (an interacting order-parameter question, addressed in §3.1 (*A present-day constraint, and the rotational sector's distinct status*) and [Ch. 19 §19.3.7], where the current evidence leans toward the un-smeared regime). A local free effective operator and an un-smeared dressed condensate are consistent: the former secures the strong NN reading of this section, while the latter leaves the radiative-stability protection of §3.1 (*Scope of the emergent-Lorentz claim*) as the open item it is stated to be there.

4.9 Anomaly cancellation and hypercharges

Theorem 14 (U(1)_Y automaticity). *The existence of a U(1) gauge factor is automatic in the multi-component framework.*

Proof. Each factor of the commutant decomposes as $\mathrm{U}(n_i) = \mathrm{SU}(n_i) \times \mathrm{U}(1)_i$. For K = 6 with multiplicities (3, 2, 1): $G = [\mathrm{SU}(3) \times \mathrm{U}(1)_B] \times [\mathrm{SU}(2) \times \mathrm{U}(1)_L] \times \mathrm{U}(1)_S$. Three U(1) factors exist automatically. A general abelian charge is $Q = \alpha B + \beta L + \gamma S$. Given SU(3) × SU(2) and the minimal fermion content (5 representations per generation), the six anomaly conditions impose 4 independent constraints on the 5 hypercharge unknowns, leaving exactly one free parameter (overall normalization). There is therefore exactly one anomaly-free U(1). □

Remark (the two non-gauge U(1) combinations). The two combinations of (B, L, S) orthogonal to hypercharge cannot be gauged: gauging them would introduce non-cancelling chiral anomalies. They survive as accidental global symmetries of the emergent theory — the same status that baryon number B and lepton number L have in the standard SM construction, broken only by the Weinberg operator and non-perturbative effects. No new gauge bosons are introduced, and the framework's prediction of B and L as accidental global symmetries matches observation (proton stability; lepton number conservation up to the small Majorana mass).

Theorem 15 (Unique hypercharges). *Given* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *with fermions in fundamental or singlet representations, the six anomaly conditions [18] determine the hypercharges uniquely up to the overall normalization and the interchange $Y_u \leftrightarrow Y_d$ — the latter a relabeling of which quark species is called up-type, since the anomaly conditions are symmetric in (Y_u, Y_d); the labeling is fixed by the electric-charge identification $Q_{\mathrm{em}} = T_3 + Y$* [16]:

$$Y_Q = \tfrac{1}{6}, \quad Y_u = \tfrac{2}{3}, \quad Y_d = -\tfrac{1}{3}, \quad Y_L = -\tfrac{1}{2}, \quad Y_e = -1$$

Corollary. |q__p| = |q__e| is a theorem, not a coincidence.

4.10 Renormalizability

Theorem 16. *The Yang-Mills coupling has mass dimension [g] = (3−d)/2; dimensionless iff d = 3.*

Proof. The Yang-Mills action $S = \frac{1}{4g^2} \int d^{d+1}x\, F_{\mu\nu}^2$ requires $[g^{-2}][F^2] = [x]^{-(d+1)}$. With $[F] = [x]^{-2}$: $[g]^{-2} = [x]^{d-3}$, so $[g] = (3-d)/2$. □

Since d = 3 is derived (§3.2), the emergent QFT supports renormalizable gauge interactions with asymptotic freedom.

4.11 Spontaneous symmetry breaking and the hierarchy

Chiral symmetry (Theorem 3) combined with the center-independent gauge sector forbids explicit masses. The unique renormalizable unitarity-preserving mechanism is the Higgs [13, 14]. The minimal scalar breaking $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_{\mathrm{em}}$ while preserving $\mathrm{SU}(3)_c$ is $H = (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$.

The hierarchy problem is dissolved: in the OI framework, the QFT is emergent. The Higgs mass is set by eigenvalues of M (properties of φ), not by radiative corrections.

4.12 The filter chain

The §4–§5 construction is now a conditional chain rather than a complete derivation from (S, φ) alone. The exact pieces are the finite projection/isotropy theorem, the cubic six-link representation 6 = 3⊕2⊕1, the staggered/taste identities on the chosen observer-level branch, and the anomaly calculations conditional on the matter assignment. The bridges are the substratum-to-observer closure that yields the normalized field operator, H-link (including single-copy $m = 1$), H-cust, H-spin', H-χ', and the already recorded flavor/CP qualifications. Downstream numerical agreements remain tests of this sharpened branch and do not discharge those bridges.

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5. The Strong CP Problem

5.1 T-invariance of the wave equation

Theorem 17. *The discrete wave equation is invariant under time reversal T: φ(n, t) → φ(n, −t).*

Proof. Under T, define ψ(n, t) = φ(n, −t). Substitution gives ψ(n, t+1) = ψ(n−1, t) + ψ(n+1, t) − ψ(n, t−1): the same wave equation. □

The checkerboard partition marginalizes over spatial sites, not temporal data. It preserves T while breaking P (§4.8).

5.2 What time reversal fixes about θ

Theorem 18. *A time-reversal-invariant point of the θ-vacuum satisfies $\theta \in \{0, \pi\} \pmod{2\pi}$.*

Proof. The θ-term $\mathcal{L}_\theta = (\theta/32\pi^2)\mathrm{Tr}(F\tilde{F})$ is T-odd: $\mathrm{Tr}(F\tilde{F}) = 2\,\mathbf{E}\cdot\mathbf{B}$, and under T, $\mathbf{E} \to \mathbf{E}$, $\mathbf{B} \to -\mathbf{B}$. The emergent Lagrangian inherits T-invariance from the substratum because the partition is spatial, so time reversal commutes with the visible/hidden projection; since θ is periodic and T sends $\theta \to -\theta$, invariance gives $\theta = -\theta \pmod{2\pi}$, whose solutions are $\theta = 0$ and $\theta = \pi$. Yang–Mills theory has T-invariant points at both. □

Selecting the $\theta = 0$ branch over $\theta = \pi$ is a separate condition, **H-top**, which T symmetry does not supply.

5.3 Detailed balance, and what it does not give

Theorem 19 (Reciprocity of visible transition counts). *Let the partition be preserved by an involutive time reversal $\mathcal{T}$ ($\mathcal{T}^2 = \mathrm{id}$), with θ its induced action on visible states and θ_H its action on hidden states, and let the hidden prior be the counting measure. Then $T_{ij} = T_{\theta j, \theta i}$ for all visible i, j; when the visible states carry no T-odd data ($\theta = \mathrm{id}$) this is detailed balance, $T_{ij} = T_{ji}$. The same holds at every time scale: $T_{ij}(n) = T_{\theta j, \theta i}(n)$ for all n.*

Proof. T-invariance (Theorem 17) gives $\varphi^{-1} = \mathcal{T} \circ \varphi \circ \mathcal{T}$, hence $\varphi = \mathcal{T} \circ \varphi^{-1} \circ \mathcal{T}$. For a hidden state h with $\varphi(i, h) = (j, h')$, apply the second identity to $\mathcal{T}(j, h') = (\theta j, \theta_H h')$: $\varphi(\theta j, \theta_H h') = \mathcal{T}\varphi^{-1}(j, h') = \mathcal{T}(i, h) = (\theta i, \theta_H h)$. So $F_{ij}(h) = \pi_H(\mathcal{T}\varphi(i, h)) = \theta_H h'$ maps $\{h : \pi_V \varphi(i, h) = j\}$ into $\{h'' : \pi_V \varphi(\theta j, h'') = \theta i\}$; the same construction for the pair $(\theta j, \theta i)$ maps back, the two maps are mutually inverse, and the sets have equal cardinality. Dividing by $|\mathcal{C}_H|$ gives $T_{ij} = T_{\theta j, \theta i}$. Since φ^n is a bijection with $\varphi^{-n} = \mathcal{T}\varphi^n\mathcal{T}$, the argument applies at every n. □

Verified numerically on the wave-equation substratum: the T-violation of the visible transition counts is exactly zero for $L = 4, 6$, $q = 2, 3, 5$ and all $n = 1, \ldots, 12$.

Proposition 20 (Reciprocity does not imply a T-invariant Hamiltonian). *Reciprocity $T_{ij}(t) = T_{ji}(t)$ for all i, j, t does not force H_{eff} to be rephasing-equivalent to a real Hamiltonian.* Counterexample: the bipartite C_4 ring $H_{01} = H_{12} = H_{23} = H_{30} = z = a + ib$ with the Hermitian conjugates on the reverse links. Its spectrum is $\{\pm 2a, \pm 2b\}$, and $|U_{ij}(t)|^2 = |U_{ji}(t)|^2$ identically (verified to 6×10^{-15} over 400 times at $a = 1, b = 0.3$). Yet the gauge-invariant loop product is $H_{01}H_{12}H_{23}H_{30} = z^4$ with $\arg z^4 = 4\arctan(b/a) = 1.165827$ and $\mathrm{Im}\, z^4 = 1.092 \neq 0$. Diagonal rephasing cannot alter a loop phase and a real Hamiltonian has a real loop product, so no such equivalence exists. Wigner's theorem does not supply the missing step: it represents a given ray transformation, and does not manufacture a physical time reversal from the scalar relations $|U_{ij}|^2 = |U_{ji}|^2$; allowing an arbitrary antiunitary makes the statement vacuous, since every Hermitian H admits conjugation in its own eigenbasis. The physically relevant Θ is inherited from the substratum and fixes the visible projectors, hence has the form DK in that basis, which would force every loop product real — the C_4 example admits no such Θ. A genericity rescue via phase-locking hypotheses is not adopted: the counterexample's bipartite structure is exactly the protected sector this construction uses, so such a rescue would itself owe a theorem that the flavor sector satisfies those hypotheses.

Proof. The reciprocity of the example is exact: for every pair of sites the uniform ring has a reflection R exchanging them, and a reflection reverses the orientation of the loop, so $RHR^{-1} = \overline{H}$ and $|U_{ij}(t)|^2 = |(RU(t)R^{-1})_{ji}|^2 = |\overline{U_{ji}(t)}|^2 = |U_{ji}(t)|^2$. The obstruction is exact as well: a rephasing $H \mapsto DHD^{-1}$ leaves $H_{01}H_{12}H_{23}H_{30}$ unchanged, a real symmetric matrix has a real loop product, and an antiunitary of the form DK commuting with H forces $\overline{H} = DHD^{-1}$, hence a real loop product again. For a fully coupled three-level system the antisymmetric part of $|U_{ij}(t)|^2$ enters at $O(t^3)$ through the rephasing-invariant triangle phase $\mathrm{Im}[H_{12}H_{23}H_{31}]$, and reciprocity at all t does force that phase to vanish; on a bipartite coupling graph there are no triangles, every closed walk has even length, and the criterion is silent. □

Theorem 21 (Strong CP is not fixed by substratum T-invariance). *In the construction of §2–§4, T-invariance of the substratum bijection together with reciprocity of the visible transition counts fixes neither θ, which may be 0 or π (Theorem 18), nor $\arg\det(Y_uY_d)$, since the emergent Hamiltonian need not be T-invariant (Proposition 20). The physical combination $\bar{\theta} = \theta + \arg\det(Y_uY_d)$ is therefore not determined by the construction. Any mechanism that fixes $\arg\det(Y_uY_d) = 0$ by making Y_u and Y_d simultaneously real also forces V_{CKM} to be real orthogonal and the Jarlskog invariant to vanish.*

Proof. The first two statements are Theorem 18 and Proposition 20. For the last: two matrices real in one common basis have real orthogonal singular value decompositions, so the mismatch unitary $V_{\mathrm{CKM}} = O_u^{\mathrm{T}}O_d$ is real orthogonal and $J = 0$, a rephasing-invariant statement. □

The strong CP problem is therefore open in this construction, under two named conditions: **H-top** (select $\theta = 0$ over π) and **H-det** ($\arg\det(Y_uY_d) = 0$ while leaving the rephasing-invariant flavor quartets free). The framework makes no prediction for the neutron electric dipole moment; a measured $\bar{\theta} \neq 0$ would constrain mechanisms proposed for H-det, not the T-symmetry of the substratum, which Theorem 19 shows is compatible with any value of $\bar{\theta}$. The narrowing of the target is partly a relief: reciprocal probabilities can hide genuine loop phases, so classical detailed balance need not kill weak CP violation, and there is no algebraic conflict between $\arg\det(Y_uY_d) = 0$ and $J_{\mathrm{CKM}} \neq 0$ — take $Y_u = D_u > 0$ and $Y_d = VD_d$ with $D_d > 0$ and V a complex CKM-type $SU(3)$ matrix. Chiral grading alone cannot help: for $\mathcal{M} = \begin{pmatrix} 0 & Y \\ Y^\dagger & 0 \end{pmatrix}$, $\det \mathcal{M} = (-1)^n |\det Y|^2$ and the phase of $\det Y$ is gone. Falsification control for any proposed mechanism: if it makes Y_u and Y_d simultaneously real it has solved too much, since it also predicts $J_{\mathrm{CKM}} = 0$ (Theorem 21).

5.4 Scope: flavor-sector CP violation as solution-specific input

The CKM and PMNS CP-violating phases are treated as solution-specific inputs rather than framework-level predictions. $\bar{\theta}$ is a property of $\det(Y_u Y_d)$, the static CP-odd combination that couples to QCD instantons and would generate a neutron electric dipole moment; CKM CP violation is encoded in the Jarlskog invariant $J \approx 3 \times 10^{-5}$ and arises from the basis mismatch between the up-type and down-type Yukawa mass eigenstates. In the Standard Model the Yukawas are not required to be real in any common basis: each is real in its own mass basis, the two mass bases generically differ by a unitary transformation, and when that transformation has complex entries that cannot be absorbed by field redefinitions the CKM matrix contains an irreducible phase [16, §6.3]. The present construction leaves that route open. Reciprocity of the visible transition counts (Theorem 19) does not force the Yukawas into a common real basis (Proposition 20): the $O(t^3)$ triangle-phase criterion that would do so for a fully coupled three-level system is silent on the bipartite staggered sector of §4.8, so J is not constrained by the construction's discrete symmetries. The basis-alignment unitary is a solution-specific property of the particular bijection φ, analogous to the absolute scale of fermion masses (§7.6) and the specific eigenvalues of the Yukawa matrices.

Which phases a Yukawa sector can hide (nullity). Among the CP phases carried by a Yukawa sector, those that are physically vacuous — removable by field redefinition — are exactly the diagonal rephasings, and nothing else is spuriously counted as removable. A phase assignment θ_{ab} on generation pairs is field-redefinition-removable iff it is a coboundary $\theta_{ab} = \alpha_a - \alpha_b$ of a rephasing vector α. The claim that the removable phases exhaust the closed ones — those consistent around every triangle $\theta_{ab} + \theta_{bc} + \theta_{ca} = 0$ — is the vanishing of the first cohomology of the generation-coupling complex. For the physical sector, where all three generations couple pairwise, that complex is the filled triangle K_3, which is simply connected, so $H^1 = 0$: every closed phase configuration is a coboundary, and the only irreducible phase is the single Jarlskog invariant that survives H^1 of the mismatch between the up- and down-sectors. The result generalizes to any number of generations whose coupling graph is chordal (every cycle triangulated); it can fail only on a coupling graph with an unfilled cycle — which is exactly the bipartite situation of Proposition 20, where a loop phase survives with no triangle to constrain it.

Pre-registration. Flavor-sector CP phases are listed as solution-specific inputs in §8.3 and in the pre-registered falsification conditions ([Substratum, §6.4, "what does not falsify"]). They join the absolute fermion mass scale m_s and the specific neutrino Yukawa eigenvalues in the class of quantities the framework treats as properties of the particular φ. The framework makes no commitment to any specific value of δ_{CKM}, δ_{PMNS}, the Jarlskog J, or the Wolfenstein parameters (ρ, η), and the measurement of these quantities does not bear on the framework's structural status.

5.5 Scope: what is established and what is not

What is established. T-invariance of the wave equation (Theorem 17); $\theta \in \{0, \pi\}$ for a T-invariant emergent Lagrangian (Theorem 18); reciprocity of the visible transition counts at every time scale, in the general form $T_{ij} = T_{\theta j, \theta i}$ (Theorem 19); that reciprocity does not make the emergent Hamiltonian T-invariant (Proposition 20); and that the construction therefore does not fix $\bar{\theta}$ (Theorem 21). Nothing in these statements depends on instanton dynamics.

What is not established. Two conditions stand between the construction and a resolution of the strong CP problem, and neither is discharged here: **H-top**, the selection of $\theta = 0$ over $\theta = \pi$; and **H-det**, a mechanism fixing $\arg\det(Y_u Y_d) = 0$ at the emergent level — the level at which $\bar{\theta}$ is measured — while leaving the flavor quartets free. Downstream of them stand radiative stability against the CP-violating weak sector and a quantitative neutron-EDM prediction with an error; neither can be addressed before H-det has content. The framework's commitment on strong CP is accordingly a statement of what T-invariance does and does not fix, and the two conditions are tracked as part of the matter-sector programme of §8.9.

A reported computation, not available for verification. A computation on the emergent operator of the §4.8 reconstruction reported that an octahedral-equivariant single-hypercube operator gives a vanishing Jarlskog invariant, and that breaking the equivariance turns on J and the E_g rotational anisotropy together, with measured exponents $J \sim \varepsilon^3$ and $E_g \sim \varepsilon$, so that the breaking strength reproducing $J = 3.0 \times 10^{-5}$ would produce a fractional E_g anisotropy of order 10^{-3} in the most favourable of the sampled directions. Its driver and numerical output are not available for verification, and its framing rested on reading reciprocity as all-order reality of H_{eff}, which Proposition 20 removes. If the equivariance statement stands on its own, it is a tension for the flavor sector — the observed J would require octahedral breaking that the rotational bounds of §3.1 exclude — and it is recorded here as such, not as a result of this paper.

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6. Gauge Coupling Prediction

The derivation chain of §§3–5 determines the gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, the matter content, and the discrete symmetries. This section extends the chain to the gauge coupling strengths — the values of α_1, α_2, α_3 at the Z boson mass.

6.1 The induced coupling

The gauge field is not an independent degree of freedom — it is the state-dependent coupling matrix $M(\mathbf{n}, \hat{e}_j)$ of §3.1, determined by the matter field $\phi \in (\mathbb{Z}/q\mathbb{Z})^{K \times N}$. The gauge kinetic term is therefore *induced* by the fermion determinant: the effective action is $S_{\text{eff}}[U] = -N_f \operatorname{Tr} \ln(D^\dagger D[U])$, and the gauge propagator arises from expanding S_{eff} to second order in A_μ — where A_μ ranges over *arbitrary* background links, not only over the microscopic image $\{U[\phi]\}$. That distinction is inert for the non-abelian factors and load-bearing for hypercharge, whose microscopic image is flat (§6.5): the coefficient computed here is the ambient one, written $\alpha_{0,\text{amb}}^{-1}$, and its identification with a native hypercharge coupling carries the conditions named there.

The induced coupling is computed from the one-loop staggered vacuum polarization $\Pi_s(0)$ — the lattice momentum integral over the fermion bubble with the OI vertex $V_j = ig\mu T^\alpha \cos(k_j + p_j/2)$. Since the coupling matrix is scalar, $M = \mu \cdot I_6$ (Theorems 6–7 jointly; the gauge group derives from the condensate, Theorem 5 revised), the Dynkin index $T(R) = 1/2$ per Dirac fermion is the same for all three gauge factors. With $N_f = 6$ (one per internal component), the induced coupling is universal:

$$\frac{1}{\alpha_0} = 6 \times \Pi_s(0) \times 4\pi = 23.25$$

This is a *structural* prediction — it depends on N_f and $T(R)$, both determined by the lattice structure (§§4.5–4.6), not on the eigenvalues μ_c, μ_w of M. The universality follows from the Dynkin index equality $T_3(R) = T_2(R) = T_1(R)$, which holds because all six components transform in the fundamental representation of their respective gauge group.

Note on the perturbative form. The integrand above is hybrid in dimensional structure: the loop integration $\int d^4q$ and the lattice propagator $D(q) = \sum_\nu \sin^2(q_\nu) + m^2$ run over four directions (consistent with the substratum's 3+1D Lorentzian structure), while the vertex factors $\cos^2(q_\mu)$ are spatial-only ($\mu \in \{1, 2, 3\}$, consistent with gauge fields living on spatial links only — there is no time-direction gauge connection in the substratum). The result is a perturbative form that is neither standard 4D staggered VP (which has $\cos^2$ on all four directions) nor standard 3D PT (which would have a 3D loop integral). At one loop, this form gives a finite, well-defined integral converging to $\Pi_s \to 0.3084$ on a robust plateau. At two loops, the strict 2-loop result for the universal threshold δ_0 vanishes at large N (the dominant scalar-bubble integrand falls as $1/N^{3.09}$), which is why $\delta_0 \approx 10$ is treated as empirical input rather than derived (§6.2). A proof that this hybrid form is a consistent perturbative scheme — closing at 1-loop and giving the observed asymptotic behavior at higher loops — is open work.

6.2 The threshold and logarithmic resummation

The universal induced coupling α_0 is defined at the lattice cutoff $\mu = M_{\rm Pl}$. At this scale, gauge self-interactions — which are proportional to the quadratic Casimir C_2 — generate a group-dependent threshold. A universal (C_2-independent) threshold δ_0 also arises from higher-loop corrections to the vacuum polarization.

The expansion parameter for the C_2-dependent threshold is $C_2 \cdot g_0^2 = C_2 \times 4\pi\alpha_0 = 1.08$ (SU(2)) and 1.62 (SU(3)) — both $\mathcal{O}(1)$. Finite-order perturbation theory converges too slowly: using known Wilson-action plaquette coefficients, 3-loop PT explains only $\sim 26\%$ of the threshold at SU(3). The all-orders result is well parameterized by a geometric-series resummation of the gauge self-energy:

$$\frac{1}{\alpha_i(M_Z)} = \underbrace{23.25}_{1/\alpha_0} + \underbrace{10.0}_{\delta_0} + \underbrace{8.3 \cdot \ln(1 + 5.59 \cdot C_2 g_0^2)}_{\text{resummed gauge self-energy}} + \underbrace{\frac{b_i}{2\pi} \ln \frac{M_{\rm Pl}}{M_Z}}_{\text{SM RG running}}$$

Above $M_{\rm Pl}$, the OI induced theory has no tree-level gauge action — gauge dynamics arise entirely from the fermion determinant — so its β-function differs at 2-loop and beyond from the standard SM running below the matching scale. The threshold function structurally absorbs this discontinuity at $M_{\rm Pl}$, which is why the matching formula has the form shown rather than continuous running across the scale. This reflects the substratum-vs-emergent split (group structural at the substratum; couplings emergent through the fermion determinant).

The four components:

- $1/\alpha_0 = 23.25$: computed structurally as the ambient coefficient $\alpha_{0,\rm amb}^{-1}$ (one-loop staggered vacuum polarization, §6.1; native for the non-abelian factors, and for hypercharge conditional on H-observer-bundle and H-Y-vertex, §6.5). Analytically verified from the lattice integral $\Pi_s(0) = \int d^4q\,[\cos^2(q_\mu)/D(q) - \cos^2(q_\mu)\sin^2(q_\mu)/D(q)^2]$ with $D(q) = \sum_\nu \sin^2(q_\nu) + m^2$, spatial average over $\mu = 1, 2, 3$; converges to $\Pi_s \to 0.3084$ as the grid refines, giving $1/\alpha_0 = 6 \cdot 0.3084 \cdot 4\pi = 23.25$ on a robust plateau $m \in [0.001, 0.05]$.
- $\delta_0 = 10.0$: the C_2-independent threshold. In the current formulation, δ_0 is determined empirically by the U(1) row (see below). A 2-loop staggered VP computation was implemented as a consistency check, and subsequently rebuilt from first principles (scalar and vector self-energy insertions, vertex correction, sails/crossed-gluon, and ghost 2-loop) using the Π_s convention above, with the conversion $\delta_0 = (1/\alpha_0) \cdot (\delta\Pi_s^{(2L)}/\Pi_s)$ tied to $1/\alpha_0 = 24\pi \cdot \Pi_s$. Large-$N$ scans of the dominant scalar-bubble SE piece across $N = 6, 8, 10, 12, 14, 16$ at $m = 0.05$ show its raw integrand falls as $1/N^{3.09}$ with per-point residuals below 6%, consistent with continuum-limit value zero; the apparent "plateau" at intermediate N is a finite-N

coincidence. The assembled 2-loop sum at $N = 8$ of approximately $+72$ in δ_0 units is therefore a finite-N artifact, and strict 2-loop lattice PT in OI does not reproduce the empirical $\delta_0 \approx 10$. We therefore treat δ_0 as a universal-row matching parameter determined by the U(1) threshold (below), analogous to how Standard Model input parameters are fixed by observation rather than derived. A natural 3-loop bound provides the structural plausibility check:

3-loop bound. The non-perturbative coupling at the staggered VP scale gives an effective expansion parameter $\sim g_0^2/\pi \approx 0.17$. Standard lattice perturbation theory in this regime has successive-order ratios of 0.20–0.30. Taking the structural input $1/\alpha_0 = 23.25$ as the leading scale and applying one power of the 2-loop perturbative ratio gives an expected threshold of order $0.2\text{–}0.3 \times 23.25 \approx 4.6\text{–}7.0$, and an expected 3-loop correction of order $0.2\text{–}0.3 \times 4.6\text{–}7.0 \approx 1\text{–}2$. The empirical value $\delta_0 \approx 10$ exceeds this natural 2-loop order by a factor consistent with the accumulation through third order — not tight enough to serve as a derivation, but consistent with typical non-asymptotic lattice-PT behavior.

Empirical determination via the U(1) row. The U(1) row (where $C_2 = 0$, eliminating the gauge self-energy contribution entirely) determines $\delta_0 = 10.02$ from the observed $1/\alpha_1(M_Z) = 59.00$, given the structural input $1/\alpha_0 = 23.25$ and SM RG running. This empirical determination is consistent with the natural 3-loop expectation (order of magnitude).

Numerical observation. The empirically-fixed value $\delta_0 = 10.02$ coincides to 0.2% with the hypercharge-squared trace $\mathrm{Tr}[Y^2] = 10$ of three generations of SM fermions in standard convention ($Q = T_3 + Y$): Q_L contributes $6 \cdot (1/6)^2 = 1/6$, u_R contributes $3 \cdot (2/3)^2 = 4/3$, d_R contributes $3 \cdot (1/3)^2 = 1/3$, L_L contributes $2 \cdot (1/2)^2 = 1/2$, e_R contributes $1 \cdot 1^2 = 1$, summing to 10/3 per generation, or 10 for three generations. The analogous group-theoretic sums for the other two factors are $\sum_R T(R) = 6$ for both SU(2) and SU(3) (with matter in fundamentals, three generations), coinciding with $N_f = 6$ (the single-copy H-link carrier count). A naive group-specific interpretation $\delta_0^{U(1)} = \mathrm{Tr}[Y^2]$, $\delta_0^{SU(N)} = \sum_R T(R)$ is ruled out within the paper's framework: it would require (A, B) yielding $A \cdot B \approx 90$, inconsistent with the independently-derived $A \cdot B = 48 \pm 1.5$ (§6.2.1). An anomaly-coefficient identification also fails structurally: $\mathrm{Tr}[Y^2]$ is the fermion-only contribution to b_Y via the (2/3) factor in the one-loop β-function formula, not a member of the anomaly polynomial — the four standard SM gauge-anomaly invariants ($\sum Y^3$, $\sum Y$, $\sum_{\text{doublets}} Y$, $\sum_{\text{triplets}} T \cdot Y$) all cancel for SM matter content. A β-function matching-scale offset $\delta_0 = (b_1/2\pi)\ln(M_{\text{Pl}}/\mu_{\text{match}})$ reproduces $\delta_0 = 10.02$ at $\mu_{\text{match}} \approx 2.6 \times 10^{12}$ GeV in the U(1) row, but the same mechanism applied at the same μ_{match} to the SU(2) and SU(3) rows yields $\delta_0 = -7.74$ and -17.11 respectively, contradicting the framework's universality requirement on the C_2-independent threshold. The universal $\delta_0 = 10.02$ is therefore required by self-consistency of the framework; whether the precise $\mathrm{Tr}[Y^2] = 10$ coincidence for the U(1) row is a numerical accident or reflects higher-order structural physics — substratum-level mode-counting on the cubic lattice, non-perturbative effects in the discrete substratum measure (§6.5), or lattice topological invariants beyond the framework's currently-developed scope — remains open and is treated as numerical pending such structural mechanisms.

- $A \cdot \ln(1 + B \cdot C_2 g_0^2)$ with $(A, B) = (8.3, 5.59)$: the resummed gauge self-energy contribution. The logarithmic form is motivated by geometric-series resummation of the gauge self-energy $1/(1-\Sigma)$ in the induced gauge theory; the effective exponent $p \approx 0.42$ at the OI coupling is the local slope of the logarithm at $C_2 g_0^2 \sim 1.3$, not a fundamental constant. In the current formulation, A and B are determined by matching to the observed SU(2) and SU(3) couplings at M_Z, given the structurally computed $1/\alpha_0$ and the empirically fixed δ_0. The leading coefficient $A \cdot B$ is independently derived from 2-loop lattice perturbation theory in the induced gauge theory (§6.2.1), giving $A \cdot B \approx 48$ under the $1/N^2$ leading-power assumption, against the fitted 46.4; the extrapolation is leading-power-dependent and the comparison inconclusive (§6.2.1).
- SM RG running: the one-loop β-function coefficients $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ over $\ln(M_{\mathrm{Pl}}/M_Z) = 39.4$.

6.2.1 First-principles derivation of $A \cdot B$

The leading coefficient of the C_2-dependent threshold in the expansion of $A \ln(1 + B \cdot C_2 g_0^2)$ — namely the product $A \cdot B$ multiplying $C_2 g_0^2$ at first order — is extracted from the 1-loop gauge self-energy $\Sigma_{\mu\nu}(p)$ in the induced gauge theory. The induced propagator $D_{\mathrm{ind}}(k) = 1/(k_{\mathrm{lat}}^2 \cdot \Pi_s)$ and the Wilson-plaquette 3-gluon vertex (the leading continuum-limit piece of the full induced Γ_3) enter the bubble diagram; a Faddeev–Popov ghost loop with the induced propagator restores transversality. The transverse self-energy is extracted via

$$\Pi_T(p^2) = \frac{\Sigma^{11}(p) - \Sigma^{00}(p)}{\hat{p}^2} \qquad \text{for } p \text{ along the 0-direction,}$$

with linear extrapolation $p^2 \to 0$ at each lattice size N (three external momenta $\lambda = 1, 2, 3$ at the smallest accessible lattice values).

Conversion formula. The relation between Π_T and the coupling shift is

$$\delta(1/\alpha) = \frac{4\pi \cdot |\Pi_T|}{b_0^{\mathrm{capture}}} \qquad \text{(per } C_2\text{, coupling-stripped),}$$

where b_0^{capture} is the fraction of the textbook pure-YM 1-loop β-function coefficient $b_0 = 11/3$ that is reproduced by the finite-lattice computation. This conversion is validated by a standard-QCD cross-check (replacing $D_{\text{ind}} \to 1/\hat{k}^2$, no Π_s factor), which reproduces the textbook Π_T to 100% at $N = 28, m = 0.05$ via the formula.

Lattice data. Computing at each lattice size N using each N's own $\Pi_s(N)$ and $b_0^{\text{capture}}(N)$ (measured from a standard-QCD cross-check at the same N and m), the resulting per-N values of $A \cdot B$ at $m = 0.05$ are:

N	$\Pi_s(N)$	$b_0^{\text{capture}}(N)$	$A \cdot B(N)$
20	0.3451	0.523	22.14
24	0.3255	0.460	28.55
28	0.3174	0.427	33.31
32	0.3135	0.408	36.92
36	0.3115	0.398	39.61
40	0.3104	0.392	41.70

Extrapolation to $N \to \infty$. Three independent estimators applied to the $A \cdot B(N)$ sequence give consistent results:

Estimator	$A \cdot B(\infty)$
Linear $1/N^2$ fit on $(N = 32, 36, 40)$	50.1 (residual std 0.05)
Shanks acceleration, level 2 (last term)	48.0
Shanks acceleration, level 3 (last term)	47.9
Shanks acceleration, level 4 (last term)	47.9

The extrapolated value is model-dependent beyond the estimator spread above, and the comparison to the fit is conditional on the assumed leading finite-size power. Fitting the six data points with a leading $1/N$ correction gives $A \cdot B(\infty) \approx 61$ and fits the sequence an order of magnitude better (rms residual 0.06 against 0.67 for the pure $1/N^2$ model over all six points); a $1/N^2 + 1/N^3$ model fits best of all (rms 0.02) and gives ≈ 52. Only the pure-$1/N^2$ extrapolation and the Shanks family (which presupposes near-geometric convergence) land near 48; the largest computed value is 41.7, so every candidate sits 15–45% above the data's endpoint. The physically motivated default for staggered actions is $O(a^2) = O(1/N^2)$ artifacts, which is the ground for preferring the 48-family — but the induced-propagator normalization $\Pi_s(N)$ and the $b_0^{\text{capture}}(N)$ division can each introduce $O(1/N)$ pieces, and the data in this range in fact prefer a leading $1/N$ term. The comparison against the fitted 46.4 is therefore *conditional on the* $1/N^2$ *leading-power assumption*: under it, agreement at the few-percent level; under the better-fitting $1/N$ model, a $\sim 30\%$ disagreement.

The cross-check is **inconclusive pending either substantially larger N or a derivation of the leading finite-size power of the induced theory**.

Uncertainty budget.

- $A \cdot B$ extrapolation to $N \to \infty$ (Shanks vs $1/N^2$ fit spread): $\pm 3\%$
- Subleading contributions. The F^3 correction has been propagated through the bubble: the sensitivity is $|\Delta(A \cdot B)/(A \cdot B)| \approx 8.4 \cdot |c_{F^3}|$, extracted from a c_{F^3} sweep at $N = 8, 12, 16, 20$ with $1/N^2$ extrapolation. Under the continuum 1-loop fermion-induced value $c_{F^3} \approx -1/(360\pi^2) \approx -2.8 \times 10^{-4}$, the shift is $\approx 0.24\%$ — negligible within the extrapolation uncertainty. Under a conservative upper bound (naive $N_f = 6$ flavor multiplication, $c_{F^3} \approx -1.7 \times 10^{-3}$), the shift is $\approx 1.4\%$. A raw lattice value $c_{F^3} \approx -0.034$ extracted separately is in a different normalization (a $\sim 120\times$ discrepancy vs the continuum Wilson coefficient was noted in that extraction) and is not directly applicable in the bubble integration without a normalization-matching step. The dominant remaining subleading contribution is lattice V_{4g} cos-factor corrections to the tadpole's p^2-coefficient (absent in the continuum V_{4g} used here), bracketed at $\sim$ 3–12% (see Open work in §6.2.1).
- Convention and gauge-fixing prescription in the induced theory: $\pm 2\%$, bounded by the standard-QCD cross-check's 100% agreement.

The combined uncertainty is dominated by the extrapolation (statistical-like, $\pm 3\%$). Including estimated subleading contributions gives a total uncertainty $\lesssim 6\%$ at the lower end of the V_{4g} bracket (under which the computed value and the fitted value agree); the upper end of the bracket would widen the combined uncertainty correspondingly. The central extrapolated value $A \cdot B \approx 48$ carries the extrapolation-only uncertainty of the $1/N^2$ model, and is conditional on that leading power (§6.2.1).

Methodological notes. The Wilson-plaquette 3-gluon vertex is the leading piece of the full induced Γ_3; the full induced vertex is transverse by Ward identity (verified numerically at $N = 20$ to 1%), while the Wilson-plaquette piece alone requires the ghost contribution to restore transversality. The tadpole with continuum V_{4g} contributes only to the gauge-violating mass term and does not enter Π_T at leading order; lattice V_{4g} cos-factor corrections contribute to Π_T at the few-percent level and are currently subsumed into the systematic uncertainty (see Open work below). The OI framework has no formal Lagrangian gauge-fixing (the gauge field is induced rather than independent), but the induced propagator being a scalar in color and Lorentz is consistent with effective Feynman gauge, which the ghost prescription assumes.

Reproducibility. The full calculation runs in ~ 40 minutes on a single CPU core at $N \leq 40$. The implementation (`reproduce_AB.py` and supporting files) is archived with the code repository referenced in §8.

Open work. Separating A from B individually requires the next-order coefficient of $(C_2 g_0^2)^2$ in the expansion of the gauge self-energy. In the OI induced theory, this coefficient is structurally suppressed at strict 2-loop across all color channels. The C_A^2 channel receives only negligible Faddeev-Popov ghost contributions, verified numerically at approximately 0.2% of $A \cdot B$ via a direct 2-loop ghost self-energy computation. The mixed $C_A \cdot C_F \cdot T_F \cdot N_f$ channel, arising from crossed-topology fermion-line diagrams (vertex correction, sails), also approaches zero at continuum limit: large-N scans of the vertex-correction integrand show values decreasing as $1/N^{\sim 5}$ (raw: $9.96, 2.59, 0.64, 0.046$ at $N = 6, 8, 10, 12$), consistent with Ward-identity cancellation against the bubble/seagull self-energy pieces. Induced 3- and 4-gluon vertices first appear at 3-loop order (no tree-level gauge self-interactions exist in the induced theory). A clean separation of A from B therefore requires 3-loop lattice perturbation theory or non-perturbative methods, both beyond the current scope. Under the one-parameter constraint $A \cdot B = 48$ alone (§6.2.1), fitting the remaining freedom to the observed SU(2) and SU(3) couplings gives $(A, B) \approx (8.07, 5.95)$, numerically close to the factorization $(A, B) = (I_{\text{loop}}, N_f) = (8, 6)$, where $N_f = 6$ is the structural fermion count from $K = 2d$ (Theorem 6) and $I_{\text{loop}} = 8$ is the bare 1-loop lattice integral factor in the OI Π_s-normalized scheme (recovered from $A \cdot B = N_f \cdot 2T(R) \cdot I_{\text{loop}}$ with $2T(R) = 1$ for fundamentals). This factorization is compatible with the framework's universality requirement on A: I_{loop} is a property of the lattice scheme (geometry plus Π_s normalization), independent of which gauge factor the threshold is being evaluated for. The identification $A = \dim \operatorname{adj} \mathrm{SU}(3)$ — paired with the numerical match $\dim \operatorname{adj} \mathrm{SU}(3) = 8$ — is incompatible with universality (it would make A group-specific) and is a numerical coincidence rather than a structural identification. Both factorizations are consistent with the unconstrained fit $(8.3, 5.59)$ within the $\pm 3\%$ systematic uncertainty on $A \cdot B$. No rigorous derivation of individual A and B is yet available: the structural origin of $I_{\text{loop}} = 8$ on the $d = 3$ cubic lattice (candidates include $2^d = 8$ BZ corners, 4 staggered tastes $\times$ 2 sublattices, or another lattice-geometric factor) is an open follow-up; identifying it would upgrade $A \cdot B = 48$ to a structural derivation of (A, B) individually. The F^3 correction has been propagated (see uncertainty bullet above); the remaining systematic component is dominated by lattice V_{4g} cos-factor corrections. Two independent derivations of the lattice V_{4g} tadpole contribution have been carried out — one based on Wilson-extension of the continuum pair-topology decomposition, the other on per-topology Lorentz structure following Peskin Eq. 16.6 with Wilson cosines per Capitani Eq. 5.28. They differ structurally in the placement of cosine factors and yield Π_T contributions whose magnitudes differ by a factor ~ 14 at the integrand level (traced analytically to the $2\cos(p_\mu) \sum_\gamma \cos(k_\gamma)$ piece present in the per-topology form but absent from the simpler pair-topology form). The corresponding shift on $A \cdot B$ is bracketed at $\sim$ 3–12% depending on which form is normalized correctly; both are honest derivations and neither matches the textbook continuum tadpole formula directly at $p = 0$. Closing this ambiguity requires either a verified reference Wilson V_{4g} implementation or a fully-worked textbook 1-loop lattice Yang-Mills gluon self-energy computation as an external anchor; both are beyond current scope. The $\pm$4–5% residual systematic quoted above accommodates the lower end of this bracket; the upper end ($\sim$ 12%) would widen the systematic correspondingly. The central extrapolated value $A \cdot B \approx 48$, derived from $1/N^2$ extrapolation of the two-loop gauge self-energy and conditional on that leading power (§6.2.1), is independent of this V_{4g} subleading correction.

Resolution of the cosine-placement ambiguity (per-topology confirmed). The structural ambiguity above has been resolved at the level of cosine placement by direct comparison against the explicit Wilson four-gluon vertex of lattice perturbation theory (Capitani, *Lattice Perturbation Theory*, Phys. Rept. **384** (2003) 71, Eq. (5.28), attributed therein to Rothe, *Lattice Gauge Theories: An Introduction*, 1997). The vertex's cosine factors carry difference arguments spanning all four legs (of the form $\cos(a(q-s)_\mu/2)$ etc.); on the tadpole contraction, where two legs carry the loop momentum k and two the external momentum p, such an argument becomes $\cos(a(p \mp k)_\mu/2) = \cos(ap_\mu/2)\cos(ak_\mu/2) + \sin(ap_\mu/2)\sin(ak_\mu/2)$, which produces external$\times$loop cosine cross-terms at the vertex. The lattice four-gluon vertex thus carries loop-momentum cosine dependence at the vertex, unlike the fermion seagull (whose tadpole vertex depends only on external momentum), so the $2\cos(p_\mu)\sum_\gamma \cos(k_\gamma)$ cross-term is a feature of the gauge tadpole rather than a retained artifact. The per-topology form retains this cross-term; the pair-topology form omits it. The comparison therefore selects the per-topology form at the level of cosine placement (not magnitude): on this resolution the residual V_{4g} systematic sits at the upper ($\sim$ 12%) end of the bracket above. The exact $A \cdot B$ shift remains to be computed by inserting Eq. (5.28) into the tadpole integral in the present Π_s-normalized scheme (a finite, well-defined task), but does not affect the central extrapolated value $A \cdot B \approx 48$, which is independent of the V_{4g} correction.

6.3 The prediction

Group	$1/\alpha_0$	δ_0	$A \cdot \ln(1+Bx)$	RG	Predicted	Observed
U(1)	23.25	10.02	0.00	+25.73	**59.00**	59.00
SU(2)	23.25	10.02	16.17	−19.88	**29.57**	29.57
SU(3)	23.25	10.02	19.13	−43.93	**8.47**	8.48

Parameter count. The chain contains one structurally computed input ($1/\alpha_0 = 23.25$) and three fitted parameters of the threshold function (δ_0, A, B) matched to the three observed couplings. Three parameters against three observables produces zero residual by construction; the zeroes in the right column are a consequence of the counting, not evidence of predictive power in themselves. The predictive content of §6 consists of the one computed value, two structural constraints on the threshold function, and a consistency check.

(i) The computed value $1/\alpha_0 = 23.25$. The one-loop staggered vacuum polarization, evaluated analytically in §6.1. No free parameters. This is the foundational structural input of the ambient calculation, and the non-abelian factors would falsify the framework if observation required a significantly different value there. Its reading as a *native* hypercharge prediction inherits H-observer-bundle and H-Y-vertex for the U(1) row (§6.5): the determinant is constant over the induced abelian configurations, so the number is the ambient coefficient $\alpha^{-1}_{0,\mathrm{amb}}$ of the staggered gauge theory until the observer sector is shown to carry a curved hypercharge connection *and* to couple it with the vertex assumed in §6.1. The computation and its value are unaffected by those conditions; what they fix is whose coupling the number is. The non-abelian factors do not inherit them.

(ii) Structural constraint — universality of δ_0. The C_2-independent part of the threshold is by definition group-independent: a correction with no C_2 dependence cannot distinguish the three gauge factors. This is a structural constraint on the functional form of the threshold, not a fit. Its force is visible in the parameter count: without the universality constraint, the threshold function would carry five free parameters (three independent $\delta_0^{U(1)}$, $\delta_0^{SU(2)}$, $\delta_0^{SU(3)}$ plus A, B) matched to three observables — underdetermined, with the apparent match of the right column becoming empty. The universality reduces the count from five to three and makes the three-observable fit non-trivial: the fit could have been incompatible with a single δ_0 across rows.

(iii) Structural constraint — the log-resum form. The C_2-dependent threshold is parameterized as $A \ln(1 + B \cdot C_2 g_0^2)$, motivated by a geometric-series resummation of the gauge self-energy $1/(1 - \Sigma)$ with $\Sigma \propto C_2 g_0^2$. This constrains the C_2 dependence to a two-parameter family rather than an unrestricted function. The step from $1/(1 - \Sigma)$ to the logarithm is not derived in the current work; the log form should be read as a constrained parameterization consistent with the resummation structure, not as a derived result. The leading coefficient $A \cdot B$ (the $(C_2 g_0^2)^1$ term in the expansion) is computed from 2-loop lattice PT in the induced gauge theory (§6.2.1), giving $A \cdot B \approx 48$ under the $1/N^2$ leading-power assumption, against the fitted 46.4; the extrapolation is leading-power-dependent and the comparison inconclusive (§6.2.1). Separating A from B requires computing the next-order $(C_2 g_0^2)^2$ coefficient and is left for future work.

(iv) Consistency check with the 2-loop VP. The U(1) row fixes $\delta_0 = 10.02$ from observation, given (i). A first-principles 2-loop staggered VP computation was implemented as a cross-check, but on examination uses a scheme-locked Π_s convention. The consistency of a correctly-implemented 2-loop value with empirical δ_0 is an open question; the U(1)-row determination is the primary source of δ_0 in the current framework.

Status of the rows. U(1) uses one fitted parameter (δ_0) against one observable; the structural content is (i) together with the consistency check of (iv). SU(2) and SU(3) use two additional fitted parameters (A, B) against two additional observables; their structural content is carried by the universality constraint (ii) and the log-form constraint (iii). The product $A \cdot B$ (the leading coefficient in the $C_2 g_0^2$ expansion) is independently derived in §6.2.1 from 2-loop lattice PT in the induced gauge theory, giving $A \cdot B \approx 48$ under the $1/N^2$ leading-power assumption (inconclusive across finite-size models, §6.2.1) against the fit 46.4.

6.4 What is structural and what is solution-specific

§8.3 distinguishes structural predictions (properties of the equivalence class) from solution-specific properties (determined by the particular φ). The coupling calculation clarifies this distinction.

Structural: The universal induced coupling $\alpha_{0,\text{amb}}^{-1} = 23.25$ is structural — it depends on $N_f = 6$ (conditional on single-copy H-link, Theorem 6) and $T(R) = 1/2$ (fundamental representation), not on the eigenvalues of M. Structural here is a statement about the ambient staggered calculation; for hypercharge its reading as the native coupling additionally carries H-observer-bundle and H-Y-vertex (§6.5). The logarithmic form of the C_2 dependence is structural — it follows from the perturbative structure of the gauge self-energy at any coupling. The SM β-functions are structural.

Solution-specific: The eigenvalues μ_c, μ_w of M determine which modes are massless (Theorem 4: $\cos\omega = \mu \cos k$), hence the mass spectrum, but they do NOT affect the gauge coupling at the cutoff — the induced coupling depends on the Dynkin index, which is eigenvalue-independent. The gauge couplings at M_Z therefore depend on the eigenvalues only indirectly, through the RG running (which involves particle masses that set decoupling thresholds). For the leading-order prediction above, this indirect dependence is subleading.

Non-unification. The couplings do not unify at M_{Pl}. The common value $1/\alpha_0 = 23.25$ is the induced coupling before gauge self-interactions; the C_2-dependent threshold splits the couplings at the cutoff itself. This is not a failure of unification — it is the correct structure of an induced gauge theory, where all factors share the same fermion-induced coupling but have different gauge self-energies.

6.5 The confinement question

A natural question is whether the purely induced theory — with no fundamental gauge action — is in the deconfined phase required for the perturbative matching to be valid. Direct Monte Carlo simulation of $Z = \int DU_{\text{Haar}} \,|\det D(U)|^{2N_f}$ reveals that it confines: $\langle P \rangle \approx 0$ for all gauge groups (SU(3), SU(2), compact U(1)), all lattice volumes ($L = 2$–8), all fermion masses ($m = 0.01$–1.0), and all anisotropy parameters ($\mu = 0.5$–2.0). Cold-start simulations confirm monotonic decline of $\langle P \rangle$ from 1.0 toward 0. The fermion determinant cannot sustain weak coupling against the gauge field entropy.

This confinement is an artifact of the Haar measure. In the OI framework, the gauge field $M(\mathbf{n}, \hat{e}_j)$ is not drawn from the Haar measure on $\mathrm{SU}(N)$ — it is determined by the matter field $\phi \in (\mathbb{Z}/q\mathbb{Z})^{K\times N}$ (§3.1). The induced measure is the pushforward of the uniform measure on $(\mathbb{Z}/q\mathbb{Z})^{K\times N}$, which concentrates gauge configurations near the identity at large q. The rate depends on which reading of "uniform" is taken. Counting non-trivial link values, whose fraction scales as $1/q$, gives $\beta_{\text{eff}} \sim \ln q$; but that counting presumes the site-independent field, whereas the on-shell measure is the smooth $\Box^{-1}$ field of §4.2, for which the exponential map's $2\pi/q$ amplitude gives $1 - \langle P \rangle \sim q^{-4}$ and hence $\beta_{\text{eff}} \sim q^4$ (verified numerically for SU(3) over $q = 20$–200, and consistent with the plaquette-matching of §6.6).

The code that produced the archived numerical checks in this section and in §7.5 is no longer extant. The construction itself is documented — the measure is uniform on the substratum field, pushed through the on-shell exponential map — and its q^{-4} plaquette-deficit scaling has been independently reproduced (fitted exponent -4.11 over $q = 15$–127). The specific archived values have not been reproduced. A pre-registered re-derivation has been carried out on the directly sampled construction. At the archived point (4^4, $mL = 3$, matched $\langle P \rangle$) the directly sampled ensemble differs from the Wilson comparator in Z_S by $+2.6\% \pm 0.15\%$, beyond the pre-registered 2% tolerance, reproducibly across independent streams. The universality reported in this section is thereby established only for ensembles generated by Wilson dynamics on the image; for the directly sampled measure it is empirically violated at this point. Since q is gauge (§2.7) and the physical coupling is q-independent after renormalization, neither form is load-bearing — $\beta = 11.1$ below is fixed by α_0, not by q — but the two differ by orders of magnitude in where they place a given β_{eff}, and only the second is consistent with §4.2. Direct simulation of the 1D state-dependent coupling (§3.1, explicit construction) confirms that the wave equation creates no gauge correlations beyond random matter, but the $(\mathbb{Z}/q\mathbb{Z})$ constraint itself acts as an effective gauge kinetic term.

Since q is gauge (§2.7), the physical coupling is q-independent after renormalization. The perturbative calculation (§6.1) works in the q-independent sector and correctly determines the physical coupling. The Haar-measure Monte Carlo explored a confined phase that does not correspond to the OI framework's gauge measure.

Status of the induced-measure identification. The step this argument turns on — that the substratum's uniform measure pushes forward to a near-identity (rather than Haar) gauge measure — is a load-bearing commitment whose status should be stated at the rigor level of §A.3. The free-sector correspondence (the matter/pseudofermion measure equivalence, §4.2 Remark) is established at Level 2; the interacting statement used here, that the pushforward under the explicit map $\phi \mapsto M(\mathbf{n}, \hat{e}_j) \in \mathrm{SU}(N)$ concentrates near the identity, is not at that level, because the explicit map is under-determined at several enumerable joints (the real-to-complex step, the unitarization prescription) and the natural candidate map $M = \mathrm{polar}(\phi(\mathbf{n})^\dagger \phi(\mathbf{n} + \hat{e}_j))$ has been found numerically to produce a *Haar* (hence confining) pushforward, not the near-identity one this section requires. The near-identity claim is therefore a conjecture about the correct map, not a derived property of the natural candidate; the concentration "at large q" holds for an un-centered variant whose canonicity is not established. This has a direct consequence for the falsification-capable lattice work: the A^2 Monte Carlo and any production gauge-sector computation sample a specific induced measure, so their verdict is conditional on the measure identification being the near-identity one rather than the candidate's Haar — a dependency that the pre-registration of those computations should carry explicitly. Resolving which map is correct, and whether its pushforward is near-identity or Haar, is the open structural task on which the coupling sector's derived-versus-fitted status ultimately rests.

A candidate resolution. The under-determination isolated above lives at two enumerable joints — the amplitude of the matter measure and the map prescription — and both admit a natural resolution, set down here for scrutiny as a candidate rather than asserted as closed. First, the measure feeding the pushforward is the uniform measure *on the energy shell*, not the independent-uniform measure on $(\mathbb{Z}/q\mathbb{Z})^{K\times N}$: by the sphere-Gaussian projection of §4.2 the on-shell measure is the smooth Gaussian with covariance $[\Box_{\mathrm{lat}}^{-1}]_{xy}$ — the same $O(1)$-amplitude field the pseudofermion equivalence already requires — so the two readings of "uniform" give opposite physics, and only the on-shell/smooth one is consistent with §4.2. Second, the canonical map from the $\mathbb{Z}/q\mathbb{Z}$-valued matter data to the group is the exponential $M(\mathbf{n}, \hat{e}_j) = \exp(\frac{2\pi i}{q}\,\mathrm{Lie}\,[\phi(\mathbf{n}+\hat{e}_j) - \phi(\mathbf{n})])$ — the composite of the canonical roots-of-unity homomorphism $\mathbb{Z}/q\mathbb{Z} \to \mathrm{U}(1)$ with the Lie exponential $\mathfrak{g} \to G$ — which respects the $2\pi/q$ scale that the scale-invariant polar decomposition ($\mathrm{polar}(cA) = \mathrm{polar}(A)$) discards; this is why the polar candidate returns Haar irrespective of amplitude. On the smooth $\Box^{-1}$ field the exponential map yields a *near-identity* pushforward carrying weak, non-zero, octahedrally-equivariant non-abelian field strength — the $\beta_{\mathrm{eff}} \sim \ln q$ regime this section requires — while the polar candidate's Haar result is recovered only on the (inconsistent) independent-uniform field. This has been checked numerically for SU(2) and SU(3): $\langle P\rangle \to 1$ as $q \to \infty$, with the deviation from unity vanishing as the field strength while remaining strictly positive. The identification's *kinematic* half — which measure, and which map — is thereby raised to theorem level, conditional on the additive-automorphism (amplitude-scale) gauge principle it invokes, which [Substratum §4, Theorem 24, *Status*] classifies as an adjoined operational input rather than a theorem (§2.7); the map-uniqueness therefore inherits that principle's open status. The measure is fixed by the sphere-Gaussian projection of §4.2. The map is fixed by a uniqueness statement grounded in the framework's own gauge principle. The physical gauge is the alphabet's additive-automorphism group — not arbitrary relabellings of $\mathbb{Z}/q\mathbb{Z}$, since a nonlinear relabelling changes the fluctuation dispersion and so is not a symmetry of the observables — so the induced connection must be a homomorphism of the matter field's additive ($\mathbb{Z}/q\mathbb{Z}$) structure, the structure the wave equation evolves. Under that requirement the exponential is the *unique* map up to the choice of generator (the homomorphisms $\mathbb{Z}/q\mathbb{Z} \to \mathrm{U}(1)$ being exactly the

q-th roots of unity), and the polar candidate, a function of the multiplicative bilinear $\phi^\dagger\phi$ rather than the additive difference, is excluded because it is not such a homomorphism. The same map is selected independently by the continuum limit: q is a resolution parameter, so a sensible $q \to \infty$ limit is required, and the exponential's rescaled connection $(q/2\pi)\log M$ tends to the continuum gauge field q-independently, whereas the polar/Haar map is confining at every q with no such limit. The substantive residual is now the *dynamical* statement, that the interacting measure — the smooth pushforward reweighted by $\det(D^\dagger D)^{N_f}$ — retains this near-identity concentration. Because that reweighting is extensive, it does not follow from the classical pushforward result and must be argued directly. It does follow from a free-energy comparison. The links are unitary, so the staggered spectrum is bounded and $|\ln\det(D^\dagger D)|$ is bounded by cN with $c = O(1)$ *independent of* q: the determinant's total influence on the measure is extensive but q-blind. The prior, by contrast, is not: disordering the links requires field differences of order q, which the on-shell Gaussian penalizes by $\sim Nq^2/\pi^2$. The determinant therefore cannot pull the field out of the near-identity region once q exceeds an $O(1)$ threshold — numerically $q \gtrsim 9$ at $N = 4^4$, $m = 0.1$, $N_f = 6$ — and at the physical alphabet size the prior wins by a factor $\sim q^2$. The determinant moreover does not oppose the identity in the first place: at the same parameters $\ln\det(D^\dagger D)$ is $+260$ at $U = I$ against -85 on Haar-random links, so it *reinforces* the near-identity concentration rather than competing with it, which is why the Haar-measure confinement of the induced theory reported above is a statement about the gauge entropy of the Haar measure, absent here, and not about the determinant. The residual is the rigor of the comparison rather than its direction: the constants are measured at one small volume and mass, and a first-order transition at intermediate q is not excluded, though none appears. Whether the near-identity measure additionally carries the chiral condensate the A^2 amplitude requires (§7.5) is a separate question, not settled here. The abelian row carries a named condition rather than a gap, and the condition is sharper than a statement about classical field strength. The induced abelian connection is a lattice gradient, so it is pure gauge — and because the matter field is $\mathbb{Z}/q\mathbb{Z}$-valued the gauge function $G_x = \exp(2\pi i\,\phi_x/q)$ is single-valued, leaving no vortex sector either. The consequence reaches the fermion determinant: the staggered operator at an induced abelian link satisfies $D[U[\phi]] = G^\dagger D[I]\,G$ with G diagonal and unitary, so $D^\dagger D$ is unitarily similar to its free value and every spectral quantity built from it — $\mathrm{Tr}\ln D^\dagger D$ included — is *constant* over the induced U(1) configurations. The induced coupling computed above is therefore a second derivative of the determinant with respect to an arbitrary background link, taken in directions the microscopic image does not contain; reading it as a native U(1) coupling requires that the physical or coarse-grained gauge sector carry genuine transverse connection degrees of freedom, and not only the pure-gradient image $\{U[\phi]\}$ — the requirement recorded below as **H-observer-bundle** together with **H-Y-vertex**. That condition is more demanding than it first appears, because the obstruction is one of holonomy rather than of lattice geometry. A coboundary connection telescopes around *every* cycle of *any* graph — the state-dependent long-edge graphs of §3.1 included — so a path-ordered product between two sites is path-independent and returns a coboundary of the same G: blocking is a fixed point of the property rather than a route out of it, and averaging over paths that are all equal leaves smeared blocking in the same position. The non-abelian rows cannot supply what is missing either, since the abelian factor is central in the direct sum and no commutator has a component along it; and reweighting by $\det(D^\dagger D)^{N_f}$ redistributes a measure over a support rather than enlarging one, and is in any case constant over these configurations. The general form of this is shorter than any of the individual arguments: the induced configurations are gauge transforms of the *trivial* connection, $U[\phi] = g \cdot I$ with $g_x = \omega^{\phi_x}$, so they lie in a single gauge orbit; any map C that depends on the link alone and is gauge-natural, $C(g \cdot U) = \Gamma(g) \cdot C(U)$, sends one orbit into one orbit; and holonomy is a gauge invariant. So the holonomy of $C(U[\phi])$ is *constant* over the whole induced manifold — whether or not C fixes the trivial connection — and S_{eff} is constant with it. The escape is therefore forced to be a construction that is not a function of the microscopic link at all, and it must supply curvature from somewhere else. The architecture presently identifies one such source: the geometry of the observer state bundle. The no-go excludes any construction that is a function of the link alone; it is not a census of every observer-level variable that might carry curvature, and the identification below is accordingly the one candidate the construction currently supplies rather than the only one there could be. If the observer carries a local line $|\psi_x\rangle$, the Bargmann connection $W_{xy} = \langle\psi_x|\psi_y\rangle / |\langle\psi_x|\psi_y\rangle|$ has holonomy $\arg\langle\psi_0|\psi_1\rangle\langle\psi_1|\psi_2\rangle\langle\psi_2|\psi_0\rangle$, which is generically nonzero — at $\psi_0 = (1,0)$, $\psi_1 = (1,1)/\sqrt{2}$, $\psi_2 = (1,i)/\sqrt{2}$ it is $\arg[(1+i)/4] = \pi/4$. What the framework currently supplies is not of that kind: the condensate stabilizer $\Sigma = aP_{T_1} + bP_E + cP_{A_1}$ of Theorem 5 is a fixed, site-independent combination, and a site-independent state gives a real positive loop product and hence zero Berry phase identically. The variation would have to come from the map into $\mathscr{H}_{\text{obs}}$ of §4.7.1.1, which is where the construction is presently undefined. What the abelian row owes is therefore two conditions rather than one. **H-observer-bundle**: the trace-out produces a local observer-state bundle with nonzero projective curvature in the hypercharge channel. **H-Y-vertex**: that observer connection enters the observer fermion operator with the same compact nearest-neighbour vertex and charge normalization used in §6.1. The second does not follow from the first — some curved observer connection existing says nothing about whether its vacuum-polarization coefficient is the value computed there. Two qualifications bound this. The non-abelian factors are not in the same position: there the telescoped exponents do not exponentiate to a pure gauge, the induced plaquette carries curvature at second order in $2\pi/q$, and the determinant varies across the induced configurations. And the statement concerns hypercharge at the cutoff, where the electroweak symmetry is unbroken: the electromagnetic U(1) of the broken phase is generated by $Q = T^3 + Y/2$, whose T^3 part lies in $\mathfrak{su}(2)$ and does carry curvature, so it is not transverse-free. The one residual that remains separate is whether the near-identity measure additionally carries the chiral condensate the A^2 amplitude requires (§7.5), which is not settled here. For audit purposes the bridge's status under the [appendix-A §A.3] scheme is recorded explicitly: the kinematic half (measure and map) is derived-conditional (solid modulo the amplitude-scale operational input of [Substratum §4, Theorem 24, *Status*]); the classical near-identity pushforward and the determinant free-energy comparison are numerically supported at one volume and mass (sketch-grade); the condensate transport is a documented gap. The induced measure is therefore a derived-conditional commitment, not a theorem, and any verdict of the A^2 falsification computation is a verdict on the conjunction of the amplitude closure with this bridge; any successor run order accordingly carries plaquette $\langle P\rangle$ and chiral-condensate measurements as first-class observables, so that a wrong-ensemble failure is empirically separable from a wrong-amplitude failure. The 2026-07-14 reconnaissance halt makes this concrete: the $\beta = 0$ induced-measure ensembles proved near-identity-leaning ($\langle P\rangle \approx 0.62$–$0.65$, ordering as m falls) but chirally symmetric (anti-GMOR trend, cutoff-null vector channel) — empirical support for the concentration half of this bridge and an empirical caution on the condensate half. Whether the physical induced measure carries the chiral condensate is therefore the live gating question for both this bridge and the A^2 observable itself.

Note on the dynamical-fermion MC. A separate set of dynamical-fermion HMC simulations is used to extract empirical inputs to the §7.5 mass relations (notably the chiral condensate $\Sigma \approx 0.20$ and the matching-scale verification $m^{\mathrm{MC}}_{\mathrm{match}} = 0.1234 \pm 0.0002$). These simulations use the standard Haar measure on $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ together with the standard staggered pseudofermion Berezin measure, run in the weak-coupling regime $\beta \in \{11.10, 7.40, 3.70\}$. Whether these standard-measure observables agree with the corresponding observables in the framework's induced $(\mathbb{Z}/q\mathbb{Z})$-pushforward measure is open work. At the level of Σ specifically, both measures concentrate near the identity at weak coupling — Haar via $\beta \gg 1$ giving $U \approx I + O(1/\sqrt{\beta})$, and the $(\mathbb{Z}/q\mathbb{Z})$-pushforward via $q \gg 1$ giving the framework's "fraction of nontrivial link values scales as $1/q$" — so the leading-order Σ values plausibly agree, with the empirical agreement of $m^{\mathrm{MC}}_{\mathrm{match}}$ with λg_0^2 at 1.4% consistent with this scenario. The agreement can be established directly, and a check at accessible parameters supports it. Matching the plaquette between the two ensembles fixes the $q \leftrightarrow \beta$ correspondence empirically, and Z_S may then be compared at matched $\langle P\rangle$ — a non-trivial test, since two measures agreeing on one observable need not agree on another. On a 4^4 lattice at $mL = 3$, the induced $(\mathbb{Z}/q\mathbb{Z})$-pushforward and the standard Haar–Wilson ensemble trace the same Z_S-versus-$\langle P\rangle$ curve to within 0.4–0.9% across the full range from strong coupling ($\langle P\rangle \approx 0.14$) to $\langle P\rangle \approx 0.82$, the latter being the plaquette of $\beta = 11.1$ itself: the induced action is Wilson-universal for this observable at the coupling the mass relations use, so the induced measure inherits the standard-measure Z_S. The same matching locates $\beta = 11.1$ at $q \approx 11$, rather than at the exponentially large value a $\beta_{\mathrm{eff}} \sim \ln q$ reading would place it; since q is gauge (§2.7) this fixes where the correspondence sits in the parameterization, not a physical alphabet size. The two families of residual correction separate cleanly under this test. The $O(1/q)$ corrections are the discreteness of the $(\mathbb{Z}/q\mathbb{Z})$-valued field, isolated by comparing the integer-valued field against its continuum idealization: at the §4.2 amplitude the field occupies only some eight letters whatever q may be — q setting the exponential map's $2\pi/q$ scale rather than the field's range, as its gauge status requires — and rounding renormalizes the $q \leftrightarrow \beta$ map by an $O(1)$ factor, inflating $1 - \langle P \rangle$ by some ten to twenty percent, while leaving the Z_S-versus-$\langle P \rangle$ relation itself intact to 0.01–0.14%, the deviation shrinking toward weak coupling. Discreteness is thus absorbed into the correspondence and does not touch the universal relation. The $O(1/\beta)$ corrections — the induced action's departure from the Wilson form — are what remains, and they are the larger of the two by roughly an order of magnitude, at the half-percent level in Z_S at matched $\langle P \rangle$. Three caveats bound all of this: the check is at one volume, one mass, and modest statistics without error bars, so the half-percent figure is an upper bound that may be partly statistical, and universality at the production parameters ($m \approx 0.12$, $L = 32$–64) is inferred rather than demonstrated; and the located q is contingent on the field-amplitude normalization of §4.2.

6.6 Status of the calculation

The chain in §6.3 reproduces all three SM gauge couplings at M_Z through the structural input $1/\alpha_0 = 23.25$ (computed analytically from the one-loop staggered VP, §6.1), the universal threshold $\delta_0 = 10.02$ (empirically fixed by the U(1) row; a 2-loop VP rebuild indicates δ_0 is not reproduced by strict 2-loop PT and is accordingly treated as empirical input), and the C_2-dependent threshold parameters $(A, B) = (8.3, 5.59)$ (matched to the observed SU(2) and SU(3) couplings at M_Z, with the product $A \cdot B = 46.4$ independently computed at ≈ 48 under the $1/N^2$ leading-power assumption, inconclusive across finite-size models; see §6.2.1). The structural predictions in this sector are: (i) the specific value $1/\alpha_0 = 23.25$, (ii) the universality of δ_0 across gauge groups, and (iii) the independent derivation of $A \cdot B$, consistent with the fitted value to $\sim 3\%$ under the $1/N^2$ leading-power assumption but inconclusive across finite-size models (§6.2.1). Separating A from B individually is open.

6.7 No-GUT structural prediction: a three-layer synthesis

§§4.6 and 6.4 individually establish components of a unified structural prediction that the framework makes against grand-unified scenarios. This subsection brings these components into one place and articulates them as a falsifiable forward-prediction set.

Layer A (group-theoretic). The Standard Model gauge group SU(3) $\times$ SU(2) $\times$ U(1) emerges directly from the cubic-commutant decomposition of the 6 nearest-neighbor link directions (§4.6, Theorem 7), with no intermediate large gauge group at any scale. The decomposition $6 = T_1(3) \oplus E(2) \oplus A_1(1)$ is single-step: three eigenspaces of the coupling matrix M, three commutant factors. The framework therefore has *no GUT stage* — no SU(5), SO(10), E_6, or Pati-Salam intermediate gauge structure at any energy. This contrasts with all standard string-phenomenology routes to the SM, each of which traverses some intermediate large gauge group: heterotic chains $E_8 \to E_6 \to SO(10) \to SU(5) \to SM$; F-theory $SU(5) \to SM$ via flux; type IIA Pati-Salam $SU(4)\times SU(2)_L \times SU(2)_R \to$ SM.

Layer B (coupling-level). The gauge couplings do not unify at any scale. §6.4 establishes this as a structural feature of induced gauge theory: the universal $1/\alpha_0 = 23.25$ at the cutoff is shared across all three gauge factors before gauge self-interactions, but the C_2-dependent threshold $A \ln(1 + B \cdot C_2 g_0^2)$ splits the couplings *at the cutoff itself*, with C_2 different across SU(3), SU(2), U(1). Below M_{Pl}, pure SM β-functions run the (already-split) couplings to M_Z, with no intermediate gauge structure modifying the running. Above M_{Pl}, the induced theory has no tree-level gauge action and the running differs from standard SM. The framework therefore predicts that the three SM gauge couplings remain non-meeting under any RG running consistent with SM matter content.

Layer C (mechanistic). The gauge couplings emerge from the fermion determinant via $S_{\text{eff}}[U] = -N_f \ln \det(D^\dagger D)$, with no fundamental gauge action at the substratum level (§6.1). This induced-gauge-theory structure is structurally distinct from: (a) standard QFT (gauge action is fundamental input, couplings are renormalizable parameters); (b) string-theoretic compactifications (gauge couplings depend on moduli — compactification volumes, dilaton vacuum expectation values, brane positions — with values determined by flux stabilization). In OI, the gauge couplings are forced by $N_f = 6$ (conditional on single-copy H-link, Theorem 6) and $T(R) = 1/2$ (fundamental representation), with no moduli parameters connecting predicted values to other values. This is a stronger statement than "no GUT": it asserts that the framework's gauge couplings exist as quantities through a specific mechanism (fermion determinant) that string compactifications do not share.

Forward-prediction consequences. The three layers jointly predict:

(i) *No proton decay through GUT-mediated channels.* Standard SU(5) GUT proton decay through dimension-6 operators with suppression scale $M_{\text{GUT}} \sim 10^{16}$ GeV gives $\tau_p \sim 10^{34}$–10^{36} years. In OI's framework, the only available cutoff is $M_{\text{Pl}} \sim 10^{19}$ GeV; dimension-6 baryon-number-violating operators are suppressed by M_{Pl}^2, giving $\tau_p \sim M_{\text{Pl}}^4/m_p^5 \sim 10^{45}$–$10^{46}$ years. This is roughly 10 orders of magnitude beyond Hyper-Kamiokande's projected $\sim 10^{35}$ year sensitivity.

(ii) *No GUT-mechanism magnetic monopoles.* The 't Hooft-Polyakov monopole production at GUT phase transitions does not occur in OI's framework. The cosmological monopole problem (which inflation was historically invoked to dissolve) does not arise. Electroweak-scale monopoles remain possible at $\sim M_W/\alpha_W \sim 10^4$ GeV, with production rates set by the (crossover) electroweak phase transition; OI inherits the standard SM treatment of these.

(iii) *No GUT-chain cosmic strings.* Topological defects from SO(10) $\to$ SM chains with non-trivial π_1 of intermediate vacuum manifolds do not occur in OI. Standard cosmic-string searches non-detection is consistent with OI's prediction.

(iv) *Discriminating prediction: shared origin of the three gauge couplings.* The three SM gauge couplings emerge from a single fermion-determinant computation with $N_f = 6$ and $T(R) = 1/2$; they are not independent moduli. This is the OI-specific structural feature distinguishing the framework from string-with-no-GUT scenarios, in which gauge couplings depend on independent compactification data. Empirically, the framework's matching of α_3, α_2, α_1 at M_Z to the observed values via the §6.3 chain confirms this structural prediction at the few-percent level.

Empirical status. The framework's predictions are jointly tested by: (a) continued non-observation of proton decay at Super-Kamiokande and forthcoming Hyper-Kamiokande (proton lifetime bounds tightening past 10^{35} years would support OI; detection would falsify it); (b) continued non-observation of magnetic monopoles in cosmic-ray and condensed-matter searches (consistent with OI); (c) absence of cosmic string signatures in CMB and gravitational-wave observations (consistent with OI); (d) the precision matching of α_3, α_2, α_1 at M_Z via the universal-induced-coupling chain (already empirically supported at few-percent level).

What this synthesis does not do. The three-layer prediction does not uniquely identify OI among no-GUT frameworks. Any framework with no GUT and Planck-scale-suppressed dimension-6 operators makes prediction (i); any framework without intermediate symmetry breaking makes (ii) and (iii). What is OI-specific is prediction (iv): the shared-origin structural relation among the three gauge couplings. The full no-GUT prediction set is therefore: confirmation of (i)-(iii) supports OI but is also consistent with other no-GUT frameworks; (iv) plus the universal $1/\alpha_0 = 23.25$ value is the OI-distinguishing content. Refutation of any of (i)-(iv) would be evidence against OI; (i)-(iii) would also be evidence against the broader no-GUT class.

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7. Quantitative Predictions

The derivation chain of §§4–6 fixes the structural content of the Standard Model — gauge group, matter content, discrete symmetries, and gauge couplings. This section presents the quantitative predictions for the remaining SM observables: the CKM matrix elements, the charged-lepton mass relations, the fermion mass chain (six masses from one input), the PMNS angles, the Higgs mass, and the bottom-to-tau mass ratio. All predictions follow from the cubic-lattice geometry. The structural-vs-bijection-level distinction (developed further in §8.3) applies throughout: the framework predicts the *angular* and *ratio* structure of fermion mass relations from cubic-group representation theory, but the overall mass scale of each chain is set by the specific bijection φ and is taken as one empirical input per chain. The predictions in this section therefore have the form "structural relations among observables, with a small number of empirical inputs setting the overall scales." Subsection 7.6 collects the full prediction table.

7.1 The CKM matrix

The Cabibbo angle.

The three T_1 sectors — the candidate generations — occupy BZ corners: $X_j = \pi\, e_j$. The distance between adjacent corners is $|X_i - X_j| = \pi\sqrt{2}$, a geometric constant of the cubic lattice. The Cabibbo angle equals the inverse of this distance:

$$\boxed{\lambda = \frac{1}{\pi\sqrt{2}} = 0.22508}$$

Observed [20]: $\lambda = 0.22500 \pm 0.00067$. Match: 0.12σ (0.04%).

The mixing matrix element between generations i and j is the continuum fermion propagator at the inter-generation momentum: $|M_{ij}| = |S(X_j - X_i)| = 1/|X_j - X_i| = 1/(\pi\sqrt{2})$. The $1/|q|$ form (not $1/|q|^2$) arises because the taste-changing transition preserves chirality.

Chirality preservation of the taste-changing vertex. This is established in the staggered fermion literature. Mason et al. [32] show explicitly that taste-changing transitions in staggered quarks have spinor structure given by a combination of γ_μ and $\gamma_{5\mu} = \gamma_5\gamma_\mu$ (their Section 2.1: "Each quark line has an odd number of gamma-matrices and therefore the spinor structure must be a combination of γ_μ, $\gamma_{5\mu}$"). Both γ_μ (vector current) and $\gamma_5\gamma_\mu$ (axial current) commute with the projection operators $P_{L,R} = (1 \mp \gamma_5)/2$ in the standard sense — they map $L \to L$ and $R \to R$ rather than $L \leftrightarrow R$ — so any combination of them is chirality-preserving. This is also forced by the unbroken $U(1)_\epsilon$ chiral symmetry of the staggered formulation: a chirality-flipping inter-generation vertex would require an explicit mass insertion violating $U(1)_\epsilon$, which is forbidden in the massless limit.

In the massless limit, the chirality-preserving propagator structure $\gamma \cdot q/q^2$ contracted with the chirality-preserving vertex gives $\mathrm{Tr}[\gamma \cdot S(q)] \propto 1/|q|$. Hence $|M_{ij}| \propto 1/|X_j - X_i|$ rather than $1/|X_j - X_i|^2$. The $1/|q|^2$ form (chirality-flipping with mass insertion) gives $\lambda^2 = m_d/m_s$ — the GST relation [21] follows as a corollary (self-energy requires two propagator insertions, so the down-strange mass ratio inherits the squared scaling). The observer's continuum theory has no BZ periodicity, so the propagator at $|q| = \pi\sqrt{2}$ is smooth.

Status of the identification (named hypothesis). The chain above contains three links of different strength, and it is important to grade them separately. The corner geometry ($|X_i - X_j| = \pi\sqrt{2}$ for adjacent generations) and the chirality-forced $1/|q|$ power (established via the staggered-fermion literature) are solid. The load-bearing link is the identification itself — that the dimensionless mixing-matrix element *equals* the taste-changing propagator magnitude at the inter-corner momentum, in lattice units, with prefactor exactly unity. This is a hypothesis, not a derivation: a mixing angle is a property of the relative misalignment of two mass matrices, and no derivation from the substrate Yukawa diagonalization to "mixing element = free propagator magnitude, coefficient one" is currently given; the unit prefactor (no coupling, wavefunction, or normalization factor) and the selection of the adjacent-corner distance (rather than another invariant of the same corner set) are commitments of the hypothesis rather than consequences of the geometry. The resolving computation is the taste-changing two-point function between T_1 corner states with explicit external-leg normalization, evaluated on the $d = 3$ staggered lattice — the same machinery class as the $1/\alpha_0$ integral of §6.1 and the $\mathcal{N}$ integral of §7.2 — whose output is the dimensionless coefficient c_λ in $\lambda = c_\lambda/(\pi\sqrt{2})$. The hypothesis commits to $c_\lambda = 1$; the empirical match at 0.04% confirms the commitment but does not derive it, and a computed c_λ away from unity by more than the experimental precision falsifies the identification. Taken by itself, this link would rest at "geometry and power forced, identification hypothesized and empirically confirmed." The *Structural narrowing* below discharges the load-bearing part of the resolving computation: the unit-prefactor condition is shown to be an algebraic identity of the staggered spin-taste map, so the structural level of the identification is closed and the only residual is the emergent-theory dressing. The prediction's status is therefore "geometry and power forced, unit prefactor a structural identity, emergent dressing pending" — established at the structural level rather than merely hypothesized, and a complementary open link to (rather than the same footing as) the Koide amplitude's dynamical-closure caveat (§7.2).

Structural narrowing of the hypothesis. Three exact results reduce what the resolving computation must decide. First, the free-level taste-changing amplitude vanishes identically: the free staggered spectrum is exactly eight-fold taste-degenerate at every momentum and taste-diagonal in its eigenbasis, so there is no free-level object for the identification to refer to — c_λ is an interaction-order quantity. Second, the generation states are taste *eigenstates* — superpositions across Brillouin-zone corners — rather than single-corner modes; the $\pi\sqrt{2}$ geometry survives at the operator level (the chirality-preserving taste-changing operators are exactly the two-flip class), but the external legs of the resolving computation must be taste eigenprojections. Third, the $1/|q|$ power arises as the energy denominator of second-order transition perturbation theory, and the candidate dispersions at the transfer momentum discriminate sharply: the substrate boson kernel gives $\lambda = 1/(2\sqrt{2}) = 0.3536$, excluded by the measured Cabibbo angle at $\approx 190\sigma$; the substrate fermion kernel is singular there (the transfer momentum is itself a Brillouin-zone corner); the emergent continuum dispersion gives $\lambda = 1/(\pi\sqrt{2})$ exactly. The hypothesis $c_\lambda = 1$ is therefore equivalent to the statement that the intermediate states of the mixing perturbation theory are the emergent continuum fermions — an emergent-EFT statement by necessity rather than by choice — and the remaining content of the resolving computation is the one-loop coefficient: whether the cubic-group Yukawa vertex and the taste-eigenprojected, residue-normalized external legs multiply to exactly unity. A scoping computation of the branch-restricted transfer operator found it to be $(1/\sqrt{3})$ times a unitary — all four singular values equal $1/\sqrt{3}$, the $(1 - A^2, A^2)$ projection split appearing in the transfer's branch structure — sharpening the gate question to whether the on-shell residue normalization absorbs this factor exactly. The full on-shell computation answers it: between physical legs — the $(1 + \gamma_0)/2$-projected spin⊗taste states of the staggered-to-Dirac reconstruction, at the massless point the free fermion sector occupies — every spin-resolved taste-changing element of the site-local Yukawa transfer has unit modulus exactly; the vertex acts as a taste permutation with unit weight in each physical channel, and the $1/\sqrt{3}$ of the scoping pass is identified as a branch/spin conflation that the physical projection removes. The gate as defined — vertex and residue-normalized legs multiplying to exactly unity — is therefore answered affirmatively at its structural level, and the answer is algebraic rather than numerical: under the spin-taste reconstruction the two-flip transfer maps to a unitary monomial ($\gamma_0\gamma_1\gamma_3$ in the spin slot for the axes-$(1, 2)$ transfer, times a taste monomial) — a generalized permutation whose nonzero matrix elements have unit modulus identically — and this monomial commutes with γ_0, so the physical energy projection preserves the structure. The unity condition is thus an algebraic identity of the staggered spin-taste map, with the finite computation serving as its witness. What remains beyond this structural level is the emergent-theory dressing.

The Wolfenstein A parameter.

The A_1 (Higgs) taste sits along the democratic direction $\hat{h} = (1, 1, 1)/\sqrt{3}$. Each generation axis e_j makes an angle θ with $\hat{h}$ where $\cos\theta = 1/\sqrt{3}$. The CKM mixing is driven by the perpendicular component:

$$\boxed{A = \sin\theta = \sqrt{2/3} = 0.8165}$$

Observed [20]: $A = 0.826 \pm 0.012$. Match: 0.8σ (1.2%).

Combined: $|V_{cb}| = A\lambda^2 = \sqrt{2/3}/(2\pi^2) = 0.04136$ (observed: 0.0408 ± 0.0014, 0.4σ).

Structural identity. $A^2 = (d - 1)/d = C_2(T_1)/d$ is a representation-theoretic identity for vector representations, not a numerical coincidence. The off-democratic projection-squared $1 - 1/d$ (from $\hat{e}_i \cdot \hat{h}$ geometry) and the cubic-group quadratic Casimir $C_2 = d - 1$ for the vector representation are the same quantity. For $d = 3$ this gives $A^2 = 2/3 = C_2/d$ directly. This identity underlies the structural status of the natural cubic-lattice perturbation amplitude $\sqrt{C_2}/d = \sqrt{2}/3$ and the consequent Layer 1 prediction of $\sin^2\theta_{13}$ (§7.3).

Where A is protected and where it is not. The geometric value $A = \sqrt{2/3}$ is exact — it is the sine of a fixed lattice angle, independent of any bijection or dynamics. Its two downstream uses, however, carry different correction status, and the distinction determines which comparisons test the geometry alone. In the reactor angle, A enters *protected*: $\sin\theta_{13} = A^2\lambda$ is exact at $O(\lambda^2)$ because the second-order correction Δ_{13} is independent of the A_2 generator (§7.3), so that prediction tests the geometric A with no Layer-2 contamination. In $|V_{cb}| = A\lambda^2$, by contrast, A enters *unprotected*: the $(2,3)$ element's leading contribution arises at $O(\lambda^2)$ through the second-order coefficient b_{23}, so identifying the measured Wolfenstein A with the geometric value is a leading-order statement whose correction is the same open Layer 2(a)-tractable object as the Condition-2 ratio (§7.3, §8.3) — sharpening the $|V_{cb}|$-side prediction beyond $\sqrt{2/3}$ is that derivation, not a separate task. The observed deviation, $(0.826 - 0.8165)/0.8165 \approx 1.2\%$, is smaller than the natural correction scale $\lambda^2 \approx 5\%$, consistent with a suppressed b_{23} correction; the suppression is observed, not derived. A global-fit A drifting outside $\sqrt{2/3}\,(1 \pm \lambda^2)$ would strain the leading-order identification, while the protected $\sin\theta_{13}$ comparison would remain a clean test of the geometry regardless.

The down-to-strange mass ratio.

The Gatto–Sartori–Tonin relation [21] $\sin\theta_C \approx \sqrt{m_d/m_s}$ gives:

The GST relation [21] is an external empirical relation in flavor physics with known higher-order corrections. The framework's prediction is structural in λ (the cubic-lattice Cabibbo derivation above); the conversion to m_d/m_s inherits GST's empirical status. To this approximation:

$$\boxed{\frac{m_d}{m_s} = \lambda^2 = \frac{1}{2\pi^2} = 0.05066}$$

Observed [20]: $m_d/m_s = 0.0503 \pm 0.0007$ (FLAG 2024 Nf=2+1+1). Match: 0.56σ.

The Jarlskog invariant.

The Jarlskog invariant $J = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*)$ measures CP violation in the quark sector. In the Wolfenstein parametrization, $J \approx A^2\lambda^6\eta$. Using $\lambda = 1/(\pi\sqrt{2})$ and $A = \sqrt{2/3}$:

$$J = \frac{\eta}{12\pi^6}$$

where $\eta = 0.348$ [20] is the solution-specific CP-violating parameter. This gives $J = 3.02 \times 10^{-5}$ (observed: $(3.08 \pm 0.13) \times 10^{-5}$, 0.5σ). The structural suppression factor $12\pi^6 \approx 11{,}537$ is purely geometric.

CP-violating parameters.

The Wolfenstein parameters ρ and η require complex CKM entries. On the OI lattice, complex phases arise from the specific bijection φ and are likely solution-specific rather than structural.

7.2 Charged lepton masses and the fermion mass chain

Structural and bijection-level content. The three lepton generations come from the three triplet (T_1) staggered tastes, related at the structural level by the cubic symmetry O. The framework predicts the *angular position* $\theta_0 = C_2/d^2 = 2/9$ in the $\mathbb{Z}_3$-symmetric mass parametrization (a property of the cubic-group representation theory of T_1), but not the overall splitting magnitude (set by the bijection-level φ, taken as one empirical mass input per chain). Given the structural angular prediction and one mass input, the Koide relation determines the remaining masses; the same structural-vs-bijection-level distinction applies to the down-quark chain (below) and the cross-sector relations (below).

The Koide parameter.

The Koide relation [22] $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ holds to 0.001%. On the OI lattice the masses take the form $\sqrt{m_k} = \mu(1 + A\cos(\theta_0 + 2k\pi/3))$. The $\mathbb{Z}_3$ symmetry of the T_1 representation — cyclic permutation of the three BZ corners — fixes the *phases* to equal angular spacing $2k\pi/3$; it does not by itself fix the amplitude A, and the Koide value $Q = \frac{1}{3} + \frac{A^2}{6} = 2/3$ is the specific statement $A^2 = 2$. The amplitude shares the angle's structural origin: $A^2 = C_2(T_1) = l(l+1) = 2$, the quadratic Casimir of the generation representation, entering at the eigenvalue level as the amplitude-squared while the same Casimir fixes the phase $\theta_0 = C_2/d^2$ at the angle level (next paragraph) — the amplitude-versus-eigenvalue correspondence used for m_u/m_d below. Both Koide degrees of freedom thus descend from the single invariant $C_2(T_1) = 2$. The status of the amplitude is that of the angle's normalization (below), not stronger: the Casimir form is structural and $A^2 = 2$ matches observation exactly, but it is not yet derived from the dynamics — one-loop perturbative taste-breaking gives $A^2 \approx 0.34$, undershooting $C_2 = 2$, so establishing that the substrate dynamics produces $A^2 = C_2$ non-perturbatively is an open computation ([Ch. 19 §19.3.1]); its observable class is constrained in advance by an exact shift-symmetry identity — the ensemble single-quark two-point function is exactly eight-fold corner-degenerate at every coupling (driver `oi_lattice_code/fermion/shift_identity_check.py`), so quark-level taste splitting vanishes identically and any dynamical measurement is necessarily composite (multi-fermion channels). The relation is therefore "structural form forced, amplitude argued from the Casimir and empirically confirmed" — the same footing as θ_0, weaker than a full derivation and stronger than a fit.

The Koide angle.

The T_1 representation of SO(3) has angular momentum $l = 1$ and quadratic Casimir $C_2 = l(l+1) = 2$. The mass splitting within the T_1 triplet is an O-invariant quadratic form on the generation space. The unique such invariant is the quadratic Casimir C_2, which measures the strength of the anisotropy at each BZ corner. Normalized by d^2 (the lattice bandwidth in generation space):

$$\boxed{\theta_0 = \frac{C_2}{d^2} = \frac{2}{9} = 0.22222}$$

Observed: $\theta_0 = 0.222225 \pm 0.000006$ — the three charged-lepton pole masses determine (μ, A, θ_0) exactly (PDG 2024/25; uncertainty from $m_\tau = 1776.93 \pm 0.09$ MeV). Match: 0.001% (0.4σ). For the A_1 (Higgs) taste, $l = 0$ gives $\theta_0 = 0$ — no generational splitting, consistent with a single Higgs field.

Uniqueness of the normalization. The Koide angle is dimensionless, so it must be a pure-number ratio of structural constants of the $(d = 3, T_1)$ data: angular momentum l, Casimir $C_2 = l(l+1)$, dimension $\dim(T_1) = 2l+1$, lattice dimension d, and the dimensions of related representations and tensor products. Enumerating the natural ratios with the right order of magnitude:

Ratio	Value	Match to $\theta_0 = 0.222225$
$C_2/d^2 = 2/9$	0.22222	**0.001%** ✓
$\dim(E)/\dim(T_1 \otimes T_1) = 2/9$	0.22222	**0.001%** ✓
$C_2/(d(d+1)) = 2/12$	0.16667	25% off
$l/d^2 = 1/9$	0.11111	50% off
$\dim(T_2)/\dim(T_1 \otimes T_1) = 3/9$	0.33333	50% off
$(2l+1)/(d(d+1)) = 3/12$	0.25000	12% off
$1/d^2 = 1/9$	0.11111	50% off
$l/(2d) = 1/6$	0.16667	25% off
$C_2/(2d) = 2/6$	0.33333	50% off
$\dim(E)/\dim(\mathrm{Sym}^2(T_1)) = 2/6$	0.33333	50% off

Only 2/9 matches. The Casimir construction $C_2(T_1)/d^2$ is the physically correct derivation: $C_2(T_1) = 2$ measures the anisotropy strength of the T_1 generation representation (it is the unique $O(3)$-invariant quadratic scalar on the generation space), and $d^2 = 9$ is the natural geometric normalization from the cubic lattice bandwidth squared. The numerical coincidence with $\dim(E)/\dim(T_1 \otimes T_1) = 2/9$ arises because $\dim(E) = C_2(T_1) = 2$ for O acting on its vector representation — a genuine structural feature of this group. However, the operator space for the Hermitian mass-splitting matrix δM is $\mathrm{Sym}^2(T_1)$ of dimension 6 (not $T_1 \otimes T_1$ of dimension 9, which includes antisymmetric operators irrelevant to symmetric mass splittings), and $\dim(E)/\dim(\mathrm{Sym}^2(T_1)) = 1/3$, not 2/9. The dim-ratio interpretation using the full $T_1 \otimes T_1$ space is therefore an arithmetic artifact; the Casimir construction is the structural derivation, with the factor $d^2 = 9$ understood as the geometric bandwidth rather than an operator-space dimension.

An independent 2026 construction [42] arrives at the same two numbers by mechanisms outside the $(d = 3, T_1)$ structural data enumerated above — the $\sqrt{2}$ amplitude from a rank-one ("self-dual") coherence condition and $|\theta_0| = 2/9$ as the Berry-phase spectral sum $(3/2\pi^2)\sum_{3\nmid n} n^{-2} = 2/9$. The coincidence of independent routes on both numbers is recorded; the Casimir construction remains this framework's derivation.

Among dimensionally consistent $(d = 3, T_1)$ structural ratios enumerated above, $\theta_0 = C_2/d^2 = 2/9$ is the unique match to observation, and it has a clean physical interpretation: the anisotropy strength of the generation representation, normalized by the lattice bandwidth.

Connection to the §7.1 structural identity. The natural cubic-lattice perturbation amplitude on T_1 is $\sqrt{C_2}/d = \sqrt{2}/3$ (Casimir scaling over bandwidth). Second-order perturbation theory on the eigenvalues then gives $\theta_0 \sim (\sqrt{C_2}/d)^2 = C_2/d^2 = A^2/d$, where $A^2 = (d-1)/d = C_2/d$ is the structural identity of §7.1 for vector representations. The bandwidth-squared normalization $(1/d^2)$ reflects this PT scaling: amplitudes scale as $\sqrt{C_2}/d$, eigenvalue shifts as the square of the amplitude.

A first-principles derivation of the specific $1/d$ amplitude scaling (vs. alternatives like $1/(2d)$) from the cubic-lattice Yukawa structure remains an open task. It is important to be precise about what is forced and what is open here, because the distinction bears directly on whether the cluster of predictions sharing this normalization (Wolfenstein A, the Koide angle θ_0, $\sin^2\theta_{13}$, and m_u/m_d) is structurally derived or empirically fitted. Three things are forced and one is open. *Forced:* (a) that the relevant invariant is the quadratic Casimir $C_2(T_1) = 2$ — this is representation theory of the cubic group on its vector representation, not a choice; (b) that the eigenvalue shift scales as the square of the amplitude — this is the standard order-counting of second-order perturbation theory, not a choice; (c) that the bandwidth normalization is a power of d — this is dimensional, fixed by the lattice geometry. *Open:* whether the amplitude normalization is $1/d$ (giving $C_2/d^2 = 2/9$) or an alternative such as $1/(2d)$ (giving $C_2/(4d^2) = 1/18$). The open element is therefore a single discrete choice among a short enumerable list of geometrically admissible normalizations, not a continuous fit parameter. The framework commits to $1/d$ on the structural ground that the amplitude is the Casimir scaled by the single-axis bandwidth d (not $2d$), and this commitment is falsifiable: the $1/(2d)$ alternative predicts $\theta_0 = 1/18 = 0.0556$, which misses the observed 0.2227 by a factor of four and is excluded by data. The empirical match at 0.02% does not *select* the normalization from a fit; it *confirms* a structural commitment that was constrained, before any fit, to a discrete choice the framework resolves on geometric grounds. The geometric ground for $1/d$ over $1/(2d)$ is argued rather than proved from the Yukawa structure; closing this is the Layer 2(a)-tractable computation referenced in §7.3. One structural component of that closure can be stated now. The per-axis weight that controls the taste-off-diagonal normalization — the Brillouin-zone average $w = \langle \sin^2 q_1 / \sum_k \sin^2 q_k \rangle$ that appears in the second-order self-energy — evaluates to exactly $1/d$, and does so as a consequence of cubic symmetry alone: the d axis-fractions are equal by the cubic point group and sum to unity by construction, so each equals $1/d$ independently of the propagator's detailed form (verified across the massless staggered weight, the propagator-squared self-energy weight, and the Klein-Gordon $\sin^2(q/2)$ form, all giving $1/d$ to numerical precision). This forces the *per-axis* $1/d$ scaling structurally rather than leaving it to geometric analogy, and it is the mechanism behind why $1/d$ and not $1/(2d)$: the loop's axis-resolved weight is the symmetric fraction $1/d$, with no factor of two available from the axis decomposition. The overall coefficient relating this per-axis weight to the full perturbation-theory amplitude — the $m \to 0$ limit of $\mathcal{N}$ below and its prefactor, which the symmetry argument does not fix — has now been computed (driver `oi_lattice_code/fermion/n_ratio_pt.py`, anchored end-to-end: the 8×8 corner operator is derived numerically from the position-space staggered operator rather than assumed, the derived corner matrices satisfy the Clifford algebra to machine precision, and the sum-rule identity $d\,w + m^2 \langle 1/D \rangle = 1$ holds to 10^{-16}). The computation resolves a genuine underdetermination in the specification en route: the matrix-element readings of the taste-off-diagonal vertex vanish identically — the free staggered propagator has no two-flip corner entries, the same structural zero as the free-level taste-changing amplitude of §7.1 — leaving the substrate-eigenvalue reading of the following subsection as the unique non-vanishing one-loop construction; under it the per-axis object is $\hat{V}_a (\hat{D}^\dagger \hat{D})^{-1} \hat{V}_a = (\sin^2 k_a / D)\, \mathbf{1}$ exactly, Schur-diagonal, with prefactor exactly one. The residual is thereby discharged: the cluster's status is "structural form forced, per-axis normalization symmetry-forced, amplitude prefactor computed ($= 1$)" — the magnitude normalization is derived, conditional only on the substrate criterion of the following subsection. This closure promotes the *magnitude* normalization only; the bijection-specific directional content of the flavor structure remains Layer 2(b) by construction (it transforms under substratum state-relabeling and is not fixed by any substrate-only computation), so closure sharpens the magnitude rather than promoting the cluster as a whole.

Specification of the resolving computation. So that the open task is precise rather than gestured, the calculation that would settle $1/d$ vs $1/(2d)$ is the following, statable as a self-contained lattice perturbation-theory problem independent of the rest of the framework. Take the staggered fermion action on the $d = 3$ simple-cubic lattice with the OI taste structure (the T_1 vector representation on the three Cartesian link directions), and introduce the flavor-breaking perturbation as the second-order taste-mixing vertex induced by the cubic-group-symmetric Yukawa coupling. The quantity to compute is the ratio of the off-diagonal second-order self-energy on the T_1 representation to the perturbation amplitude squared:

$$\mathcal{N} \;=\; \lim_{m\to 0} \frac{\Sigma^{(2)}_{T_1,\text{off-diag}}(0)}{\left(g\,\sqrt{C_2}\right)^2},$$

where $\Sigma^{(2)}$ is the one-loop (second-order in the Yukawa) taste-off-diagonal self-energy evaluated at zero external momentum, computed as the lattice momentum integral over the staggered propagator $D(q) = \sum_\nu \sin^2 q_\nu + m^2$ with the cubic-symmetric taste vertices on the T_1 corners. The prediction is that this ratio evaluates to $\mathcal{N} = 1/d^2 = 1/9$ (the framework's $1/d$ amplitude normalization), as against $1/(2d)^2 = 1/36$ (the $1/(2d)$ alternative). (The following subsection refines this comparison: the decisive alternative is the *time-direction* value $\mathcal{N} = 1/16$ — i.e. whether the taste loop carries a time direction — not $1/36$; the $1/(2d)$ framing here is superseded by that analysis.) The two candidates differ by a factor of four and are not close, so the computation is decisive at modest numerical precision: a lattice evaluation of $\Sigma^{(2)}$ converging to within, say, 20% of either candidate settles the normalization. This is the same class of computation as the $1/\alpha_0 = 23.25$ staggered vacuum-polarization integral of §6.1 (which converges on a robust plateau), differing only in the vertex structure (taste-off-diagonal rather than gauge-current), so it is tractable with the same lattice-PT machinery already validated in §6.1. The framework predicts $\mathcal{N} = 1/9$; a specialist running this integral either confirms it — promoting the *magnitude* normalization $C_2/d^2 = 2/9$ from argued to derived — or refutes it (falsifying the $1/d$ normalization and with it the $\theta_0 = 2/9$ magnitude). This computation has now been run to closure (same driver as above): under the construction the specification's own refinement selects, $\mathcal{N} = w^2 = 1/d^2 = 1/9$ exactly at $m \to 0$, with the finite-mass plateau approaching from below as the sum-rule identity requires ($w^2 = 0.10906$ at $m = 0.1$, 0.11056 at $m = 0.05$) — inside the 20% band called decisive above, and exact rather than numerical by the identity. The computation requires no input not already fixed by the lattice geometry, which is what makes the magnitude question Layer 2(a)-tractable rather than open-ended. It is important to delimit what such a closure would and would not establish: the quantity $\mathcal{N}$ is a magnitude with a gauge-invariant interpretation, and its closure promotes the magnitude of θ_0 only. It does not, and cannot, fix the bijection-specific directional content of the broader flavor structure (which pair-ratio carries a given coefficient, the generation-label assignment, the CP phase), which remains Layer 2(b): these quantities transform non-trivially under substratum state-relabeling and are therefore not determined by any substrate-only computation, gauge-invariant by construction. The lattice computation specified here is thus a magnitude-closure target, not a route to deriving the full cluster; the directional/label content stays bijection-conditional by design.

Structure of $\mathcal{N}$: the dimensionality of the taste loop is the load-bearing choice.

The forced second-order scaling makes $\mathcal{N}$ the square of a per-axis weight $w = \langle \sin^2 q_i/D \rangle$, and the staggered sum rule $\sum_\nu \sin^2 q_\nu = D - m^2$ makes w an exact rational, so $\mathcal{N} = w^2$ is fixed entirely by which directions the loop runs over. The decisive feature is that there is no intermediate value. The time direction carries exactly the same share of the sum rule as each spatial one, $\langle \sin^2 q_0/D_4 \rangle = 1/4$, so a four-dimensional loop gives $w = 1/4$ and $\mathcal{N} = 1/16$ — and this holds *even when the taste vertex is purely spatial*, because the time direction still claims its quarter of the sum rule regardless of the vertex. Only a loop with no time direction at all gives $w = 1/d = 1/3$ and $\mathcal{N} = 1/d^2 = 1/9$. The two outcomes are far apart in observable terms: $1/9$ gives $\theta_0 = C_2/d^2 = 2/9 = 0.2222$ (observed 0.2227), whereas $1/16$ gives $\theta_0 = 1/8 = 0.125$, off by nearly a factor of two. The genuine open question is therefore not "$1/d$ versus $1/(2d)$" but whether the splitting is computed *with or without a time direction*; and the comparison to the four-dimensional $1/\alpha_0$ integral misleads on exactly this point, since $1/\alpha_0$ is a genuinely four-dimensional running-coupling quantity while θ_0 is not.

The framework's own architecture selects the time-free computation. The generation masses are eigenvalues of the substrate coupling structure — properties of the semisimple part of the deterministic bijection (the μ_k of the mass-generation mechanism) — fixed at the $d = 3$ spatial substrate, before the emergent time direction exists. The Koide angle is the splitting pattern of those substrate eigenvalues, so its second-order shift is an instantaneous $d = 3$ quantity carrying no time loop, giving $w = 1/3$ and $\mathcal{N} = 1/9$. On this reading the eigenvalue-perturbation description above is the correct one; the "$\Sigma^{(2)}$ self-energy" phrasing of the resolving computation must be understood as this substrate eigenvalue calculation and *not* as a four-dimensional emergent self-energy, which would carry the time direction and yield $1/16$. The two phrasings are otherwise in tension precisely on the dimensionality, and it is the substrate-eigenvalue origin of the masses that resolves it. The same criterion, stated once for both sectors: quantities that are *spectral* — eigenvalues of the substrate coupling structure, involving no intermediate-state propagation — are computed at the $d = 3$ substrate (the mass ratios and the Koide angle, as here); quantities that are *transitional* — matrix elements between distinct eigenstates, mediated by intermediate-state propagation — carry the emergent continuum kinematics on the propagating line (the mixing elements, §7.1, where the substrate alternatives are respectively vanishing, excluded, and singular). Spectrum is substrate; transition is emergent.

The open element is correspondingly sharper, and part of it can now be settled directly. The three sectors form the irreducible vector representation T_1, so by Schur's lemma any cubic-invariant (taste-blind) emergent dressing acts on them as a multiple of the identity — an affine map on the mass eigenvalues — which rescales the democratic mass component but leaves the splitting phase θ_0 exactly invariant. The generic four-dimensional Higgs-loop mass renormalization is of this taste-blind kind (the tree-level Yukawa is generation-independent, Theorem 10, and the Higgs occupies the singlet taste), so it provably cannot shift θ_0;

only cubic-symmetry-*breaking* structure can. That breaking is the substrate coupling-matrix eigenvalue structure (the μ_k of M, Theorem 4) — the same substrate-eigenvalue origin the mass mechanism uses elsewhere — so $\theta_0 = C_2/d^2$ introduces no assumption beyond that mechanism. What Schur's lemma does *not* exclude is a loop-level cubic-symmetry-breaking emergent operator: cubic symmetry alone does not forbid the radiative generation of such an operator (the same operator-mixing phenomenon that governs the radiative-stability question for emergent Lorentz invariance, §4.5), though none arises at leading order.

A second class escapes the Schur step without being cubic-symmetry-breaking: a dressing that is cubic-invariant and acts on the mass matrix as a superoperator rather than as a multiple of the identity on T_1. Cubic invariance forces such a map to be a scalar on each irreducible component of $\mathrm{Sym}^2(T_1) = A_1 \oplus E \oplus T_2$ and permits those scalars to differ; a correction proportional to the tree mass matrix is the special case in which they coincide. For the one-loop staggered taste kernel, $\lambda_{T_2} = 0$ exactly and $\lambda_E/\lambda_{A_1} \approx 0.22$–$0.24$ in three and four dimensions. On the $\sqrt{m}$ pattern such a dressing leaves θ_0 invariant: the angle is a direction within a two-dimensional irreducible component, on which an equivariant map acts as a scalar. On the mass pattern the $\sqrt{\cdot}$ nonlinearity breaks that invariance, with $d\theta_0/\theta_0 = \kappa\,(1 - \lambda_E/\lambda_{A_1})\,(\delta m/m)$ and $\kappa \approx -35$ at the physical Koide point, the magnitude reflecting the near-degeneracy $\sqrt{m_e} \ll \sqrt{m_\mu}$. The 0.2% agreement of $\theta_0 = 2/9$ bounds the taste-structured part of an emergent mass-level dressing by $\delta m/m \lesssim 8 \times 10^{-5}$; a second-order lepton-Yukawa loop supplies $\sim 10^{-6}$. The open element is a loop-level cubic-*breaking* operator. The status is thus "$\mathcal{N} = 1/9$ protected by representation theory against all taste-blind emergent dressing and fixed by the substrate eigenvalue structure, with the sole residual a possible loop-level cubic-breaking contribution that is shared with the emergent-Lorentz radiative-stability problem rather than specific to the flavor sector" — closer to "derived" than "argued geometrically."

Mass predictions.

With $Q = 2/3$, $\theta_0 = 2/9$, and $m_\tau = 1776.93$ MeV as input:

Mass	Predicted	Observed	Deviation
m_e	0.51098 MeV	0.51100 MeV	0.003%
m_μ	105.656 MeV	105.658 MeV	0.002%

The quark sector does not satisfy $Q = 2/3$: $Q_{\mathrm{down}} = 0.731$, $Q_{\mathrm{up}} = 0.849$. The down-quark deviation is numerically described by $Q_{\mathrm{down}} \approx (2/3)(1 + \alpha_s(2\,\mathrm{GeV})/\pi)$ to 0.15%, a suggestive regularity that matches the leading-order form of a color correction but for which a first-principles derivation from framework primitives is not currently available. Two observations argue against treating this form as structural: (i) the generation $\mathbb{Z}_3$ symmetry that produces $Q_{\mathrm{lepton}} = 2/3$ acts on the staggered-taste space (BZ corners), while SU(3) color acts on the internal T_1 (K-component) space — a direct breaking of the former by the latter would require a cross-coupling between the two T_1 representations beyond what §4.7.1.2 establishes; and (ii) the 0.15% match holds only in the PDG mixed scheme (light quarks at $\mu = 2$ GeV, b quark at $\mu = m_b$); in physically consistent single-scale schemes $Q_{\text{down}} \approx 0.755$–$0.760$ and the $(1 + \alpha_s/\pi)$ form misses by 4–5%. The observed numerical match is reported below as empirical input to the m_b determination, not as a derived structural relation. A full derivation would require either the link-carrier cross-coupling mentioned above, or a taste-decomposed Coleman–Weinberg analysis yet to be carried out.

The bottom quark mass from the down-sector Koide.

Taking $Q_{\text{down}} = (2/3)(1 + \alpha_s(2\,\text{GeV})/\pi)$ as a phenomenological input (above), the Koide formula determines m_b from m_d and m_s. Using $m_d/m_s = 1/(2\pi^2)$ (§7.1) and $m_s = 93.4$ MeV as the single input:

$$\boxed{m_b = 4144 \text{ MeV} \quad (\text{obs: } 4180 \pm 30,\ 0.9\%)}$$

The up-to-down mass ratio.

The Koide angle $\theta_0 = C_2/d^2 = 2/9$ determines the inter-generation hierarchy. Its square root determines the intra-doublet splitting:

$$\boxed{\frac{m_u}{m_d} = \sqrt{\theta_0} = \sqrt{2/9} = 0.4714}$$

The PDG value [20] is $m_u/m_d = 0.474 \pm 0.056$. Match: 0.05σ. Both arise from the same quadratic Casimir $C_2 = 2$ of the T_1 representation, acting in different channels: the inter-generation hierarchy enters at the *eigenvalue* level (Koide angle parametrizing the three masses in the T_1 representation, where $\theta_0 = C_2/d^2$ enters the eigenvalue formula directly through second-order energy-level structure), while the intra-doublet m_u/m_d ratio enters at the *amplitude* level (mass matrix element between the up and down components within a doublet). The square-root relation reflects the standard QM scaling: amplitudes $\sim V$, eigenvalues $\sim V^2$, so ratios at the amplitude level scale as the square root of ratios at the eigenvalue level.

Status: solution-specific (Layer 2(b)). The same-perturbation hypothesis underlying this derivation — that *one* perturbation V controls both the cubic-group T_1 Casimir splitting and the SU(2)-doublet up-down splitting — requires alignment between the SU(2) Yukawa structure and the cubic-group T_1 perturbation that is not forced by representation theory. Operator counting on the first-generation Yukawas Y_u^{11}, Y_d^{11} shows that each receives contributions from A_1 (singlet) and E (generation-dependent diagonal) channels of $T_1 \otimes T_1$ with four independent Wilson coefficients $(a_u^{(1)}, a_u^{(E)}, a_d^{(1)}, a_d^{(E)})$ in total. The ratio Y_u^{11}/Y_d^{11} depends on these four coefficients; no representation-theory argument forces it to equal $\sqrt{C_2}/d$. Per the §8.3 four-layer classification, this places $m_u/m_d = \sqrt{\theta_0}$ at Layer 2(b): the empirical match at 0.05σ indicates the framework's bijection has the relevant alignment property, but identifying that alignment as forced by symmetry (Layer 2(a)) requires explicit substrate-level Yukawa derivation that has not been performed. A 1-loop OI-vertex computation, while useful, would calculate the alignment for the specific bijection rather than show it is forced.

Mechanism specificity. Within the §8.3 sub-classification of Layer 2(b), m_u/m_d is *single-parameter direction selection*: the up-down asymmetry in OI's structure comes from the Higgs's choice of internal V_2-axis to condense along (which by gauge invariance is equivalent at the Lagrangian level for any direction, but is fixed once chosen by the bijection). The specific value $\sqrt{2/9}$ then follows from how the cubic structure projects this single-axis choice into the up vs down components — a one-parameter family of bijection-dependent values, not a multi-parameter Wilson coefficient ambiguity. The 0.05σ empirical match therefore constrains the bijection's V_2-axis selection more tightly than generic Layer 2(b) classification suggests.

The mass chain: six masses from one input.

The structural relations link all light fermion masses through a single input (m_s):

$$m_s \xrightarrow{\lambda^2} m_d \xrightarrow{\sqrt{\theta_0}} m_u \xrightarrow{Q_{\mathrm{down}}} m_b \xrightarrow{Z_S} m_\tau \xrightarrow{\theta_0} m_e,\, m_\mu$$

Mass	Formula	Predicted	Observed	Match
m_d	$m_s/(2\pi^2)$	4.73 MeV	4.67 ± 0.48	1.3%
m_u	$m_d \times \sqrt{2/9}$	2.20 MeV	2.16 ± 0.49	0.1σ
m_b	Koide(Q_{down})	4144 MeV	4180 ± 30	0.9%
m_τ	$m_b \times Z_S/4.28$	1755 MeV	1776.9	1.2%
m_e	Koide(θ_0, m_τ)	0.505 MeV	0.511	1.2%
m_μ	Koide(θ_0, m_τ)	104.4 MeV	105.7	1.2%

Six masses from one input, all within $\sim$ 1–1.3%. The remaining input m_s sets the overall mass scale.

The top quark mass.

The tree-level Yukawa couplings are taste-independent within each generation (Theorem 10 with standard staggered-lattice Yukawa structure [12]) and are normalized to unity at the compositeness scale M_{Pl} as the lattice-to-continuum matching convention: in the normalized wave equation (coefficient +1 on nearest-neighbor terms, §4.1), the staggered fermion-to-scalar overlap integral at a single site evaluates to an O-invariant Clebsch-Gordan coefficient normalized to unity. For y_t, the relation between this convention and the observed top mass is open rather than closed. Under standard one-loop SM running, the IR quasi-fixed point sits at $y_t(m_t) \approx 1.25$ ($m_t \approx 215$–220 GeV), not at unity; the normalization $y_t(M_{\text{Pl}}) = 1$ flows down to $y_t(m_t) \approx 1.19$, i.e. $m_t \approx 207$ GeV — a 20% excess over observation — and the attractor basin $y_{\text{UV}} \in [0.5, 5]$ focuses only to $m_t \approx 177$–221 GeV, a 25% spread. The observed top corresponds to $y_t(M_{\text{Pl}}) \approx 0.4$–$0.5$, not unity. What is empirically exact to 0.9% is the *weak-scale* statement $y_t(m_t) \approx 1$; the compositeness-scale normalization plus IR attraction does not deliver it. A derivation of $y_t = 1$ at the electroweak matching scale specifically — or of $y_t(M_{\text{Pl}}) \approx 1/2$, which runs to ≈ 177 GeV at one loop — would close the gap; neither is currently available. The relation $m_t = v/\sqrt{2}$ is accordingly classified as a retrodiction of the known weak-scale $y_t \approx 1$ coincidence with the structural mechanism open (§7.6), not as a parameter-free structural prediction. Since all other fermion-mass predictions use either the taste-independence equality $y_b = y_\tau$ (m_b/m_τ) or Koide relations referencing observed masses, the tree-level Yukawa convention does not enter any other quantitative prediction:

$$\boxed{m_t = \frac{v}{\sqrt{2}} = 174.1\ \text{GeV} \quad (\text{obs: } 172.5 \pm 0.3,\ 0.9\%)}$$

7.3 PMNS mixing angles

Tribimaximal (TBM) mixing — $\sin^2\theta_{12} = 1/3$, $\sin^2\theta_{23} = 1/2$, $\sin^2\theta_{13} = 0$ — arises from $A_4 \subset O$ [23]. The deviations from TBM are controlled by three structural ingredients: (i) the overall scale $\lambda^2 = 1/(2\pi^2)$, the squared Cabibbo angle; (ii) the sum rule $2\Delta_{12} + \Delta_{23} = 0$ associated with a U-parity grading (μ-τ exchange antisymmetry) of the charged-lepton perturbation and motivated by the projection geometry below; and (iii) the ratio $\Delta_{13}/\Delta_{23} = A^4 = 4/9$ following from the Wolfenstein $A = \sqrt{2/3}$ of §7.1 via a second-order projection onto the Higgs democratic direction. These three ingredients determine the PMNS corrections:

$$\sin^2\theta_{12} = \frac{1}{3} - \frac{1}{4\pi^2} = 0.3080$$

$$\sin^2\theta_{23} = \frac{1}{2} + \frac{1}{2\pi^2} = 0.5507$$

$$\sin^2\theta_{13} = \frac{4}{9} \cdot \frac{1}{2\pi^2} = \frac{4}{18\pi^2} = 0.02252$$

Angle	Predicted	Observed	Match
$\sin^2\theta_{12}$	0.3080	0.3085 ± 0.0073 (post-JUNO global fit [27])	0.07σ
$\sin^2\theta_{23}$	0.5507	$0.561^{+0.012}_{-0.015}$ (NuFIT 6.0, NO)	0.74σ
$\sin^2\theta_{13}$	0.02252	0.02195 ± 0.00056 (NuFIT 6.0, NO)	1.01σ

The three predictions have different structural status, made precise in the derivation below and in §8.3's four-layer classification. The reactor angle prediction $\sin^2\theta_{13}$ depends only on A_1 at second order — the A_2 coefficients cancel from $|U_{e3}|^2$ — and is therefore independent of Condition 2 (the structural relation among A_2 coefficients introduced below). The solar angle and atmospheric angle predictions $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$ depend on Cond 2 jointly: they are equivalent (via the sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$) to the relation $b_{23} = (4/3)(b_{12}+b_{13})$. After the structural decomposition of "Status of Cond 2" below, this relation has an A_1 component closed at Layer 1 (sum rule preserved by A_1 alone via the $\cos^2\theta_{13}$ dressing) and an A_2 component whose substrate ratio $b_{23}/(b_{12}+b_{13}) = 4/3 = 2A^2$ was Layer 2(a)-tractable until its defining substrate calculation was performed (§7.3, "Status of Cond 2") and excluded the second-order route — demoting it to Layer 2(b), with the 7/6 sum rule as its direct experimental test. Within the four-layer structural framework of §8.3, $\sin^2\theta_{13}$ is a Layer 1 unconditional structural prediction, while $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$ are layered predictions: Layer 1 form combined with the Layer 2(b) Cond 2 coefficient.

Experimental confirmation (JUNO). The Jiangmen Underground Neutrino Observatory reported its first measurement of reactor neutrino oscillations in November 2025 [24], achieving the world's most precise determination of $\sin^2\theta_{12}$: 0.3092 ± 0.0087 (a factor of 1.6 improvement over all previous measurements combined). The OI prediction $1/3-1/(4\pi^2) = 0.3080$ matches this measurement at 0.14σ. The updated global fit including JUNO data gives $\sin^2\theta_{12} = 0.3085 \pm 0.0073$ [27], matching the prediction at 0.07σ. As JUNO accumulates data over its 30-year design lifetime, the error bar is projected to reach ± 0.0014, testing the prediction at sub-percent precision.

Structural relations among the prefactors. The reactor angle has the clean primitive form

$$\sin\theta_{13} = A^2\lambda$$

where $A = \sqrt{2/3}$ is the Wolfenstein A of §7.1 and $\lambda = 1/(\pi\sqrt{2})$ is the Cabibbo angle. The A^2 factor arises from the path-summed calculation: for the A_1-

determined first-row entries of $V_\ell^\dagger$ ($-\lambda\alpha, +\lambda\alpha$ with $\alpha = \sqrt{C_2}/d = \sqrt{2}/3$, the natural cubic-lattice perturbation amplitude on T_1) combined with the TBM third column $(0, -1/\sqrt{2}, +1/\sqrt{2})$, the two non-trivial paths contribute $\lambda\alpha/\sqrt{2}$ each, summing to $\sin\theta_{13} = \sqrt{2}\alpha\lambda = (C_2/d)\lambda = A^2\lambda$ via the structural identity $A^2 = C_2/d = (d-1)/d$ for vector representations (§7.1). The result $\sin^2\theta_{13} = A^4\lambda^2 = (4/9)\lambda^2$ is Layer 1 structural; the ratio $\Delta_{13}/\Delta_{23} = A^4$ follows immediately with $\Delta_{23} = \lambda^2$ by convention.

Determination of A_1 and second-order structure. The charged-lepton rotation in the AF basis admits the expansion $V_\ell = \exp(\lambda A_1 + \lambda^2 A_2 + O(\lambda^3))$ with A_n antisymmetric. Three framework constraints — vanishing $O(\lambda)$ contributions to both Δ_{12} and Δ_{23} (pushing these deviations to $O(\lambda^2)$), plus the correct $\sin^2\theta_{13} = 4\lambda^2/9$ coefficient — uniquely determine

$$A_1 = \frac{\sqrt{2}}{3}(E_{12} - E_{13}),$$

where $(E_{ij})_{kl} = \delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}$. Geometrically, A_1 is a rotation by angle $A^2 = 2/3$ around the axis $(e_2 + e_3)/\sqrt{2}$, reproducing $\sin\theta_{13} = A^2\lambda$ as above. The structural property is that A_1 is purely U*-odd* under the μ-τ exchange generator of $S_4 \supset A_4$: $UA_1U^{-1} = -A_1$.

Decomposing the most general antisymmetric $A_2 = b_{12}E_{12} + b_{13}E_{13} + b_{23}E_{23}$ and computing $|U_{\mathrm{PMNS},ij}|^2$ to $O(\lambda^2)$ (with $\cos^2\theta_{13}$ correction included), the deviations are:

$$\Delta_{13} = \tfrac{4}{9}\lambda^2 = A^4\lambda^2, \qquad \Delta_{12} = -\tfrac{2}{3}(b_{12} + b_{13})\lambda^2, \qquad \Delta_{23} = b_{23}\lambda^2.$$

Three points: (i) Δ_{13} is independent of A_2 — the structural identity $\sin\theta_{13} = A^2\lambda$ is exact at $O(\lambda^2)$; (ii) $b_{12} - b_{13}$ does not enter any of the three angles, and is identified with δ_{CP}; (iii) under U-parity, $(b_{12} + b_{13})$ is the U-even and b_{23} the dominant U-odd coefficient of A_2, with $(b_{12} - b_{13})$ the sub-dominant U-odd combination.

The absolute normalization is fixed by the cubic-lattice realization of the charged-lepton mass matrix. The natural-normalization choice $b_{23} = 1$ — corresponding to A_2 entering the perturbation series at unit strength relative to λ^2 — combined with the structural relation

$$b_{23} = \tfrac{4}{3}(b_{12} + b_{13})$$

This condition has a direct experiment-facing form: it is equivalent to the exact sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$, since $2\Delta_{12} + \Delta_{23} = \lambda^2\,[\,b_{23} - \tfrac{4}{3}(b_{12} + b_{13})\,]$ vanishes precisely when the condition holds — the $O(\lambda^2)$ deviations cancel in that combination and in no simpler one. The sum rule is therefore the observable statement of the Condition-2 ratio (companion sum-rule paper, §3): a measured violation of 7/6 beyond $O(\lambda^4)$ falsifies the ratio directly, independently of how the individual angles are attributed.

between the dominant U-odd coefficient and the U-even combination yields

$$b_{12} + b_{13} = \tfrac{3}{4}, \qquad b_{23} = 1,$$

which translates into the angle-level sum rule

$$2\Delta_{12} + \Delta_{23} = 0.$$

Status of Cond 2: structural decomposition. The relation $b_{23} = (4/3)(b_{12} + b_{13})$ admits a decomposition into two structurally distinct components, with closure achieved at different layers.

Equivalence with TBM sum rule preservation. Combining the explicit results above:

$$2\Delta_{12} + \Delta_{23} = \lambda^2 \left[b_{23} - \tfrac{4}{3}(b_{12} + b_{13})\right]$$

so Cond 2 is *equivalent* to the statement that the TBM sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ — which holds at zeroth order from cubic geometry as $2/d + 1/(d-1) = (3d-2)/(d(d-1)) = 7/6$ for $d = 3$ — continues to hold at $O(\lambda^2)$. The Wilson-coefficient relation and the angle-level sum rule encode the same content under different normalizations.

Layer 1 closure of the A_1 component. With $A_2 = 0$, direct calculation gives $2|U_{e2}|^2 + |U_{\mu 3}|^2 = 7/6 - (14/27)\lambda^2 + (\cos^2\theta_{13}$ corrections). Dividing by $\cos^2\theta_{13} = 1 - (4/9)\lambda^2$ — which itself follows from $\sin^2\theta_{13} = A^4\lambda^2$, a Layer 1 quantity — exactly cancels the $-14/27$ offset:

$$\left(2\sin^2\theta_{12} + \sin^2\theta_{23}\right)\Big|_{A_2=0} = \tfrac{7}{6} + O(\lambda^4).$$

The TBM sum rule is therefore *structurally protected* against the A_1 perturbation by cubic geometry alone — no substrate input required for this component. The cancellation depends on A_1's Layer 1-forced form: the magnitude $\sqrt{2}/3$ is fixed by $\sin\theta_{13} = A^2\lambda$, and the U-odd character is fixed by the requirement $\Delta_{12}, \Delta_{23} = O(\lambda^2)$ rather than $O(\lambda)$. With those two Layer 1 constraints, the A_1^2-induced $-14/27$ contribution and the $\cos^2\theta_{13}$-induced $+14/27$ contribution cancel automatically.

Layer 2(a)-tractable residual question for the A_2 component. What remains is the question of whether A_2 preserves the sum rule (which A_1 alone has structurally protected) or breaks it. Cond 2 reduces to: *why does* $b_{23}/(b_{12} + b_{13}) = 4/3 = 2A^2 = 4(d-1)/(2d)$*?* This is a single numerical ratio with a structural reading from the squared off-democratic projection for the vector representation, sharply specified at the level of substrate Wilson coefficients. The question admits definite outcomes from substrate calculations targeting this specific ratio (e.g., one-loop staggered-fermion calculation on the cubic lattice testing whether the $C_2(T_1)$ Casimir structure produces the $4(d-1)/(2d)$ coefficient). This is Layer 2(a)-tractable rather than the prior "arbitrary T_2-direction alignment" formulation.

The geometric reading of $4/3$. The relation admits the interpretation

$$\tfrac{4}{3} = 2A^2 = 2\sin^2\!\big(\angle(e_i, \hat{h})\big)$$

decomposing as a *partner-count* $\times$ *per-partner projection*: the U-even combination $E_{12}+E_{13}$ couples generation e_1 to its two partners $\{e_2, e_3\}$, each notionally contributing factor A^2 from the $\hat{h}$-perpendicular projection geometry — total $2 \times A^2 = 4/3$. The dominant U-odd combination E_{23} couples e_2 to its single partner e_3, with no analogous multiplicity, giving relative weight 1. The structural identity $A^2 = (d-1)/d = C_2(T_1)/d$ for vector representations is what makes $4/3$ a representation-theoretic quantity rather than a numerical coincidence. However, the load-bearing claim of the geometric reading is the *operator alignment* — that the substrate's T_2-direction in $\Lambda^2(T_1)$ space is aligned such that the partner-count $\times$ projection picture holds. Operator counting on $T_1 \otimes T_1 = A_1 \oplus E \oplus T_1 \oplus T_2$ shows that the T_2 direction of the substrate's second-order corrections is *not forced by representation theory alone* — the natural Higgs-democratic spurion $\xi_h \propto (1,1,1)/\sqrt{3}$ does not produce the relation; the relation requires a specific cubic-lattice Yukawa structure that has not been independently derived.

The empirical match $2\sin^2\theta_{12} + \sin^2\theta_{23} = 1.178 \pm 0.020$ (using post-JUNO global fit and NuFIT 6.0 atmospheric) tests the bijection's specific operator structure at the few-percent level on the structural coefficient, conditional on the atmospheric octant. Promoting the residual question to Layer 2(a) requires demonstrating that the cubic-lattice substrate Yukawa structure produces the specific $4(d-1)/(2d)$ ratio in the T_2-channel; the closure path is the one-loop staggered calculation referenced above.

Positive support: explicit substrate calculation. An explicit calculation of the natural cubic-symmetric staggered Yukawa structure on T_1 BZ corners shows that both the leading A_h-mediated and subleading T_2-corner-mediated channels produce equal off-diagonal $Y^{(2)}_{ab}$ entries (by cubic symmetry of BZ corner separations and uniform staggered phase prefactors). Equivalently, the natural-form substrate Yukawa correction satisfies $\delta M^{\mathrm{nat}}_{ij} = \delta M_0$ (generation-democratic) for all off-diagonal pairs (i,j).

This produces b^{sub} ratios $b_{12} : b_{13} : b_{23} = m_\tau/m_\mu : 1 : 1$ exactly in the heavy-tau limit. The hierarchy follows from a Layer 1 mass-matrix algebraic identity: at order λ^2 the off-diagonal entries of $M_\ell M_\ell^\dagger$ are

$$m_{ij}^{(2)} = (m_i + m_j)\,\delta M_{ij}$$

(from $M_{ik}^{(0)}\delta M_{jk}^* + \delta M_{ik} M_{jk}^{(0)*}$ with $M^{(0)} = \mathrm{diag}(m_a)$ at LO and symmetric δM). Combined with the natural form $\delta M_{ij} = \delta M_0$ and the b normalization $b_{ij} = m_{ij}^{(2)}/m_{\max(i,j)}^2$, this gives $b_{ij} \propto (m_i + m_j)/m_{\max(i,j)}^2 \approx 1/m_{\max(i,j)}$ — the linear inverse-mass scaling that produces the $m_\tau/m_\mu \approx 17$ hierarchy in b_{12}/b_{13}. The factor 17 is therefore not a numerical happenstance but a derived consequence of (i) the natural form's generation-democratic substrate Yukawa and (ii) the linear $(m_i + m_j)$ algebraic factor in the second-order off-diagonal mass-squared.

The framework's bijection therefore has structure beyond the natural cubic-symmetric form. Cond 2's symmetric U-even split $b_{12}^{\mathrm{sub}} = b_{13}^{\mathrm{sub}}$ requires the substrate amplitudes to obey a different scaling:

$$\delta M_{ij}^{\mathrm{sub}} \propto m_{\max(i,j)}$$

— linear in the heavier external mass (rather than democratic). This is the substrate-level structural pattern Cond 2 imposes: not "alignment" in some abstract operator space, but a definite scaling law for the off-diagonal Yukawa correction. The required substrate matrix-element hierarchy (from Cond 2 plus the symmetric U-even split $b_{12}^{\mathrm{sub}} = b_{13}^{\mathrm{sub}} = 3/8$) is

$$m_{12}^{(2)} : m_{13}^{(2)} : m_{23}^{(2)} = \tfrac{3}{8} m_\mu^2 : \tfrac{3}{8} m_\tau^2 : \tfrac{10}{9}(m_\tau^2 - m_\mu^2)$$

— sharply specified, quantitative, and falsifiable. Any first-principles substrate calculation of $m_{ij}^{(2)}$ produces a definite outcome on whether this hierarchy emerges.

Status under emergent quantum mechanics. In standard effective field theory, where quantum mechanics is taken as fundamental, the U-even and dominant U-odd second-order coefficients $b_{12}+b_{13}$ and b_{23} are independent Wilson coefficients of distinct cubic-invariant operators. No EFT-level symmetry relates them. In the OI framework, where quantum mechanics is itself emergent — a tool for describing the marginal statistics of an embedded observer over a deterministic substrate (companion paper [Main]) — EFT Wilson coefficients are not independent inputs but derived quantities arising from substrate-level mode counting, and constraints among them are *expected* whenever multiple EFT operators share a common substrate-level origin. This places OI in the broader research program of emergent QM developed by 't Hooft [30] (cellular automaton interpretation), Adler [31] (trace dynamics from matrix models), and Barandes [28] (stochastic-quantum correspondence). The relation $b_{23} = (4/3)(b_{12} + b_{13})$ is consistent with such a substrate-level origin; the empirical match indicates this is realized in the framework's bijection. The current Layer 2(a)-tractable status — sharply specified as the $4(d-1)/(2d)$ ratio with structural numerology — reflects that the substrate-level mechanism producing this specific coefficient has not yet been derived, while the question of whether the cubic-lattice substrate Yukawa produces it is concretely tractable rather than open at the level of arbitrary operator alignment.

Sharpening the open task: the cross-term contribution. The substrate-level question admits a concrete formulation. In the AF basis where the charged-lepton mass matrix $M_\ell M_\ell^\dagger$ is diagonal at LO, second-order perturbative diagonalization gives the A_2 coefficients in terms of the substrate's direct second-order off-diagonal corrections $m_{ij}^{(2)}$ *plus* cross-terms involving products of first-order amplitudes. Direct computation in the framework's $A_1 = (\sqrt{2}/3)(E_{12} - E_{13})$ ansatz yields, in the $m_e^2 \to 0$ limit:

$$b_{12} = \frac{m_{12}^{(2)}}{m_\mu^2}, \quad b_{13} = \frac{m_{13}^{(2)}}{m_\tau^2}, \quad b_{23} = -\frac{1}{9} - \frac{m_{23}^{(2)}}{m_\tau^2 - m_\mu^2}.$$

The $-1/9$ contribution to b_{23} is *hierarchy-independent* and is forced by A_1's cubic-group structure ($a_{12} \cdot a_{13} = -2/9$ from the explicit A_1 form). It arises with no substrate input at second order. The relation Cond.~2 at the observable level, $b_{23} = (4/3)(b_{12} + b_{13})$, therefore translates to the following constraint on the substrate's direct second-order corrections:

$$b_{23}^{\mathrm{sub}} = \tfrac{4}{3}(b_{12}^{\mathrm{sub}} + b_{13}^{\mathrm{sub}}) + \tfrac{1}{9}.$$

The $+1/9$ augmentation compensates for the $-1/9$ cross-term contribution from A_1. The Layer 2(a) closure question reduces to: *does the cubic-lattice Yukawa structure produce direct second-order corrections* $m_{ij}^{(2)}$ *that satisfy this augmented relation?* With $b_{23}^{\mathrm{sub}} = 10/9$ and $b_{12}^{\mathrm{sub}} + b_{13}^{\mathrm{sub}} = 3/4$ in the framework's natural normalization, the required substrate matrix-element hierarchy is $m_{12}^{(2)} : m_{13}^{(2)} : m_{23}^{(2)} = (3/8)m_\mu^2 : (3/8)m_\tau^2 : (10/9)(m_\tau^2 - m_\mu^2)$ (assuming symmetric U-even split).

Progress on the sharpened task (scoping-level, inputs declared). Three results from the resolving computation's scoping phases bear on this formulation. First, the symmetric U-even split assumed above now has structural support: the geometric second-order taste channel (two single-flip hops through the heavy non-T_1 corners) is exactly democratic — equal coefficients for all three pairs — as a consequence of cubic symmetry, computed over the full range of heavy-corner gap assignments. Second, the dressing rule is forced rather than chosen: reproducing A_1's mass-independent angles requires the mass-matrix off-diagonals to carry the column (heavier) mass, $M_{ij} = T_{ij}\, m_j$ — the taste transition living in the left sector — under which the heavier-mass-squared scaling of the required $m_{ij}^{(2)}$ hierarchy is automatic. Third, with these two results the augmented relation reduces to a single condition on the substrate's heavy-corner spectrum: the democratic two-hop amplitude must equal $-1/15$ (in the $m_\mu/m_\tau \to 0$ limit), equivalently $|1/\Delta_S - 1/\Delta_{T_2}| = 1/(30\rho)$, where Δ_S, Δ_{T_2} are the denominators of the singlet- and T_2-class exchange channels — the wave-sector corner frequencies of Theorem 8b, since the taste-changing messenger at a corner-difference momentum propagates at that corner's frequency — and ρ the spin-to-chirality projection. The gaps are not free: Theorem 8b fixes the corner-class couplings at $\mu = (1, 1/3, -1/3, -1)$ — universality-class numbers — so the rest frequencies $\arccos\mu$ determine Δ_S and Δ_{T_2} outright (with opposite signs: the singlet sits below the generation corners, the T_2 class above). The condition then fixes the required projection: $\rho \approx -0.015$ in frequency units or -0.030 in mass-squared units — small rather than order unity, satisfiable but not yet derived. The remaining content of Condition 2 is thereby a single number: the spin-to-chirality projection with proper on-shell normalization — the same machinery as the c_λ computation of §7.1. That machinery resolves the question, and the resolution is negative — algebraically, in the same identity class as the §7.1 result. In the complete $(\gamma_S \otimes \xi_T)$ channel decomposition, the two-hop operator is a *single pure monomial* ($\gamma_0\gamma_1\gamma_3$ in the spin slot for the axes-$(1,2)$ transfer, times a taste monomial), and a monomial is orthogonal to every other channel: its weight in each mass-channel spin structure — scalar, γ_0, and both γ_5-graded partners — vanishes identically. The projection ρ is therefore not small but exactly zero, against the required -0.015 to -0.03: the substrate dressing chain is retired. Condition 2 accordingly stands as what the construction always declared it to be — a stated condition, with the 7/6 sum rule as its direct experimental test — and the derivation-mechanism space is now mapped: first-order-squared effects are excluded by the spectrum-dominance computation, and the geometric second-order route is excluded by monomial orthogonality. One scalar route survives the mapping and is recorded without being pursued here: the $O(\lambda^3)$ direct–two-hop interference is spin-scalar (the square of the transfer monomial is proportional to the identity — explicitly $(\gamma_0\gamma_1\gamma_3)^2 = -1$ in the derived corner algebra), a different construction that would require its own chain and carries an additional factor of λ. Specifying that chain requires two elements the present hop model does not fix: the operator-level rule by which the direct and two-hop insertions compose in the mass-squared matrix, and the on-shell normalization of the composite. Declared inputs of this reduction: the hop model of the taste geometry, the projection ρ, and the neglect of the $O(\lambda^3)$ interference; one convergence caveat is recorded — the $(1,2)$ channel carries a hierarchy-enhanced correction $\lambda^2 m_\tau^2/m_\mu^2 \approx 0.7$ at physical λ, so its strict-λ^2 extraction is the least robust of the three. The limitation extends beyond that channel. The value of $m_{23}^{(2)}$ at which the sum rule holds exactly converges, as $\lambda \to 0$, to $b_{23}^{\rm sub} = \frac{4}{3}(b_{12}^{\rm sub} + b_{13}^{\rm sub}) + \frac{1}{9}$, and in the $m_\mu/m_\tau \to 0$ limit the required democratic two-hop amplitude converges to $-1/15$; the deviation from these grows linearly in λ with a large coefficient, reaching order unity relative to the target at physical λ. The substrate-level form of Condition 2 holds as a $\lambda \to 0$ statement, and a test at physical λ requires exact diagonalization rather than the strict second-order extraction. The Layer 2(a) question is thereby reduced, not closed: its remaining content is the heavy-corner gap determination.This is concrete, quantitative, and falsifiable — any first-principles substrate calculation of $m_{ij}^{(2)}$ can be tested against this hierarchy.

The observed $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ (observed: 1.178 ± 0.020 using post-JUNO global fit and NuFIT 6.0 atmospheric) therefore follows from the cubic-lattice structure conditional on the substrate-level relation above. The empirical match distinguishes this prediction from TM1 or TM2 partial-TBM patterns (which do not produce this combination as an exact sum rule); the match indicates the framework's bijection has the alignment property required by Cond 2 — equivalently, that the substrate-level mechanism producing the $4(d-1)/(2d)$ amplitude ratio is realized in the bijection.

The quoted value uses the NuFIT 6.0 normal-ordering best fit $\sin^2\theta_{23} \approx 0.561$, the upper octant. Under the lower-octant solution ($\sin^2\theta_{23} \approx 0.455$) the same combination is ≈ 1.07, roughly 5σ below $7/6$, and the natural-form alternative discussed below fails in the opposite direction. A lower-octant determination excludes the prediction.

7.4 The Higgs mass

The Higgs is the A_1 taste — a composite scalar. Its quartic self-coupling at the compositeness scale has no tree-level contribution: $\lambda(M_{\text{Pl}}) = 0$. The SM quartic is generated entirely by RGE running. The observed $\lambda(M_{\text{Pl}}) \approx -0.013 \pm 0.020$ [25] is consistent with zero at 0.6σ. The structural prediction is $\lambda(M_{\text{Pl}}) = 0$; the numerical value of m_H in GeV is derived from this structural boundary condition combined with the measured m_t as a one-input parameter.

Using the 3-loop SM RGE [25], $\lambda(M_{\text{Pl}}) = 0$ gives $m_H \approx 129$–132 GeV for $m_t = 172$–173 GeV. The observed $m_H = 125.10 \pm 0.14$ GeV is 4–7 GeV below; the gap is sensitive to m_t ($\partial m_H/\partial m_t \approx -1.1$ GeV/GeV). The CMS measurement $m_t = 170.5 \pm 0.8$ GeV [26] would bring the prediction to $\sim 128 \pm 2$ GeV. The structural claim $\lambda(M_{\text{Pl}}) = 0$ is therefore confirmed; the specific m_H value in GeV is consistent with observation within the current m_t uncertainty rather than predicted at sub-GeV precision.

Higgs near-criticality. The observed $\lambda(M_{\text{Pl}}) \approx 0$ is part of a broader empirical pattern: the SM Higgs sector sits at the boundary between vacuum stability and metastability, with $\lambda \approx 0$ and $\beta_\lambda \approx 0$ both occurring near the Planck scale (Buttazzo et al. [25]). This "near-criticality" is treated as a remarkable coincidence in standard SM analyses requiring beyond-the-Standard-Model explanation. The Multiple Point Principle (Bennett-Froggatt-Nielsen [34], Froggatt-Nielsen [35]) postulated $\lambda(M_{\text{Pl}}) = 0$ as a fine-tuning condition; this prediction matched the observed Higgs mass before its discovery. Subsequent BSM model-building (singlet scalar dark matter, vector-like leptons, strong dynamics extensions, 2HDM realizations) [36, 37, 38] has produced extensions enforcing $\lambda(M_{\text{Pl}}) = 0$ as a boundary condition postulated from outside the SM. The framework's prediction $\lambda(M_{\text{Pl}}) = 0$ is therefore the structural realization within a derived framework of a boundary condition that an active BSM literature has been postulating for three decades.

The asymptotic-safety convergence. Wetterich (asymptotic-safety prediction [39]) showed that if the SM is valid up to the Planck mass and gravity contributes an irrelevant fixed-point structure to the running, the observed $M_H \approx 125$ GeV predicts $M_t \approx 171$ GeV. This is structurally the same prediction as the framework's, viewed from the opposite direction: from observed m_H to predicted m_t. The convergence between two structurally distinct frameworks (asymptotic safety vs OI) predicting the same near-criticality boundary condition strengthens the empirical content of the prediction.

Empirical confirmation status of independent lattice composite-Higgs work. Aoki et al. [40] observed via lattice simulations of $N_f = 8$ QCD that the singlet-taste channel produces a light composite scalar (technidilaton candidate) consistent with the observed 125 GeV Higgs mass. The framework's specific mechanism — Higgs as composite scalar in the singlet staggered taste with chiral protection — has independent empirical support from lattice gauge theory studies of strongly-coupled gauge theories.

Joint boundary-condition structure: $y_t(M_{\mathrm{Pl}}) = 1$ and $\lambda(M_{\mathrm{Pl}}) = 0$. The framework fixes two boundary conditions at the compositeness scale — the top Yukawa $y_t(M_{\mathrm{Pl}}) = 1$ from the staggered fermion-to-scalar overlap normalization (§7), and the Higgs quartic $\lambda(M_{\mathrm{Pl}}) = 0$ from the composite-Higgs structure (this section) — but their empirical statuses differ and must not be conflated. The quartic condition $\lambda(M_{\mathrm{Pl}}) = 0$, combined with the observed m_t and the Higgs vev $v = 246$ GeV (Tier 3 input), predicts $m_H \approx 128$–132 GeV via SM 3-loop RGE running [25] — consistent with the observed 125.10 ± 0.14 GeV to ~ 2–3σ (reduced to $\lesssim 1\sigma$ for the CMS $m_t = 170.5 \pm 0.8$ GeV [26]). The Yukawa condition $y_t(M_{\mathrm{Pl}}) = 1$, by contrast, does *not* reproduce the observed top mass: one-loop running carries it toward the IR quasi-fixed point at $y_t(m_t) \approx 1.19$, i.e. $m_t \approx 207$ GeV — a $\sim 20\%$ excess (§7.2) — while the observed $m_t = 172.7$ GeV corresponds to $y_t(M_{\mathrm{Pl}}) \approx 0.4$–$0.5$, not unity ($\lambda$ does not enter the one-loop y_t β-function, so the quartic condition cannot rescue it). The weak-scale coincidence $m_t = v/\sqrt{2}$ ($y_t(m_t) \approx 1$) is therefore a retrodiction with the fixed-point mechanism open (§7.2, §7.6), not a forward prediction from the compositeness-scale normalization. The structural content that survives is the single boundary condition $\lambda(M_{\mathrm{Pl}}) = 0$ — the same condition the MPP literature *postulates*, here *derived* from the composite-Higgs (singlet-taste) construction.

Unified matching at the compositeness scale. The framework's substratum produces a unified matching picture at the single scale M_{Pl}: gauge coupling matching ($1/\alpha_0 = 23.25$ and the universal threshold $\delta_0 = 10$, both independent of gauge group; §6), Higgs quartic matching ($\lambda(M_{\mathrm{Pl}}) = 0$; this section), top Yukawa matching ($y_t(M_{\mathrm{Pl}}) = 1$; this section), and bare Higgs mass matching (Veltman condition $m_H^{\mathrm{bare}}(M_{\mathrm{Pl}}) = 0$; §8.8). All four matching conditions hold at the same scale — the substratum's intrinsic UV cutoff — because they all arise from the same compositeness structure: emergent SM couplings inherit their UV boundary conditions from the substratum's structural features (group-theoretic, taste, chiral) at the compositeness scale. This is structurally analogous to (and significantly more constrained than) conventional composite-Higgs scenarios at $f \sim$ TeV, where multiple matching conditions are imposed at the compositeness scale but typically require additional assumptions (custodial symmetry, partial compositeness ansatz, top partners). The framework's single-scale unified matching with no additional assumptions is a substantive structural feature: the same compositeness scale yields the gauge couplings (§6), the Higgs quartic and mass (§7.4 and §8.8), the top Yukawa (§7), and the bare Higgs mass (§8.8) without independent matching parameters for each. The Buttazzo near-criticality coincidence — $\lambda(M_{\mathrm{Pl}}) \approx 0$, $\beta_\lambda(M_{\mathrm{Pl}}) \approx 0$, $y_t(M_{\mathrm{Pl}})$ at minimum, Veltman condition near-satisfied — is therefore not four separate coincidences but four consequences of a unified compositeness-scale matching structure.

7.5 The bottom-to-tau mass ratio

Tree-level result.

The tree-level Yukawa coupling is taste-independent [12]: $y_b = y_\tau$. With two-loop SM RGE [25]: $m_b/m_\tau|_{\mathrm{tree}} = 4.276$ (with $\sim 1\%$ residual systematic from higher-loop corrections and EW-threshold matching), indicating a substantial non-perturbative correction.

Non-perturbative correction.

The correction is the scalar density renormalization $Z_S(m) = \langle\bar{\psi}\psi\rangle_{\mathrm{int}}/\langle\bar{\psi}\psi\rangle_{\mathrm{free}}$, computed on SU(3) gauge backgrounds at $\beta = 11.1$ as a function of bare mass m. The denominator $\langle\bar{\psi}\psi\rangle_{\mathrm{free}}$ is evaluated analytically on the same lattice with the eight BZ-corner modes excluded from the momentum sum (`scripts/analyze_zs.py`); this finite-L regulator differs from the standard staggered convention (corner modes included) by $\sim 2\%$ at $L = 32$, $\sim 0.3\%$ at $L = 64$, and converges to it at $L \to \infty$. The prediction: $m_b/m_\tau = 4.28/Z_S(m_{\mathrm{match}})$.

Monte Carlo simulations at $L = 16$ (30 configs) and $L = 32$ (50 configs), scanning 30 masses from $m = 0.005$ to 0.50, reveal that $Z_S(m)$ is monotonically decreasing in the volume-converged region ($mL \gtrsim 3$). At $m = 0.10$: Z_S converges from 1.70 (L=16) to 1.92 (L=32) to 1.94 (L=64).

Chiral condensate formation.

Comparison of $L = 16$ and $L = 32$ confirms spontaneous chiral symmetry breaking: Z_S at small m grows $8\times$ between volumes, and the apparent peak shifts from $m = 0.087$ (L=16) to $m = 0.024$ (L=32, $Z_S = 2.828 \pm 0.008$), tracking the finite-volume boundary $mL \sim 1$. The chiral condensate $\Sigma \approx 0.20$ (from linear extrapolation in the converged region at $L = 32$).

The matching mass.

The matching scale where lattice dynamics connects to the perturbative Yukawa description is set by the product of the two relevant dimensionless parameters:

the taste-breaking amplitude $\lambda = 1/(\pi\sqrt{2})$ and the unified gauge coupling $g_0^2 = 4\pi\alpha_0 = 4\pi/23.25$:

$$m_{\mathrm{match}} = \lambda \times g_0^2 = \frac{1}{\pi\sqrt{2}} \times \frac{4\pi}{23.25} = \frac{4}{23.25\sqrt{2}} = 0.1217$$

Physically, λ controls inter-generation mixing (taste-breaking) while g_0^2 controls the non-perturbative condensate (confinement). The matching scale is derived in the taste-decomposed Coleman–Weinberg potential by tracing three ingredients through the framework's existing structure:

(i) The b-quark is identified with the democratic eigenstate $\hat{h} = (|e_1\rangle + |e_2\rangle + |e_3\rangle)/\sqrt{3}$ of the cubic-symmetric taste-breaking operator (this follows from the §7.1 Wolfenstein $A = \sqrt{2/3}$ identification and the §7.2 Koide derivation, both of which place the heaviest fermion along $\hat{h}$).

(ii) The taste-breaking operator in the X-corner basis has the cubic-symmetric form $M_{\mathrm{taste}} = m_{\mathrm{off}} \cdot (J - I)$ (after trace removal), giving the diagonal element in the b-quark eigenstate $c_{bb} = \langle b|M_{\mathrm{taste}}|b\rangle = 2m_{\mathrm{off}}$.

(iii) The off-diagonal element m_{off} is identified with the BZ-distance geometric primitive of §7.1: $m_{\mathrm{off}} = 1/|X_i - X_j| = \lambda$.

Combining these gives $c_{bb} = 2\lambda$. The scalar density self-energy takes the form $\Pi_{\mathrm{taste}}(m) = K \cdot c_{bb} \cdot g_0^2 \cdot \Sigma/m$ for some loop kinematic coefficient K, and matching at the natural condensate-scale threshold $T = \Sigma$ gives $m_{\mathrm{match}} = 2K\lambda \cdot g_0^2$.

Structural form: $K = 1/(n_{\mathbf{gen}} - 1)$. The factor 2 in $c_{bb} = 2\lambda$ has a representation-theoretic origin parallel to the Wolfenstein $A^2 = (d-1)/d$ identity of §7.1. For the democratic eigenstate $\hat{h} = (e_1 + ... + e_{n_{\mathrm{gen}}})/\sqrt{n_{\mathrm{gen}}}$ in n_{gen} generations:

$$J\hat{h} = n_{\mathrm{gen}} \cdot \hat{h}, \quad \langle\hat{h}|J|\hat{h}\rangle = n_{\mathrm{gen}}, \quad \langle\hat{h}|I|\hat{h}\rangle = 1$$

giving

$$c_{bb} = \langle\hat{h}|M_{\mathrm{taste}}|\hat{h}\rangle = (n_{\mathrm{gen}} - 1) \cdot m_{\mathrm{off}} = (n_{\mathrm{gen}} - 1) \cdot \lambda.$$

The $(n_{\mathrm{gen}} - 1)$ factor counts the off-democratic "partners" of $\hat{h}$ in the generation rep — the inverse of the Wolfenstein A structural quantity $\sqrt{(d-1)/d}$ applied to the same off-democratic projection.

Substituting into the matching condition gives $m_{\mathrm{match}} = K \cdot (n_{\mathrm{gen}} - 1) \cdot \lambda \cdot g_0^2$. The framework's prediction $m_{\mathrm{match}} = \lambda \cdot g_0^2$ (with unit prefactor: the proportional form $m_{\mathrm{match}} \propto \lambda g_0^2$ is fixed by power-counting in the taste-decomposed CW potential with the natural threshold convention $T = \Sigma$, while the specific dimensionless coefficient is currently MC-verified to 1.4% rather than derived in closed form — see below) therefore requires:

$$\boxed{K = \frac{1}{n_{\mathrm{gen}} - 1}}$$

For $n_{\text{gen}} = 3$ — itself forced by the cubic-group decomposition of the 6-link representation $T_1 \oplus E \oplus A_1$ giving 3 fermion generations on T_1 (Theorem 7) — this gives $K = 1/2$.

This is a representation-theoretic structural identity, not a numerical happenstance. $K = 1/2$ is the $n_{\text{gen}} = 3$ value of a generation-count-dependent identity $K = 1/(n_{\text{gen}}-1)$, paralleling the family of $(d-1)$-based identities the framework already employs:

Identity	Formula	$d = 3$ / $n_{\text{gen}} = 3$ value	Source
Wolfenstein A^2	$(d-1)/d$	2/3	Off-democratic projection (§7.1)
Cond 2 ratio	$4(d-1)/(2d) = 2A^2$	4/3	Squared off-democratic projection (§7.3)
K **(this work)**	$1/(n_{\mathbf{gen}} - 1)$	1/2	**Inverse off-democratic partner count**

Unifying identity. Wolfenstein $A^2 = (d-1)/d$ and $K = 1/(d-1)$ are dual aspects of the same off-democratic structure on a d-dimensional vector representation: A^2 is the projection magnitude per unit dimension, K is inverse partner count. Their product is the per-component weight:

$$K \cdot A^2 = \frac{1}{d-1} \cdot \frac{d-1}{d} = \frac{1}{d}.$$

For $d = 3$: $A^2 = 2/3$, $K = 1/2$, $K \cdot A^2 = 1/3$. The framework's $(d-1)$-based identity family is internally consistent and unified, not a list of independent coincidences.

Consistency check on $n_{\mathbf{gen}} = 3$. Given the matching condition $K \cdot (n_{\text{gen}}-1) = 1$ (forced by $m_{\text{match}} = \lambda g_0^2$ with prefactor unity from power counting and confirmed by MC at 1.4%) and the value $K = 1/2$ inherited from the standard staggered Z_S calculation (Argument 1; plausibility evidence pending the explicit one-loop OI derivation), $n_{\text{gen}} - 1 = 2$ — consistent with the framework's three-generation structure. This is a *consistency check* rather than an independent over-determination: $n_{\text{gen}} = 3$ is itself fixed independently by the cubic decomposition $T_1 \oplus E \oplus A_1$ (Theorem 7), and the matching consistency uses an inherited (not independently derived) value of K. The framework's commitment to three candidate generations (physical reading under H-spin') rests primarily on the cubic decomposition; the matching-condition consistency adds a self-consistency confirmation rather than an independent forcing.

Layer stratification of $K = 1/2$. The structural identity admits the following decomposition (analogous to §7.3's Cond 2 stratification):

Component	Required value	Layer status
Democratic eigenstate $\hat{h}$ as O-singlet	from cubic-group rep theory	**Layer 1** (rep theory)
$c_{bb} = (n_{\text{gen}} - 1) \cdot \lambda$	from $\langle \hat{h} \| J - I \| \hat{h} \rangle = n_{\text{gen}} - 1$ algebraic identity	**Layer 1** (algebraic)
Power counting forces $m_{\text{match}} \propto \lambda \cdot g_0^2$	from CW dimensional analysis	**Layer 1** (already noted in §7.5)
$n_{\text{gen}} = 3$	from cubic decomposition $T_1 \oplus E \oplus A_1$	**Layer 0/1** (Theorem 7)
Loop calculation gives $K = 1/(n_{\text{gen}} - 1)$ explicitly	from OI-specific taste-decomposed CW one-loop	**Layer 2(a)-tractable** (residual gap)

The first four components are Layer 1 closed: they together fix $K = 1/(n_{\text{gen}} - 1) = 1/2$ as the value the loop calculation *must* yield for the framework's matching prediction to hold. The Layer 2(a)-tractable residual is whether the OI's specific taste-decomposed CW one-loop calculation actually produces this kinematic prefactor — addressed by Argument 1 below (inheritance from standard staggered) and the Residual gap subsection.

The framework's prediction $m_{\text{match}} = \lambda \cdot g_0^2$ requires $K = 1/2$. Given the algebraic identity $c_{bb} = (n_{\text{gen}} - 1)\lambda$ (Layer 1, clean), the matching condition $m_{\text{match}} = K \cdot c_{bb} \cdot g_0^2$ forces $K = 1/(n_{\text{gen}} - 1) = 1/2$ as a structural identity whenever the matching condition itself holds. The matching condition $m_{\text{match}} = \lambda g_0^2$ is supported by the following arguments, MC-verified to 1.4% precision:

Argument 1 (Staggered convention). The standard staggered fermion action uses the symmetric kinetic term $S_{\text{kin}} = \frac{1}{2} \sum_{x,\mu} \eta_\mu(x) \bar{\chi}(x) [U_\mu(x)\chi(x + \hat{\mu}) - U_\mu^\dagger(x - \hat{\mu})\chi(x - \hat{\mu})]$, with the gauge vertex $V_\mu = (g_0/2)\eta_\mu(x)[U_\mu + U_\mu^\dagger]$. Patel & Sharpe [33] derive the standard one-gluon vertex $V_\mu \propto -ig\cos(q'_\mu + k_\mu/2)(\gamma_\mu \otimes 1)$ (their Eq. 17) — the same form as the OI vertex $V_j = ig_0\mu T^\alpha \cos(k_j + p_j/2)$ — and the two-gluon (tadpole) vertex with explicit factor $-i\frac{1}{2}g^2$ (their Eq. 18) tracing directly to the symmetrization. The full one-loop renormalization of staggered scalar density bilinears computed there and in subsequent work [Constantinou-Costa-Panagopoulos and others] is a specific known number consistent with $K = 1/2$ in the framework's decomposition.

Scope of the inheritance argument. The algebraic identity $\cos^2(k_\mu/2) = (1 + \cos(k_\mu))/2$ is dimension-independent per direction, but its "1/2" content propagates into the lattice integral only when the $\cos(k_\mu)$ residual vanishes by symmetry. The relevant symmetry is the π-shift $k_\mu \to \pi - k_\mu$: integrands depending on k only through $\sin^2(k_\mu)$ (i.e., the staggered fermion propagator) are π-shift symmetric and the residual integrates to zero exactly, leaving the 1/2 factor;

integrands involving the lattice gluon propagator $\hat{q}^2 = 4\sum \sin^2(k_\mu/2)$ are *not* π-shift symmetric (under the shift, $\sin^2(k_\mu/2) \to \cos^2(k_\mu/2)$), and the residual does not vanish in isolation. The actual one-loop Z_S integrand involves both fermion and gluon propagators, so the simple $\cos^2 \to 1/2$ algebra does not by itself constitute the analytical derivation of $K = 1/2$; the residual $\cos(k_\mu)$ pieces from the vertex-correction diagram combine with analogous pieces from the fermion self-energy and tadpole diagrams to produce the published staggered Z_S. In the analogous condensate-ratio construction the residual does not cancel: in four dimensions the residual fraction is finite and convergent at ≈ 0.11 (tadpole-normalization bracket 0.11–0.23), and the three-dimensional analogue falls toward the chiral limit without converging at accessible volumes. $Z_S^{(1)}$ is a well-defined finite number with that residual included, and the $\cos^2 \to 1/2$ symmetrization is a numerical approximation at the ten-percent level rather than an identity. The inheritance argument requires only the published value. The object evaluated is this framework's condensate-ratio Z_S; a scheme difference from the published-literature object is not excluded. The inheritance argument therefore reduces to: the published staggered $Z_S^{(1)}$ is a specific number consistent with $K = 1/2$, and OI uses the same vertex, action, and gauge fixing as the published calculations, so OI's Z_S should reproduce the same number. This is plausibility evidence, not analytical derivation; the matching mass μ enters the OI calculation as an overall factor (it sets the scale at which the matching is performed, not the dimensionless loop coefficient), so it does not modify K by inspection.

Consistency check 2 (Symmetric averaging — definitional). The taste-breaking insertion can appear before or after the gauge exchange in the fermion loop. By Hermitian conjugation symmetry these orderings contribute equally; if the full amplitude is *defined* as the average $\Pi^{\text{full}}_{\text{taste}} = \frac{1}{2}[\Pi^{(L)} + \Pi^{(R)}]$, the prefactor 1/2 appears explicitly and equals $\Pi^{(L)}$. Defining Π^{full} as the sum (rather than the average) gives no factor; the 1/2 here is therefore a convention that aligns with Argument 1's normalization, not an additional first-principles derivation.

Consistency check 3 (Empirical exclusion of natural fractions). Among the natural fractions $K \in \{1/4, 1/2, 1, 2\}$, the observed m_b/m_τ excludes all but $K = 1/2$:

K	m_{match}	m_b/m_τ	Discrepancy
1/2	0.122	2.36	0.4% ✓
1	0.243	≈ 3.17	35% ✗
1/4	0.061	≈ 1.78	24% ✗
2	0.487	$\gg 4$	excluded ✗

Intermediate fractions ($K \in \{1/3, 2/3, 3/4\}$) give m_{match} between 0.08 and 0.18, with corresponding m_b/m_τ in the 1.8–3.2 range; no fraction outside $\{1/2\}$ falls within $\sim 5\%$ of observation. This is empirical fitting on a moderately-restricted set rather than a first-principles derivation, but the resulting selection is sharp.

$K = 1/2$ is therefore *consistent with* the algebraic identity $c_{bb} = (n_{\text{gen}} - 1)\lambda$ and the matching condition $m_{\text{match}} = \lambda g_0^2$, with the latter MC-verified to 1.4% precision. The Layer 1 content is the algebraic identity for c_{bb}; the matching condition has the *form* of a Layer 1 structural relation (dimensional analysis plus the "natural threshold" convention $T = \Sigma$) but the specific dimensionless coefficient is currently established by MC rather than by closed-form analytical derivation. Argument 1 (inheritance from standard staggered convention), Consistency check 2 (definitional symmetric averaging), and Consistency check 3 (empirical exclusion of natural-fraction alternatives) provide plausibility evidence for the matching condition rather than analytical closure.

Within the four-layer framing (§8.3), $K = 1/2$ is MC-verified at 1.4% precision; the analytical residual is a controlled derivation of $m_{\text{match}} = \lambda g_0^2$ from the taste-decomposed Coleman–Weinberg potential. A strict one-loop calculation cannot supply it at the required precision: in the condensate-ratio construction the one-loop coefficient evaluates to 0.446 (tadpole-normalization bracket 0.393–0.446), 11–21% below 1/2, at a coupling where $Z_S - 1 = 0.81$ — a truncation error an order of magnitude larger than the 1.4% the MC verification achieves. Closure requires the resummed staggered-χPT chain (the K_S items above) or a non-perturbative identity; the task is Layer 2(a) in scope but not one-loop in method. Given the matching condition, $K = 1/(n_{\text{gen}} - 1)$ follows by structural identity from $c_{bb} = (n_{\text{gen}} - 1)\lambda$; the matching condition itself awaits analytical closure. The stratification matches §7.3 Cond 2: a Layer 1 representation-theoretic core ($c_{bb} = (n_{\text{gen}} - 1)\lambda$ here; the geometric projection structure there) plus a Layer 2(a)-tractable substrate-loop verification residual.

Residual gap. With Argument 1 understood as plausibility evidence rather than independent analytical derivation, the explicit analytical task is the resummed or non-perturbative evaluation of the three-diagram structure — vertex correction, fermion self-energy, and tadpole with the OI vertex $V_j = ig_0\mu T^\alpha \cos(k_j + p_j/2)$, combined via the relevant Ward identity. A strict one-loop evaluation of that structure cannot close the gap: its truncation error at the physical coupling exceeds the MC precision by an order of magnitude (coefficient evaluation above). Two supporting routes remain: (a) a Ward-identity argument that the residual $\cos(k_\mu)$ pieces cancel across the three diagrams in the Π_{taste} scheme proper — in the condensate-ratio construction the measured residual does not cancel (Scope paragraph above), which any such argument must accommodate; (b) tighter MC measurement to confirm $K = 1/2$ at sub-percent precision and bound the higher-order corrections explicitly.

The per-vertex $\cos^2$ algebraic identity $\cos^2(k/2) = (1 + \cos(k))/2$ is dimension-independent per direction; the 1/2 constant piece survives any dimensional change. Whether this 1/2 propagates cleanly through the full one-loop integral depends on the π-shift cancellation discussed under "Scope of the inheritance argument" above — a question of integral structure rather than dimensionality. Four bridge gaps in the SχPT inheritance are addressed below; their current status:

(i) *Substratum measure reduction.* **Level 2 closed via pseudofermion correspondence** (cf. §4.2 *Remark*). The standard MC code uses the bosonic pseudofermion representation, not Grassmann variables directly. The substratum (Gaussian on real scalar bijections at large L) and pseudofermion (bosonic Gaussian auxiliary fields) measures are in the same class; their covariances differ by a factor of 4 absorbable into normalization, with Theorem 2 ($D_{st}^2 = -\frac{1}{4}\Box_{\text{lat}}$) providing the explicit match. The Level 3 residual is cycle ergodicity for OI's specific bijection orbit structure: pure linear lattice dynamics is integrable rather than chaotic (the frequencies $\omega(\vec{k}) = 2\sin(|\vec{k}|/2)$ are not linearly independent over $\mathbb{Q}$), so full-shell sampling requires either effective non-linearity from the discretization rule (the standard interpretation for lattice-substrate frameworks) or revision of the Gaussianization argument to sub-torus sampling. We adopt the former implicitly; rigorous treatment of OI's specific orbit structure is open work.

(ii) *3D operational MC vs 4D SχPT dimensional bridge.* **Decomposes into Part A (time-direction factorization, Level 1 closed) and Part B (measure identification, Level 2 closed via inheritance from gap (i)).** Part A: For ρ stationary under φ and T the framework's involution with $T\varphi T = \varphi^{-1}$ (§5), $(\rho \circ T) \circ \varphi = \rho \circ (T\varphi) = \rho \circ (\varphi^{-1}T) = \rho \circ T$, so $\rho \circ T$ is φ-invariant; by P-indivisibility (Main Theorem 4), the stationary distribution is unique, hence $\rho \circ T = \rho$. Static observables therefore factorize through the time direction independent of which time slice. Part B is the same investigation as gap (i) viewed operationally: closing gap (i) automatically closes gap (ii) Part B. With gap (i) Level 2 closed, gap (ii) is Level 2 closed framework-essentially.

(iii) and (iv) *Bridge theorem:* **Closed at Level 1 framework-essentially via the explicit theorem below.**

Bridge Theorem (3D-4D staggered correspondence). Let $\chi(x)$ be the staggered field on $\mathbb{Z}^3$ with bijection time-evolution $\varphi : S \to S$. Then:

(a) *Mode-space isomorphism.* The 16-dim space spanned by 3D BZ corners $\times$ sublattice grading is isomorphic to the 4D Dirac $\times$ taste space:

$$\{(\eta, s) : \eta \in \{0,1\}^3, s \in \{0,1\}\} \xrightarrow{\sim} V_{\text{Dirac}}^{(4D)} \otimes V_{\text{taste}}$$

via $(\eta_1, \eta_2, \eta_3, s) \mapsto (\eta_0 = s, \eta_1, \eta_2, \eta_3)$. Counting: $2^3 \times 2 = 16 = 4 \times 4$. (Level 1.)

(b) *Clifford algebra identification.* The 4D chirality operator corresponds to spatial sublattice grading: $\gamma_5 \leftrightarrow \varepsilon_{3D}$. The 4D time-direction Dirac matrix is derived from spatial structure: $\gamma_0 \leftrightarrow \pm\varepsilon_{3D} \cdot \gamma_1\gamma_2\gamma_3$. (Level 1 for γ_5 identification; Level 2 for γ_0 derivation, sign conventions not separately verified.)

(c) *Chirality equivalence.* For static observables in the bijection's stationary distribution, ε_{3D} and ε_{4D} classify modes equivalently. Their ratio is $\varepsilon_{4D}/\varepsilon_{3D} = (-1)^{x_0}$, constant on each time slice; chirality classification differs by overall sign convention only. (Level 1 for the algebraic identity; full operator-theoretic interpretation involving the taste structure remains at Level 3 but is not framework-essential — operational chirality assignment is the framework-essential claim.)

(d) *Symmetry compatibility.* Staggered $U(1)$ shift symmetry (underlying the two-loop structure of K_S) and cubic group O action are preserved by the isomorphism of (a). (Level 2.)

Proof, by part with explicit rigor levels.

Part (a) — Level 1. The map $(\eta_1, \eta_2, \eta_3, s) \mapsto (\eta_0 = s, \eta_1, \eta_2, \eta_3)$ is a bijection between $\{0,1\}^3 \times \{0,1\}$ and $\{0,1\}^4$ by direct counting ($2^3 \times 2 = 16 = 2^4$). The map respects the natural cubic-group action on spatial indices since the spatial coordinates η_1, η_2, η_3 are preserved unchanged, and $\eta_0 = s$ corresponds to the sublattice grading which is cubic-invariant.

Part (b), $\gamma_5 \leftrightarrow \varepsilon_{3D}$ — Level 1. Two properties to verify: $\varepsilon_{3D}^2 = 1$ and $\{\varepsilon_{3D}, \gamma_i\} = 0$ for $i = 1, 2, 3$. The first is direct: $\varepsilon_{3D}(x)^2 = ((-1)^{x_1+x_2+x_3})^2 = 1$. The second follows from $\varepsilon_{3D}(x \pm \hat{i}) = -\varepsilon_{3D}(x)$ via direct computation: $(\gamma_i \varepsilon_{3D}\chi)(x) = \eta_i(x)[\varepsilon_{3D}(x+\hat{i})\chi(x+\hat{i}) - \varepsilon_{3D}(x-\hat{i})\chi(x-\hat{i})]/2 = -\varepsilon_{3D}(x) \cdot \eta_i(x)[\chi(x+\hat{i}) - \chi(x-\hat{i})]/2 = -(\varepsilon_{3D}\gamma_i\chi)(x)$. Both γ_0 and γ_5 satisfy these two properties, so further identification is required: in chiral basis, γ_0 is off-diagonal in sublattice blocks while γ_5 is diagonal with opposite signs. Sublattice grading ε_{3D} acts diagonally with opposite signs on even/odd sublattices — that is the γ_5 pattern, not γ_0. Therefore $\gamma_5 \leftrightarrow \varepsilon_{3D}$.

Part (b), γ_0 derivation — Level 2. The 4D Clifford algebra contains the identity $\gamma_0 = \pm\gamma_5\gamma_1\gamma_2\gamma_3$ (up to a sign convention depending on the metric signature and the specific Clifford basis convention). Substituting the $\gamma_5 \leftrightarrow \varepsilon_{3D}$ identification gives $\gamma_0 \leftrightarrow \pm\varepsilon_{3D} \cdot \gamma_1\gamma_2\gamma_3$. The sign convention depends on the specific Clifford basis convention and is not separately verified here.

Part (c), chirality equivalence — Level 1. Direct algebraic identity: $\varepsilon_{4D}(x_0, \vec{x}) = (-1)^{x_0+x_1+x_2+x_3} = (-1)^{x_0} \cdot (-1)^{x_1+x_2+x_3} = (-1)^{x_0} \cdot \varepsilon_{3D}(\vec{x})$. For static observables (evaluated at fixed x_0), the prefactor $(-1)^{x_0}$ is a constant — observables computed via ε_{3D} and via ε_{4D} classify modes equivalently up to overall sign convention.

Part (d) — Level 2. The staggered $U(1)$ shift symmetry transforms $\chi(x) \to \omega(x)\chi(x)$ with sign factors depending on sublattice and BZ corner. Under the isomorphism of (a), this transformation acts on the corresponding Dirac $\times$ taste basis vectors compatibly — full verification with explicit taste-index tracking is at Level 2 and not written out here. The cubic group O action commutes with the isomorphism by part (a). □

Operator-theoretic interpretation involving full taste structure (Level 3 — not framework-essential). The operational identification $\gamma_5 \leftrightarrow \varepsilon_{3D}$ is the framework-essential claim and is Level 1 closed in part (b). A full operator-theoretic interpretation involving the 4-dim taste rep ξ_5 and showing ε_{4D} corresponds to $\gamma_5 \otimes \xi_5$ in the spin-taste decomposition is at Level 3 (sketch only) and is not required for the framework's use of the bridge theorem.

Implication for the framework's $d = 3$ commitment. The bridge theorem reveals that the entire 4D Clifford structure emerges from 3D Clifford + sublattice grading, with γ_0 derived rather than independent. There is no input from time-direction structure; time enters only as bijection-evolution index x_0, and the Clifford-algebra role is played entirely by spatial structures. This means structural bandwidth quantities like $\theta_0 = C_2/d^2 = 2/9$ and $A^2 = (d-1)/d = 2/3$ must use spatial $d = 3$, not spacetime $d = 4$. The framework's commitment to $d = 3$ is structurally forced (not chosen) by the time-emergence picture.

The same gaps apply to the other §7 mass-cluster predictions (D11, m_τ, etc.) that share the SχPT machinery; the closure status above propagates to those predictions automatically. None of them changes the closure path for K_S, which remains the two-loop SχPT calculation specializing Panagopoulos–Spanoudes [Panagopoulos-Spanoudes 2017] to the OI cubic-group setting.

MC consistency check: the empirical matching scale extracted from the Monte Carlo data is $m_{\text{match}}^{\text{MC}} = 0.1234 \pm 0.0002$, which agrees with the predicted $\lambda \cdot g_0^2 = 0.1217$ within 1.4% (compatible with NLO truncation of the leading-order chain above). Alternative choices of coefficient — e.g., $K = 1$ giving $m_{\text{match}} = 2\lambda g_0^2 \approx 0.24$ — are strongly excluded by the MC data (would give $m_b/m_\tau \approx 3.17$ vs. observed 2.35).

The prediction.

$$\boxed{\frac{m_b}{m_\tau} = \frac{4.28}{Z_S(\lambda g_0^2)} = \frac{4.28}{1.813} = 2.361}$$

Observed: $m_b/m_\tau = 2.352 \pm 0.017$. The central-value match at $L = 32$ is 0.4% (0.5σ), with a realistic systematic uncertainty of $\sim$ 1–2% from finite volume, the regulator choice noted above, and higher-loop SM RGE corrections to $R = 4.28$. The $L \to \infty$ extrapolation in either lattice regulator gives $Z_S \approx 1.83$ and $m_b/m_\tau \approx 2.34$, consistent with observation at $\sim 0.7\sigma$. The $L = 32$ run (50 configs) gives $Z_S(0.122) = 1.813 \pm 0.001$ (statistical only) by cubic interpolation; finite-volume systematics are not exhaustively quantified in the released code (see `README.md` §Code, point 3). The inputs λ, g_0^2, and $R = 4.28$ are determined by the lattice structure; the matching-scale identification $m_{\text{match}} = \lambda \cdot g_0^2$ follows from power-counting in the taste-decomposed CW potential. The dimensional form $m_{\text{match}} \propto \lambda \cdot g_0^2$ is forced: λ controls inter-generation taste-breaking and g_0^2 controls the gauge-induced condensate, and these are the only two dimensionless framework primitives entering the leading-order CW potential. The scalar prefactor (the K coefficient) is the separate Layer 2 question addressed by Arguments 1–3 above and the Residual gap subsection.

7.6 Summary table of all predictions

Twenty-two quantitative observables from a $d = 3$ cubic lattice with spacing $\epsilon = 2\, l_P$, classified by input status: **S** = parameter-free structural (no fitted input; the cubic substratum itself is empirically selected, §8.3); **L** = layered (structural form combined with a Layer-2(b) substrate-specific coefficient or Layer-3 empirical input — see §8.3 four-layer framing); **M** = one-input mass/parameter chain; **E** = uses explicit empirical input; **R** = retrodiction; **P** = phenomenological/ansatz input.

Prediction	Formula	Value	Observed	Match	Input
$1/\alpha_1(M_Z)$	lattice + RGE	59.00	59.00	$< 0.1\%$	**E**[a] (fixes δ_0)
$1/\alpha_2(M_Z)$	lattice + RGE	29.57	29.57	$< 0.1\%$	**R**[a]
$1/\alpha_3(M_Z)$	lattice + RGE	8.47	8.47	$< 0.1\%$	**R**[a]
λ (Cabibbo)	$1/(\pi\sqrt{2})$	0.2251	0.2250 ± 0.0007	0.04%	**S**
A (Wolfenstein)	$\sqrt{2/3}$	0.8165	0.826 ± 0.012[h]	0.8σ	**S**
m_d/m_s	$1/(2\pi^2)$	0.0507	0.0503 ± 0.0007	0.56σ	**L** (structural λ; GST empirical)
$\lvert V_{cb}\rvert$	$\sqrt{2/3}/(2\pi^2)$	0.0414	0.0408 ± 0.0014[h]	0.4σ	**S**
J (Jarlskog)	$\eta/(12\pi^6)$	3.02×10^{-5}	$(3.08 \pm 0.13) \times 10^{-5}$[h]	0.5σ	**E** (empirical η)
Koide Q	$2/3$	0.66667	0.66666	$< 0.01\%$	**S**
Koide angle θ_0	$C_2/d^2 = 2/9$	0.22222	0.222225	0.001%	**S**
m_e (from m_τ)	$\theta_0 = 2/9$	0.51096 MeV	0.51100 MeV	0.007%	**M** (m_τ)
m_μ (from m_τ)	$\theta_0 = 2/9$	105.652 MeV	105.658 MeV	0.006%	**M** (m_τ)

Prediction	Formula	Value	Observed	Match	Input
Q_{down}	$(2/3)(1+\alpha_s/\pi)$	0.7303	0.7314	0.15% (scheme-dep)	**P**
m_b (from m_s)	Koide(Q_{down})	4144 MeV	4180 ± 30	0.9%	**L**[d]
m_u/m_d	$\sqrt{2/9}$	0.4714	0.465 ± 0.024	0.27σ	**L**[e]
$\sin^2\theta_{12}$	$1/3 - 1/(4\pi^2)$	0.3080	0.3085 ± 0.0073 (Capozzi 2025 post-JUNO global)	0.07σ	**L**[f]
$\sin^2\theta_{23}$	$1/2 + 1/(2\pi^2)$	0.5507	$0.561^{+0.012}_{-0.015}$ (NuFIT 6.0, NO)	0.74σ	**L**[f]
$\sin^2\theta_{13}$	$4/(18\pi^2)$	0.02252	0.02195 ± 0.00056 (NuFIT 6.0, NO)	1.01σ	**S**
$\lambda(M_{\text{Pl}}) = 0$	composite Higgs	0	-0.013 ± 0.020	0.6σ	**S**
m_H (from $\lambda(M_{\text{Pl}}) = 0$, m_t)	RGE running	129–132 GeV	125.10 ± 0.14	m_t-consistent	**M** (m_t)
m_b/m_τ	$4.28/Z_S(\lambda g_0^2)$	2.361	2.352	0.5σ	**L**[b]
m_t	$v/\sqrt{2}$ (weak-scale $y_t \approx 1$; mechanism open)	174.1 GeV	172.5 ± 0.3	0.9%	**R**[g]

Footnotes:

- [a] The three SM gauge couplings at M_Z are reproduced with one empirical input in this sector (§6.3): the U(1) coupling fixes the universal threshold $\delta_0 = 10.02$, and the SU(2) and SU(3) entries are retrodictions using the C_2-dependent threshold parameterization $(A, B) = (8.3, 5.59)$ matched to the observed couplings. The structural input $1/\alpha_0 = 23.25$ is computed analytically from the one-loop staggered vacuum polarization. A 2-loop staggered VP rebuild indicates δ_0 is not reproduced by strict 2-loop PT and is accordingly treated as empirical input (see §6.2). The structural predictions in this sector are: (i) the specific value $1/\alpha_0 = 23.25$, (ii) the universality of δ_0 across gauge groups, and (iii) the 1-loop analytic computation of $A \cdot B$, whose extrapolated value is consistent with the fitted 46.4 under the $1/N^2$ leading-power assumption but is leading-power-dependent (range $\approx$ 48–62 across finite-size models; §6.2.1) and is inconclusive pending larger lattices. Deriving (A, B) individually from first-principles lattice perturbation theory would upgrade the SU(2) and SU(3) entries from retrodictions (**R**) to strict predictions (**S**).

- [b] *Layered prediction (m_b/m_τ).* The matching scale $m_{\text{match}} = \lambda \cdot g_0^2$ is partly structural in the taste-decomposed Coleman–Weinberg potential (§7.5): the b-quark is identified with the democratic eigenstate $\hat{h}$ of the cubic-symmetric taste-breaking operator (from §7.1, §7.2 representation theory), giving diagonal taste-breaking $c_{bb} = (n_{\text{gen}} - 1)\lambda = 2\lambda$ — Layer 1 algebra. Combining with the natural matching condition at condensate scale Σ and a loop kinematic coefficient K, the prediction $m_{\text{match}} = \lambda g_0^2$ requires $K = 1/(n_{\text{gen}} - 1) = 1/2$ — a structural identity that follows by definition *given* the matching condition; the matching condition itself is MC-verified to 1.4%, and analytical derivation of its specific dimensionless coefficient is the Layer 2(a)-tractable residual (an explicit one-loop calculation on the OI cubic lattice or a Ward-identity argument). Within $K \in \{1/4, 1/2, 1, 2\}$, $K = 1$ is excluded at 35% and $K = 1/4$ at 24%; see §7.5 for the structural detail. The MC-extracted value $m_{\text{match}}^{\text{MC}} = 0.1234 \pm 0.0002$ agrees with the derived $\lambda g_0^2 = 0.1217$ within 1.4% (NLO truncation). The central-value match $m_b/m_\tau = 0.5\sigma$ is at $L = 32$; systematic uncertainty including finite-volume and regulator effects is $\sim$ 1–2% (§7.5). Within the four-layer framing (§8.3), the prediction is MC-verified at the matching-condition level with the Layer 2(a)-tractable analytical closure pending.

- [d] *Layered conditional structural prediction (m_b via Q_down).* The structural form (Koide formula applied to the down-sector with input $Q_{\text{down}} + m_d/m_s + m_s = 93.4$ MeV) is Layer 1. The Q_{down} value ($= (2/3)(1 + \alpha_s/\pi) \approx 0.731$) is phenomenological per §7.2 — see the Q_{down} subsection for full discussion of why this is treated as empirical input rather than derived. The m_d/m_s input inherits §7.1's Cabibbo λ^2 via the GST relation. The m_s input is one M-chain mass scale. The empirical match at 0.9% is excellent given the layered structure.

- [g] *Retrodiction (m_t).* One-loop SM running gives an IR quasi-fixed point at $y_t \approx 1.25$, and the normalization $y_t(M_{\text{Pl}}) = 1$ flows to $m_t \approx 207$ GeV; the 0.9% match is the weak-scale $y_t(m_t) \approx 1$ coincidence, whose structural derivation is open (§7.2).

- [h] *Data source.* The observed values for A, $|V_{cb}|$, and J follow a single global-fit source per observable, quoted in §7.1 and used identically here. The observed Wolfenstein A is itself fit-family dependent (central values span approximately 0.79–0.83 across PDG, CKMfitter, and UTfit determinations, a spread larger than any single fit's quoted uncertainty); the prediction $\sqrt{2/3} = 0.8165$ lies inside that spread, so the honest match statement for A is consistency within the fit-family spread rather than a single-σ figure. The J entry additionally carries a parametrization caveat: η is extracted by global fits within the Wolfenstein parametrization using the fitted A and λ, so inserting the fitted η alongside the framework's structural A and λ mixes parametrizations; the J row tests the combination $A^2\lambda^6$ only to the extent the extraction of η is insensitive to that mixing.

- [e] *Layered prediction (m_u/m_d), Layer 1 + Layer 2(b).* The structural form $m_u/m_d = \sqrt{\theta_0} = \sqrt{2/9}$ uses the same quadratic Casimir $C_2 = 2$ of the T_1 representation that produces the Koide angle $\theta_0 = 2/9$, with the square root following from amplitude-vs-eigenvalue PT scaling between intra-doublet and inter-generation channels (§7.2). Operator counting on the first-generation Yukawas shows that the relation requires alignment between SU(2)-doublet structure and cubic-group T_1 perturbation that is bijection-specific (Layer 2(b)). The empirical match at 0.05σ indicates the framework's bijection has the alignment property; promoting to Layer 2(a) requires explicit substrate-level Yukawa derivation that has not been performed.
- [f] *Layered prediction (sin²θ_12, sin²θ_23), Layer 1 + Layer 2(b).* The Layer 0 component (TBM zeroth-order value) and Layer 1 component (λ^2 scale) are unconditional structural. The values follow given Condition 2 of §7.3 (the relation $b_{23} = (4/3)(b_{12} + b_{13})$ among A_2 Wilson coefficients). The §7.3 structural decomposition closes the A_1 component at Layer 1 (the TBM sum rule is structurally protected against the A_1 perturbation by the $\cos^2\theta_{13}$ dressing alone) and reduces the A_2 component to the substrate ratio $b_{23}/(b_{12} + b_{13}) = 4/3 = 4(d-1)/(2d)$ for $d = 3$, with the structural reading from the squared off-democratic projection of vector representations. This ratio was Layer 2(a)-tractable until its defining substrate one-loop calculation was performed (§7.3, "Status of Cond 2") and excluded the second-order route, demoting it to Layer 2(b); the 7/6 sum rule is its direct experimental test, and only the recorded $O(\lambda^3)$ construction remains as a possible derivation. The natural Higgs-democratic spurion does not produce this ratio (the natural form gives $b_{12} : b_{13} = m_\tau/m_\mu \approx 17 : 1$ from Layer 1 mass-matrix algebra applied to a generation-democratic substrate Yukawa, per §7.3 "Positive support"). The empirical match ($\sin^2\theta_{12}$ at 0.07σ to the post-JUNO global fit; sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ at 0.6σ) tests the bijection's specific T_2-channel structure. See §7.3 "Status of Cond 2: structural decomposition" and "Positive support: explicit substrate calculation."

Parameter-accounting summary: Of the 22 observables tabulated above, **7 are unconditional parameter-free structural predictions** (**S**, Layer 0 or Layer 1: Cabibbo λ, Wolfenstein A, $|V_{cb}|$, Koide Q, Koide angle θ_0, $\sin^2\theta_{13}$, $\lambda(M_{\rm Pl}) = 0$). Two further parameter-free structural results — the ambient induced coupling $\alpha_{0,\mathrm{amb}}^{-1} = 23.25$ (native for the non-abelian factors; for hypercharge under H-observer-bundle and H-Y-vertex, §6.5) and the $A \cdot B$ product — belong to the same class but are *not* observables in this table: they enter the chain as structural inputs to the coupling sector (footnote [a]), and $A \cdot B$ is in any case leading-power-dependent and inconclusive pending larger lattices. Earlier statements of this accounting counted those two among "nine of the twenty-two"; the observables and the inputs are counted separately here. m_d/m_s is classified **L** because its conversion from the structural λ runs through the Gatto–Sartori–Tonin relation, an external empirical relation whose status the chain inherits (§7.1); the top mass is classified **R** because one-loop SM running gives $m_t \approx 207$ GeV from $y_t(M_{\mathrm{Pl}}) = 1$, so the 0.9% match is a retrodiction of the weak-scale $y_t \approx 1$ coincidence with the mechanism open (§7.2); **6 are layered predictions** (**L**, Layer 1 form combined with one Layer-2 substrate-specific coefficient or Layer-3 empirical input: m_d/m_s via GST as above, $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$ via Cond 2 (Layer 2(b) per the §7.3 routing computation), m_u/m_d via amplitude-vs-eigenvalue alignment (Layer 2(b)), m_b inheriting Q_{down}'s phenomenological status, m_b/m_τ via $K = 1/(n_{\mathrm{gen}} - 1)$ representation-theoretic identity (Layer 1 closed per §7.5) with Layer 2(a)-tractable substrate-loop verification residual); 3 are retrodictions (**R**: the SU(2) and SU(3) couplings, where (A, B) are matched to observation rather than computed, and the top mass per §7.2); 1 fixes the universal threshold δ_0 from the U(1) coupling (**E**); 3 form a one-input mass/parameter chain (**M**); 1 uses an explicit empirical input (**E**, the Jarlskog phase η). No parameters are fit to the predicted quantities in any of the **S** or **L** classes; the layered classifications reflect explicit dependence on stated structural inputs (Cond 2 of §7.3 — Layer 2(a)-tractable; the open derivations referenced in §7.2 and §7.5) or phenomenological inputs (Q_{down}) rather than free parameter fitting. The C class (conditional structural pending derivation) is currently empty.

In total, 9 unconditional + 6 layered = 15 predictions in the structural-or-near-structural class.

In addition: the SM gauge group; threefold taste multiplicity — three candidate generations, physical reading under H-spin', chirality under H-χ', spectrum selection open; strong CP open — T-invariance fixes $\theta \in \{0, \pi\}$ and reciprocity, not $\bar{\theta}$ (§5, H-top and H-det); Majorana neutrinos; normal mass ordering — all consistent with data. The matter-sector chain, open links marked:

$$\text{taste} \xrightarrow{\text{H-spin'}} \text{Lorentz spin} \xrightarrow{\text{H-}\chi\text{ (sharpened)}} \text{weak chirality} \to \text{selected light spectrum} \to \text{three SM generations},$$

established through the first node's label content (Theorem 9); each subsequent arrow is a named open input. Per-link statuses: $1 \oplus 3$ taste structure — proved (Theorem 8); H-spin' — free kernel refuted and the 4D-staggered route settled negative (`hspin_kernel_probes.py`, `hspin4d_probes.py`); the condensate-dressed form open (§4.7); H-χ' — open in sharpened form (§4.8; the minimal embedding is certified vector-like; mechanism class nonempty, dynamical selection open — `hchiprime_probes.py`); light-spectrum selection — open;

three physical generations — conditional on the above; strong CP — unresolved (§5.5).

Remaining open: (i) m_s (the overall mass scale — requires taste-decomposed Coleman–Weinberg potential); (ii) m_c (no clean structural relation found; the up-sector hierarchy involves top Yukawa backreaction); (iii) CP-violating phases (solution-specific); (iv) neutrino masses (solution-specific, but structurally constrained: Majorana, normal ordering).

7.7 The coincidence bound

The unconditional structural (**S**) and layered (**L**) retrodictions of §7.6 — fifteen in total — are not independent draws: several derive from the same cubic-group quadratic Casimir $C_2(T_1) = 2$ (the Koide amplitude $A^2 = C_2$, the Koide angle $\theta_0 = C_2/d^2$, $\sin^2\theta_{13}$, and the m_e, m_μ chain from m_τ), and several derive from the same one-loop lattice integral structure (Cabibbo λ, Wolfenstein A, $|V_{cb}|$, m_d/m_s). Conservatively treated, the **S** set contains *two* structurally independent prediction families at the tightest experimental precision: the Casimir family — anchored by the Koide relation $Q = 2/3$ via $A^2 = C_2$ — and the Cabibbo angle $\lambda = 1/(\pi\sqrt{2})$ (one-loop lattice integral on $\mathbb{Z}^3$). The two are tested by structurally distinct features of the lattice (Casimir eigenvalue structure vs. spatial integral). Each family carries an identified open link: the Casimir family carries the dynamical-closure caveat of §7.2 (the amplitude $A^2 = C_2$ is structural and exact but not yet derived from the dynamics, which at one loop undershoots), and the Cabibbo family carries the propagator-identification link of §7.1 (the identification of the mixing element with the unit-normalized taste-changing propagator), whose structural level §7.1 discharges as an algebraic identity of the staggered spin-taste map, leaving the emergent-theory dressing as its residual open element. Neither pillar should be ranked above the other pending its respective closure.

Cubic-group T_1-Casimir consistency test. Beyond the two-independent-predictions framing, the cubic-group T_1-Casimir structure $C_2 = 2$ for vector reps on the $d = 3$ lattice produces a six-prediction simultaneous-match consistency check:

Prediction	Formula	Predicted	Observed	Match
Wolfenstein A^2	$C_2/d = (d-1)/d = 2/3$	0.6667	0.683	1.2%
$\sin^2\theta_{13}$	$\propto A^4 = (C_2/d)^2 = 4/9$	0.0225	0.0224	0.4%
Koide θ_0	$C_2/d^2 = 2/9$	0.2222	0.2222	0.02%
m_u/m_d	$\sqrt{C_2}/d = \sqrt{2}/3$	0.471	0.474	0.6%
$\lvert V_{cb}\rvert$	$A\lambda^2$	0.04136	0.0408	0.4σ

Prediction	Formula	Predicted	Observed	Match
PMNS sum rule	$7/6 = 1 + 2A^2/3$ via Cond 2	1.167	1.178	0.6σ

All six derive from the structural identity $A^2 = C_2/d$ for vector representations on $d = 3$ lattice, and all six match observation at sub-percent or sub-σ precision simultaneously. This is one successful test of the framework's cubic-group structure with six observables — methodologically a single-test framing rather than six independent successes — but the simultaneous coherence at this precision provides strong evidence for the underlying C_2/d identity.

The framework's empirical case therefore rests on two independent structural ingredients passing simultaneous tests: (a) Cabibbo $\lambda = 1/(\pi\sqrt{2})$ from BZ geometry, tested directly and through its appearances in $|V_{cb}|$, $\sin^2\theta_{13}$, and m_d/m_s via GST; and (b) cubic-group T_1-Casimir $C_2/d = 2/3$, tested via the six-observable simultaneous match above. Both ingredients are Layer 1.

On quantifying the joint match, and why no probability is quoted. A joint null probability of the natural form — each observed value treated as an independent uniform draw against a fixed formula — answers the wrong question for this package. The observed values were fixed and known before the framework existed — the Cabibbo angle since 1963, the Koide relation since 1981 (a celebrated pre-existing empirical regularity that the framework explains rather than discovers) — and the structural expressions were identified against them. The correct null must therefore account for the multiplicity of candidate structural expressions available to any such search, a multiplicity this paper itself documents: the normalization analysis of §7.2 enumerates ten candidate rational structures of comparable naturalness, the K coefficient of §7.5 is selected within $\{1/4, 1/2, 1, 2\}$, and the $1/d$-versus-$1/(2d)$ and $d = 3$-versus-$d = 4$ forks each expanded the candidate set during the framework's development. With tens of natural closed-form candidates per observable, per-mille agreement of *some* candidate is not a per-mille-probability event, and multiplying per-observable probabilities is not licensed. No honest closed-form null is available without an explicit model of the expression-search space, which is not attempted here.

What the retrodictive package supports. Three statements carry the section's content. First, the matches are *concentrated*: the two tightest (λ at 0.04%, θ_0 at 0.02%) sit an order of magnitude or more inside the spacing of their own candidate families as enumerated in §7.2, so formula-multiplicity alone does not comfortably absorb them — though this is a qualitative observation, not a probability. Second, the six-observable C_2/d simultaneous-consistency table above is a genuine single structural test with no per-observable freedom, and it passes; its evidential weight is that of one test, as already stated. Third, and decisively, the framework's exact, freedom-free forms convert into *pre-registered forward* exposure — the JUNO design-lifetime precision on $\sin^2\theta_{12}$, the DESI

Y5 ν measurement, the lattice thresholds of the A^2 computation — and it is these forward tests, not any retrodictive probability, that carry the framework's empirical case. A reader should weight the retrodictive package as a compression claim (the Standard Model's flavor structure is expressible in cubic-lattice invariants to the stated precisions) and reserve confirmation-strength language for the forward set.

7.8 Forward predictions and the retrodiction-vs-prediction distinction

The coincidence bound of §7.7 quantifies how concentrated the framework's *retrodictive* content is. It does not, and cannot, address a distinct and more fundamental concern: every quantity in §7.1–§7.5 is a known value, and a framework developed with knowledge of its targets will tend to reproduce them whether or not its underlying structure is correct, because the development process has access to the answers. No internal analysis — no coincidence bound, no audit, no re-derivation — distinguishes "the structure forced these values" from "the known values guided the construction of the structure." Only a prediction of a value *not known at construction time* breaks this symmetry. This subsection separates the framework's genuinely forward content from its retrodictions and states it as such, because the forward content is the only part that bears on the retrodiction-vs-prediction distinction.

The JUNO match (a retrodiction with a pre-registered forward conversion). The solar mixing angle value $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.3080$ matches JUNO's first oscillation result (November 2025) at 0.14σ and the post-JUNO global fit (0.3085 ± 0.0073, Capozzi et al.) at 0.07σ. The derivation postdates the November-2025 measurement, so the match is retrodictive and belongs to the compression claim of §7.7; the post-hoc-selection concern of this subsection applies to it in full. What the exact, freedom-free form supplies is the framework's cleanest forward conversion: the value admits no adjustment, so JUNO's design-lifetime precision (± 0.0014) will test the same number at a $6\times$ tighter error — a pre-registered commitment whose outcome the framework cannot absorb.

Registered forward predictions (not yet measured to testing precision). The same construction forces the following, stated here as pre-registered commitments whose values are fixed by the framework before the measurements that will test them:

1. *The PMNS sum rule* $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$. Currently confirmed at 0.6σ but at the precision of present atmospheric-angle data. A direct measurement of the combination at the 0.005 level — within reach of DUNE and T2HK — tests this at a precision that no current fit reaches. The framework commits to 7/6 exactly; a measured value displaced from 7/6 by more than experimental error falsifies the structural relation.
2. *The atmospheric angle* $\sin^2\theta_{23} = 1/2 + 1/(2\pi^2) = 0.5507$. The framework commits to a specific octant ($\theta_{23} > 45°$) and a specific value. Present data

($\sim 1.1\sigma$) does not resolve the octant; DUNE/T2HK will. An ultimate measurement in the lower octant, or displaced from 0.5507 beyond error, falsifies the prediction.

3. *The reactor angle* $\sin^2\theta_{13} = 4/(18\pi^2) = 0.02252$, unconditional structural (Layer 1). Already consistent with current data; tightening reactor measurements provide a continuing test against a value the framework cannot adjust.
4. *Continued JUNO precision on* $\sin^2\theta_{12}$. JUNO's 30-year design lifetime projects ± 0.0014 on $\sin^2\theta_{12}$, a $6\times$ improvement. The framework's 0.30798 is tested at sub-percent precision against a value it committed to before this precision exists; a 2σ ultimate disagreement falsifies it.

Distinction from a parameter-fitted account. A framework that accommodated the known mixing angles by fitting would retain latitude to absorb a future measurement that moved against it. The four predictions above have no such latitude: they are forced by the same cubic-group structure that produces the retrodictions, with no free parameter to adjust if the data moves. The retrodiction-vs-prediction concern of §7.7 is therefore not resolved by the retrodictions themselves; it is resolved, to the extent it can be resolved, by the registered predictions above as they are measured. The framework's correctness is not internally decidable from the known-value matches; it is decidable, in either direction, by the forward predictions, which are stated in advance and carry fixed values.

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8. Discussion

The cosmological application — including the derivation of $\hbar$, the Bekenstein-Hawking area law, the dissolution of the cosmological constant problem, and the dynamical dark energy prediction — is developed in [GR]. The substratum-level reconstruction theorem and the synthesis claim that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same (S, φ) are developed in [Substratum].

8.1 Structural realism

The structural reading aligns with ontic structural realism but does not require it. What the framework proves is that (S, φ) is a *complete description* of reality up to gauge equivalence (the reconstruction theorem, developed in [Substratum]). Whether the structure *is* reality (the OSR commitment) or *describes* a reality that exists independently is a question the framework identifies as gauge for physics — provably undecidable by any measurement an embedded observer can perform — while remaining potentially substantive in mathematical foundations or other modes of inquiry not bound by embedded physical observation. The "stuff" of the universe, in any reading, is fully characterized by the abstract structure of a bijection on a finite set with bounded coupling.

The structural-realism stance articulated here operates at one specific level of the framework's hierarchical structure: the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ within OI's specific universality class, identified in [Structure §2.3] as Level D $\times$ Level G3. The framework's broader structural realism extends to other levels — the universality-class level (Level C $\times$ Level G4) where partial-trace observational features are universal across structural classes, and the foundational levels (A and B) where the observation axiom and observer-admission criterion apply. The complete-description claim of this paper is class-specific: it asserts that within OI's universality class, (S, φ) modulo $\mathcal{G}_{\text{sub}}$ is the complete description. The full multi-level articulation is in [Structure §2.3].

8.2 Observer genericity

Theorem 22 (Genericity of observers). *Let φ be the wave equation (or any energy-conserving dynamics) on a connected bounded-degree coupling graph with diameter $D \geq 4$. Then for any connected subgraph V with $|V| \leq N/3$, the partition (V, H) satisfies C1, with the C3 capacity floor holding for the realized process by the data-processing bound $I^* \leq \log_2 |\mathcal{C}_H|$ ([Main §3.4]); with record persistence (C2) and history readback (C4) — the record surviving to a readback the coupling performs — as explicit additional hypotheses, the partition realizes the full C1–C4 memory architecture.*

Proof. C1: G_φ is connected and V is a proper subgraph, so $\partial\text{V} \neq \emptyset$. The wave equation couples nearest neighbors; any edge in ∂V produces non-trivial dynamical coupling.

C2 is *not* established by energy conservation. Reversibility and phase-space-volume preservation exclude fundamental information loss, but phase mixing within the Hamiltonian hidden sector can destroy the *accessibility* of the particular record written by V before readback — precisely the possibility [Main, §3.4]'s physical-memory theorem and mixing-hypothesis analysis are formulated to handle. Record persistence therefore enters as an explicit hypothesis for this realization.

C3: $|H| = N - |V| \geq 2N/3$, so $\log_2 |\mathcal{C}_H| = |H| \log_2 q \geq (2N/3) \log_2 q$. For the process this realization actually generates, [Main, §3.4]'s data-processing bound gives $I^* \leq \log_2 |\mathcal{C}_H|$: the capacity floor holds for the realized process. No stronger saturation claim is made — cumulative KI_0 accumulation would require a no-reuse/compression hypothesis, which is not assumed ([Main, §3.4]). □

Remark (C2/C4 status). The proof establishes C1 structurally and the C3 capacity floor for the realized process. Record persistence (C2) and history readback (C4) — the persistence and load-bearing memory conditions ([Main §3.4]; storage, persistence, and capacity alone do not imply memory, per the certified counterexample class) — enter as explicit hypotheses: energy conservation does not imply that the particular written record survives *accessible* to readback, since phase mixing can destroy record accessibility without fundamental information loss. The stationarity-window analysis below supplies the natural route to the two genericity lemmas — persistence via light-cone survival of deposited boundary data until τ_{return}, readback via its return to V — which, if proved, would restore the unconditional four-condition form; neither is proved here.

Remark (validity window). Under the persistence hypothesis, the emergent Hamiltonian has a finite *stationarity window*: the time before information deposited at the V–H boundary propagates through H and returns to V at a different boundary point. The wave equation has an exact light cone (no exponential tails), so the minimum return time is $\tau_{\text{return}} = \min_{b,b' \in \partial V} \text{dist}_H(b, b')$. For core-shell partitions with core radius r on a lattice of side L: $\tau_{\text{return}} = (L - 2r)/v$. After τ_{return}, the reduced law remains universally Q_{fb}-representable — representability does not depend on C2; what changes is the stationarity of the effective-Hamiltonian approximation, the emergent Hamiltonian becoming time-dependent. Numerical verification: for L = 20–320, τ_{return} ranges from 6 to 318 steps, confirming the light-cone bound.

Corollary. Any finite deterministic system with energy-conserving dynamics and bounded-degree coupling necessarily contains subsystems with non-trivial coupling (C1) and the C3 capacity floor for the realized process, whose reduced internal description universally admits the fixed-basis quantum representation; with record persistence (C2) and history readback (C4) as explicit hypotheses — pending their respective genericity lemmas — the memory-bearing non-Markovian phenomenology follows. The coupled, capacious partition is a mathematical consequence; the memory-bearing observer requires the persistence and readback hypotheses.

8.3 What the framework determines and what the specific bijection determines

The framework's predictions fall into two categories, and the distinction matters for assessing their status.

Structural predictions are properties of the equivalence class $[(\text{S}, \varphi)]/\mathcal{G}_{\text{sub}}$ — they hold for every bijection satisfying the six structural properties. These include: SU(3) × SU(2) × U(1) with multiplicities (3, 2, 1), three generations (chirality conditional on H-χ', §4.8) with identical gauge quantum numbers, one Higgs doublet (1, 2, +1/2), unique hypercharges from anomaly cancellation, $\hbar = \text{c}^3\varepsilon^2/(4\text{G})$, the Bekenstein-Hawking formula, and the universal induced coupling $1/\alpha_0 = 23.25$ (§6.1). These are theorems. They do not depend on which specific φ describes our universe.

Solution-specific properties are determined by the particular bijection φ within the equivalence class. These include: gauge couplings at M_Z (set by the eigenvalues μ_c, μ_w of M, the threshold δ_0, and the resummation parameters A, B — see §6); fermion mass *scales* (set by taste-breaking in φ at $\mathcal{O}(\epsilon^2)$ — though many mass *ratios*, including the Koide chain $m_e/m_\mu/m_\tau$, m_d/m_s, and m_u/m_d, are structural, see §7.2); basis-alignment unitaries between up-type and down-type Yukawa eigenstates (and hence the Jarlskog invariant, δ_{CKM} via the empirical η, and δ_{PMNS}); and individual Wilson coefficients of the cubic-invariant operators contributing to the Yukawa sector beyond whatever structural relations (such as Condition 2 of §7.3, a stated structural input whose second-order derivation route is excluded by the monomial-orthogonality result there, and which the 7/6 sum rule tests directly) constrain them. The CP phases are compatible with §5: the construction fixes neither $\bar{\theta}$ nor the basis-alignment structure that produces the Jarlskog invariant, and reciprocity of the transition counts leaves both free (§5.4).

The framework's predictions stratify into four layers, made precise here for downstream reference:

Layer 0 — Gauge / discrete structure. Properties of the equivalence class $[(S,\varphi)]/\mathcal{G}_{\text{sub}}$ that follow from C1–C4 + cubic-group representation theory (§4.5–§4.7). Hold for every bijection. Examples: the gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ with multiplicities $(3,2,1)$, three generations (chirality conditional on H-χ' in its sharpened form (the minimal embedding certified vector-like), §4.8), anomaly-cancelling hypercharges, Majorana neutrinos with normal mass ordering, the TBM zeroth-order pattern ($\sin^2\theta_{12} = 1/3$ etc.), the Higgs as A_1 taste, the universal induced coupling $1/\alpha_0 = 23.25$ from $N_f = 6$ and $T(R) = 1/2$.

Layer 1 — Structural form of corrections. Properties of the *form* of the perturbative expansion around the zeroth-order structural prediction. Follow from cubic-group representation theory + uniqueness constraints. Hold for every bijection. Examples: the Cabibbo angle scale $\lambda^2 = 1/(2\pi^2)$ (Brillouin geometry, §7.1), the Wolfenstein $A = \sqrt{2/3}$ (off-democratic projection), the Koide angle $\theta_0 = C_2/d^2 = 2/9$ (cubic-group quadratic Casimir over bandwidth squared), the form $V_\ell = \exp(\lambda A_1 + \lambda^2 A_2 + \cdots)$, A_1's unique determination, $\sin^2\theta_{13} = (4/9)\lambda^2$ (independent of A_2 at $O(\lambda^2)$), the Higgs quartic boundary condition $\lambda(M_{\text{Pl}}) = 0$, the universality of δ_0 across gauge groups (§6.3).

Layer 2 — Wilson-coefficient relations. Relations among the Wilson coefficients of cubic-invariant operators in the substrate Lagrangian. Three sub-cases:

- *Layer 2(a) — operator-level structural.* A finite family of cubic-invariant operators produces coefficient relations that hold for every bijection.
- *Layer 2(a)-tractable — sharply specified residual question.* A coefficient ratio reduces to a sharply specified structural quantity (with structural reading from representation theory) not yet derived from substrate dynamics, but with concrete substrate calculations available to test it. Promotes to 2(a) once the calculation is performed and confirms the ratio. The intermediate case between 2(a) and 2(b) (see "Diagnostic" below).
- *Layer 2(b) — solution-specific.* Coefficient values are free parameters of the bijection.

Cond 2 ($b_{23} = (4/3)(b_{12} + b_{13})$) admits the structural decomposition of §7.3: an A_1 component closed at Layer 1 (sum rule preserved by A_1 alone via $\cos^2\theta_{13}$

dressing) and an A_2 component whose ratio $b_{23}/(b_{12}+b_{13}) = 4(d-1)/(2d) = 4/3$ now sits at Layer 2(b): the sharply specified substrate calculation that defined its 2(a)-tractable period has been performed (§7.3) and excludes the second-order route, so the ratio is a solution-specific input unless the recorded $O(\lambda^3)$ construction is some day pursued and succeeds. The natural Higgs-democratic spurion does not produce this ratio (the natural form gives $b_{12} : b_{13} = m_\tau/m_\mu \approx 17 : 1$ from Layer 1 mass-matrix algebra applied to a generation-democratic substrate Yukawa, per §7.3 "Positive support"); the closure path is a one-loop staggered substrate calculation (§7.3 "Status of Cond 2"). Similarly, the intra-doublet vs inter-generation channel decomposition for $m_u/m_d = \sqrt{\theta_0}$ (§7.2) requires alignment between SU(2) Yukawa structure and cubic-group T_1 perturbation that is bijection-specific (Layer 2(b)). The $K = 1/2$ coefficient in the m_b/m_τ derivation (§7.5) follows by structural identity $K = 1/(n_{\text{gen}} - 1)$ *given* the matching condition $m_{\text{match}} = \lambda g_0^2$ (which is MC-verified to 1.4%); the residual Layer 2(a)-tractable question is whether the OI's specific taste-decomposed Coleman-Weinberg one-loop calculation derives this matching condition's specific dimensionless coefficient from first principles, rather than relying on MC verification of the matching scale.

Diagnostic for distinguishing Layer 2(a) from 2(b): When a Layer-2-looking ratio reduces to a structural identity for the relevant representation, it can promote to Layer 1. For example, $A^2 = (d-1)/d = C_2/d$ is a representation-theoretic identity for vector reps (§7.1), not a numerical coincidence; the off-democratic projection-squared and the Casimir-over-dimension ratio are the same quantity. This identity is what makes $\sin^2\theta_{13} = A^4\lambda^2$ Layer 1 structural, even though the calculation goes through PMNS matrix-element products. Conversely, when a Layer-2 ratio requires alignment between distinct operators or transfer between sectors, the natural classification is Layer 2(b) until the alignment is shown to be forced by substrate structure. The intermediate case — a ratio that reduces to a sharply specified structural quantity not yet derived from substrate dynamics, but with concrete substrate calculations available to test it — is Layer 2(a)-tractable. The Cond 2 ratio $b_{23}/(b_{12}+b_{13}) = 4(d-1)/(2d)$ was the canonical example until its defining calculation was performed (§7.3) and excluded the second-order route — demoting it to 2(b) and demonstrating the class's exit rule in the negative direction; the current 2(a)-tractable residents are the D11 mechanism, the D10 K-value, and the δ_0 threshold questions.

Sub-classifications of Layer 2(b) by mechanism. Within Layer 2(b), the framework distinguishes three mechanisms by which a coefficient becomes bijection-specific: (i) *single-parameter direction selection* — a single direction in some internal space is selected by the bijection (D11: which V_2-axis the Higgs condenses along determines m_u/m_d); (ii) *multi-parameter operator alignment* — multiple Wilson coefficients in distinct operator channels align in a specific pattern (Cond 2 was the canonical example before its §7.3 structural decomposition; the decomposition's residual ratio, briefly Layer 2(a)-tractable, has since been computed and demoted to 2(b); sub-classification (ii) currently has no Layer-2(b)-resident example); (iii) *cross-layer transfer* — a structural relation transfers between distinct group-theoretic layers (spatial vs internal T_1). The mechanism specificity is informative: single-parameter selection constrains the bijection more tightly than generic Wilson-coefficient ambiguity, and the empirical match becomes a tighter test of the bijection's structure.

Layer 2 values are discretization-determined, not free Wilson coefficients. The bijection φ is constrained by linear dynamics, T-invariance, P-indivisibility, and cycle ergodicity (cf. §4.2 *Remark*, §7.5). Within this constrained class, the bijection space is parametrized by *discretization-scheme choices* (field quantization, lattice size, timestep, boundary conditions, rounding rule), not by abstract permutation freedom. Once the discretization scheme is fixed, the bijection is essentially unique. As a consequence, Layer 2 coefficients (Cond 2 ratio at 2(b) status, D11 mechanism, D10 K-value, δ_0 threshold — all at Layer 2(a)-tractable or 2(b) status) are *all* functions of the same underlying discretization choices — they are not independent. This sharpens what Layer 2 means: the framework's empirical matches at sub-percent / sub-σ precision (Cond 2 at 0.6σ on the sum rule, m_u/m_d at 0.6%) are tighter tests of discretization scheme than they appear as tests of "generic bijection alignment." Cross-prediction consistency is therefore a meaningful test within structural clusters; the cubic-group T_1-Casimir cluster of §7.7 is one such successful test.

Selection problem. The framework characterizes a constrained class of substratum bijections (above) and derives empirical predictions from any element of this class. *Selection within the class* — which specific discretization scheme realizes our universe — is not addressed. This is the framework's analog of "why these laws of physics?" in standard physics, but the question is more pointed because the framework explicitly characterizes a meta-space (discretization-scheme space) within which the selection occurs. We treat this as an open question: the framework is a theory of what follows from a given discretization scheme, not a theory of which scheme is realized.

Derivation status: the three tiers. A recurring question about any framework that "derives the Standard Model from one object" is whether the derivation is genuine or whether the answer was quietly inserted in the construction — the tautology objection (stated in full at [Substratum §3, Remark on substantive content]). The objection does not admit a single yes/no answer, because the framework's results do not all have the same logical status. They separate cleanly into three tiers, and naming the tier is the honest form of the claim.

Tier 1 — exact representation-theoretic core plus named carrier bridge. Given the simple-cubic signed-link orbit, the character table forces $6 = T_1(3) \oplus E(2) \oplus A_1(1)$ with no free multiplicities. This is genuinely non-tautological and independent of QM. Turning that geometric representation into the physical SM carrier, however, inherits H-link (single-copy for $K = 6$) and H-cust; the physical-generation reading additionally inherits H-spin'. The anomaly and Brillouin-zone calculations remain sharp conditional consequences of that branch rather than proofs of H-link itself.

Tier 2 — empirically selected (the inputs that do the forcing). Several quantities that the chain treats as fixed are in fact pinned by observation, not by the substratum: the spatial dimension $d = 3$ (argued at self-consistency strength by filters E5–E7 of §3.2 — propagating gravity requires $d \geq 3$ and stable matter requires $d \leq 3$, with the boundary-entropy concordance an independent, non-anthropic filter that also excludes $d \geq 4$ by equalling unity only at $d = 3$, and so substitutes for the anthropic stable-matter step in the circularity rebuttal of §3.2; the argument's declared exposure is that the filters consume d-dependent emergent physics, so this is a self-consistency selection rather than an axiomatic derivation, independently anchored empirically by the observed gauge group); the simple-cubic Bravais structure itself (Theorem 7b selects it *because* an SU(3) factor is observed; §4.6); the universal threshold δ_0 (fixed by the U(1) row, §6.2); and the absolute mass scales (Tier 3 / solution-specific inputs to the mass chains). For these the direction of inference runs from the observed world back to the commitment. In particular, "the Standard Model from a cubic lattice" and "the lattice is cubic because SU(3) is observed" are the same fact stated in two directions; the framework treats the cubic structure as a commitment forced by the observed gauge group rather than a free choice (§4.6), and that candor belongs in the headline, not only in a remark.

Tier 3 — definitional stipulation. Some load-bearing moves are choices about what counts as physical structure rather than facts the bijection forces: coupling-degree minimization as the *definition* of "site" and of the internal factorization (§2.4, §4.5 — a system with a different K corresponds to a different bijection, not to the same one factored differently, so the definition is principled but is still a definition); the nearest-neighbor restriction (flagged as an assumption beyond bounded coupling degree, §4.1 *Remark (scope)*); and, at the foundation, the C1–C4 architecture as the model of observational limitation ([Main §1.3]).

The upshot is that the framework is best described as a *maximally tight compression* of the Standard Model onto the cubic-lattice commitment: Tier 1 shows that, once the commitment is granted, the SM's discrete structure is largely not independent — a single representation-theoretic object — which is a genuine and unusual claim about the unity of physics; while Tiers 2–3 show that the commitment itself is empirically anchored and partly stipulated, so the framework is not deriving physics from a parameter-free first principle. The tautology question is therefore not "open pending computation" — no lattice calculation moves it, because the status of each link is already determined by the construction as written. It is resolved by this classification: non-tautological at Tier 1, tautological-in-the-relevant-sense at Tier 2, definitional at Tier 3.

Organizing principle behind the tiers: constitution versus consequence. The split between what lands in Tier 1 (derived) and what lands in Tiers 2–3 (input or stipulated) is not arbitrary; it tracks the distinction between the substratum's *constitution* and its *consequences*. Tier 1 is exactly the set of consequences that follow, by representation theory of the cubic group on the link directions together with the trace-out, once the substratum is fixed. Tiers 2–3 are the constitution itself — the spatial dimension, the Bravais type, the rotational isotropy, the periodicity, the definition of a site. The rule is that the framework derives the consequences of a constitution but not the constitution, with one qualification on tractability grounds: a consequence the framework cannot yet compute is also booked as input (the threshold δ_0 of §6.2 is a consequence of the coupling-running held as input only because it is not yet derived, not a constitutional feature). So the inputs comprise the constitution together with any not-yet-tractable consequences, and the derived results are the tractable consequences. The structural reason an embedded observer can derive consequences but not constitution — and the reason the boost sector of §3.1 is more tractable than the rotational sector — is developed in the book's open-problems chapter (Ch. 19 §19.3.7): the observation map is realized at a horizon that selects a rest frame, so it makes contact with the time-distinguishing (boost) structure but not with any spatial-direction-distinguishing (rotational) structure, the latter being pure constitution. This is an organizing observation about the derivability boundary, not a formal theorem; notably, the representation-theoretic Tier-1 results would follow for any analyst of the fixed lattice without reference to an observer, so derivability is not claimed to coincide with any single relational property.

Formal statement of the determination (Proposition). The contentful half of the organizing principle can be stated precisely. **Proposition.** *Fix the substratum constitution C — the spatial dimension, the Bravais type, the wave-equation dynamics with its nearest-neighbor and linearity stipulations, and the alphabet — and the trace-out map T. Then the emergent gauge group, the candidate-sector count ($1 \oplus 3$; its reading as three physical generations is conditional on H-spin', §4.7), the anomaly-free hypercharge assignment given the sector pattern, and the Cabibbo angle are each determined as functions of (C, T)* — by the octahedral representation theory of §4.5–4.7 together with the chiral/taste structure of §5 and §7. This is the conjunction of the structural results already proved above, restated as a single determination claim; it carries no new proof burden. What it deliberately does *not* assert is that C is underivable or that "derivable ⇔ consequence": it states only that the listed Tier-1 quantities are functions of (C, T). The epistemic observation that C is *input* to the framework is separate, and is not part of the proposition.

Why the constitution/consequence split is a definition with a census, not a theorem. The biconditional "derivable ⇔ consequence-of-constitution" is not a theorem in either reading: under the natural definition of "consequence" (that which is derivable from C) it is tautological and contentless, and under any observer-independent definition of "consequence" it is false — the threshold δ_0 is an independently characterizable consequence of the coupling-running that is nonetheless held as input, on tractability grounds. The contentful objects are therefore (i) the *census* — the explicit assignment of each quantity to constitution or consequence, with the intractable-consequence exceptions (δ_0, the absolute mass scales) flagged — and (ii) the rest-frame-selection lemma of [Main §3.5], which explains the one structurally interesting case, the boost/rotation asymmetry, via the fact that the observation map selects a timelike direction but no spatial one. The honest summary is that the framework derives the tractable consequences of a constitution it must otherwise take as given, and that the boundary between the two is organized — not by a single formal property, but by this constitution/consequence distinction together with the rest-frame lemma that explains its one non-obvious instance.

Relation to the framework's hierarchy of structural realism. The Layer 0–Layer 3 classification refines the within-paper structure of which predictions are derivable from the cubic-lattice substratum and which are bijection-specific. This is a finer-grained classification than the framework's broader observation × gauge hierarchy (per [Structure §2]). In hierarchy terms, all SM predictions live at Level D × Levels G1–G2 — they are predictions of OI's specific universality-class representative on the emergent QFT. Within this single intersection, Layer 0–Layer 3 distinguishes which predictions are class-specific structural theorems (Layer 0 / Layer 1) vs. which depend on the specific representative within the class (Layer 2 / Layer 3). The hierarchy framing locates this paper's predictions within the framework's overall geography; the layered framing distinguishes the derivability status of each prediction within that location.

Rigor-level discipline. Closure claims in the framework are accompanied by explicit rigor levels: Level 1 (verified at the level of an explicit chain of equations, suitable for referee submission), Level 2 (verified up to standard moves of the field, where each step is "obvious to anyone in the field" without being written out), or Level 3 (sketch identifying the right approach, *not closure*). Multi-part theorems may have different rigor levels for different parts. The bridge theorem of §7.5 is an example: parts (a) and (c)-direct are Level 1 closed; parts (b)-γ_5 and the framework-essential γ_5-chirality identification are Level 1 closed; parts (b)-γ_0-derivation and (d) are Level 2; the full operator-theoretic interpretation involving taste structure is Level 3 but is not framework-essential. Honest annotation of rigor level is part of the framework's methodology.

Layer 3 — Solution-specific properties. Properties that depend on the particular bijection φ. Examples: absolute fermion mass scales (e.g., m_τ, m_t, m_s as input scales in the M-chain), basis-alignment unitaries, CP-violating phases (δ_{CKM} via empirical η, δ_{PMNS} identified with $b_{12} - b_{13}$ in §7.3), eigenvalues μ_c, μ_w of the coupling matrix M that set the gauge couplings.

The §7.6 summary table classifies each prediction by its layered status: items marked **S** are unconditional structural (Layer 0 or Layer 1, fully derived); items marked **L** are layered (combine multiple layers, typically Layer 0 or 1 + Layer 2(b) substrate-specific coefficient or Layer 3 empirical input). Items marked **R**, **P**, **M**, **E** denote retrodictions, phenomenological inputs, one-input mass/parameter chains, and explicit empirical inputs respectively, as detailed in §7.6. The C class (conditional structural pending derivation) is currently empty: Cond 2 — the only candidate for C in v1.9.5 — has been classified Layer 2(b) following operator counting, moving the dependent predictions to L; the §7.3 routing computation subsequently confirmed this classification by excluding the second-order derivation route.

These are analogous to the mass M in the Schwarzschild solution: GR derives the functional form of the metric but does not predict which M describes a particular black hole. The OI framework derives the SM's structure but does not predict which φ describes our universe.

The "mass hierarchy" — the five-order-of-magnitude spread of fermion masses — falls in the second category. It is not a gap in the framework; it is a property of the specific solution. The framework makes the correct structural prediction: three generations with *identical* gauge quantum numbers (from cubic symmetry) but *non-degenerate* masses (because the specific φ breaks the cubic symmetry through its Yukawa structure). The hierarchy problem proper — why the Higgs mass is so much lighter than the Planck scale — is dissolved: the Higgs mass is an eigenvalue of M (a property of φ), not a radiative correction within the emergent QFT (§4.11).

Remaining open problems. (i) *Lattice-continuum comparison* — the structural predictions are exact theorems at the lattice level and do not require a continuum limit. The lattice is the fundamental description ($\epsilon = 2l_p$ is physical, not a regulator), so there is no $\epsilon \to 0$ limit to take. The comparison to experiment — which is interpreted through continuum perturbation theory — introduces corrections of order $(E/M_{\mathrm{Pl}})^2 \sim 10^{-32}$ at LHC energies, far below any foreseeable experimental sensitivity. The Yang-Mills mass gap problem (proving rigorous existence of the continuum limit for Yang-Mills theory) is a famous open problem in mathematics, but it does not affect any prediction of the OI framework: the lattice theory is complete as stated, and the lattice is the fundamental description, not a regularization. The same $(E/M_{\mathrm{Pl}})^2$ suppression sets the framework's structural commitment on Lorentz-invariance violation: linear LIV is forbidden at the substrate level (the cubic-lattice dispersion relation admits no $\mathcal{O}(k^3)$ term), and the gauge-invariant quadratic coefficient evaluates to $\omega^2 = k^2[1-(4/45)(E/M_{\mathrm{Pl}})^2+...]$ — subluminal, with the specific 4/45 fixed by cubic coordination and the statistical-isotropy gauge average. Pre-registered as Class A in [Substratum, §6.4].

8.4 Neutrino masses from taste-breaking

The staggered fermion construction gives 4 tastes decomposing as $1(A_1)+3(T_1)$ under the cubic group O (§4.7). The singlet taste is identified as the Higgs sector — not a right-handed neutrino. Neutrinos therefore acquire mass only through the dimension-5 Weinberg operator $(LLHH)/\Lambda$, making them Majorana particles.

The mass matrix. The neutrino mass matrix in the generation basis $\{e_x, e_y, e_z\}$ (the T_1 representation) is $M_\nu = m_0 I_3 + \delta M$, where δM is the taste-breaking correction. Under the cubic group, the symmetric matrix δM decomposes as $\mathrm{Sym}^2(T_1) = A_1 \oplus E \oplus T_2$. The A_1 part shifts all masses equally (absorbed into m_0). The E part (2-dimensional) controls the mass splittings via two parameters (a, b). The T_2 part (3-dimensional) controls the mixing angles.

Structural predictions (Tier 2). Four predictions follow from the representation theory alone, independent of the specific φ:

(i) *Majorana neutrinos.* No gauge-singlet neutrino representation is available in the spectrum — the singlet taste is the Higgs. Neutrinoless double beta decay ($0\nu\beta\beta$) should be observed.

(ii) *Normal mass ordering.* The taste-breaking in T_1 generically gives the largest correction to the state most coupled to all three lattice directions — the [111] direction in taste space. This corresponds to $m_3 > m_2 > m_1$.

(iii) *Large PMNS mixing angles.* The charged lepton mass matrix (Dirac, general 3×3) and the neutrino mass matrix (Majorana, symmetric 3×3) have different symmetry structures under the cubic group. The structural mismatch produces large angles — unlike the quark sector, where both up-type and down-type matrices are general 3×3 with similar dominant structure, giving small CKM angles.

(iv) *Hierarchical spectrum.* The taste-breaking splittings are the same order as the common mass m_0. A quasi-degenerate spectrum ($m_1 \approx m_2 \approx m_3 \gg \Delta m$) requires fine cancellation between m_0 and the splittings, which is not generic. The framework naturally produces Σm_ν near the minimum allowed by the mass splittings.

Observational status. DESI DR2 combined with CMB data gives $\Sigma m_\nu < 0.064$ eV with preference for normal ordering and $m_{\text{lightest}} < 0.023$ eV — consistent with all four structural predictions. The minimum sum for normal ordering is $\Sigma m_\nu^{\min} \approx 0.059$ eV; the DESI+CMB bound is only 8% above this minimum, consistent with the hierarchical prediction. The observed PMNS angles ($\theta_{12} \approx 34°$, $\theta_{23} \approx 49°$, $\theta_{13} \approx 9°$) are all large compared to CKM angles, as predicted.

Solution-specific properties (Tier 3). The absolute mass scale (set by Λ_{seesaw}), the ratio $\Delta m^2_{32}/\Delta m^2_{21} \approx 33$ (set by the relative magnitudes of the E-component parameters a/b), and the CP-violating phase δ_{CP} (which corresponds to the unconstrained free parameter $b_{12} - b_{13}$ in the $O(\lambda^2)$ charged-lepton structure of §7.3) are properties of the particular φ. The PMNS *angles* themselves are structural (§7.3): TBM from cubic-group representation theory plus U-parity-graded corrections from the charged-lepton sector, with the Cabibbo angle λ as the structural scale.

8.5 Position relative to mainstream observer-essentiality programs

The Standard Model derivation chain developed in this paper sits within a broader framework whose foundational claim — that observation cannot be treated as an external operation on physics — has been independently arrived at by several mainstream programs since 2022. Chandrasekaran-Longo-Penington-

Witten [CLPW 2022, JHEP 02 (2023) 082] showed that incorporating an observer in QFT in a gravitational subregion promotes the von Neumann algebra of observables from type III to type II_∞. Maldacena's recent dS observer cancellation [arXiv:2412.14014] removes the unphysical phase of the de Sitter sphere partition function exactly when an observer-with-clock is properly incorporated. Harlow-Usatyuk-Zhao [JHEP 02 (2026) 108] and the AAIL construction [arXiv:2501.04305] show that the closed-universe Hilbert-space dimension is determined by the observer's degrees of freedom, with refined scaling $d_{\mathrm{Ob}} \cdot e^{2S_0}$. Slagle-Preskill [Phys. Rev. A 108, 012217 (2023)] demonstrated emergent quantum mechanics at the boundary of a classical lattice model, establishing the logical possibility of QM-from-classical at the cost of an exponentially long extra spatial dimension and explicit stochasticity. The present framework is the unique member of this convergence in which the observer-essentiality content is derived from a finite *deterministic* substratum — no extra spatial dimension, no fundamental stochasticity — and in which the resulting characterization theorem produces an explicit empirical record (the twenty-two structural predictions of §7) that the comparison programs do not currently match. The convergence positions the framework's foundational stance within mainstream observer-emergent physics; the SM derivation chain provides the specific empirical content that distinguishes it. The full positioning relative to this body of work is developed in [Main, §4.4].

8.6 Pattern vs. output: what transports across SM-reproducing classes

The §4 construction separates an exact geometric theorem from a conditional physical reading. The six simple-cubic signed links decompose uniquely as $(3, 2, 1)$ under the cubic group. **Under H-link single-copy + H-cust**, the same blocks act on the physical carrier and the condensate-stabilizer/background-independence route produces the stated gauge structure. Without those bridges the exact result is the link representation, not the physical Standard-Model gauge group.

The *commutant gauge-invariance pattern* — that the visible-sector gauge group is the commutant of a substratum-level symmetry under the trace-out — operates at the universality-class level (Level G4 $\times$ Level C of [Structure §2]). Per the four-feature audit in [Structure §14.4], this pattern transports to other SM-reproducing structural classes through different machinery: heterotic string compactifications realize it through E_8 broken by Calabi-Yau holonomy and bundle data; Type II / D-brane compactifications realize it through D-brane stack gauge groups with chiral matter from intersection or magnetization; F-theory realizes it through singularity structure of elliptic fibrations with hypercharge from flux configurations. The pattern is shared across all SM-reproducing universality-class members; the framework-specific machinery differs.

The *specific output* — the SM gauge group, matter multiplicities and downstream quantitative branch — is OI-specific and conditional on H-link/H-cust together with the matter-sector hypotheses recorded above. Other universality-class members may reach the same output through different structure; the cubic $(3, 2, 1)$ link decomposition is the exact OI-specific group-theoretic core.

The distinction matters for falsification. Failure of the exact six-link character decomposition would falsify the group-theoretic calculation itself. Failure of H-link or H-cust would instead falsify the proposed identification of that geometry with the physical gauge carrier while leaving the observer-isotropy theorem and the geometric $(3, 2, 1)$ result intact. Downstream numerical claims that consume $K = 6$ are therefore conditional tests of the single-copy H-link branch.

8.7 Substratum-level baryon-number conservation and the substratum-emergent operator distinction

The framework's matter sector (§4.7) places fermions on links with carrier $V_K \otimes V_K^*$, where the V_3 factor distinguishes quarks (with a V_3 component) from leptons (without). At the substratum level, baryon number B is essentially the count of V_3 factors weighted by $1/3$. This subsection shows that B is exactly conserved by the substratum dynamics, derives the consequences for the emergent EFT, and articulates the substratum-emergent operator distinction that organizes the framework's treatment of non-perturbative SM features.

Substratum-level B conservation. Two structural features of the substratum dynamics force exact B conservation:

(i) *Bilinear coupling structure.* The lattice wave equation couples each link's field $\phi(\mathbf{n})$ to its neighbor $\phi(\mathbf{n}+\hat{e}_j)$ via the bilinear $\bar{\phi}(\mathbf{n})M(\mathbf{n}, \hat{e}_j)\phi(\mathbf{n}+\hat{e}_j)$. Bilinear couplings can move excitations between sites and produce particle-antiparticle pairs (preserving B) but cannot change the *count* of fermions in each gauge-charge sector through unilateral creation or annihilation.

(ii) *Block-diagonal structure of M.* By Theorem 7, the coupling matrix M is block-diagonal in the $V_3 \oplus V_2 \oplus V_1$ decomposition: $M = \mathrm{diag}(\mu_c I_3, \mu_w I_2, \mu_y)$. Off-block matrix elements that would mix V_3 with V_1 are absent; the dynamics therefore never converts a quark-like state (V_3 component) into a lepton-like state (V_1 component) directly.

These two features jointly imply that the substratum dynamics preserves the count of V_3-fermion factors modulo pair production, which is the substratum-level statement that B is exactly conserved *in the gauge/kinetic sector.* Baryon violation through non-gauge operators — the condensate and Yukawa sectors, and higher-dimension operators — is a separate accounting item, not excluded here.

The substratum-emergent operator distinction. The Standard Model has $B + L$ violation through non-perturbative sphaleron processes, finite at high temperatures and exponentially suppressed at zero temperature [9]. This appears to conflict with substratum-level exact B conservation. The resolution lies in recognizing that B at the substratum level and B at the emergent EFT level are distinct operators related by the trace-out:

(a) *Substratum-level* B is the count of V_3-fermion factors on the cubic lattice. This is exactly conserved by the bilinear, block-diagonal substratum dynamics, as shown above.

(b) *Emergent-level* B is the standard SM operator on coarse-grained fermions, which is acted on by sphaleron processes in the emergent quantum field theory. Sphalerons are emergent classical solutions of the electroweak Lagrangian; they exist as non-perturbative features of the emergent EFT but have no direct substratum-level analog.

The trace-out from substratum to emergent EFT introduces non-perturbative quantum-mechanical phenomena (instantons, sphalerons, θ-vacuum structure) that the deterministic substratum dynamics does not have. The induced gauge action $S_{\text{eff}}[U] = -N_f \ln \det(D^\dagger D)$ inherits topological structure from the fermion determinant via the Atiyah-Singer index theorem on the lattice; configurations with non-trivial Chern-Simons number produce the standard sphaleron and instanton physics in the emergent EFT.

Substratum-level B conservation and emergent-level $B + L$ violation therefore coexist consistently: the trace-out maps the conserved substratum operator onto an emergent operator that is conserved at the bilinear renormalizable level but violated by emergent non-perturbative effects.

Generalization to other non-perturbative SM features. The substratum-emergent operator distinction generalizes to a systematic mapping. The substratum has exact discrete symmetries (T-invariance, gauge-invariance, block-diagonal coupling, bipartite checkerboard structure) at the bilinear level. The trace-out introduces emergent non-perturbative effects (sphalerons, instantons, chiral anomalies, θ-vacuum structure) through standard mechanisms operating in the emergent QFT. The mapping has the following pattern across the framework:

Substratum-level structural precursor	Emergent-level non-perturbative feature	Trace-out role
T-invariance of the wave equation (§5)	$\theta \in \{0, \pi\}$ in QCD; $\bar{\theta}$ not fixed	Narrows parameter value; H-top, H-det open
Topology of induced SU(2) gauge action (via fermion-determinant index)	Sphalerons, $B + L$ violation	Permits standard mechanism
Bipartite checkerboard (Theorem 12)	Chiral fermions, anomaly structure	Produces emergent structure
Block-diagonal M (Theorem 7)	Substratum B conservation	Inherits to emergent B at bilinear level

Substratum-level structural precursor	Emergent-level non-perturbative feature	Trace-out role
Cubic-commutant gauge structure + matter content (§4.7.1.2)	Anomaly cancellation	Forced by structural derivation
Chiral structure of substratum + commutant matter content (Theorems 13, 15)	't Hooft anomaly matching	Structurally automatic
Fermion determinant + induced gauge action (§6.1)	Anomalous dimensions, non-perturbative renormalization	Determined by fermion-determinant structure (matches standard QFT at one- and two-loop; higher-order open)
Induced gauge action with running coupling (§6.1)	Conformal/trace anomaly $T^\mu_\mu \propto \beta(g) F^{\mu\nu} F_{\mu\nu}$	Standard form, emerges automatically from induced gauge structure
Topology of induced SU(3) gauge action (instantons)	't Hooft determinant interaction; η' mass via Witten-Veneziano	Standard mechanism (open derivation in framework)

The framework's treatment of the strong CP problem (§5) is not yet an instance of this pattern — substratum T-invariance narrows θ to $\{0, \pi\}$ and gives reciprocal transition counts, but does not fix $\bar{\theta}$; it would become one if H-top and H-det were supplied. The pattern is: substratum-level symmetry forces a value of an emergent parameter that would otherwise be free. The substratum-level B conservation derived above is another instance: substratum-level structural feature provides exact conservation that maps to bilinear-renormalizable conservation at the emergent level, with non-perturbative violations through emergent processes only.

Empirical implications. The composite picture supports the framework's existing predictions and clarifies their structural origins:

(i) *Proton decay rate.* With substratum-level B conservation and emergent-level B violation only through Planck-suppressed dimension-6 operators (since OI has no GUT scale per §6.7), proton lifetime is $\tau_p \sim M_{\rm Pl}^4/m_p^5 \sim 10^{45}$–$10^{46}$ years.

(ii) *Baryogenesis mechanism is open.* Sphaleron-mediated $B + L$ violation operates normally in the emergent EFT. But the framework's spectrum contains **no right-handed neutrino** (§4.9, §8.4: the singlet taste is the Higgs, and anomaly cancellation forces exactly five representations per generation), so standard type-I thermal leptogenesis — which requires heavy right-handed Majorana neutrinos decaying out of equilibrium — does not directly apply. The lepton asymmetry must instead arise from whatever ultraviolet physics generates the dimension-5 Weinberg operator (§8.4), which the framework does not currently fix. The full quantitative baryogenesis prediction — connecting that scale to the observed baryon asymmetry $\eta = n_B/n_\gamma \approx 6 \times 10^{-10}$ — is open work.

(iii) *Cleanness of anomaly structure.* The substratum-derived matter content (§4.7.1.2 + Theorem 15) automatically satisfies all SM anomaly conditions; anomaly cancellation is a derived consequence rather than an imposed consistency condition.

(iv) *Strong CP, sphalerons, instantons coexist.* The framework permits all standard non-perturbative SM phenomena while retaining exact substratum-level discrete symmetries; the trace-out is what allows this to be consistent.

(v) *'t Hooft anomaly matching.* The framework's chiral structure derivation (Theorem 13) and anomaly cancellation derivation (Theorem 15) together imply 't Hooft anomaly matching automatically: the chiral anomalies of the emergent SM fermions match those required by the substratum's bipartite-lattice plus commutant structure. This is not an added constraint but a structurally automatic feature, consistent with the framework's broader pattern of deriving anomaly cancellation rather than imposing it.

(vi) *Open work in the non-perturbative sector.* Three features in the extended table are flagged as open: higher-loop anomalous dimensions matching to all orders (the framework's matching at M_Z works to few-percent at one and two loop; higher-order non-perturbative effects could differ from standard QFT); the trace anomaly's explicit derivation from induced gauge action (likely automatic but not verified); the η' mass via 't Hooft determinant interaction in OI's induced theory (standard mechanism applies but the explicit derivation has not been worked through in the framework).

The substratum-level discrete symmetries (T-invariance, gauge invariance, block-diagonal coupling, bipartite checkerboard structure) and the emergent-level non-perturbative effects (sphalerons, instantons, chiral anomalies, θ-vacuum structure) coexist consistently under this mapping. The substratum operators are exactly conserved at the bilinear level; the emergent operators inherit the conservation modulo non-perturbative violations that follow from standard QFT mechanisms operating in the emergent EFT.

8.8 Composite-Higgs chiral protection and the unified naturalness pattern

The framework's treatment of three open problems of the Standard Model — strong CP, the hierarchy problem, the cosmological constant — is organized by a common structural pattern: a substratum-level structural feature forces, or narrows, an emergent parameter value which would otherwise be naturally at the cutoff. The hierarchy problem and the cosmological constant are the two instances; strong CP is the pattern's missing instance, since §5 narrows θ without fixing $\bar{\theta}$. This subsection articulates the pattern and develops its hierarchy-problem instance (composite-Higgs chiral protection) alongside the treatment of the cosmological constant ([GR §6]).

The unified naturalness pattern. Standard quantum field theory has three classical naturalness problems where dimensionless parameters of the SM should naturally be $O(1)$ at the cutoff but are observed to be many orders of magnitude smaller: (i) the strong CP angle $\bar{\theta}$ is observed to be $< 10^{-10}$ but the natural cutoff value is $O(1)$; (ii) the Higgs mass-squared ratio m_H^2/M_{Pl}^2 is observed to be $\sim 10^{-34}$ but the natural cutoff value is $O(1)$; (iii) the cosmological constant ratio Λ/M_{Pl}^4 is observed to be $\sim 10^{-122}$ but the natural cutoff value is $O(1)$. In standard SM, each requires either fine-tuning or anthropic selection or new physics (axion, supersymmetry, dark energy mechanism) to resolve.

OI's framework dissolves all three through the same mechanism: a substratum-level structural feature (discrete symmetry, conservation law, or geometric constraint) forces the emergent parameter to a specific value that escapes the naturalness puzzle. The mechanism is one instance per problem:

(i) *Strong CP* (§5, Theorems 17–21): substratum-level T-invariance of the wave equation narrows θ to $\{0, \pi\}$ and gives reciprocal visible transition counts at every scale, but reciprocity does not make the emergent Hamiltonian T-invariant, so $\arg\det(Y_u Y_d)$ and hence $\bar{\theta}$ are not fixed. The structural feature is T-invariance; the parameter it would protect is $\bar{\theta}$; the instance is open under H-top and H-det, and is the weakest of the three.

(ii) *Hierarchy problem* (developed below): the staggered-taste chiral symmetry pairing $\mu = +1$ singlet with $\mu = -1$ singlet provides composite-Higgs-style chiral protection. The structural feature is the singlet-taste chiral symmetry; the protected emergent parameter is m_H/f where f is the compositeness scale.

(iii) *Cosmological constant* ([GR §6]): the 10^{122} discrepancy between Λ_{obs} and M_{Pl}^4 is dissolved as the trace-out's information compression ratio between the substratum's full information content and the observer-accessible coarse-grained projection. The structural feature is the cosmological-horizon partition; the protected emergent parameter is Λ/M_{Pl}^4.

The three are not three separate fine-tuning miracles but three instances of the same pattern: *substratum-level structural symmetries dissolve naturalness problems by forcing emergent parameter values that would otherwise be free.* This is a substantive structural feature of the framework that has not been articulated as a unified theme.

Hierarchy-problem instance: composite-Higgs chiral protection. The framework predicts the Higgs as a composite scalar in the singlet staggered taste ([Theorem 11]). Standard composite-Higgs scenarios in beyond-the-SM phenomenology (Kaplan-Georgi 1984; Agashe-Contino-Pomarol 2005) dissolve the hierarchy problem by realizing the Higgs as a pseudo-Nambu-Goldstone boson of an approximate global symmetry, with mass protected by Goldstone's theorem against cutoff-scale corrections. OI's framework structurally implements an analog of this mechanism.

The relevant symmetry is the *singlet-taste chiral symmetry*. The framework's staggered-fermion construction (§4.8) produces four staggered tastes labeled by the lattice momentum eigenvalues $\mu \in \{+1, +1/3, -1/3, -1\}$. The chirality structure (Theorem 13) pairs the $\mu = +1$ singlet taste with the $\mu = -1$ singlet taste through the bipartite checkerboard. At the substratum level, the symmetry pairing these tastes is *exact* — a discrete chiral symmetry of the wave equation acting on the singlet-taste sector. In the emergent EFT, this symmetry is *spontaneously broken* by the Higgs vev, which provides asymmetric mass contributions to the $\mu = +1$ and $\mu = -1$ singlet tastes.

The Higgs is the order parameter of this spontaneous breaking. Concretely, the Higgs lives in the E projection of the singlet taste ([Theorem 11]), which is a 2-dimensional complex irrep of the cubic group. Under the chiral symmetry, the E projection transforms in a specific way that makes the Higgs's vev the natural order parameter for breaking the chiral pairing.

By Goldstone's theorem, the Higgs mass is bounded by the *explicit* breaking of the chiral symmetry — that is, by gauge couplings and Yukawa couplings squared times the chiral-breaking scale, not by the cutoff M_{Pl}. The hierarchy problem is dissolved at the level of *cutoff sensitivity*: the Higgs mass-squared does not receive $O(M_{\mathrm{Pl}}^2)$ corrections, only $O(g^2 f^2/16\pi^2)$ corrections where g is a gauge or Yukawa coupling and f is the chiral-symmetry-breaking scale.

Scope and caveats. Chiral protection bounds the Higgs mass by explicit breaking but does not automatically explain why the Higgs vev v sits 17 orders of magnitude below the compositeness scale f. In OI, f is set by the substratum lattice scale and is effectively M_{Pl}, so the ratio $v/f \sim 10^{-17}$ remains. Standard composite-Higgs scenarios put f at $\sim$ TeV to resolve this; OI does not have an analogous intermediate scale (per [Structure §13.3] Q10). The full hierarchy resolution therefore consists of two parts: (a) chiral protection against cutoff corrections (developed here), and (b) the structural origin of $v \ll f$ (open).

Honest characterization: the framework structurally implements (a). For (b), the framework currently treats v as solution-specific (Tier 3) input, with the small value $v \sim 246$ GeV being a feature of the specific bijection φ rather than a derived structural constant. This is consistent with the framework's Tier 3 honesty about parameters not forced by structure ([Structure §13.3] Q10's negative resolution). Possible structural origins for (b) — including trace-out-induced fractional-power suppression connecting the Higgs hierarchy to the cosmological-constant compression — are speculative directions for future investigation; the framework does not currently commit to them.

Linkage to the seesaw scale. The same singlet-taste chiral symmetry breaking that gives the Higgs its mass also breaks the lepton-number-like pairing in the singlet sector. The emergent Weinberg operator $(LH)(LH)/\Lambda_{\text{seesaw}}$ that generates Majorana neutrino masses (§8.4) has its suppression scale Λ_{seesaw} structurally linked to the same chiral-breaking scale. The seesaw hierarchy ($\Lambda_{\text{seesaw}}/M_{\text{Pl}} \sim 10^{-3}$) and the Higgs hierarchy ($v/M_{\text{Pl}} \sim 10^{-17}$) are therefore not independent — both should be derivable from the same trace-out-induced mechanism, with the dimension-dependent suppression giving different effective scales for different operators (dimension-3 Majorana, dimension-4 Higgs, dimension-5 Weinberg). The quantitative relationship is the generic Weinberg-operator estimate $\Lambda_{\text{seesaw}} \sim v^2/m_\nu \sim 10^{16}$ GeV — *not* a type-I seesaw, since the framework's spectrum contains no right-handed neutrino to integrate out (§4.9, §8.4); Λ_{seesaw} is the suppression scale of the dimension-5 Weinberg operator, whose ultraviolet completion the framework does not fix. Whether this scale is structurally linked to the Higgs sector's chiral-breaking scale or is a solution-specific (Tier 3) input is not settled: [Structure §13.3] Q10 finds no derived intermediate scale, so it is currently treated as a Tier-3 input (§8.4).

Empirical signatures. Standard composite-Higgs scenarios predict modified Higgs couplings to SM particles by factors of $\sqrt{1 - v^2/f^2}$ where f is the compositeness scale, with deviations from SM scaling as v^2/f^2. In OI, with $f \sim M_{\text{Pl}}$, $v^2/f^2 \sim 10^{-34}$ — far below any conceivable experimental sensitivity. The framework's prediction is therefore no observable deviations from SM Higgs couplings, consistent with current LHC data. This is not a sharp empirical discriminator from standard SM phenomenology, which also predicts SM-like Higgs couplings within current sensitivities.

Status. The composite-Higgs chiral protection mechanism is *structurally identified* in OI's framework through the singlet-taste chiral symmetry breaking. A fully rigorous derivation — explicit identification of the broken global symmetry's generators, identification of the Higgs as a specific Goldstone direction in the coset, computation of the explicit-breaking corrections to the Higgs mass — is open work. The structural pattern is identified; the theorem-level derivation is deferred.

Derived components and Tier 3 inputs. The framework derives: the singlet-taste chiral symmetry at the substratum level (Theorem 13); the Higgs as the E-projection order parameter of the singlet taste (Theorem 11); spontaneous breaking of the chiral pairing by the Higgs vev in the emergent EFT; the Goldstone bound $\delta m_H^2 \lesssim g^2 f^2/(16\pi^2)$ per coupling; cutoff insensitivity of m_H^2, with no $O(M_{\text{Pl}}^2)$ corrections; and the compositeness scale $f \sim M_{\text{Pl}}$ (no intermediate scale exists structurally, per [Structure §13.3]). The framework does not derive the Higgs vev $v \sim 246$ GeV itself, which is Tier 3 (solution-specific to the bijection φ). The Higgs mass value $m_H \sim 125$ GeV given v follows from $\lambda(M_{\text{Pl}}) = 0$ and RG running (§7.4).

The derived/open boundary is therefore: *the framework derives why m_H is not at the cutoff* — chiral protection ensures cutoff-insensitivity — but *does not derive why v is at the electroweak scale rather than at the compositeness scale.* The first is the substantive content of the hierarchy problem in standard SM phenomenology: why do quantum corrections not push m_H to the cutoff despite the absence of a protecting symmetry? The framework provides a structural answer. The second is a residual question about the Higgs vev's specific magnitude, which the framework treats as Tier 3 alongside other solution-specific parameters such as CP phases and generation-specific Yukawa magnitudes.

Quantitative content of the partial resolution. Standard SM without chiral protection predicts $\delta m_H^2 \sim M_{\mathrm{Pl}}^2/(16\pi^2) \sim 10^{37}$ GeV^2 from quadratically-divergent loop corrections, requiring fine-tuning of the bare Higgs mass to ~ 34 decimal places to obtain $m_H^2 \sim (125\,\mathrm{GeV})^2$. The framework's chiral protection mechanism predicts $\delta m_H^2 \sim g^2 f^2/(16\pi^2)$ from explicit-breaking corrections only, with no quadratic-divergence dependence on the cutoff. The two predictions differ by a factor of $\sim (f/v)^2 g^{-2} \sim 10^{34}$ in the size of the required fine-tuning. The framework's claim is that m_H does not require 34 decimal places of fine-tuning to be at the electroweak scale; it requires only as much fine-tuning as the Higgs vev's specific value itself. The factor of 10^{34} is the substantive empirical content of "partial resolution" versus "no resolution," even when v is taken as solution-specific input.

Status of v as solution-specific. Some parameters must be solution-specific in any finite-bijection framework — the bijection φ has freedom that is not fixed by structural constraints, and the values realized in nature are features of the specific φ rather than of the structural class. The framework identifies which parameters are structural and which are solution-specific. The Higgs vev $v \sim 246$ GeV sits in the solution-specific category alongside CP phases, generation-specific Yukawa magnitudes, and other parameters whose values are not forced by the derivation chain in §§3–6.

Custodial symmetry is not required at $f \sim M_{\mathrm{Pl}}$. Conventional composite-Higgs models at $f \sim$ TeV require an engineered custodial $SU(2)_C$ symmetry to suppress T-parameter / ρ-parameter deviations that scale as $v^2/f^2 \sim 10^{-2}$. At $f \sim M_{\mathrm{Pl}}$, $v^2/f^2 \sim 10^{-34}$ and all such deviations are automatically negligible regardless of whether custodial symmetry is present in the framework's structure. The framework therefore predicts no observable composite-Higgs precision-electroweak deviations from SM, consistent with current LEP and LHC data to the 0.1% level, without needing to engineer custodial structure.

Empirical signatures and discriminating from conventional composite Higgs. Standard composite-Higgs scenarios at $f \sim$ TeV predict Higgs-coupling deviations from SM scaling as $v^2/f^2 \sim 10^{-2}$ — potentially detectable at the HL-LHC. The framework's $f \sim M_{\mathrm{Pl}}$ predicts deviations $v^2/f^2 \sim 10^{-34}$, structurally identical to SM at all practical experimental precision. Future Higgs-trilinear coupling measurements at HL-LHC (15% precision, [41]) and FCC-hh (4–8% precision, [41]) will progressively constrain conventional composite-Higgs models. If no deviations are observed up to FCC-hh sensitivity, the framework's high-f stance is the surviving regime; the conventional low-f composite-Higgs hypothesis would be progressively disfavored. The framework's prediction for the Higgs trilinear coupling $\kappa_3 \equiv \lambda_3/\lambda_{3,\text{SM}} = 1 - O(10^{-34})$ is the same as SM to all relevant precision, and this prediction will be tested experimentally over the next 20 years.

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9. Conclusion

This paper has shown that the Standard Model's gauge group, matter content, and discrete symmetries are determined by a finite bijection on a $d = 3$ cubic lattice — itself empirically anchored (the dimension fixed by the §3.2 filters and the cubic Bravais structure forced by the observed gauge group, §4.6; classified as Tier-2 input in §8.3, not derived) — through the lattice wave equation and a chain of theorems running from center independence to anomaly cancellation. The chain is proved step-by-step at the structural level, with the flagged inputs (H-χ'; H-spin'; light-spectrum selection) entering where marked. Where it makes contact with measured numbers, the agreement is striking: the Cabibbo angle from a single Brillouin-zone distance to 0.04%, the Koide angle from the cubic-group quadratic Casimir to 0.02%, the down-to-strange mass ratio $m_d/m_s = 1/(2\pi^2)$ at 0.56σ, all three PMNS angles within 1.1σ — with $\sin^2\theta_{12}$ matching the November-2025 JUNO measurement at 0.14σ (a retrodiction) — six fermion masses from a single empirical input to better than 1%, and the SM gauge couplings at M_Z to 0.1% and 0.3% (the SU(2) and SU(3) values being retrodictions matched through the threshold parameters, not first-principles predictions; §6.3). For strong CP the paper shows that substratum T-invariance gives $\theta \in \{0, \pi\}$ and reciprocal transition counts but not a T-invariant emergent Hamiltonian, so $\bar{\theta}$ is not fixed by the construction; the problem is open under two named conditions, H-top and H-det (§5).

The framework is sharply falsifiable by experiments now in progress or under construction. JUNO's first results (November 2025) are matched at 0.14σ, and the subsequent post-JUNO global fit at 0.07σ, by the parameter-free value $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$ — a retrodiction; sub-percent precision over the 30-year design lifetime provides the pre-registered forward test at the level required to distinguish it from competing tribimaximal variants. The n2EDM experiment at PSI is targeting neutron EDM sensitivity at $10^{-27}\, e \cdot \text{cm}$; the framework predicts exactly zero. Neutrinoless double beta decay searches will test the Majorana-neutrino prediction. The continuing absence of any new gauge bosons, fundamental scalars, or supersymmetric partners at the LHC and its successors corroborates structural predictions of the framework, which forbids all such states.

The cosmological application of the same framework — including the derivation of $\hbar$ from the cosmological horizon and the Bekenstein-Hawking area law including the 1/4 coefficient (both conditional on the horizon conditions recorded there), the dissolution of the cosmological constant problem at the descriptive level, and dark energy in running-vacuum form (the functional form the local expansion shared by every smooth model; the content the Type II channel, with magnitude not derived at theorem level but consistent with the Bertini et al. 2025 direct DESI-DR2-era fit at 2σ) — is developed in [GR]. The substratum-level reconstruction theorem, the substratum gauge group, and the synthesis claim that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same (S, φ) are developed in [Substratum]. Together, this four-paper synthesis ([Main], [SM], [GR], [Substratum]) makes a single claim: the Standard Model, general relativity, and the arrow of time are three projections of one finite deterministic construction — the visible-sector projection developed here, the boundary-level thermodynamic limit developed in [GR], and the horizon-clock-plus-observer-selection projection developed in [Substratum §5.4]. The fifth core paper [Structure] articulates the framework's two-dimensional hierarchical structural realism and develops its relationship to other unification programs through universality classes of embedded observers.

Appendix A: The Trace-Out as a Jordan-Chevalley Projection

This appendix proves that the OI trace-out extracts the semisimple part of the evolution matrix's Jordan-Chevalley decomposition, erasing the nilpotent monodromy. Results are stated for the scalar wave equation on a ring of L sites over $\mathbb{F}_q$; the multi-component extension incorporates the gauge group.

Setup

The wave equation $x(n, t+1) = x(n-1, t) + x(n+1, t) - x(n, t-1) \pmod{q}$ has phase space $(\mathbb{Z}/q\mathbb{Z})^{2L}$ and evolution matrix $F = \left(\begin{smallmatrix} 0 & I \\ -I & A \end{smallmatrix}\right)$ where A is the circulant adjacency matrix.

Theorem A.1 (Period formula)

For q prime with $\gcd(L, q) = 1$*:* $\mathrm{ord}(F \bmod q) = qL$ *if q is odd, and L if q = 2.*

Proof. For each Fourier mode k, the 2×2 block B_k has characteristic polynomial $t^2 - \lambda_k t + 1$. At parabolic modes ($\lambda_k = \pm 2$), the Jordan form gives $B_k^n = \alpha^n \left(\begin{smallmatrix} 1 & n\alpha^{-1} \\ 0 & 1 \end{smallmatrix}\right)$, contributing order q (or 2q). The diagonalizable eigenvalues contribute orders dividing L. Therefore $\mathrm{ord}(F) = qL$. For q = 2: the nilpotent part is automatically killed. □

Theorem A.2 (Jordan-Chevalley decomposition)

Define $N = (F^L - I)/L \bmod q$*. Then: (i) N is nilpotent with* $N^2 = 0$ *and rank(N) = 2. (ii)* $F_u = I + N$ *is unipotent with* $F_u^q = I$*. (iii)* $F_{\mathrm{ss}} = F \cdot (I - N)$

is semisimple with $F_{\mathrm{ss}}^L = I$. *(iv)* $F = F_{\mathrm{ss}} \cdot F_u$ *and* $[F_{\mathrm{ss}}, N] = 0$.

Proof. F^L has the form $I + LN'$ where N' arises from the Jordan blocks: $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)^L = \left(\begin{smallmatrix}1&L\\0&1\end{smallmatrix}\right)$ contributes a rank-1 nilpotent, and similarly for $\alpha = -1$ (since $(-1)^L = 1$ for even L). So $N = N'$ has $N^2 = 0$ (each block is rank-1 nilpotent on a 2D subspace) and $\mathrm{rank}(N) = 2$ (two parabolic modes). Since $N^2 = 0$: $(I+N)^{-1} = I - N$, giving $F_{\mathrm{ss}} = F(I-N)$. Then $F_{\mathrm{ss}}^L = F^L(I-N)^L = (I+LN)(I-LN+...) = I \bmod q$ since terms involving LN cancel modulo q (using $N^2 = 0$). Commutativity follows from N being supported on the parabolic eigenspaces, which are F-invariant. ☐

Verified computationally for L = 4, 6, 8, 10, 12 and q = 3, 5, 7, 11, 13.

Theorem A.3 (q-independence of the Weil-Deligne conductor)

The Weil-Deligne conductor $\mathfrak{f}_{\mathrm{WD}} = \mathfrak{f}_{\mathrm{ss}}(L) + \mathrm{rank}(N) = \mathfrak{f}_{\mathrm{ss}}(L) + 2$ *is q-independent when* $\gcd(L, q) = 1$, *where* $\mathfrak{f}_{\mathrm{ss}}(L) = \sum_\alpha (\mathrm{ord}(\alpha) - 1)$ *is computed from the eigenvalue orders of* F_{ss}, *all dividing L.*

Proof. F_{ss} has order L; its eigenvalues are L-th roots of unity. For $\gcd(L, q) = 1$, the L-th roots in $\bar{\mathbb{F}}_q$ are isomorphic to those in $\mathbb{C}$, so their orders match. ☐

L	$\mathfrak{f}_{\mathrm{ss}}(L)$	$\mathfrak{f}_{\mathrm{WD}}$	$\mathrm{NM}^2 = 3L/4$
4	14	16	3.00
6	30	32	4.50
8	70	72	6.00
10	106	108	7.50
12	130	132	9.00

Both $\mathfrak{f}_{\mathrm{ss}}$ and NM^2 are q-independent encodings of the same semisimple eigenvalue data — one via multiplicative orders (integers), one via fourth moments of magnitudes (reals).

Theorem A.4 (Additive decomposition over gauge irreps)

For the K-component wave equation with coupling matrix $M = \mathrm{diag}(\mu_1 I_{n_1}, \dots, \mu_r I_{n_r})$*:*

$$\mathfrak{f}_{\mathrm{WD}}(M) = \sum_{i=1}^{r} n_i \cdot \mathfrak{f}_{\mathrm{WD}}(\mu_i)$$

In particular, for K = 6 with $M = \mathrm{diag}(\mu_c I_3, \mu_w I_2, 1)$*:* $\mathfrak{f}_{\mathrm{WD}} = 3\mathfrak{f}_{\mathrm{WD}}(\mu_c) + 2\mathfrak{f}_{\mathrm{WD}}(\mu_w) + \mathfrak{f}_{\mathrm{WD}}(1)$, *with the same multiplicities* $(3, 2, 1)$ *that determine* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

Proof. The multi-component evolution matrix is block-diagonal: the a-th component has its own $2L \times 2L$ block F_{μ_a} with eigenvalues depending only on

μ_a. Both F_{ss} and N inherit the block-diagonal structure, so $\mathfrak{f}_{\mathrm{ss}}$ and $\mathrm{rank}(N)$ decompose additively. □

Verified for all $(\mu_c, \mu_w) \in \{1, \dots, q-1\}^2$ at q = 3, 5, 7 with L = 4. Every case matches exactly.

The projection

The OI trace-out (marginalization over the hidden sector) produces the reduced observable description — universally admitting the quantum representation — which depends on the coupling eigenvalues μ_k via the NM formula $\mathrm{NM}^2 = 3\langle\mu^4\rangle$ (a consequence of the stochastic-quantum correspondence applied to the wave equation's Fourier decomposition). These μ_k are properties of F_{ss} — the semisimple part of the dynamics. The nilpotent monodromy N contributes nothing to the emergent description: it affects only the off-diagonal Jordan block entries, which are erased by the coarse-graining over the hidden sector.

The trace-out therefore performs the Jordan-Chevalley projection $(F_{\mathrm{ss}}, N) \mapsto F_{\mathrm{ss}} \mapsto \{\mu_k\} \mapsto \mathrm{NM}^2$: it extracts the semisimple part, encodes it via magnitudes rather than orders, and organizes it by the representation theory of the partition. The nilpotent monodromy — genuine mathematical structure present in the full dynamics — is invisible to the embedded observer. This is the precise sense in which physics is the semisimple shadow of mathematics: the observer sees only the diagonalizable spectral data, projected by the trace-out and organized by the gauge group's representation structure.

Code and Data Availability

The lattice Monte Carlo computations reported in this paper were performed using a custom C implementation. Specifically:

- The scalar density renormalization $Z_S(m)$ in §7.5 was computed on SU(3) gauge backgrounds at $\beta_3 = 11.1$, scanning 30 bare masses from $m = 0.005$ to 0.50. Volumes $L = 16$ (30 configurations) and $L = 32$ (50 configurations) were the primary ensembles, with $L = 64$ used for convergence checks. The reported value $Z_S(0.122) = 1.813 \pm 0.001$ is the cubic interpolant of the $L = 32$ measurements. The chiral condensate $\Sigma \approx 0.20$ is the linear extrapolation in the volume-converged region ($mL \gtrsim 3$).

The source code under `papers/oi_lattice_code/`, together with the run drivers and analysis scripts, is archived on Zenodo with concept DOI 10.5281/zenodo.19060318, which resolves to the latest version; specific per-release DOIs are minted at release time. Paths are relative to the repository root, which is also the root of the archived snapshot. The code is also available on GitHub at https://github.com/amaybaum/incompleteness and is released under the MIT License. Gauge configurations at the OI coupling $\beta_3 = 11.1$

and the raw $Z_S(m)$ measurement outputs underlying the §7.5 prediction are available from the author on request.

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